



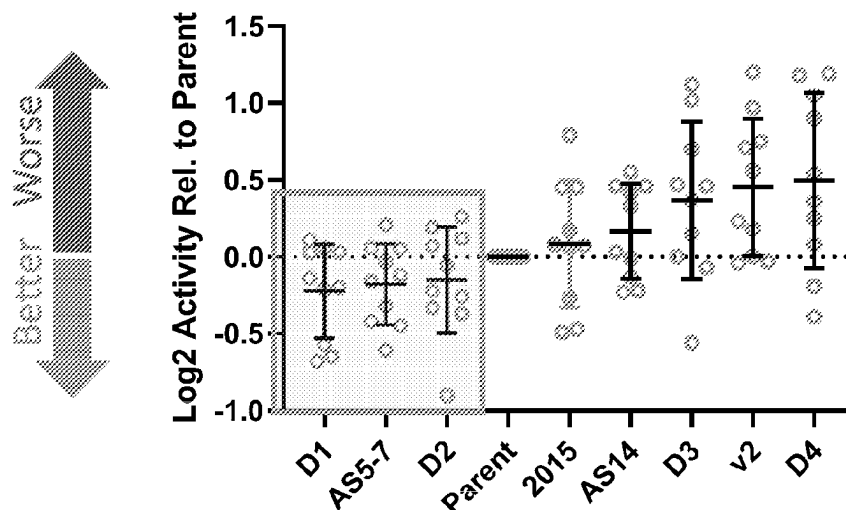
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(19) **United States**(12) **Patent Application Publication**
SCHLEGEL et al.(10) **Pub. No.: US 2025/0283071 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **MODIFIED DOUBLE STRANDED
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plication No. 63/153,983, filed on Feb. 26, 2021,
provisional application No. 63/156,476, filed on Mar.
4, 2021, provisional application No. 63/161,313, filed
on Mar. 15, 2021, provisional application No. 63/164,
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2310/3515 (2013.01)(57) **ABSTRACT**

One aspect of the present invention relates to double-stranded RNA (dsRNA) agent capable of inhibiting the expression of a target gene. Other aspects of the invention relate to pharmaceutical compositions comprising these dsRNA molecules suitable for therapeutic use, and methods of inhibiting the expression of a target gene by administering these dsRNA molecules, e.g., for the treatment of various disease conditions.

Specification includes a Sequence Listing.**Transfection (10 nM)**

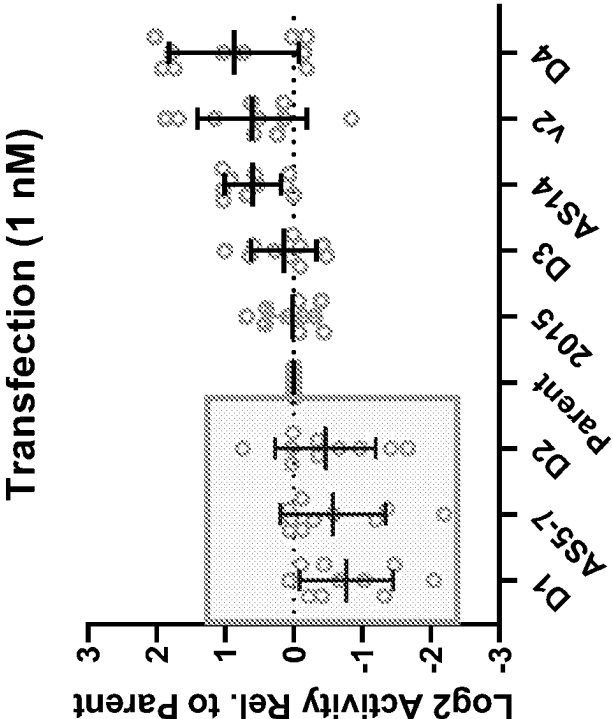


FIG. 1A

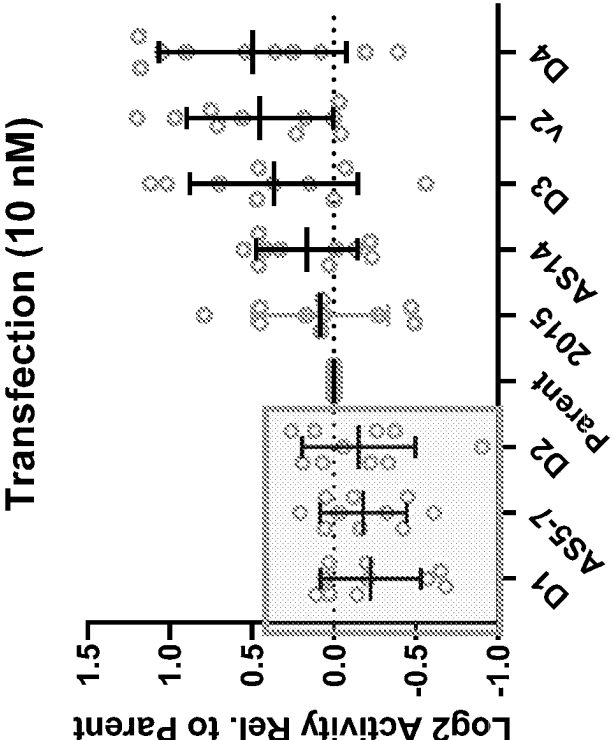


FIG. 1B

← Better
Worse →

FIG. 2A

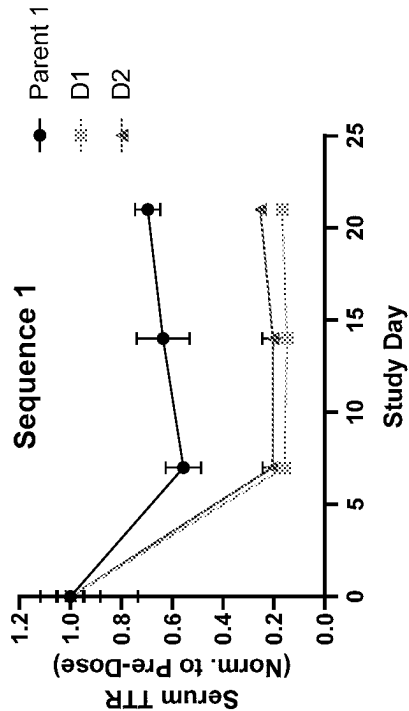


FIG. 2B

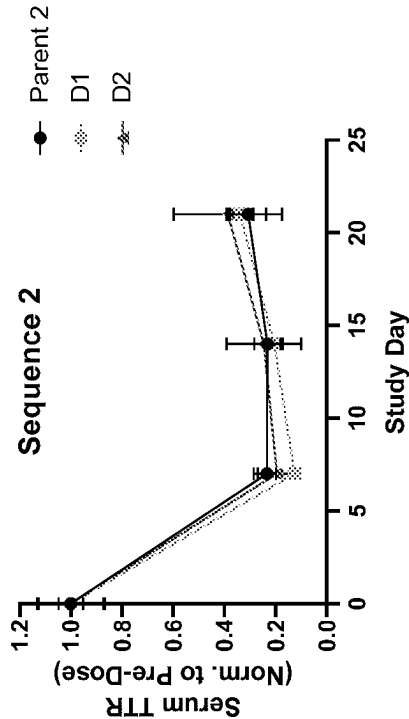


FIG. 2C

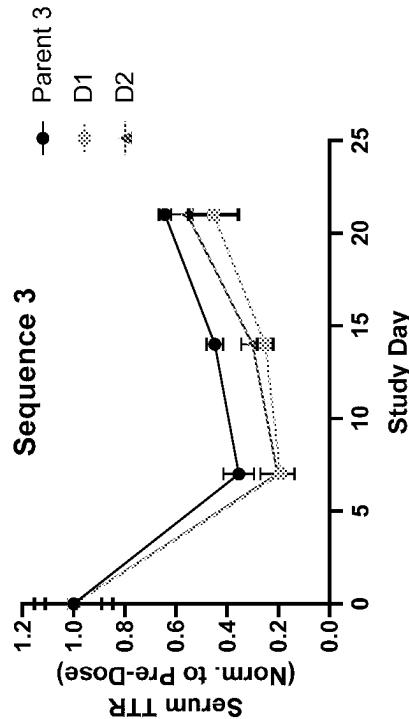
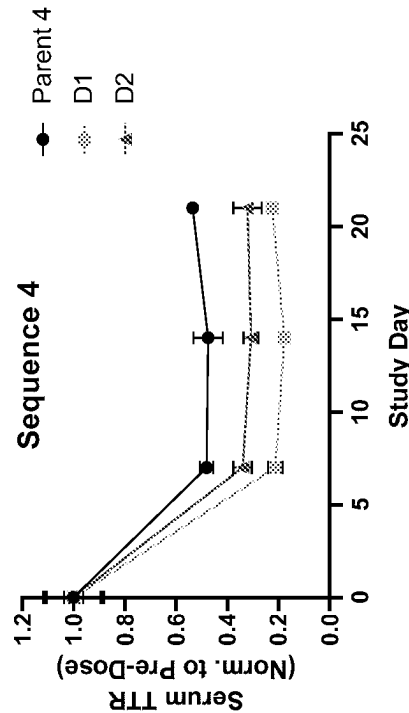


FIG. 2D



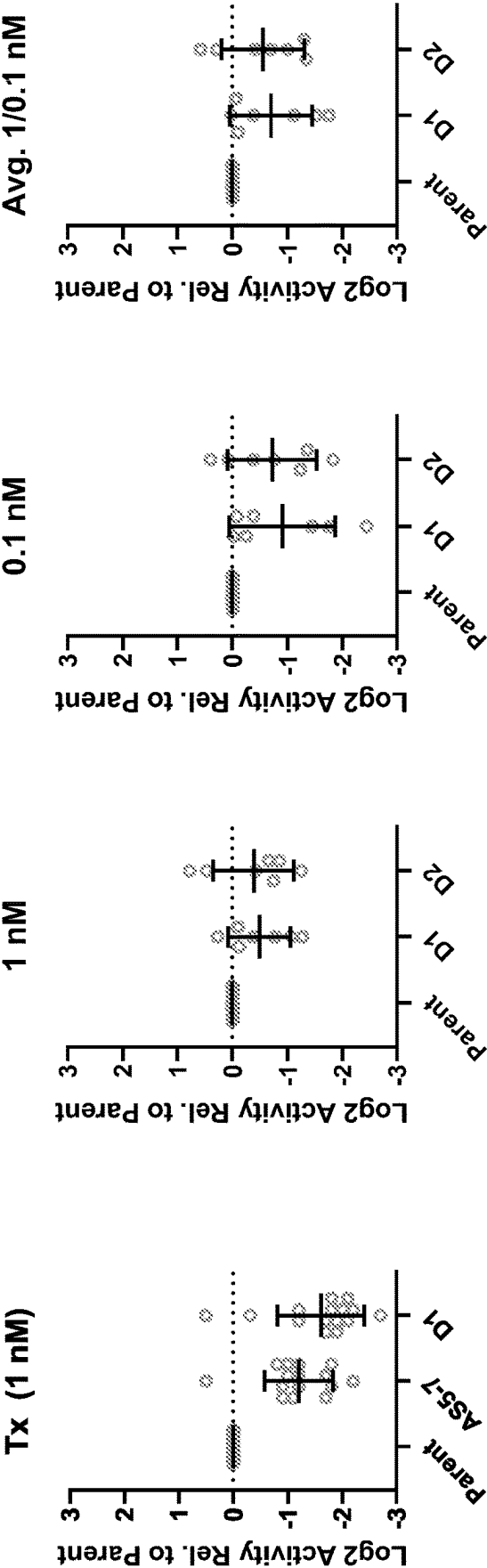


FIG. 3A

FIG. 3B

FIG. 3C

FIG. 3D

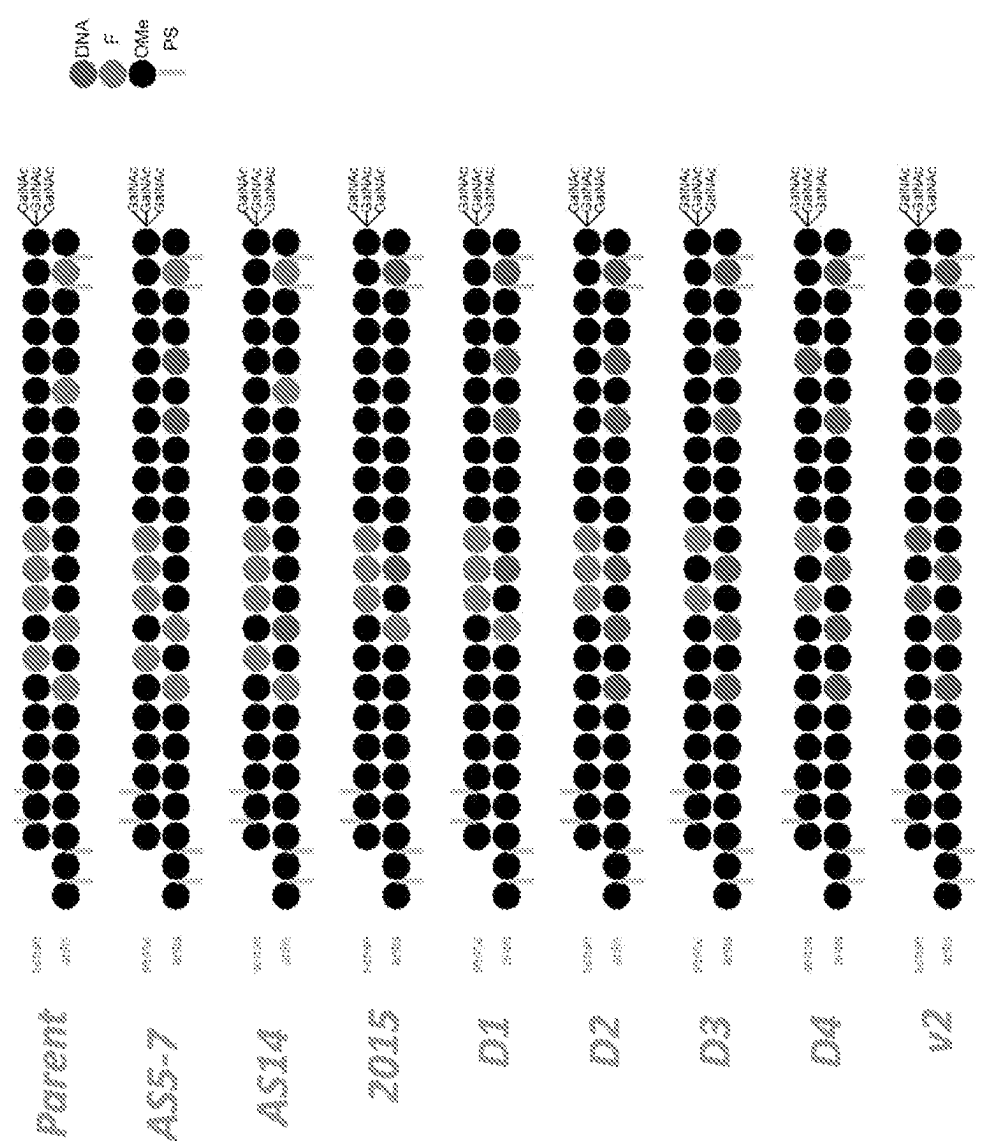


FIG. 5A

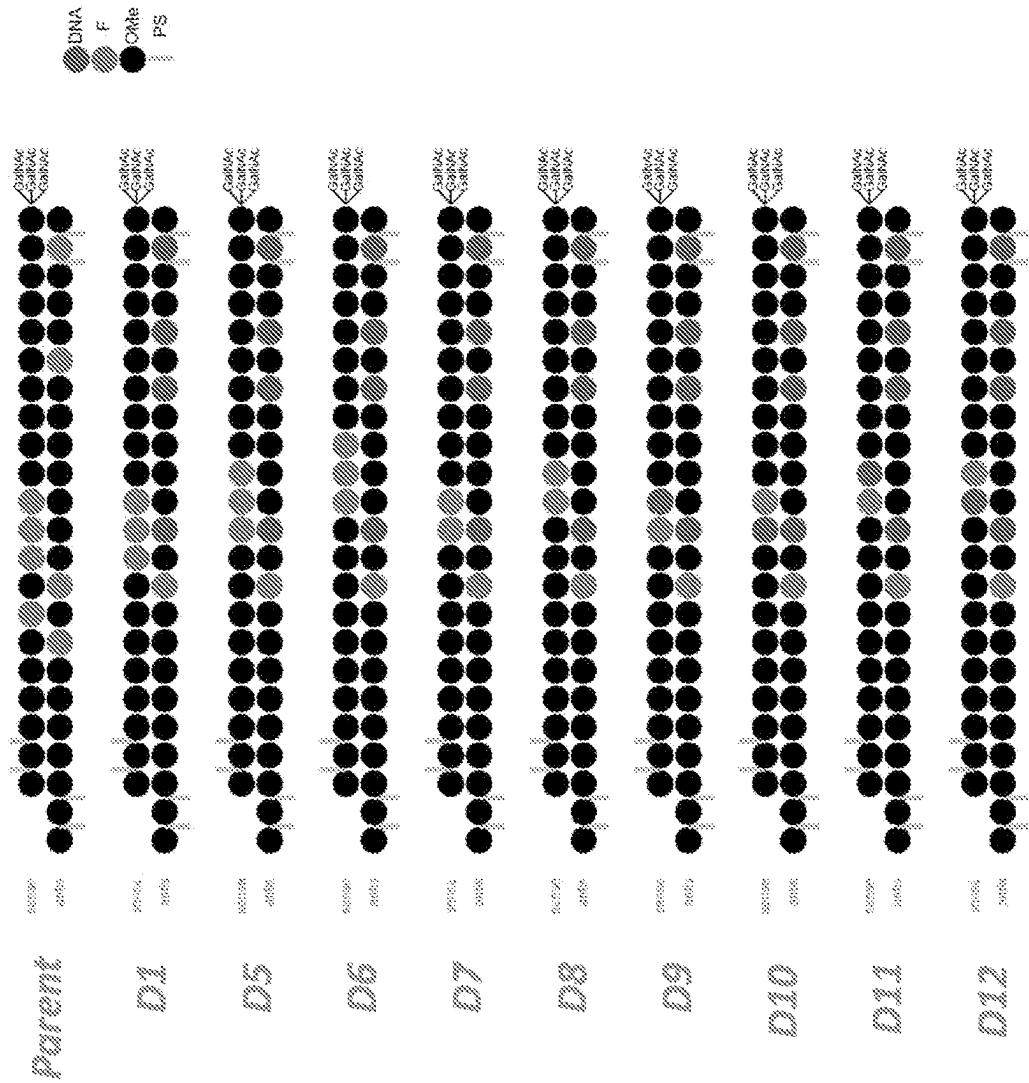


FIG. 5B

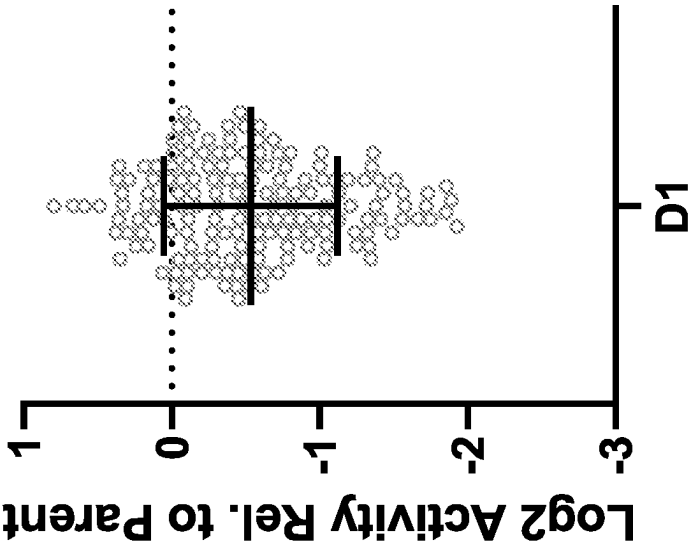


FIG. 6B

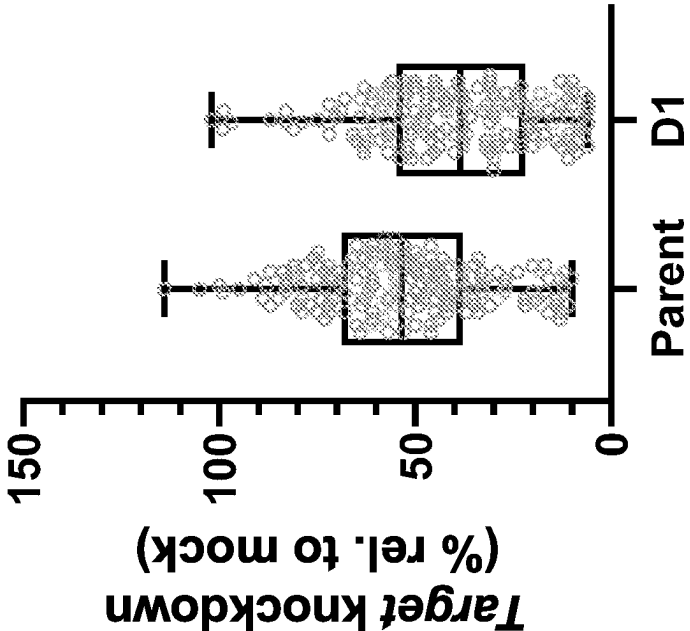


FIG. 6A

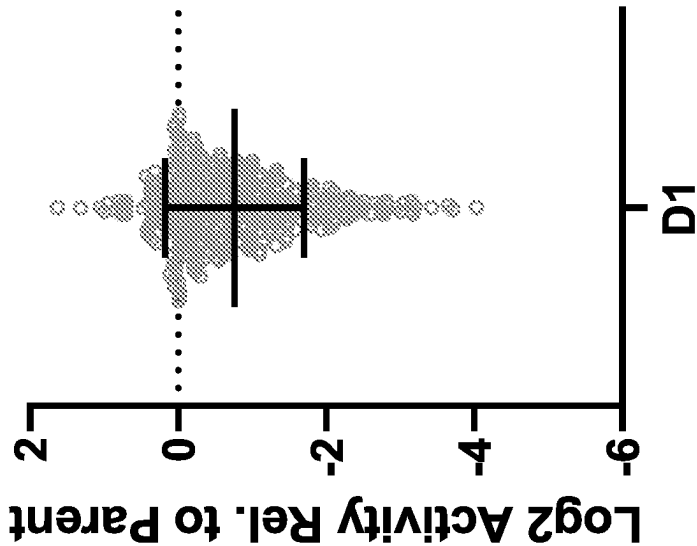


FIG. 7B

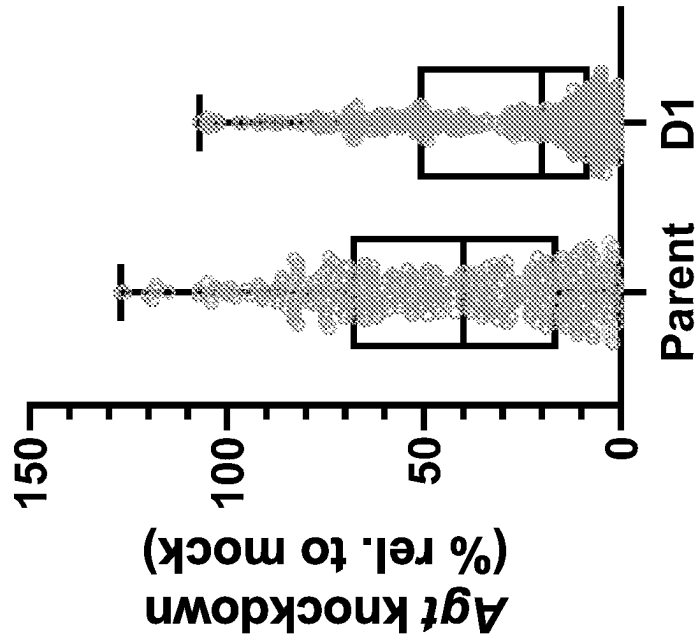


FIG. 7A

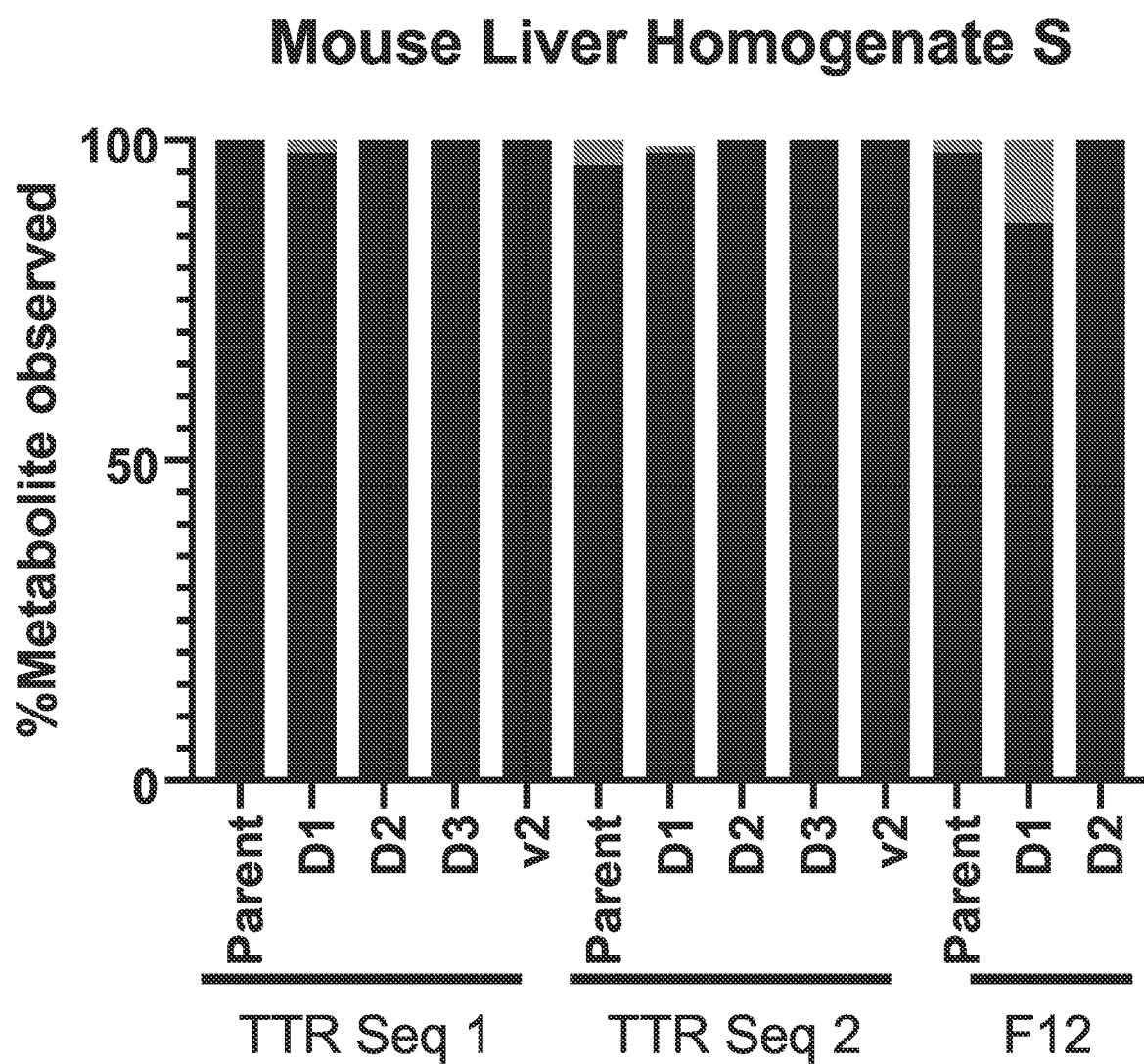


FIG. 8A

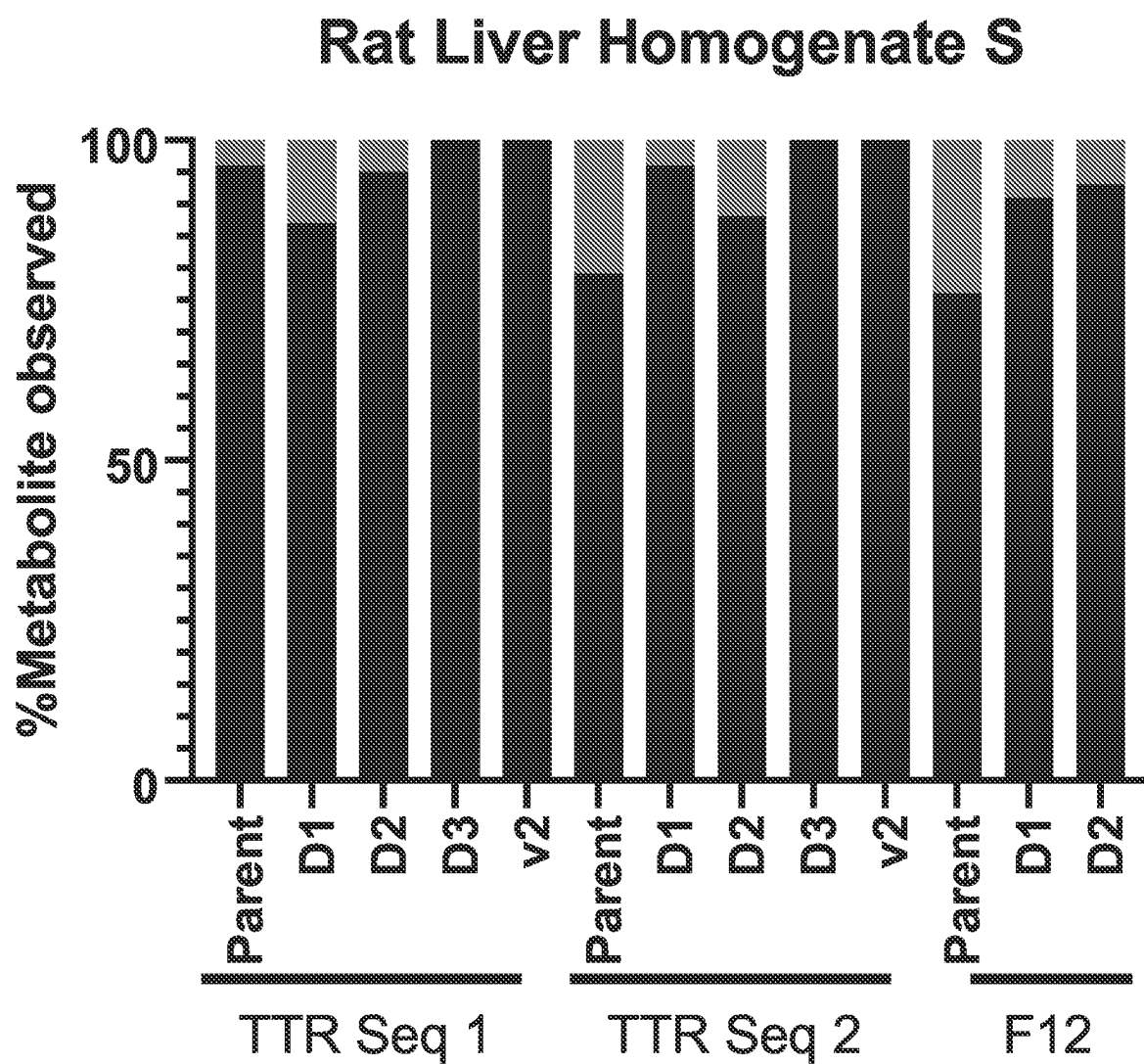


FIG. 8B

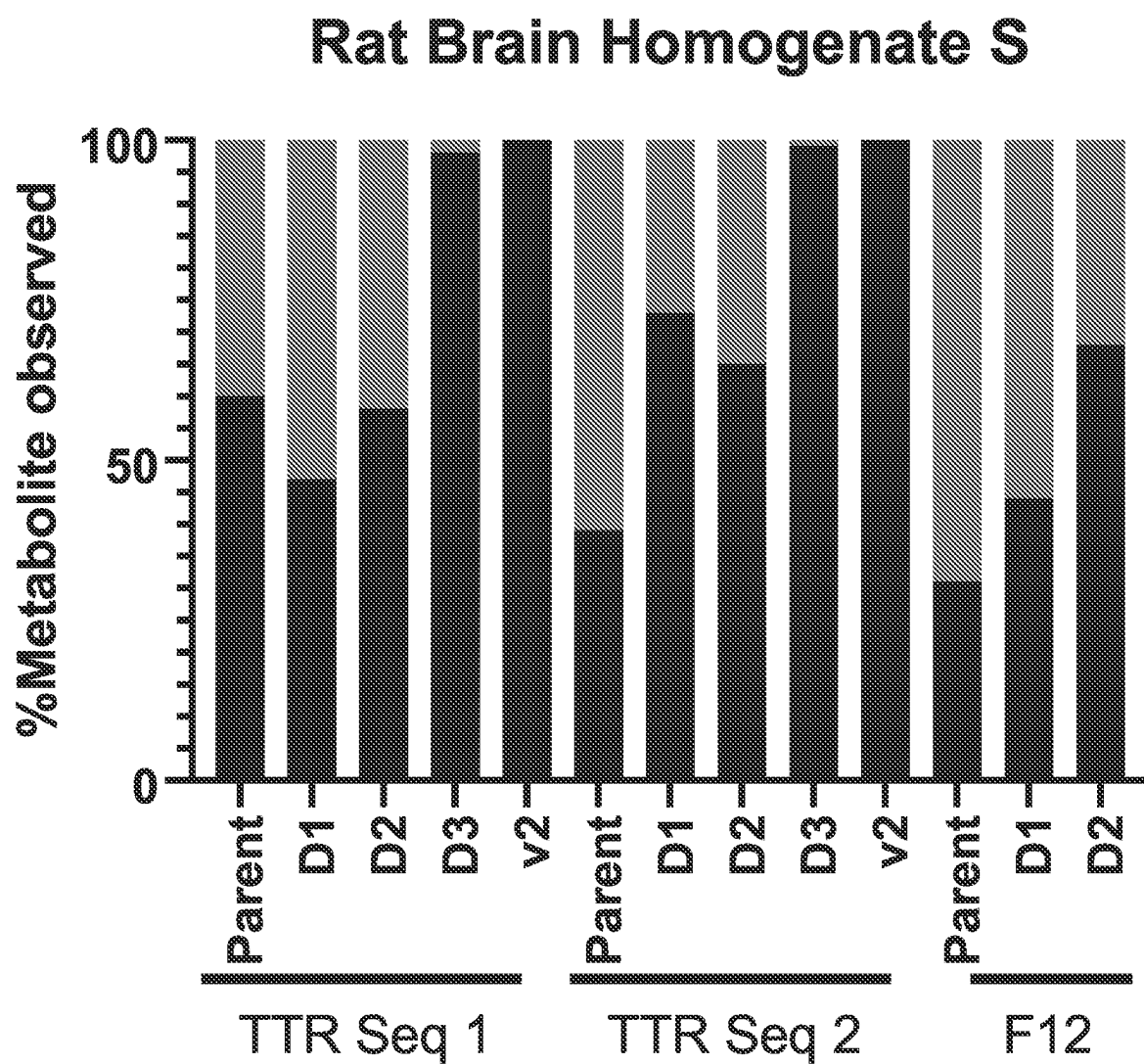


FIG. 8C

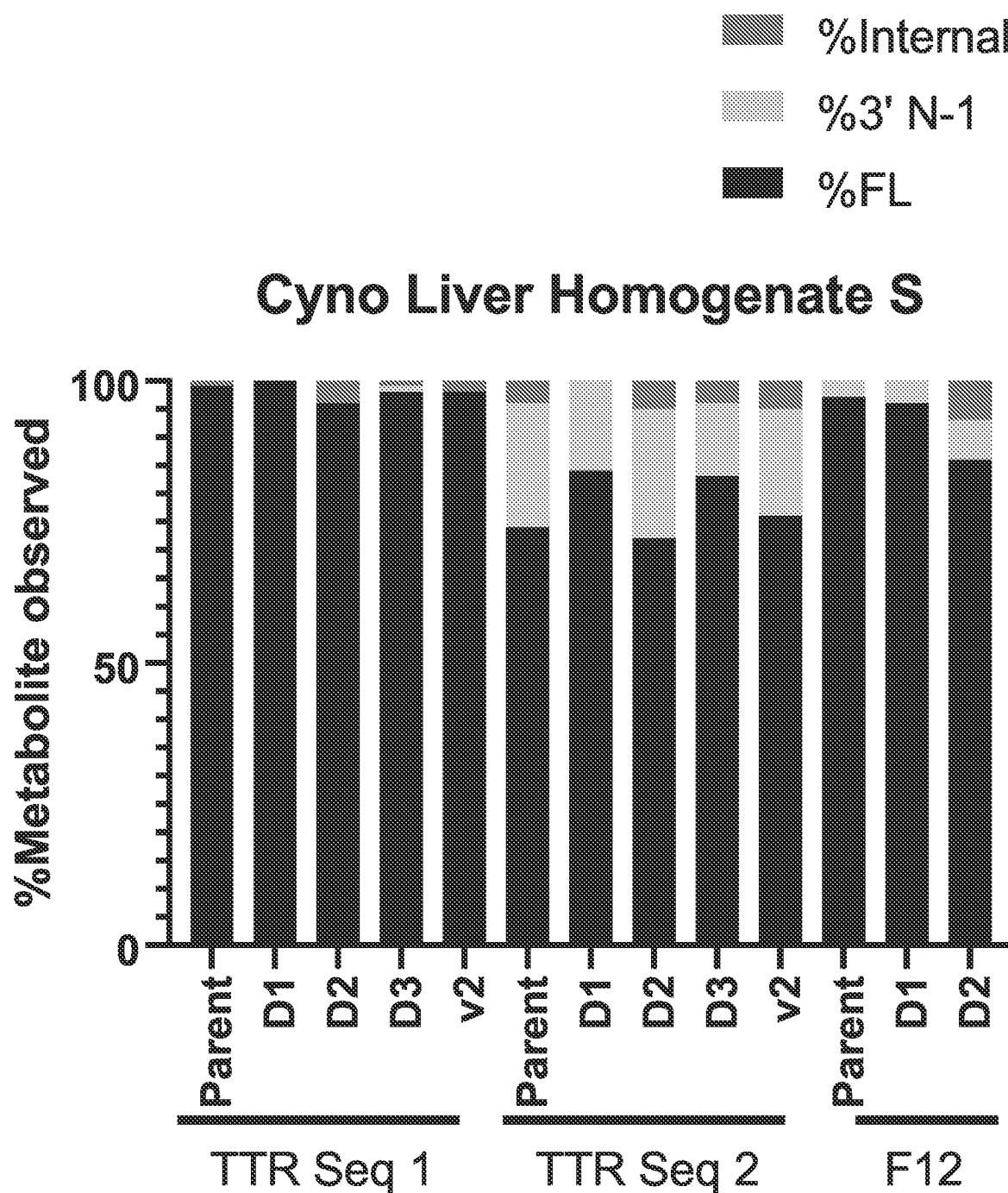


FIG. 8D

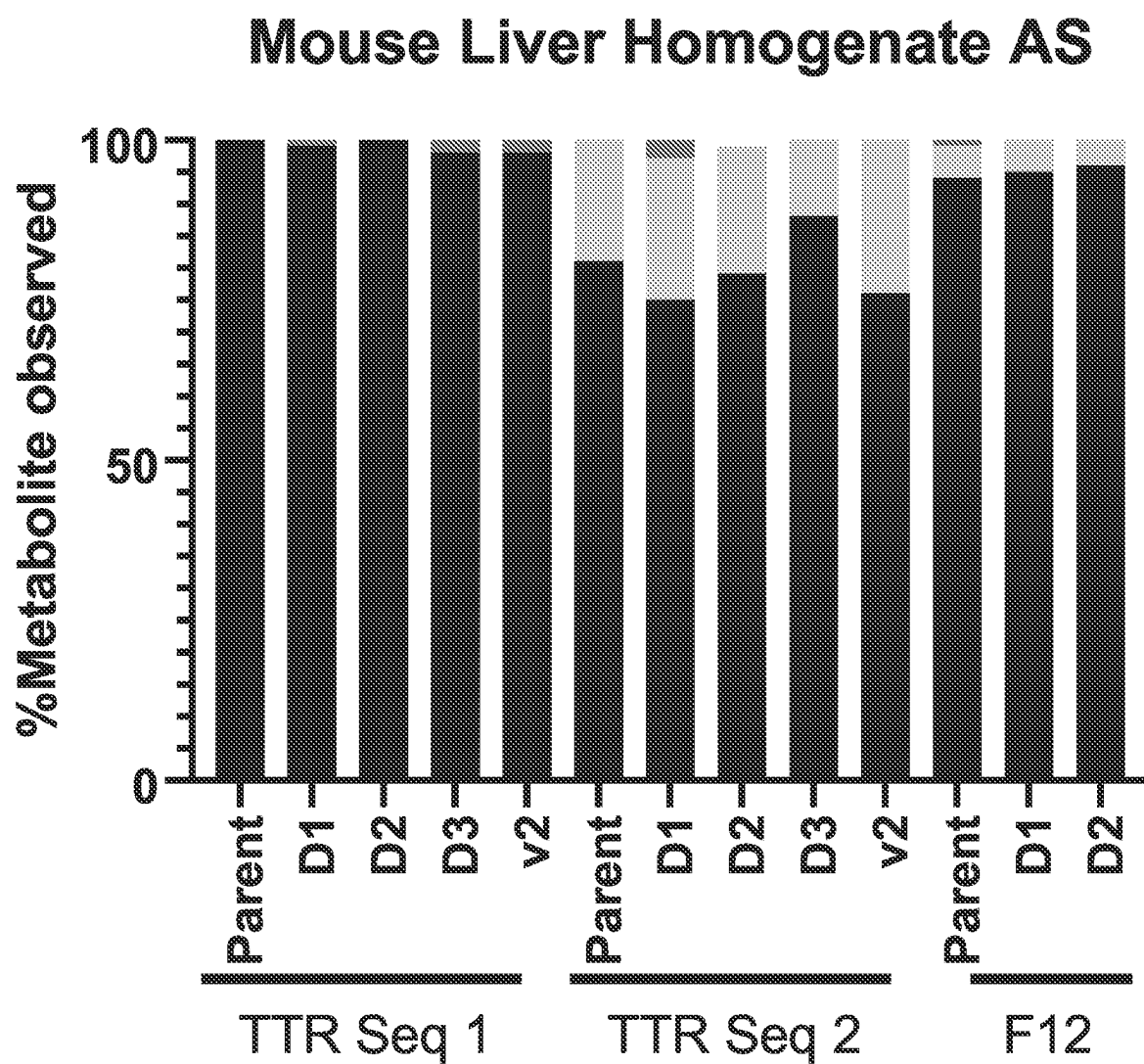


FIG. 8E

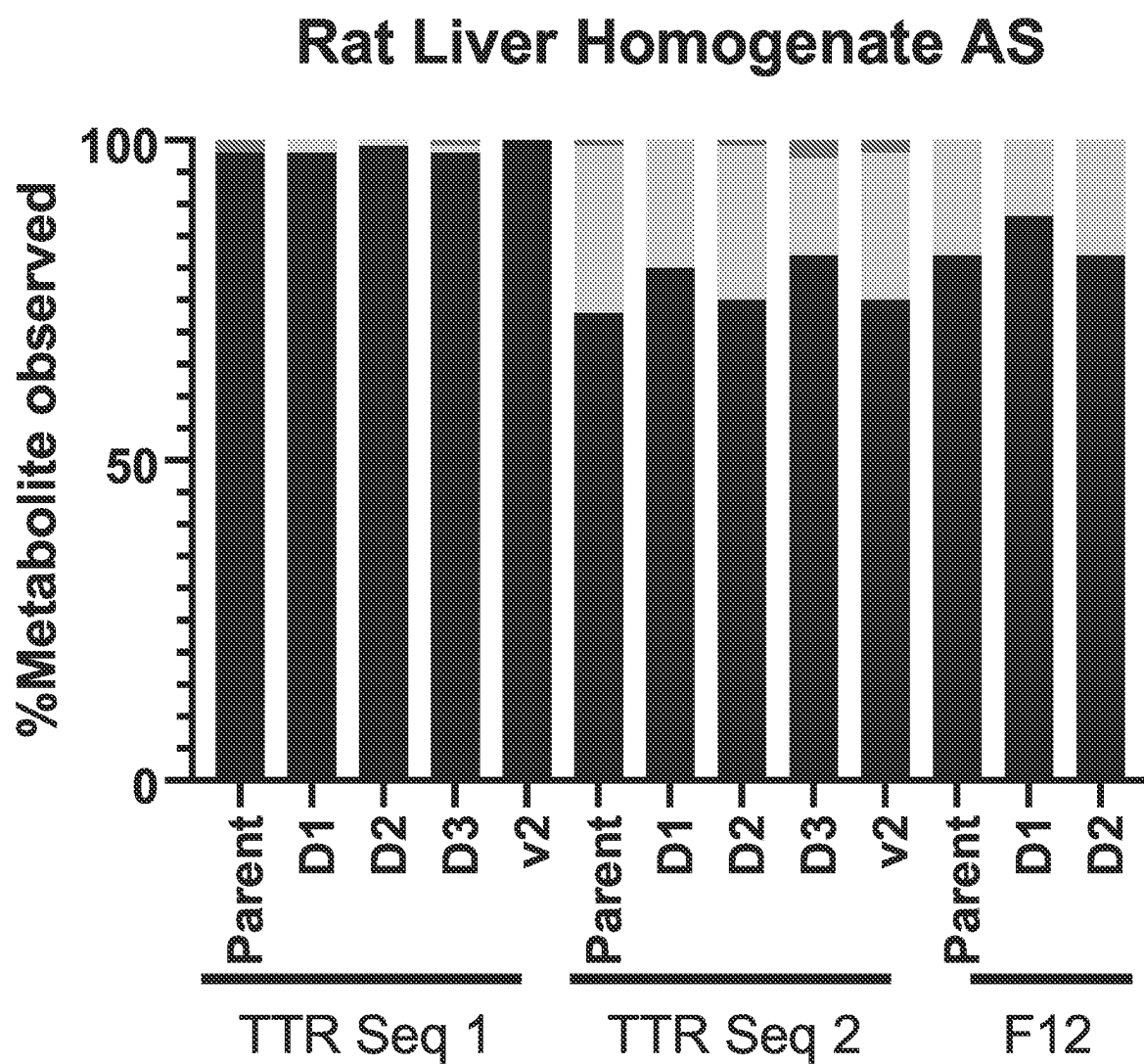


FIG. 8F

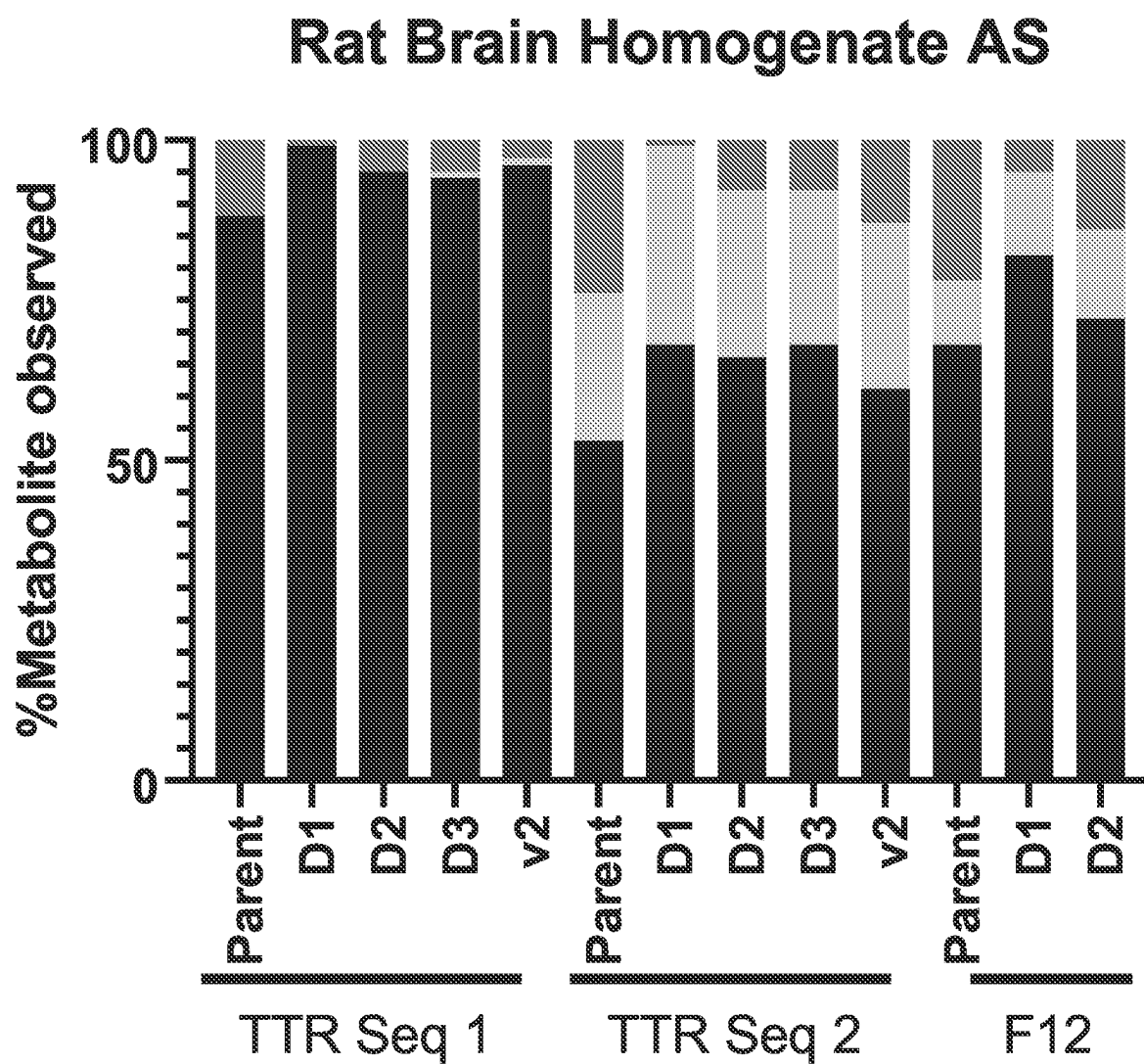


FIG. 8G

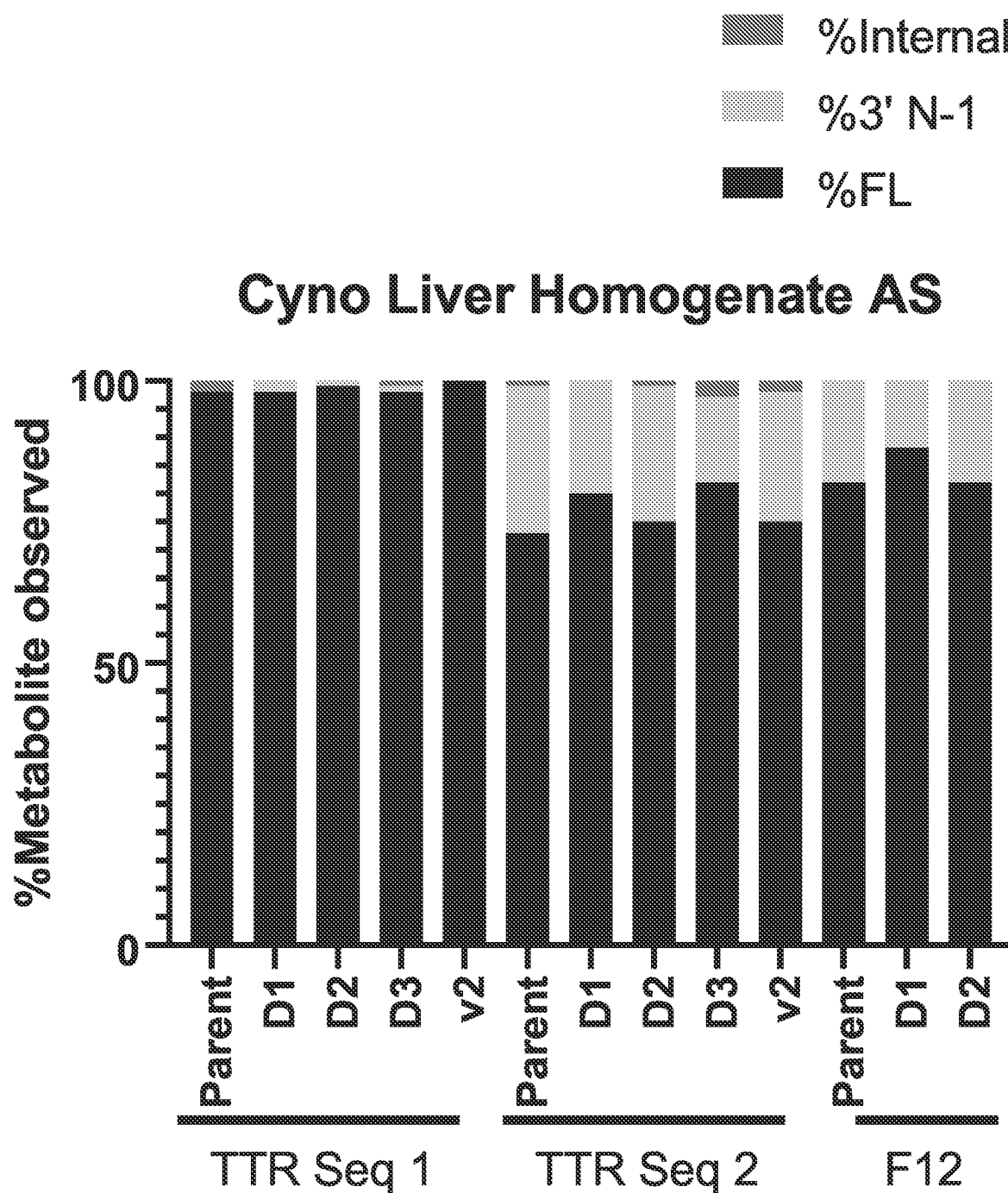


FIG. 8H

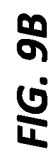
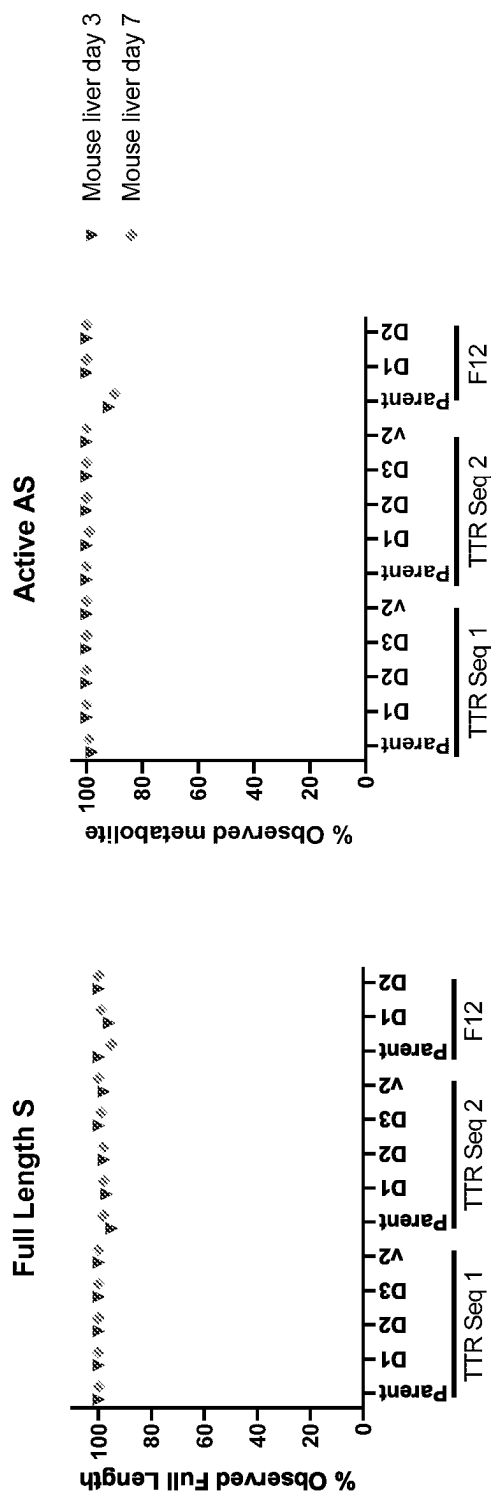


FIG. 10B

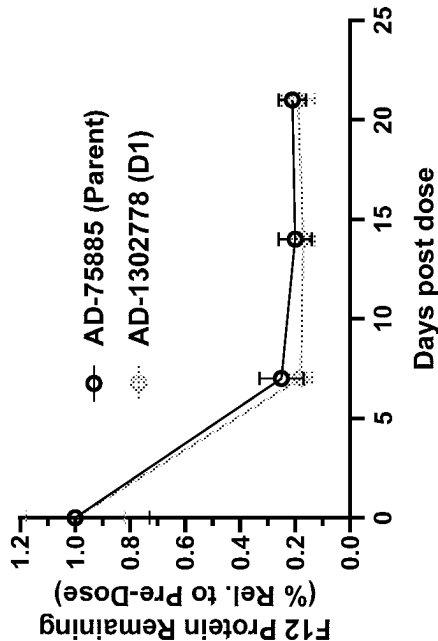


FIG. 10A

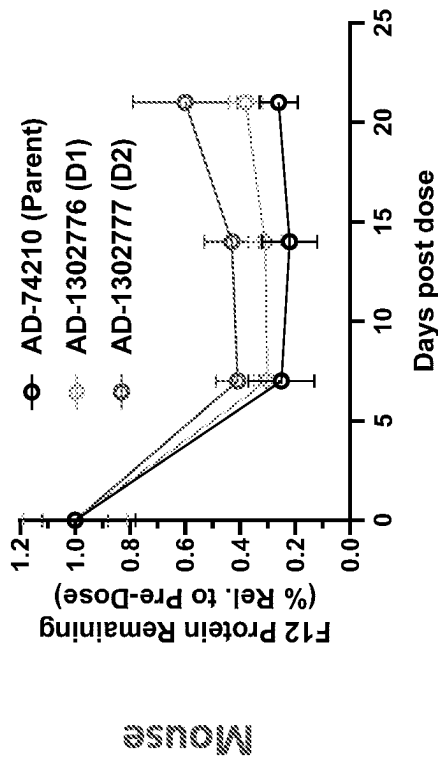


FIG. 10D

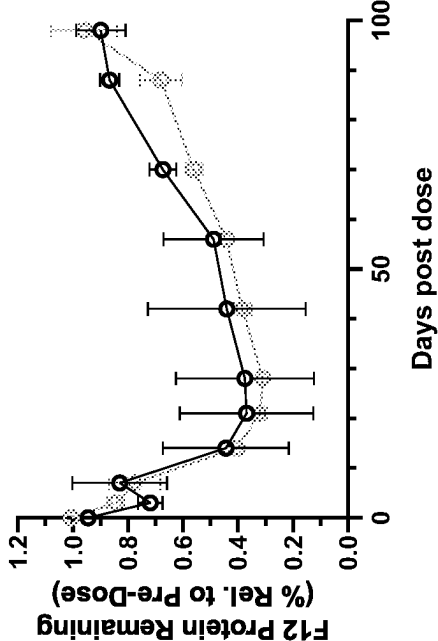
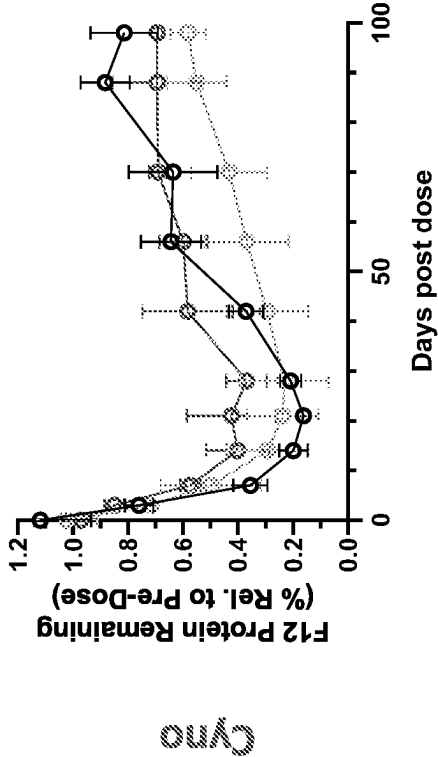


FIG. 10C



MODIFIED DOUBLE STRANDED OLIGONUCLEOTIDES

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a 35 U.S.C. § 371 National Phase Entry Application of International Patent Application No. PCT/US2021/057016 filed on Oct. 28, 2021, which designated the U.S., and which claims benefit under 35 U.S.C. § 119(e) of U.S. Provisional Application No. 63/140,714 filed, Jan. 22, 2021, U.S. Provisional Application No. 63/146,115 filed Feb. 5, 2021, U.S. Provisional Application No. 63/148,991 filed, Feb. 12, 2021, U.S. Provisional Application No. 63/153,983 filed Feb. 26, 2021, U.S. Provisional Application No. 63/156,476 filed Mar. 4, 2021, U.S. Provisional Application No. 63/161,313 filed Mar. 15, 2021, U.S. Provisional Application No. 63/164,467 filed Mar. 22, 2021, U.S. Provisional Application No. 63/179,607 filed Apr. 26, 2021, and U.S. Provisional Application No. 63/181,748 filed Apr. 29, 2021, the contents of each of which are incorporated herein by reference in their entireties.

SEQUENCE LISTING

[0002] The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Dec. 10, 2021, is named O51058-O99300WOPT_SL.txt and is 837,545 bytes in size.

FIELD OF THE INVENTION

[0003] The invention relates to dsRNA molecules having particular motifs that are advantageous for inhibition of target gene expression, as well dsRNA agent compositions, suitable for therapeutic use. Additionally, the invention provides methods of inhibiting the expression of a target gene by administering these dsRNA agents, e.g., for the treatment of various diseases.

BACKGROUND

[0004] RNA interference or “RNAi” is a term initially coined by Fire and co-workers to describe the observation that double-stranded RNAi (dsRNA) can block gene expression (Fire et al. (1998) *Nature* 391, 806-811; Elbashir et al. (2001) *Genes Dev.* 15, 188-200). Short dsRNA directs gene-specific, post-transcriptional silencing in many organisms, including vertebrates, and has provided a new tool for studying gene function. RNAi is mediated by RNA-induced silencing complex (RISC), a sequence-specific, multi-component nuclease that destroys messenger RNAs homologous to the silencing trigger. RISC is known to contain short RNAs (approximately 22 nucleotides) derived from the double-stranded RNA trigger, but the protein components of this activity remained unknown.

[0005] There remains a need in the art for effective nucleotide or chemical motifs for dsRNA molecules, which are advantageous for inhibition of target gene expression. This invention is directed to that effort.

SUMMARY

[0006] This invention provides effective nucleotide or chemical motifs for dsRNA molecules, which are advanta-

geous for inhibition of target gene expression, as well as RNAi compositions suitable for therapeutic use.

[0007] Inventors have discovered inter alia that double stranded RNA (dsRNA) molecules having a 2'-fluoro nucleotide at least at position 10 of the sense strand unexpectedly and surprisingly have improved in vitro potency, i.e., increased RNA interference (RNAi) activity. Accordingly, in one aspect provided herein is a double stranded RNA (dsRNA) molecule comprising a sense strand and an antisense strand, each strand independently having a length of 15 to 35 nucleotides, and wherein the sense strand comprises a 2'-fluoro nucleotide at position 10, counting from 5'-end of the sense strand.

[0008] It is noted that the sense strand can further comprises one or more, e.g., 1, 2, 3, 4 or 5 additional 2'-fluoro nucleotides. Accordingly, in some embodiments, the sense strand comprises 1, 2, 3, 4, or 5 additional 2'-fluoro nucleotides. The additional 2'-fluoro nucleotides can be located anywhere in the sense strand. Thus, in some embodiments, the sense strand further comprises a 2'-fluoro nucleotide at position 8, 9, 11 or 12, counting from 5'-end of the sense strand. For example, the sense strand further comprises a 2'-fluoro nucleotide at position 9, counting from 5'-end of the sense strand. In other words, the sense strand comprises a 2'-fluoro nucleotide at positions 9 and 10, counting from 5'-end of the sense strand. In another example, the sense strand further comprises a 2'-fluoro nucleotide at position 11, counting from 5'-end of the sense strand. For example, the sense strand comprises a 2'-fluoro nucleotide at positions 10 and 11, counting from 5'-end of the sense strand.

[0009] In some embodiments, the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, counting from 5'-end of the sense strand. In some other embodiments, the sense strand comprises a 2'-fluoro nucleotide at positions 8, 9 and 10, counting from 5'-end of the sense strand. In yet some other embodiments, the sense strand comprises a 2'-fluoro nucleotide at positions 10, 11 and 12, counting from 5'-end of the sense strand.

[0010] In some embodiments of any one of the aspects, the sense strand does not comprise a 2'-fluoro nucleotide at position 7, counting from 5'-end of the sense strand. For example, the sense strand comprises a 2'-OMe nucleotide at position 7, counting from the 5'-end of the sense strand.

[0011] The antisense strand of the dsRNA molecules described herein can comprise one or more 2'-deoxy, e.g., 2'-H nucleotides. For example, the antisense strand comprises 1, 2, 3, 4, 5, 6 or more 2'-deoxy nucleotides. In some embodiments, the antisense strand comprises 2, 3, 4, 5 or 6 2'-deoxy nucleotides. The 2'-deoxy nucleotides can be located anywhere in the antisense strand. For example, the antisense strand comprises a 2'-deoxy nucleotide at 1, 2, 3, 4, 5 or 6 of positions 2, 5, 7, 12, 14 and 16, counting from 5'-end of the antisense strand. In some embodiments, the antisense comprises a 2'-deoxy nucleotide at positions 2 and 12, counting from 5'-end of the antisense strand. In some embodiments, the antisense comprises a 2'-deoxy nucleotide at positions 5 and 7, counting from 5'-end of the antisense strand. In some embodiments, the antisense comprises a 2'-deoxy nucleotide at positions 2, 5, 7 and 12, counting from 5'-end of the antisense strand.

[0012] In some embodiments, the antisense can comprise one or more, e.g., 1, 2, 3, 4, 5 or more of 2'-fluoro

nucleotides. For example, the antisense strand can comprise a 2'-fluoro nucleotide at position 14, counting from 5'-end of the antisense strand.

[0013] In some embodiments, the antisense strand comprises a 2'-fluoro nucleotide at position 14 and a nucleotide other than a 2'-deoxy or 2'-fluoro at position 16, counting from 5'-end of the antisense strand. For example, the antisense strand comprises a 2'-fluoro nucleotide at position 14 and a nucleotide other than a 2'-deoxy or 2'-fluoro at position 16, counting from 5'-end of the antisense strand.

[0014] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2 and 12 and a 2'-fluoro nucleotide at position 14, counting from 5'-end of the antisense strand. In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2 and 12, a 2'-fluoro nucleotide at position 14, and a nucleotide other than a 2'-deoxy or 2'-fluoro at position 16, counting from 5'-end of the antisense strand. For example, the antisense strand comprises a 2'-deoxy nucleotide at positions 2 and 12, a 2'-fluoro nucleotide at position 14, and a 2'-OMe nucleotide at position 16, counting from 5'-end of the antisense strand.

[0015] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at position 14, counting from the 5'-end of the antisense strand, and the sense strand comprises a nucleotide other than a 2'-fluoro at position 7, counting from 5'-end of the sense strand. For example, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 12 and 14, counting from the 5'-end of the antisense strand, and the sense strand comprises a 2'-fluoro nucleotide at position 10 and a nucleotide other than a 2'-fluoro at position 7, counting from 5'-end of the sense strand.

[0016] In various embodiments, the dsRNA molecule has a double stranded (duplex) region of between 19 to 25 base pairs. For example, the dsRNA molecule has a duplex region of 20, 21, 22, 23 or 24 basepairs. In some particular embodiments, the dsRNA molecule has a double duplex) region of 20, 21 or 22 base pairs.

[0017] In some embodiments, the dsRNA molecule comprises a ligand. For example, the sense strand of the dsRNA molecule comprises a ligand. Exemplary ligands include, but are not limited to, ASGPR ligands and ligands comprising a lipophilic group.

[0018] The dsRNA molecule can comprise one or more, e.g., 1, 2, 3, 4, 5, 6, 7, 8 or more phosphorothioate linkages. The phosphorothioate linkages can be present only in one of the strands or in both strands of the dsRNA. For example, the sense strand can comprise 1, 2, 3 or 4 phosphorothioate linkages. In another non-limiting example, the antisense strand can comprise 1, 2, 3, 4, 5 or 6 phosphorothioate linkages. In some embodiments, the sense strand comprises 1, 2, 3 or 4 phosphorothioate linkages and the antisense independently comprises 1, 2, 3, 4, 5, or 6 phosphorothioate linkages. For example, the sense strand comprises 1 or 2 phosphorothioate linkages and the antisense strand comprises 1, 2, 3 or 4 phosphorothioate linkages.

[0019] In some embodiments, the sense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 5' end of the sense strand, the antisense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 5'-end of the antisense strand and the antisense further comprises at least two phosphorothioate internucleotide linkages between the first

five nucleotides counting from the 3'-end of the antisense strand. For example, the sense strand comprises phosphorothioate linkages between nucleotides 1 and 2, and between nucleotides 2 and 3, counting from 5'-end of the sense strand, and the antisense strand comprises phosphorothioate linkages and between nucleotides 1 and 2, and between nucleotides 2 and 3, counting from 5'-end of the antisense strand, and between nucleotides 1 and 2, and between nucleotides 2 and 3, counting from 3'-end of the antisense strand.

[0020] In some embodiments, the remaining nucleotides in the dsRNA are 2'-OMe nucleotides. For example, all of the remaining nucleotides in the sense strand are 2'-OMe nucleotides. In other words, the sense strand solely comprises 2'-fluoro and 2'-OMe nucleotides.

[0021] It is understood that the antisense strand has sufficient complementarity to a target sequence to mediate RNA interference. In other words, the dsRNA molecules of the invention are capable of inhibiting the expression of a target gene.

[0022] In another aspect, the invention further provides a method for delivering the dsRNA molecule of the invention to a specific target in a subject by subcutaneous or intravenous administration. The invention further provides the dsRNA molecules of the invention for use in a method for delivering said agents to a specific target in a subject by subcutaneous or intravenous administration.

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] This patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

[0024] FIGS. 1A and 1B are graphs showing dsRNAs according to exemplary embodiments of the invention have improved in vitro potency relative to the parent dsRNA molecules when dosed at 10 nM (FIG. 1A) or at 1 nM (FIG. 1B).

[0025] FIGS. 2A-2D are graphs showing that dsRNA molecules according to embodiments of the invention have improved in vivo efficacy compared to the parent molecules. Parent duplexes are AD-1181401 (Sequence 1, FIG. 2A); AD-1181410 (Sequence 2, FIG. 2B); AD-1181426 (Sequence 3, FIG. 2C); and AD-1181451 (Sequence 4, FIG. 2D).

[0026] FIGS. 3A-3D are graphs showing dsRNAs according to exemplary embodiments of the invention and targeting different targets have improved in vitro potency relative to the parent dsRNA molecules when dosed at 1 nM (FIGS. 3A and 3B) or at 0.1 nM (FIG. 3C).

[0027] FIGS. 4A-4C are graphs showing that presence of a 2'-fluoro nucleotide at position 10 of the sense strand, counting from the 5'-end of the sense strand, enhances the RNAi efficacy of the dsRNA molecule compared to the parent.

[0028] FIGS. 5A and 5B are schematic representation of some exemplary designs of dsRNA molecules according to embodiments of the invention.

[0029] FIGS. 6A and 6B are graphs showing improved in vitro target knockdown (FIG. 6A) and log 2 activity (FIG. 6B) of exemplary dsRNA according to some embodiments of the disclosure relative to exemplary parent dsRNA molecules.

[0030] FIGS. 7A and 7B are graphs showing improved in vitro target knockdown of AGT (FIG. 6A) and log 2 activity (FIG. 6B) of exemplary dsRNA according to some embodiments of the disclosure relative to exemplary parent dsRNA molecules targeting AGT.

[0031] FIGS. 8A-8H are showing similar or improved metabolic stability of sense strand (FIGS. 8A-8D) and antisense strand (FIGS. 8E-8H) of exemplary dsRNA molecules in mouse liver homogenate (FIGS. 8A and 8E), rat liver homogenate (FIGS. 8B and 8F), rat brain homogenate (FIGS. 8C and 8G), and cynomolgus liver homogenate (FIGS. 8D and 8H). Parent duplexes are AD-1181401(TTR Seq 1); AD-1181410(TTR Seq 2); and AD-74210(F12).

[0032] FIGS. 9A and 9B are showing similar or improved metabolic stability of exemplary dsRNA molecules in mouse. Parent duplexes are AD-1181401(TTR Seq 1); AD-1181410(TTR Seq 2); and AD-74210(F12).

[0033] FIGS. 10A-10D are graphs showing that dsRNA molecules according to embodiments of the invention have improved in vivo efficacy and/or duration in non-human primates, mice (FIGS. 10A and 10B) and in cynomolgus monkeys (FIGS. 10C and 10D) compared to the parent molecules AD-74210(FIGS. 10A and 10C) and AD-75885 (FIGS. 10B and 10D).

DETAILED DESCRIPTION

[0034] In one aspect, the invention provides a double-stranded RNA (dsRNA) agent capable of inhibiting expression of a target gene. Without limitations, the dsRNA agents of the invention can be substituted for the dsRNA molecules and can be used in RNA interference based gene silencing techniques, including, but not limited to, in vitro or in vivo applications.

[0035] Generally, the dsRNA molecule comprises a sense strand (also referred to as passenger strand) and an antisense strand (also referred to as guide strand). Each strand of the dsRNA molecule can range from 15-35 nucleotides in length. For example, each strand can be between, 17-35 nucleotides in length, 17-30 nucleotides in length, 25-35 nucleotides in length, 27-30 nucleotides in length, 17-23 nucleotides in length, 17-21 nucleotides in length, 17-19 nucleotides in length, 19-25 nucleotides in length, 19-23 nucleotides in length, 19-21 nucleotides in length, 21-25 nucleotides in length, or 21-23 nucleotides in length. Without limitations, the sense and antisense strands can be equal length or unequal length. For example, the sense strand and the antisense strand independently have a length of 18, 19, 20, 21, 22, 23, 24 or 25 nucleotides.

[0036] In some embodiments, the antisense strand is of length 15-35 nucleotides. In some embodiments, the antisense strand is 15-35, 17-35, 17-30, 25-35, 27-30, 17-23, 17-21, 17-19, 19-25, 19-23, 19-21, 21-25, 21-25, or 21-23 nucleotides in length. For example, the antisense strand can be 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34 or 35 nucleotides in length. In some embodiments, the antisense strand is 19, 20, 21, 22, 23, 24 or 25 nucleotides in length. For example, the antisense strand is 21, 22, 23, 24 or 25 nucleotides in length. In some particular embodiments, the antisense strand is 22, 23 or 24 nucleotides in length. For example, the antisense strand is 23 nucleotides in length.

[0037] Similar to the antisense strand, the sense strand can be, in some embodiments, 15-35 nucleotides in length. In some embodiments, the sense strand is 15-35, 17-35, 17-30,

25-35, 27-30, 17-23, 17-21, 17-19, 19-25, 19-23, 19-21, 21-25, 21-25, or 21-23 nucleotides in length. For example, the sense strand can be 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34 or 35 nucleotides in length. In some embodiments, the sense strand is 17, 18, 19, 20, 21, 22, 23, 24 or 25 nucleotides in length. For example, the sense strand is 19, 20, 21, 22 or 23 nucleotides in length. In some particular embodiments, the sense strand is 20, 21 or 22 nucleotides in length. For example, the sense strand is 21 nucleotides in length.

[0038] In some embodiments, the sense strand can be 15-35 nucleotides in length, and the antisense strand can be independent from the sense strand, 15-35 nucleotides in length. In some embodiments, the sense strand is 15-35, 17-35, 17-30, 25-35, 27-30, 17-23, 17-21, 17-19, 19-25, 19-23, 19-21, 21-25, 21-25, or 21-23 nucleotides in length, and the antisense strand is independently 15-35, 17-35, 17-30, 25-35, 27-30, 17-23, 17-21, 17-19, 19-25, 19-23, 19-21, 21-25, 21-25, or 21-23 nucleotides in length. For example, the sense and the antisense strand can be independently 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34 or 35 nucleotides in length. In some embodiments, the sense strand and the antisense strand are independently 17, 18, 19, 20, 21, 22, 23, 24 or 25 nucleotides in length. For example, the sense strand is 19, 20, 21, 22 or 23 nucleotides in length and the antisense strand is 21, 22, 23, 24 or 25 nucleotides in length. In some particular embodiments, the sense strand is 20, 21 or 22 nucleotides in length and the antisense strand is 22, 23 or 24 nucleotides in length. For example, the sense strand is 21 nucleotides in length and the antisense strand is 23 nucleotides in length.

[0039] The sense strand and antisense strand typically form a double-stranded or duplex region. Without limitations, the duplex region of a dsRNA agent described herein can be 12-35 nucleotide (or base) pairs in length. For example, the duplex region can be between 14-35 nucleotide pairs in length, 17-30 nucleotide pairs in length, 25-35 nucleotides in length, 27-35 nucleotide pairs in length, 17-23 nucleotide pairs in length, 17-21 nucleotide pairs in length, 17-19 nucleotide pairs in length, 19-25 nucleotide pairs in length, 19-23 nucleotide pairs in length, 19-21 nucleotide pairs in length, 21-25 nucleotide pairs in length, or 21-23 nucleotide pairs in length. In another example, the duplex region is selected from 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, and 27 nucleotide pairs in length. In some embodiments, the duplex region is 18, 19, 20, 21, 22, 23, 24 or 25 nucleotide pairs in length. For example, the duplex region is 19, 20, 21, 22 or 23 nucleotide pairs in length. In some embodiments, the duplex region is 20, 21 or 22 nucleotide pairs in length. For example, the dsRNA molecule has a duplex region of 21 base pairs.

[0040] As described herein, the dsRNA molecule of the invention can further comprise at least one, e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more 2'-deoxy, e.g., 2'-H nucleotides. For example, the dsRNA can comprise 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 2'-deoxy, e.g., 2'-H nucleotides. The 2'-deoxy nucleotide may occur on any nucleotide of the sense strand or antisense strand or both in any position of the strand. In one non-limiting example, the sense strand does not comprise a 2'-deoxy nucleotide at position 11, counting from 5'-end of the sense strand.

[0041] In some embodiments, the antisense strand the antisense strand comprises 1, 2, 3, 4, 5 or 6 of 2'-deoxy nucleotides. For example, antisense strand can comprise 2,

3, 4, 5 or 6 of 2'-deoxy nucleotides. The 2'-deoxy nucleotides can be located anywhere in the antisense strand. For example, the antisense strand comprises a 2'-deoxy nucleotide at 1, 2, 3, 4, 5 or 6 of positions 2, 5, 7, 12, 14 and 16, counting from 5'-end of the antisense strand. In one non-limiting example, the antisense strand comprises a 2'-deoxy nucleotide at 1, 2, 3 or 4 of positions 2, 5, 7, and 12, counting from 5'-end of the antisense strand.

[0042] In some embodiments, the antisense comprises a 2'-deoxy nucleotide at positions 5 and 7, counting from 5'-end of the antisense strand. For example, the antisense strand comprises a 2'-deoxy nucleotide at positions 5, 7 and 12, counting from 5'-end of the antisense strand. In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5 and 7, counting from 5'-end of the antisense strand. For example, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7 and 12, counting from 5'-end of the antisense strand. In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7, 12 and 14, counting from 5'-end of the antisense strand. For example, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7, 12, 14 and 16, counting from 5'-end of the antisense strand.

[0043] In some embodiments, the antisense comprises a 2'-deoxy nucleotide at position 2 or 12, counting from 5'-end of the antisense strand. For example, the antisense comprises a 2'-deoxy nucleotide at position 12, counting from 5'-end of the antisense strand.

[0044] In some embodiments, the sense strand comprises a 2'-fluoro nucleotide at position 10, counting from 5'-end of the sense strand, and the antisense comprises a 2'-deoxy nucleotide at positions 5 and 7, counting from 5'-end of the antisense strand. For example, the sense strand comprises a 2'-fluoro nucleotide at positions 9 and 10, counting from 5'-end of the sense strand, and the antisense comprises a 2'-deoxy nucleotide at positions 5 and 7, counting from 5'-end of the antisense strand. In another example, the sense strand comprises a 2'-fluoro nucleotide at positions 8, 9 and 10, counting from 5'-end of the sense strand, and the antisense comprises a 2'-deoxy nucleotide at positions 5 and 7, counting from 5'-end of the antisense strand.

[0045] In some embodiments, the sense strand comprises a 2'-fluoro nucleotide at positions 10 and 11, counting from 5'-end of the sense strand, and the antisense comprises a 2'-deoxy nucleotide at positions 5 and 7, counting from 5'-end of the antisense strand. For example, the sense strand comprises a 2'-fluoro nucleotide at positions 10, 11 and 12, counting from 5'-end of the sense strand, and the antisense comprises a 2'-deoxy nucleotide at positions 5 and 7, counting from 5'-end of the antisense strand.

[0046] In some embodiments of any one of the aspects, the sense strand does not comprise a 2'-fluoro nucleotide at position 7, counting from 5'-end of the sense strand. For example, the sense strand comprises a 2'-fluoro nucleotide at at least one e.g., 1, 2 or 3 of positions 9, 10 and 11 but does not comprise a 2'-fluoro nucleotide at position 7, counting from 5'-end of the sense strand. In some embodiments, the sense strand comprises a 2'-fluoro nucleotide at position 10 and a nucleotide other than a 2'-fluoro nucleotide at position 7, counting from 5'-end of the sense strand.

[0047] For example, the sense strand comprises a 2'-fluoro nucleotide position 10 and a 2'-OMe nucleotide at position 7, counting from the 5'-end of the sense strand.

[0048] In some embodiments, the sense strand comprises a 2'-fluoro nucleotide at positions 9 and 10, and a nucleotide other than a 2'-fluoro nucleotide at position 7, counting from 5'-end of the sense strand. For example, the sense strand comprises a 2'-fluoro nucleotide at positions 9 and 10, and a 2'-OMe nucleotide at position 7, counting from 5'-end of the sense strand.

[0049] In some embodiments, the sense strand comprises a 2'-fluoro nucleotide at positions 10 and 11, and a nucleotide other than a 2'-fluoro nucleotide at position 7, counting from 5'-end of the sense strand. For example, the sense strand comprises a 2'-fluoro nucleotide at positions 10 and 11, and a 2'-OMe nucleotide at position 7, counting from 5'-end of the sense strand.

[0050] In some embodiments, the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, and a nucleotide other than a 2'-fluoro nucleotide at position 7, counting from 5'-end of the sense strand. For example, the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, and a 2'-OMe nucleotide at position 7, counting from 5'-end of the sense strand.

[0051] In some embodiments, the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, counting from 5'-end of the sense strand, and the remaining nucleotides in the sense strand are 2'-OMe nucleotides. 2'-fluoro modifications (antisense strand)

[0052] The dsRNA molecules of the invention comprise one or more (e.g., one, two, three, four, five, six, seven, eight, nine, ten or more)₂'-fluoro nucleotides. Without limitations, the 2'-fluoro nucleotides all can be present in one strand. In some embodiments, both the sense and the antisense strands comprise at least one, 2'-fluoro nucleotide. The 2'-fluoro modification can occur on any nucleotide of the sense strand or antisense strand. For instance, the 2'-fluoro modification can occur on every nucleotide on the sense strand and/or antisense strand; each 2'-fluoro modification can occur in an alternating pattern on the sense strand or antisense strand; or the sense strand or antisense strand comprises both 2'-fluoro modifications in an alternating pattern. The alternating pattern of the 2'-fluoro modifications on the sense strand may be the same or different from the antisense strand, and the alternating pattern of the 2'-fluoro modifications on the sense strand can have a shift relative to the alternating pattern of the 2'-fluoro modifications on the antisense strand.

[0053] In some embodiments, the antisense strand comprises at least two (e.g., two, three, four, five, six, seven, eight, nine, ten or more)₂'-fluoro nucleotides. Without limitations, a 2'-fluoro modification in the antisense strand can be present at any positions.

[0054] In some embodiments, the antisense comprises one or more, e.g., 1, 2, 3, 4, 5 or more 2'-fluoro nucleotides. For example, the antisense strand comprises 1, 2, 3, 4, or 5 more 2'-fluoro nucleotides. In some embodiments, the antisense strand comprises 1, 2 or 3 2'-fluoro nucleotides. For example, the antisense strand comprises a single 2'-fluoro nucleotide. It is noted that a 2'-fluoro nucleotide can be located anywhere in the antisense strand. For example, a 2'-fluoro nucleotide can be at position 2 or 14, counting from 5'-end, of the antisense strand. In some embodiments, the antisense comprises a 2'-fluoro nucleotide at position 14, counting from 5'-end, of the antisense strand.

[0055] In some embodiments, the antisense comprises a 2'-fluoro nucleotide at position 14 and 2'-deoxy nucleotides

at positions 5 and 7, counting from 5'-end of the antisense strand. For example, the antisense comprises a 2'-fluoro nucleotide at position 14 and 2'-deoxy nucleotides at positions 5, 7 and 12, counting from 5'-end of the antisense strand. In a further example, the antisense comprises a 2'-fluoro nucleotide at position 14 and 2'-deoxy nucleotides at positions 2, 5, 7 and 12, counting from 5'-end of the antisense strand.

[0056] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2 and 12, counting from the 5'-end of the antisense strand. In some embodiments, the antisense strand further comprises a 2'-fluoro nucleotide at position 14, counting from the 5'-end of the antisense strand. For example, the antisense strand comprises a 2'-deoxy nucleotide at positions 2 and 12 and a 2'-fluoro nucleotide at position 14, counting from the 5'-end of the antisense strand.

[0057] In some embodiments, the antisense strand comprises a nucleotide other than a 2'-deoxy nucleotide at position 16, counting from the 5'-end of the antisense strand. In some embodiments, antisense strand comprises a nucleotide other than a 2'-fluoro nucleotide at position 16, counting from the 5'-end of the antisense strand. For example, the antisense strand comprises a 2'-deoxy nucleotide at positions 2 and 12 and a nucleotide other than a 2'-deoxy or 2'-fluoro nucleotide at position 16, counting from the 5'-end of the antisense strand. In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2 and 12 and a 2'-OMe nucleotide at position 16, counting from the 5'-end of the antisense strand.

[0058] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2 and 12, a 2'-fluoro nucleotide at position 14 and a nucleotide other than a 2'-deoxy at position 16, counting from the 5'-end of the antisense strand. In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2 and 12, a 2'-fluoro nucleotide at position 14 and a nucleotide other than a 2'-fluoro at position 16, counting from the 5'-end of the antisense strand. For example, the antisense strand comprises a 2'-deoxy nucleotide at positions 2 and 12, a 2'-fluoro nucleotide at position 14 and a nucleotide other than a 2'-deoxy or 2'-fluoro nucleotide at position 16, counting from the 5'-end of the antisense strand. In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2 and 12, a 2'-fluoro nucleotide at position 14, and a 2'-OMe nucleotide at position 16, counting from the 5'-end of the antisense strand.

[0059] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7 and 12 and a 2'-fluoro nucleotide at position 14, counting from the 5'-end of the antisense strand, and the remaining nucleotides in the antisense strand are 2'-OMe nucleotides.

[0060] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7 and 12 and a 2'-fluoro nucleotide at position 14, counting from the 5'-end of the antisense strand, the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, counting from 5'-end of the sense strand, and the remaining nucleotides in the antisense strand and the sense strand are 2'-OMe nucleotides.

[0061] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7, 12 and 14,

counting from the 5'-end of the antisense strand, and the remaining nucleotides in the antisense strand are 2'-OMe nucleotides.

[0062] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7, 12 and 14, counting from the 5'-end of the antisense strand, the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, counting from 5'-end of the sense strand, and the remaining nucleotides in the antisense strand and the sense strand are 2'-OMe nucleotides.

[0063] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7, 12 and 16, counting from the 5'-end of the antisense strand, and the remaining nucleotides in the antisense strand are 2'-OMe nucleotides.

[0064] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7, 12 and 16, counting from the 5'-end of the antisense strand, the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, counting from 5'-end of the sense strand, and the remaining nucleotides in the antisense strand and the sense strand are 2'-OMe nucleotides.

[0065] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7, 12, 14 and 16, counting from the 5'-end of the antisense strand, and the remaining nucleotides in the antisense strand are 2'-OMe nucleotides.

[0066] In some embodiments, the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7, 12, 14 and 16, counting from the 5'-end of the antisense strand, the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, counting from 5'-end of the sense strand, and the remaining nucleotides in the antisense strand and the sense strand are 2'-OMe nucleotides.

[0067] It is noted the remaining nucleotides, i.e., at positions not explicitly defined in the sense strand and/or the antisense strand can be unmodified or modified nucleotides. Accordingly, in some embodiments, the remaining nucleotides, i.e., at positions not explicitly defined in the sense strand are unmodified or modified nucleotides. For example, the remaining nucleotides, i.e., at positions not explicitly defined in the sense strand can be modified nucleotides selected from the group consisting of 2'-OMe, 2'-F, 2'-H, and an 2'-O—C10-30aliphatic group, optionally provided no more than one modified nucleotide is an 2'-O—C10-30aliphatic group.

[0068] In some embodiments, the remaining nucleotides, i.e., at positions not explicitly defined in the antisense strand are unmodified or modified nucleotides. For example, the remaining nucleotides, i.e., at positions not explicitly defined in the antisense strand can be modified nucleotides. In some embodiments, the remaining nucleotides, i.e., at positions not explicitly defined in the antisense strand can be selected from the group consisting of 2'-OMe, 2'-F, 2'-H, GNA and 3'-RNA, the 3'-RNA being optionally 3'-OH, provided no more than one modified nucleotide is GNA or 3'-RNA.

[0069] In some embodiments, the remaining nucleotides in the sense strand and/or the antisense strand are 2'-OMe nucleotides.

[0070] As described herein, the dsRNA agent can comprise one or more, e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more nucleotides comprising a modified sugar. By a "modified sugar" is meant a sugar other than 2'-deoxy (i.e. 2'-H),

2'-OH, 2'-F or 2'-OMe ribose sugar. Some exemplary nucleotides comprising a modified sugar are locked nucleic acid (LNA), HNA, CeNA, 2'-methoxyethyl, 2'-O-allyl, 2'-C-allyl, 2'-O—N-methylacetamido (2'-O-NMA), a 2'-O-dimethylaminoethoxyethyl(2'-O-DMAEOE), 2'-O-aminopropyl (2'-O-AP), and 2'-ara-F. Accordingly, in some embodiments, the dsRNA agent can comprise one or more, e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more nucleotides independently selected from the group consisting of acyclic nucleotides, locked nucleic acid (LNA), HNA, CeNA, 2'-methoxyethyl, 2'-O-allyl, 2'-C-allyl, 2'-O—N-methylacetamido (2'-O-NMA), a 2'-O-dimethylaminoethoxyethyl(2'-O-DMAEOE), 2'-O-aminopropyl(2'-O-AP), and 2'-ara-F. A nucleotide comprising modified sugar can be present anywhere in the dsRNA molecule. For example, a nucleotide comprising a modified sugar can be present in the sense strand or a nucleotide comprising a modified sugar can be present in the antisense strand. When two or more nucleotides comprising a modified sugar are present in the dsRNA molecule, they can all be in the sense strand, antisense strand or both in the sense and antisense strands.

[0071] In some embodiments, an unmodified nucleotide is a 2'-OH nucleotide comprising an unmodified nucleobase, i.e., adenine, guanine, cytosine, or uracil.

[0072] In some embodiments, the dsRNA can comprise one or more, e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more nucleotides comprising a non-natural nucleobase. By a “non-natural nucleobase” is meant a nucleobase other than adenine, guanine, cytosine, uracil, or thymine. Exemplary non-natural nucleobases include, but are not limited to, inosine, xanthine, hypoxanthine, nubularine, isoguanisine, tubercidine, and substituted or modified analogs of adenine, guanine, cytosine and uracil, such as 2-aminoadenine, 6-methyl and other alkyl derivatives of adenine and guanine, 2-propyl and other alkyl derivatives of adenine and guanine, 5-halouracil and cytosine, 5-propynyl uracil and cytosine, 6-azo uracil, cytosine and thymine, 5-uracil (pseudouracil), 4-thiouracil, 5-halouracil, 5-(2-aminopropyl) uracil, 5-amino allyl uracil, 8-halo, amino, thiol, thioalkyl, hydroxyl and other 8-substituted adenines and guanines, 5-trifluoromethyl and other 5-substituted uracils and cytosines, 7-methylguanine, 5-substituted pyrimidines, 6-azapyrimidines and N-2, N-6 and O-6 substituted purines, including 2-aminopropyladenine, 5-propynyluracil and 5-propynylcytosine, dihydrouracil, 3-deaza-5-azacytosine, 2-aminopurine, 5-alkyluracil, 7-alkylguanine, 5-alkyl cytosine, 7-deazaadenine, N6, N6-dimethyladenine, 2,6-diaminopurine, 5-amino-allyl-uracil, N3-methyluracil, substituted 1,2,4-triazoles, 2-pyridinone, 5-nitroindole, 3-nitropyrrole, 5-methoxyuracil, uracil-5-oxyacetic acid, 5-methoxycarbonylmethyluracil, 5-methyl-2-thiouracil, 5-methoxycarbonylmethyl-2-thiouracil, 5-methylaminomethyl-2-thiouracil, 3-(3-amino-3-carboxypropyl) uracil, 3-methylcytosine, 5-methylcytosine, N4-acetyl cytosine, 2-thiocytosine, N6-methyladenine, N6-isopentyladenine, 2-methylthio-N6-isopentyladenine, N-methylguanines, or O-alkylated bases. Further purines and pyrimidines include those disclosed in U.S. Pat. No. 3,687,808, those disclosed in the Concise Encyclopedia of Polymer Science and Engineering, pages 858-859, Kroschwitz, J. I., ed. John Wiley & Sons, 1990, and those disclosed by Englisch et al., *Angewandte Chemie, International Edition*, 1991, 30, 613.

[0073] In some embodiments, the non-natural nucleobase can be selected from the group consisting of inosine, xan-

thine, hypoxanthine, nubularine, isoguanisine, tubercidine, 2-(halo) adenine, 2-(alkyl) adenine, 2-(propyl) adenine, 2-(amino) adenine, 2-(aminoalkyl) adenine, 2-(aminopropyl) adenine, 2-(methylthio)-N⁶-(isopentenyl) adenine, 6-(alkyl) adenine, 6-(methyl) adenine, 7-(deaza) adenine, 8-(alkenyl) adenine, 8-(alkyl) adenine, 8-(alkynyl) adenine, 8-(amino) adenine, 8-(halo) adenine, 8-(hydroxyl) adenine, 8-(thioalkyl) adenine, 8-(thiol) adenine, N⁶-(isopentyl) adenine, N⁶-(methyl) adenine, N⁶, N⁶-(dimethyl) adenine, 2-(alkyl) guanine, 2-(propyl) guanine, 6-(alkyl) guanine, 6-(methyl) guanine, 7-(alkyl) guanine, 7-(methyl) guanine, 7-(deaza) guanine, 8-(alkyl) guanine, 8-(alkenyl) guanine, 8-(alkynyl) guanine, 8-(amino) guanine, 8-(halo) guanine, 8-(hydroxyl) guanine, 8-(thioalkyl) guanine, 8-(thiol) guanine, N-(methyl) guanine, 2-(thio) cytosine, 3-(deaza)-5-(aza) cytosine, 3-(alkyl) cytosine, 3-(methyl) cytosine, 5-(alkyl) cytosine, 5-(alkynyl) cytosine, 5-(halo) cytosine, 5-(methyl) cytosine, 5-(propynyl) cytosine, 5-(propynyl) cytosine, 5-(trifluoromethyl) cytosine, 6-(azo) cytosine, N⁴-(acetyl) cytosine, 3-(3-amino-3-carboxypropyl) uracil, 2-(thio) uracil, 5-(methyl)-2-(thio) uracil, 5-(methylaminomethyl)-2-(thio) uracil, 4-(thio) uracil, 5-(methyl)-4-(thio) uracil, 5-(methylaminomethyl)-4-(thio) uracil, 5-(methyl)-2,4-(dithio) uracil, 5-(methylaminomethyl)-2,4-(dithio) uracil, 5-(2-aminopropyl) uracil, 5-(alkyl) uracil, 5-(alkynyl) uracil, 5-(allylamino) uracil, 5-(aminoalkyl) uracil, 5-(aminoalkyl) uracil, 5-(guanidiniumalkyl) uracil, 5-(1,3-diazole-1-alkyl) uracil, 5-(cyanoalkyl) uracil, 5-(dialkylaminoalkyl) uracil, 5-(dimethylaminoalkyl) uracil, 5-(halo) uracil, 5-(methoxy) uracil, uracil-5-oxyacetic acid, 5-(methoxycarbonylmethyl)-2-(thio) uracil, 5-(methoxycarbonyl-methyl) uracil, 5-(propynyl) uracil, 5-(propynyl) uracil, 5-(trifluoromethyl) uracil, 6-(azo) uracil, dihydrouracil, N³-(methyl) uracil, 5-uracil (i.e., pseudouracil), 2-(thio) pseudouracil, 4-(thio) pseudouracil, 2,4-(dithio) pseudouracil, 5-(alkyl) pseudouracil, 5-(methyl) pseudouracil, 5-(alkyl)-2-(thio) pseudouracil, 5-(methyl)-2-(thio) pseudouracil, 5-(alkyl)-4-(thio) pseudouracil, 5-(methyl)-4-(thio) pseudouracil, 5-(alkyl)-2,4-(dithio) pseudouracil, 5-(methyl)-2,4-(dithio) pseudouracil, 1-substituted pseudouracil, 1-substituted 2-(thio)-pseudouracil, 1-substituted 4-(thio) pseudouracil, 1-substituted 2,4-(dithio) pseudouracil, 1-(aminocarbonyl-ethylethynyl)-pseudouracil, 1-(aminocarbonyl-ethylethynyl)-2-(thio)-pseudouracil, 1-(aminocarbonyl-ethylethynyl)-4-(thio) pseudouracil, 1-(aminocarbonyl-ethylethynyl)-2,4-(dithio) pseudouracil, 1-(aminoalkylaminocarbonyl-ethylethynyl)-pseudouracil, 1-(aminoalkylamino-carbonyl-ethylethynyl)-2-(thio)-pseudouracil, 1-(aminoalkylaminocarbonyl-ethylethynyl)-4-(thio) pseudouracil, 1-(aminoalkylaminocarbonyl-ethylethynyl)-2,4-(dithio) pseudouracil, 1,3-(diaz)-2-(oxo)-phenoxazin-1-yl, 1-(aza)-2-(thio)-3-(aza)-phenoxazin-1-yl, 1,3-(diaz)-2-(oxo)-phenanthiazin-1-yl, 1-(aza)-2-(thio)-3-(aza)-phenanthiazin-1-yl, 7-substituted 1,3-(diaz)-2-(oxo)-phenoxazin-1-yl, 7-substituted 1-(aza)-2-(thio)-3-(aza)-phenoxazin-1-yl, 7-substituted 1,3-(diaz)-2-(oxo)-phenanthiazin-1-yl, 7-substituted 1-(aza)-2-(thio)-3-(aza)-phenanthiazin-1-yl, 7-(aminoalkylhydroxy)-1,3-(diaz)-2-(oxo)-phenoxazin-1-yl, 7-(aminoalkylhydroxy)-1-(aza)-2-(thio)-3-(aza)-phenoxazin-1-yl, 7-(aminoalkylhydroxy)-1,3-(diaz)-2-(oxo)-phenanthiazin-1-yl, 7-(aminoalkylhydroxy)-1-(aza)-2-(thio)-3-(aza)-phenanthiazin-1-yl, 7-(guanidiniumalkylhydroxy)-1,3-(diaz)-2-(oxo)-phenoxazin-1-yl, 7-(guanidiniumalkylhydroxy)-1-(aza)-2-(thio)-3-(aza)-phenoxazin-1-yl,

7-(guanidiniumalkyl-hydroxy)-1,3-(diaz)-2-(oxo)-phenanthiazin-1-yl, 7-(guanidiniumalkylhydroxy)-1-(aza)-2-(thio)-3-(aza)-phenanthiazin-1-yl, 1,3,5-(triaz)-2,6-(diox)-naphthalene, inosine, xanthine, hypoxanthine, nubularine, tubercidine, isoguanisine, inosinyl, 2-aza-inosinyl, 7-deaza-inosinyl, nitroimidazolyl, nitropyrrolyl, nitrobenzimidazolyl, nitroindazolyl, aminoindolyl, pyrrolopyrimidinyl, 3-(methyl) isocarbostyryl, 5-(methyl) isocarbostyryl, 3-(methyl)-7-(propynyl) isocarbostyryl, 7-(aza) indolyl, 6-(methyl)-7-(aza) indolyl, imidizopyridinyl, 9-(methyl)-imidizopyridinyl, pyrrolopyridinyl, isocarbostyryl, 7-(propynyl) isocarbostyryl, propynyl-7-(aza) indolyl, 2,4,5-(trimethyl)phenyl, 4-(methyl) indolyl, 4,6-(dimethyl) indolyl, phenyl, naphthalenyl, anthracenyl, phenanthracenyl, pyrenyl, stilbenyl, tetracenyl, pentacenyl, difluorotolyl, 4-(fluoro)-6-(methyl)benzimidazole, 4-(methyl)benzimidazole, 6-(azo) thymine, 2-pyridinone, 5-nitroindole, 3-nitropyrrole, 6-(aza)pyrimidine, 2-(amino) purine, 2,6-(diamino) purine, 5-substituted pyrimidines, N²-substituted purines, N⁶-substituted purines, O⁶-substituted purines, substituted 1,2,4-triazoles, and any O-alkylated or N-alkylated derivatives thereof.

[0074] A nucleotide comprising a non-natural nucleobase can be present anywhere in the dsRNA molecule. For example, a nucleotide comprising a non-natural nucleobase can be present in the sense strand or a nucleotide comprising a non-natural nucleobase can be present in the antisense strand. When two or more nucleotides comprising a non-natural nucleobase are present in the dsRNA molecule, they can all be in the sense strand, antisense strand or both in the sense and antisense strands.

[0075] The dsRNA molecule of the invention can further comprise at least one phosphorothioate or methylphosphonate internucleotide linkage. The phosphorothioate or methylphosphonate internucleotide linkage modification may occur on any nucleotide of the sense strand or antisense strand or both in any position of the strand. For instance, the internucleotide linkage modification may occur on every nucleotide on the sense strand and/or antisense strand; each internucleotide linkage modification may occur in an alternating pattern on the sense strand or antisense strand; or the sense strand or antisense strand comprises both internucleotide linkage modifications in an alternating pattern. The alternating pattern of the internucleotide linkage modification on the sense strand may be the same or different from the antisense strand, and the alternating pattern of the internucleotide linkage modification on the sense strand may have a shift relative to the alternating pattern of the internucleotide linkage modification on the antisense strand.

[0076] In some embodiments, the dsRNA molecule comprises the phosphorothioate or methylphosphonate internucleotide linkage modification in the overhang region. For example, the overhang region comprises two nucleotides having a phosphorothioate or methylphosphonate internucleotide linkage between the two nucleotides. Internucleotide linkage modifications also may be made to link the overhang nucleotides with the terminal paired nucleotides within duplex region. For example, at least 2, 3, 4, or all the overhang nucleotides may be linked through phosphorothioate or methylphosphonate internucleotide linkage, and optionally, there may be additional phosphorothioate or methylphosphonate internucleotide linkages linking the overhang nucleotide with a paired nucleotide that is next to the overhang nucleotide. For instance, there may be at least

two phosphorothioate internucleotide linkages between the terminal three nucleotides, in which two of the three nucleotides are overhang nucleotides, and the third is a paired nucleotide next to the overhang nucleotide. Preferably, these terminal three nucleotides may be at the 3'-end of the antisense strand.

[0077] In some embodiments, the sense strand of the dsRNA molecule comprises 1-10 blocks of two to ten phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or 16 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said sense strand is paired with an antisense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0078] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of two phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, or 18 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0079] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of three phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 or 16 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0080] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of four phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 or 14 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0081] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of five phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 or 12 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an

antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0082] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of six phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0083] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of seven phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5, 6, 7 or 8 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0084] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of eight phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3, 4, 5 or 6 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0085] In some embodiments, the antisense strand of the dsRNA molecule comprises two blocks of nine phosphorothioate or methylphosphonate internucleotide linkages separated by 1, 2, 3 or 4 phosphate internucleotide linkages, wherein one of the phosphorothioate or methylphosphonate internucleotide linkages is placed at any position in the oligonucleotide sequence and the said antisense strand is paired with a sense strand comprising any combination of phosphorothioate, methylphosphonate and phosphate internucleotide linkages or an antisense strand comprising either phosphorothioate or methylphosphonate or phosphate linkage.

[0086] In some embodiments, the dsRNA molecule of the invention further comprises one or more phosphorothioate or methylphosphonate internucleotide linkage modification within 1-10 of the termini position(s) of the sense and/or antisense strand. For example, at least 2, 3, 4, 5, 6, 7, 8, 9 or 10 nucleotides may be linked through phosphorothioate or methylphosphonate internucleotide linkage at one end or both ends of the sense and/or antisense strand.

[0087] In some embodiments, the dsRNA molecule of the invention comprises one or more phosphorothioate or methylphosphonate internucleotide linkage modification within 1-10 of the internal region of the duplex of each of the sense and/or antisense strand. For example, at least 2, 3, 4, 5, 6, 7, 8, 9 or 10 nucleotides may be linked through phosphoroth-

ioate methylphosphonate internucleotide linkage at position 8-16 of the duplex region counting from the 5'-end of the sense strand; the dsRNA molecule can optionally further comprise one or more phosphorothioate or methylphosphonate internucleotide linkage modification within 1-10 of the termini position(s).

[0088] In some embodiments, the dsRNA molecule of the invention further comprises one to five phosphorothioate or methylphosphonate internucleotide linkage modification(s) within position 1-5 and one to five phosphorothioate or methylphosphonate internucleotide linkage modification(s) within the last 3 positions of the sense strand (counting from the 5'-end), and one to five phosphorothioate or methylphosphonate internucleotide linkage modification at positions 1 and 2 and one to five phosphorothioate or methylphosphonate internucleotide linkage modification within the last six positions of the antisense strand (counting from the 5'-end).

[0089] In some embodiments, the dsRNA molecule of the invention further comprises one phosphorothioate internucleotide linkage modification within position 1-5 and one phosphorothioate or methylphosphonate internucleotide linkage modification within the last six positions of the sense strand (counting from the 5'-end), and one phosphorothioate internucleotide linkage modification at positions 1 and 2 and two phosphorothioate or methylphosphonate internucleotide linkage modifications within the last six the last six positions of the antisense strand (counting from the 5'-end).

[0090] In some embodiments, the dsRNA molecule of the invention further comprises two phosphorothioate internucleotide linkage modifications within position 1-5 and one phosphorothioate internucleotide linkage modification within the last six positions of the sense strand (counting from the 5'-end), and one phosphorothioate internucleotide linkage modification at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within the last six positions of the antisense strand (counting from the 5'-end).

[0091] In some embodiments, the dsRNA molecule of the invention further comprises two phosphorothioate internucleotide linkage modifications within position 1-5 and two phosphorothioate internucleotide linkage modifications within the last four positions of the sense strand (counting from the 5'-end), and one phosphorothioate internucleotide linkage modification at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within the last six positions of the antisense strand (counting from the 5'-end).

[0092] In some embodiments, the dsRNA molecule of the invention further comprises two phosphorothioate internucleotide linkage modifications within position 1-5 and two phosphorothioate internucleotide linkage modifications within the last four positions of the sense strand (counting from the 5'-end), and one phosphorothioate internucleotide linkage modification at positions 1 and 2 and one phosphorothioate internucleotide linkage modification within the last six positions of the antisense strand (counting from the 5'-end).

[0093] In some embodiments, the dsRNA molecule of the invention further comprises one phosphorothioate internucleotide linkage modification within position 1-5 and one phosphorothioate internucleotide linkage modification within the last four positions of the sense strand (counting from the 5'-end), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phos-

[0110] In some embodiments, the sense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 5' end of the sense strand and the antisense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 5'-end of the antisense strand. For example, the sense strand comprises phosphorothioate linkages between nucleotides 1 and 2, and between nucleotides 2 and 3, counting from 5'-end of the sense strand, and the antisense strand comprises phosphorothioate linkages between nucleotides 1 and 2, and between nucleotides 2 and 3, counting from 5'-end of the antisense strand.

[0112] In some embodiments, compound of the invention comprises a pattern of backbone chiral centers. In some embodiments, a common pattern of backbone chiral centers comprises at least 5 internucleotidic linkages in the Sp configuration. In some embodiments, a common pattern of backbone chiral centers comprises at least 6 internucleotidic linkages in the Sp configuration. In some embodiments, a common pattern of backbone chiral centers comprises at least 7 internucleotidic linkages in the Sp configuration. In some embodiments, a common pattern of backbone chiral centers comprises at least 8 internucleotidic linkages in the Sp configuration. In some embodiments, a common pattern of backbone chiral centers comprises at least 9 internucleotidic linkages in the Sp configuration. In some embodiments, a common pattern of backbone chiral centers comprises at least 10 internucleotidic linkages in the Sp configuration. In some embodiments, a common pattern of backbone chiral centers comprises at least 11 internucleotidic linkages in the Sp configuration. In some embodiments, a common pattern

[illegible]

more than 7 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises at least 12 internucleotidic linkages in the Sp configuration, and no more than 6 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises at least 13 internucleotidic linkages in the Sp configuration, and no more than 6 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises at least 14 internucleotidic linkages in the Sp configuration, and no more than 5 internucleotidic linkages which are not chiral. In some embodiments, a common pattern of backbone chiral centers comprises at least 15 internucleotidic linkages in the Sp configuration, and no more than 4 internucleotidic linkages which are not chiral. In some embodiments, the internucleotidic linkages in the Sp configuration are optionally contiguous or not contiguous. In some embodiments, the internucleotidic linkages in the Rp configuration are optionally contiguous or not contiguous. In some embodiments, the internucleotidic linkages which are not chiral are optionally contiguous or not contiguous.

[0113] In some embodiments, compound of the invention comprises a block is a stereochemistry block. In some embodiments, a block is an Rp block in that each internucleotidic linkage of the block is Rp. In some embodiments, a 5'-block is an Rp block. In some embodiments, a 3'-block is an Rp block. In some embodiments, a block is an Sp block in that each internucleotidic linkage of the block is Sp. In some embodiments, a 5'-block is an Sp block. In some embodiments, a 3'-block is an Sp block. In some embodiments, provided oligonucleotides comprise both Rp and Sp blocks. In some embodiments, provided oligonucleotides comprise one or more Rp but no Sp blocks. In some embodiments, provided oligonucleotides comprise one or more Sp but no Rp blocks. In some embodiments, provided oligonucleotides comprise one or more PO blocks wherein each internucleotidic linkage in a natural phosphate linkage.

[0114] In some embodiments, compound of the invention comprises a 5'-block is an Sp block wherein each sugar moiety comprises a 2'-fluoro modification. In some embodiments, a 5'-block is an Sp block wherein each of internucleotidic linkage is a modified internucleotidic linkage and each sugar moiety comprises a 2'-fluoro modification. In some embodiments, a 5'-block is an Sp block wherein each of internucleotidic linkage is a phosphorothioate linkage and each sugar moiety comprises a 2'-fluoro modification. In some embodiments, a 5'-block comprises 4 or more nucleoside units. In some embodiments, a 5'-block comprises 5 or more nucleoside units. In some embodiments, a 5'-block comprises 6 or more nucleoside units. In some embodiments, a 5'-block comprises 7 or more nucleoside units. In some embodiments, a 3'-block is an Sp block wherein each sugar moiety comprises a 2'-fluoro modification. In some embodiments, a 3'-block is an Sp block wherein each of internucleotidic linkage is a modified internucleotidic linkage and each sugar moiety comprises a 2'-fluoro modification. In some embodiments, a 3'-block is an Sp block wherein each of internucleotidic linkage is a phosphorothioate linkage and each sugar moiety comprises a 2'-fluoro modification. In some embodiments, a 3'-block comprises 4 or more nucleoside units. In some embodiments, a 3'-block comprises 5 or more nucleoside units. In some embodi-

ments, a 3'-block comprises 6 or more nucleoside units. In some embodiments, a 3'-block comprises 7 or more nucleoside units.

[0115] In some embodiments, compound of the invention comprises a type of nucleoside in a region or an oligonucleotide is followed by a specific type of internucleotidic linkage, e.g., natural phosphate linkage, modified internucleotidic linkage, Rp chiral internucleotidic linkage, Sp chiral internucleotidic linkage, etc. In some embodiments, A is followed by Sp. In some embodiments, A is followed by Rp. In some embodiments, A is followed by natural phosphate linkage (PO). In some embodiments, U is followed by Sp. In some embodiments, U is followed by Rp. In some embodiments, U is followed by natural phosphate linkage (PO). In some embodiments, C is followed by Sp. In some embodiments, C is followed by Rp. In some embodiments, C is followed by natural phosphate linkage (PO). In some embodiments, G is followed by Sp. In some embodiments, G is followed by Rp. In some embodiments, G is followed by natural phosphate linkage (PO). In some embodiments, C and U are followed by Sp. In some embodiments, C and U are followed by Rp. In some embodiments, C and U are followed by natural phosphate linkage (PO). In some embodiments, A and G are followed by Sp. In some embodiments, A and G are followed by Rp.

[0116] Various publications describe multimeric siRNAs which can all be used with the dsRNA of the invention. Such publications include WO2007/091269, U.S. Pat. No. 7,858,769, WO2010/141511, WO2007/117686, WO2009/014887 and WO2011/031520 which are hereby incorporated by their entirety.

Ligands

[0117] A wide variety of entities can be coupled to the dsRNA agents described herein. Preferred moieties are ligands, which are coupled, preferably covalently, either directly or indirectly via an intervening tether. Generally, a ligand alters the distribution, targeting or lifetime of the molecule, e.g., a dsRNA described herein, into which it is incorporated. In some embodiments a ligand provides an enhanced affinity for a selected target, e.g., molecule, cell or cell type, compartment, receptor e.g., a cellular or organ compartment, tissue, organ or region of the body, as, e.g., compared to a species absent such a ligand. Ligands providing enhanced affinity for a selected target are also termed targeting ligands herein.

[0118] Some ligands can have endosomolytic properties. The endosomolytic ligands promote the lysis of the endosome and/or transport of the composition of the invention, or its components, from the endosome to the cytoplasm of the cell. The endosomolytic ligand may be a polyanionic peptide or peptidomimetic which shows pH-dependent membrane activity and fusogenicity. In some embodiments, the endosomolytic ligand assumes its active conformation at endosomal pH. The "active" conformation is that conformation in which the endosomolytic ligand promotes lysis of the endosome and/or transport of the composition of the invention, or its components, from the endosome to the cytoplasm of the cell. Exemplary endosomolytic ligands include the GALA peptide (Subbarao et al., *Biochemistry*, 1987, 26:2964-2972, which is incorporated by reference in its entirety), the EALA peptide (Vogel et al., *J. Am. Chem. Soc.*, 1996, 118:1581-1586, which is incorporated by reference in its entirety), and their derivatives (Turk et al., *Biochem. Biophys. Acta*, 2002,

1559:56-68, which is incorporated by reference in its entirety). In some embodiments, the endosomolytic component may contain a chemical group (e.g., an amino acid) which will undergo a change in charge or protonation in response to a change in pH. The endosomolytic component may be linear or branched.

[0119] Ligands can improve transport, hybridization, and specificity properties and can also improve nuclease resistance of the resultant natural or modified oligoribonucleotide, or a polymeric molecule comprising any combination of monomers described herein and/or natural or modified ribonucleotides.

[0120] Ligands in general can include therapeutic modifiers, e.g., for enhancing uptake; diagnostic compounds or reporter groups e.g., for monitoring distribution; cross-linking agents; and nuclease-resistance conferring moieties. General examples include lipids, steroids, vitamins, sugars, proteins, peptides, polyamines, and peptide mimics.

[0121] Ligands can include a naturally occurring substance, such as a protein (e.g., human serum albumin (HSA), low-density lipoprotein (LDL), high-density lipoprotein (HDL), or globulin); a carbohydrate (e.g., a dextran, pullulan, chitin, chitosan, inulin, cyclodextrin or hyaluronic acid); or a lipid. The ligand may also be a recombinant or synthetic molecule, such as a synthetic polymer, e.g., a synthetic polyamino acid, an oligonucleotide (e.g. an aptamer). Examples of polyamino acids include polyamino acid is a polylysine (PLL), poly L-aspartic acid, poly L-glutamic acid, styrene-maleic acid anhydride copolymer, poly(L-lactide-co-glycolide) copolymer, divinyl ether-maleic anhydride copolymer, N-(2-hydroxypropyl) methacrylamide copolymer (HMPA), polyethylene glycol (PEG), polyvinyl alcohol (PVA), polyurethane, poly(2-ethylacrylic acid), N-isopropylacrylamide polymers, or polyphosphazene. Example of polyamines include: polyethylenimine, polylysine (PLL), spermine, spermidine, polyamine, pseudopeptide-polyamine, peptidomimetic polyamine, dendrimer polyamine, arginine, amidine, protamine, cationic lipid, cationic porphyrin, quaternary salt of a polyamine, or an alpha helical peptide.

[0122] Ligands can also include targeting groups, e.g., a cell or tissue targeting agent, e.g., a lectin, glycoprotein, lipid or protein, e.g., an antibody, nanobody, or portion of an antibody of nanobody that binds to a specified cell type such as a kidney cell or a cell of the blood-brain barrier.

[0123] A targeting group can be a thyrotropin, melanotropin, lectin, glycoprotein, surfactant protein A, Mucin carbohydrate, multivalent lactose, multivalent galactose, N-acetyl-galactosamine, N-acetyl-glucosamine multivalent mannose, multivalent fucose, glycosylated polyamino acids, multivalent galactose, transferrin-targeting group, bisphosphonate, polyglutamate, polyaspartate, a lipid, cholesterol, a steroid, bile acid, folate, vitamin B12, biotin, an RGD peptide, an RGD peptide mimetic or an aptamer.

[0124] Other examples of ligands include dyes, intercalating agents (e.g. acridines), cross-linkers (e.g. psoralen, mitomycin C), porphyrins (TPPC4, texaphyrin, Sapphyrin), polycyclic aromatic hydrocarbons (e.g., phenazine, dihydrophenazine), artificial endonucleases or a chelating agent (e.g. EDTA), lipophilic molecules, e.g., cholesterol, cholic acid, adamantane acetic acid, 1-pyrene butyric acid, dihydrotestosterone, 1,3-Bis —O (hexadecyl)glycerol, geranyloxyhexyl group, hexadecylglycerol, borneol, menthol, 1,3-propanediol, heptadecyl group, palmitic acid, myristic acid,

O3-(oleoyl) lithocholic acid, O3-(oleoyl) cholenic acid, dimethoxytrityl, or phenoxazine) and peptide conjugates (e.g., antennapedia peptide, Tat peptide), alkylating agents, phosphate, amino, mercapto, PEG (e.g., PEG-40K), MPEG, [MPEG]₂, polyamino, alkyl, substituted alkyl, radiolabeled markers, enzymes, haptens (e.g. biotin), transport/absorption facilitators (e.g., aspirin, vitamin E, folic acid), synthetic ribonucleases (e.g., imidazole, bisimidazole, histamine, imidazole clusters, acridine-imidazole conjugates, Eu3+ complexes of tetraazamacrocycles), dinitrophenyl, HRP, or AP.

[0125] Ligands can be proteins, e.g., glycoproteins, or peptides, e.g., molecules having a specific affinity for a co-ligand, or antibodies e.g., an antibody, that binds to a specified cell type such as a cancer cell, endothelial cell, or bone cell. Ligands may also include hormones and hormone receptors. They can also include non-peptide species, such as lipids, lectins, carbohydrates, vitamins, cofactors, multivalent lactose, multivalent galactose, N-acetyl-galactosamine, N-acetyl-glucosamine multivalent mannose, multivalent fucose, or aptamers. The ligand can be, for example, a lipopolysaccharide, an activator of p38 MAP kinase, or an activator of NF-κB.

[0126] The ligand can be a substance, e.g., a drug, which can increase the uptake of the iRNA agent into the cell, for example, by disrupting the cell's cytoskeleton, e.g., by disrupting the cell's microtubules, microfilaments, and/or intermediate filaments. The drug can be, for example, taxon, vincristine, vinblastine, cytochalasin, nocodazole, japlakinolide, latrunculin A, phalloidin, swinholide A, indanocine, or myoservin.

[0127] The ligand can increase the uptake of the dsRNA into the cell by activating an inflammatory response, for example. Exemplary ligands that would have such an effect include tumor necrosis factor alpha (TNF-alpha), interleukin-1 beta, or gamma interferon.

[0128] In some embodiments, the ligand is a lipid or lipid-based molecule. Such a lipid or lipid-based molecule preferably binds a serum protein, e.g., human serum albumin (HSA). An HSA binding ligand allows for distribution of the conjugate to a target tissue, e.g., a non-kidney target tissue of the body. For example, the target tissue can be the liver, including parenchymal cells of the liver. Other molecules that can bind HSA can also be used as ligands. For example, naproxen or aspirin can be used. A lipid or lipid-based ligand can (a) increase resistance to degradation of the conjugate, (b) increase targeting or transport into a target cell or cell membrane, and/or (c) can be used to adjust binding to a serum protein, e.g., HSA. A lipid based ligand can be used to modulate, e.g., control the binding of the conjugate to a target tissue. For example, a lipid or lipid-based ligand that binds to HSA more strongly will be less likely to be targeted to the kidney and therefore less likely to be cleared from the body. A lipid or lipid-based ligand that binds to HSA less strongly can be used to target the conjugate to the kidney.

[0129] In a preferred embodiment, the lipid based ligand binds HSA. Preferably, it binds HSA with a sufficient affinity such that the conjugate will be preferably distributed to a non-kidney tissue. However, it is preferred that the affinity not be so strong that the HSA-ligand binding cannot be reversed.

[0130] In another preferred embodiment, the lipid based ligand binds HSA weakly or not at all, such that the conjugate will be preferably distributed to the kidney. Other

moieties that target to kidney cells can also be used in place of or in addition to the lipid based ligand.

[0131] In some embodiments, the ligand is a moiety, e.g., a vitamin, which is taken up by a target cell, e.g., a proliferating cell. These are particularly useful for treating disorders characterized by unwanted cell proliferation, e.g., of the malignant or non-malignant type, e.g., cancer cells. Exemplary vitamins include vitamin A, E, and K. Other exemplary vitamins include B vitamins, e.g., folic acid, B12, riboflavin, biotin, pyridoxal or other vitamins or nutrients taken up by cancer cells. Also included are HAS, low density lipoprotein (LDL) and high-density lipoprotein (HDL).

[0132] In another aspect, the ligand is a cell-permeation agent, preferably a helical cell-permeation agent. Preferably, the agent is amphipathic. An exemplary agent is a peptide such as tat or antennapedia. If the agent is a peptide, it can be modified, including a peptidylmimetic, invertomers, non-peptide or pseudo-peptide linkages, and use of D-amino acids. The helical agent is preferably an alpha-helical agent, which preferably has a lipophilic and a lipophobic phase.

[0133] The ligand can be a peptide or peptidomimetic. A peptidomimetic (also referred to herein as an oligopeptidomimetic) is a molecule capable of folding into a defined three-dimensional structure similar to a natural peptide. The peptide or peptidomimetic moiety can be about 5-50 amino acids long, e.g., about 5, 10, 15, 20, 25, 30, 35, 40, 45, or 50 amino acids long. A peptide or peptidomimetic can be, for example, a cell permeation peptide, cationic peptide, amphipathic peptide, or hydrophobic peptide (e.g., consisting primarily of Tyr, Trp or Phe). The peptide moiety can be a dendrimer peptide, constrained peptide or cross-linked peptide. In another alternative, the peptide moiety can include a hydrophobic membrane translocation sequence (MTS). An exemplary hydrophobic MTS-containing peptide is RFGF having the amino acid sequence AAVALLPAVL-LALLAP (SEQ ID NO: 1). An RFGF analogue (e.g., amino acid sequence AALLPVLLAAP (SEQ ID NO: 2)) containing a hydrophobic MTS can also be a targeting moiety. The peptide moiety can be a "delivery" peptide, which can carry large polar molecules including peptides, oligonucleotides, and protein across cell membranes. For example, sequences from the HIV Tat protein (GRKKRRQRRRPPQ (SEQ ID NO: 3)) and the *Drosophila* Antennapedia protein (RQIKIWFQNRRMKWKK (SEQ ID NO: 4)) have been found to be capable of functioning as delivery peptides. A peptide or peptidomimetic can be encoded by a random sequence of DNA, such as a peptide identified from a phage-display library, or one-bead-one-compound (OBOC) combinatorial library (Lam et al., Nature, 354:82-94, 1991, which is incorporated by reference in its entirety). Preferably the peptide or peptidomimetic tethered to an iRNA agent via an incorporated monomer unit is a cell targeting peptide such as an arginine-glycine-aspartic acid (RGD)-peptide, or RGD mimic. A peptide moiety can range in length from about 5 amino acids to about 40 amino acids. The peptide moieties can have a structural modification, such as to increase stability or direct conformational properties. Any of the structural modifications described below can be utilized. An RGD peptide moiety can be used to target a tumor cell, such as an endothelial tumor cell or a breast cancer tumor cell (Zitzmann et al., Cancer Res., 62:5139-43, 2002, which is incorporated by reference in its entirety). An RGD peptide can facilitate targeting of an iRNA agent to tumors of a variety of other tissues, including the lung, kidney, spleen, or

liver (Aoki et al., Cancer Gene Therapy 8:783-787, 2001, which is incorporated by reference in its entirety). Preferably, the RGD peptide will facilitate targeting of an iRNA agent to the kidney. The RGD peptide can be linear or cyclic, and can be modified, e.g., glycosylated or methylated to facilitate targeting to specific tissues. For example, a glycosylated RGD peptide can deliver an iRNA agent to a tumor cell expressing $\alpha v \beta_3$ (Haubner et al., Jour. Nucl. Med., 42:326-336, 2001, which is incorporated by reference in its entirety). Peptides that target markers enriched in proliferating cells can be used. For example, RGD containing peptides and peptidomimetics can target cancer cells, in particular cells that exhibit an integrin. Thus, one could use RGD peptides, cyclic peptides containing RGD, RGD peptides that include D-amino acids, as well as synthetic RGD mimics. In addition to RGD, one can use other moieties that target the integrin ligand. Generally, such ligands can be used to control proliferating cells and angiogenesis. Preferred conjugates of this type ligands that targets PECAM-1, VEGF, or other cancer gene, e.g., a cancer gene described herein.

[0134] A "cell permeation peptide" is capable of permeating a cell, e.g., a microbial cell, such as a bacterial or fungal cell, or a mammalian cell, such as a human cell. A microbial cell-permeating peptide can be, for example, an α -helical linear peptide (e.g., LL-37 or Ceropin P1), a disulfide bond-containing peptide (e.g., α -defensin, β -defensin or bactenecin), or a peptide containing only one or two dominating amino acids (e.g., PR-39 or indolicidin). A cell permeation peptide can also include a nuclear localization signal (NLS). For example, a cell permeation peptide can be a bipartite amphipathic peptide, such as MPG, which is derived from the fusion peptide domain of HIV-1 gp41 and the NLS of SV40 large T antigen (Simeoni et al., Nucl. Acids Res. 31:2717-2724, 2003, which is incorporated by reference in its entirety).

[0135] In some embodiments, a targeting peptide can be an amphipathic α -helical peptide. Exemplary amphipathic α -helical peptides include, but are not limited to, cecropins, lycotoxins, paradaxins, buforin, CPF, bombinin-like peptide (BLP), cathelicidins, ceratotoxins, *S. clava* peptides, hagfish intestinal antimicrobial peptides (HFIAPs), magainins, brevinins-2, dermaseptins, melittins, pleurocidin, H_2A peptides, *Xenopus* peptides, esculentin-1, and caerins. A number of factors will preferably be considered to maintain the integrity of helix stability. For example, a maximum number of helix stabilization residues will be utilized (e.g., leu, ala, or lys), and a minimum number of helix destabilization residues will be utilized (e.g., proline, or cyclic monomeric units). The capping residue will be considered (for example Gly is an exemplary N-capping residue and/or C-terminal amidation can be used to provide an extra H-bond to stabilize the helix. Formation of salt bridges between residues with opposite charges, separated by $i \pm 3$, or $i \pm 4$ positions can provide stability. For example, cationic residues such as lysine, arginine, homo-arginine, ornithine or histidine can form salt bridges with the anionic residues glutamate or aspartate.

[0136] Peptide and peptidomimetic ligands include those having naturally occurring or modified peptides, e.g., D or L peptides; a, B, or y peptides; N-methyl peptides; azapeptides; peptides having one or more amide, i.e., peptide, linkages replaced with one or more urea, thiourea, carbamate, or sulfonyl urea linkages; or cyclic peptides.

[0137] The targeting ligand can be any ligand that is capable of targeting a specific receptor. Examples are: folate, GalNAc, galactose, mannose, mannose-6P, clusters of sugars such as GalNAc cluster, mannose cluster, galactose cluster, or an aptamer. A cluster is a combination of two or more sugar units. The targeting ligands also include integrin receptor ligands, Chemokine receptor ligands, transferrin, biotin, serotonin receptor ligands, PSMA, endothelin, GCP II, somatostatin, LDL and HDL ligands. The ligands can also be based on nucleic acid, e.g., an aptamer. The aptamer can be unmodified or have any combination of modifications disclosed herein.

[0138] Endosomal release agents include imidazoles, poly or oligoimidazoles, PEIs, peptides, fusogenic peptides, polycarboxylates, polycations, masked oligo or poly cations or anions, acetals, polyacetals, ketals/polyketals, orthoesters, polymers with masked or unmasked cationic or anionic charges, dendrimers with masked or unmasked cationic or anionic charges.

[0139] PK modulator stands for pharmacokinetic modulator. PK modulator include lipophiles, bile acids, steroids, phospholipid analogues, peptides, protein binding agents, PEG, vitamins etc. Exemplary PK modulator include, but are not limited to, cholesterol, fatty acids, cholic acid, lithocholic acid, dialkylglycerides, diacylglyceride, phospholipids, sphingolipids, naproxen, ibuprofen, vitamin E, biotin etc. Oligonucleotides that comprise a number of phosphorothioate linkages are also known to bind to serum protein, thus short oligonucleotides, e.g. oligonucleotides of about 5 bases, 10 bases, 15 bases or 20 bases, comprising multiple of phosphorothioate linkages in the backbone are also amenable to the present invention as ligands (e.g. as PK modulating ligands).

[0140] In addition, aptamers that bind serum components (e.g. serum proteins) are also amenable to the present invention as PK modulating ligands.

[0141] Other ligand conjugates amenable to the invention are described in U.S. patent application Ser. No. 10/916,185, filed Aug. 10, 2004; U.S. Ser. No. 10/946,873, filed Sep. 21, 2004; USSN: U.S. Ser. No. 10/833,934, filed Aug. 3, 2007; USSN: U.S. Ser. No. 11/115,989 filed Apr. 27, 2005 and USSN: U.S. Ser. No. 11/944,227 filed Nov. 21, 2007, which are incorporated by reference in their entireties for all purposes.

[0142] In some embodiments, the dsRNA molecule can comprise two or more, e.g., 2, 3, 4 or 5 ligands. When two or more ligands are present, the ligands can all have same properties, all have different properties or some ligands have the same properties while others have different properties. For example, a ligand can have targeting properties, have endosomolytic activity or have PK modulating properties. In a preferred embodiment, all the ligands have different properties.

[0143] In some embodiments the dsRNA molecule comprises two ligands. For example, the sense strand of the dsRNA molecule comprises a first ligand attached at the 3'-end of the sense strand and a second ligand attached at the 5'-end of the sense strand. In some embodiments, the dsRNA molecule comprises two ligands linked to the sense strand, where the first ligand comprises an inverted abasic nucleotide (i.e., an abasic nucleotide linked via 3'→3' linkage) and the second ligand comprises an ASGPR ligand.

[0144] Ligands can be coupled to the dsRNA at various places, for example, 3'-end, 5'-end, and/or at an internal

position of the sense and/or antisense strand. In preferred embodiments, the ligand is attached to the sense and/or antisense strand of the dsRNA via a linker or tether. The ligand or tethered ligand can be present on a monomer when said monomer is incorporated into the growing strand. In some embodiments, the ligand may be incorporated via coupling to a "precursor" monomer after said "precursor" monomer has been incorporated into the growing strand. For example, a monomer having, e.g., an amino-terminated tether (i.e., having no associated ligand), e.g., TAP-(CH₂)_n, NH₂ may be incorporated into a growing oligonucleotide strand. In a subsequent operation, i.e., after incorporation of the precursor monomer into the strand, a ligand having an electrophilic group, e.g., a pentafluorophenyl ester or aldehyde group, can subsequently be attached to the precursor monomer by coupling the electrophilic group of the ligand with the terminal nucleophilic group of the precursor monomer's tether.

[0145] In another example, a monomer having a chemical group suitable for taking part in Click Chemistry reaction may be incorporated e.g., an azide or alkyne terminated tether/linker. In a subsequent operation, i.e., after incorporation of the precursor monomer into the strand, a ligand having complementary chemical group, e.g. an alkyne or azide can be attached to the precursor monomer by coupling the alkyne and the azide together.

[0146] The ligands can be attached to one or both strands. In some embodiments, a dsRNA described herein comprises a ligand conjugated to the sense strand. In some embodiments, a dsRNA described herein comprises a ligand conjugated to the antisense strand.

[0147] In some embodiments, the ligand is conjugated to the sense strand. As described herein, the ligand can be conjugated at the 3'-end, 5'-end or at an internal position of the sense strand. In some embodiments, the ligand is conjugated to the 3'-end of the sense strand. In some embodiments, the ligand is conjugated to the 5'-end of the sense strand. In some embodiments, the ligand is conjugated at an internal position of the sense strand. In other words, the ligand is conjugated to a non-terminal nucleotide of the sense strand. It is noted that the ligand can be conjugated to a nucleobase, sugar moiety or internucleotide linkage of the sense strand.

[0148] In some embodiments, the ligand is conjugated at the 2'-position of a nucleotide in the sense strand. For example, the ligand is conjugated at the 2'-position of a nucleotide at an internal, i.e., non-terminal position of the sense strand.

[0149] In some embodiments, ligand can be conjugated to nucleobases, sugar moieties, or internucleosidic linkages of nucleic acid molecules. Conjugation to purine nucleobases or derivatives thereof can occur at any position including, endocyclic and exocyclic atoms. In some embodiments, the 2-, 6-, 7-, or 8-positions of a purine nucleobase are attached to a conjugate moiety. Conjugation to pyrimidine nucleobases or derivatives thereof can also occur at any position. In some embodiments, the 2-, 5-, and 6-positions of a pyrimidine nucleobase can be substituted with a conjugate moiety. Conjugation to sugar moieties of nucleosides can occur at any carbon atom. Example carbon atoms of a sugar moiety that can be attached to a conjugate moiety include the 2', 3', and 5' carbon atoms. The 1' position can also be attached to a conjugate moiety, such as in an abasic residue. Internucleosidic linkages can also bear conjugate moieties.

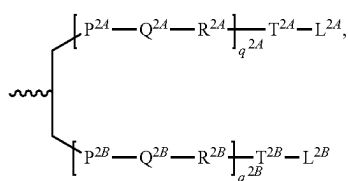
For phosphorus-containing linkages (e.g., phosphodiester, phosphorothioate, phosphorodithioate, phosphoramidate, and the like), the conjugate moiety can be attached directly to the phosphorus atom or to an O, N, or S atom bound to the phosphorus atom. For amine- or amide-containing inter-nucleosidic linkages (e.g., PNA), the conjugate moiety can be attached to the nitrogen atom of the amine or amide or to an adjacent carbon atom.

[0150] In some embodiments, the ligand is conjugated to the sense strand. As described herein, the ligand can be conjugated at the 3'-end, 5'-end or at an internal position of the sense strand. In some embodiments, the ligand is conjugated to the 3'-end of the sense strand. Further, the ligand can be conjugated to a nucleobase, sugar moiety or inter-nucleotide linkage of the sense strand.

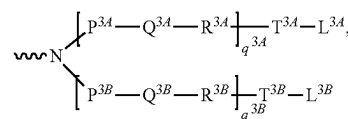
[0151] Any suitable ligand in the field of RNA interference may be used, although the ligand is typically a carbohydrate e.g. monosaccharide (such as GalNAc), disaccharide, trisaccharide, tetrasaccharide, polysaccharide.

[0152] Linkers that conjugate the ligand to the nucleic acid include those discussed above. For example, the ligand can be one or more carbohydrates, e.g., GalNAc (N-acetylgalactosamine) derivatives attached through a monovalent, bivalent or trivalent branched linker.

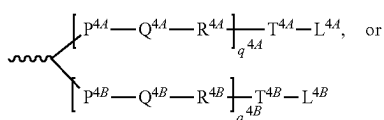
[0153] In some embodiments, the dsRNA of the invention is conjugated to a bivalent and trivalent branched linkers include the structures shown in any of Formula (IV)-(VII):



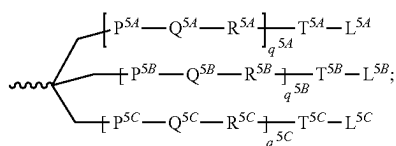
Formula (IV)



Formula (V)



Formula (VI)



Formula (VII)

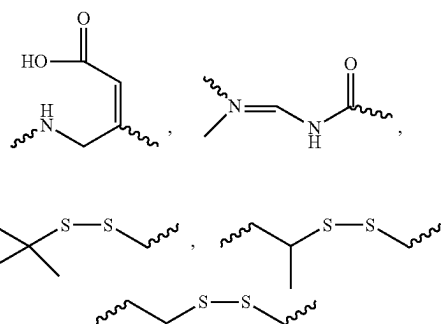
wherein:

[0154] q^{2A} , q^{2B} , q^{3A} , q^{3B} , q^{4A} , q^{4B} , q^{5A} , q^{5B} and q^{5C} represent independently for each occurrence 0-20 and wherein the repeating unit can be the same or different;

[0155] p^{2A} , p^{2B} , p^{3A} , p^{3B} , p^{4A} , p^{4B} , p^{5A} , p^{5B} , p^{5C} , T^{2A} , T^{2B} , T^{3A} , T^{3B} , T^{4A} , T^{4B} , T^{5A} , T^{5B} , T^{5C} are each independently for each occurrence absent, CO, NH, O, S, OC(O), NHC(O), CH₂, CH₂NH or CH₂O;

[0156] Q^{2A} , Q^{2B} , Q^{3A} , Q^{3B} , Q^{4A} , Q^{4B} , Q^{5A} , Q^{5B} , Q^{5C} are independently for each occurrence absent, alkylene, substituted alkylene wherein one or more methylenes can be interrupted or terminated by one or more of O, S, S(O), SO₂, N(RN), C(R')=C(R''), C=C or C(O);

[0157] R^{2A} , R^{2B} , R^{3A} , R^{3B} , R^{4A} , R^{4B} , R^{5A} , R^{5B} , R^{5C} are each independently for each occurrence absent, NH, O, S, CH₂, C(O) O, C(O) NH, NHCH(R^a) C(O), -C(O)-CH(R^a)-NH-, CO, CH=N-O,

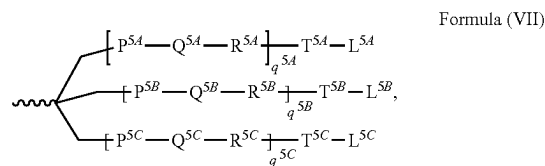


or heterocyclyl;

[0158] L^{2A} , L^{2B} , L^{3A} , L^{3B} , L^{4A} , L^{4B} , L^{5A} , L^{5B} and L^{5C} represent the ligand; i.e. each independently for each occurrence a monosaccharide (such as GalNAc), disaccharide, trisaccharide, tetrasaccharide, oligosaccharide, or polysaccharide; and

[0159] R^a is H or amino acid side chain.

[0160] Trivalent conjugating GalNAc derivatives are particularly useful for use with dsRNA agents described herein for inhibiting the expression of a target gene, such as those of Formula (VII):

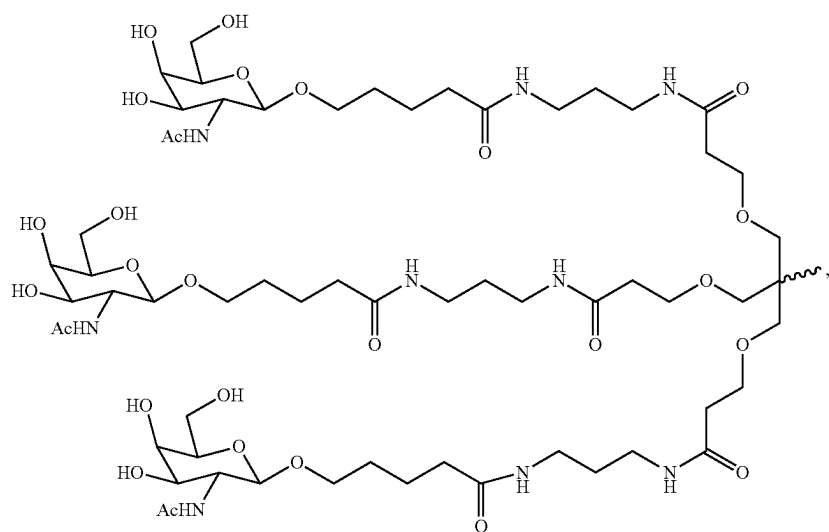


Formula (VII)

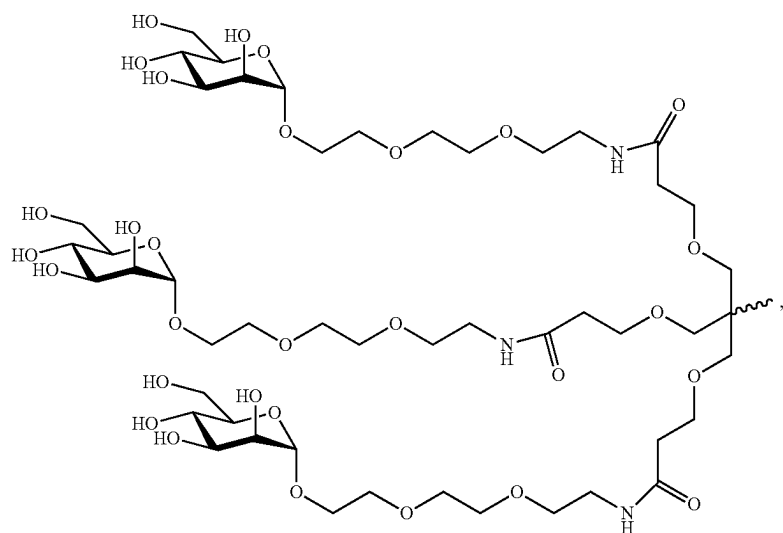
wherein L^{5A} , L^{5B} and L^{5C} represent a monosaccharide, such as GalNAc derivative.

[0161] Examples of suitable bivalent and trivalent branched linker groups conjugating GalNAc derivatives include, but are not limited to, the following compounds:

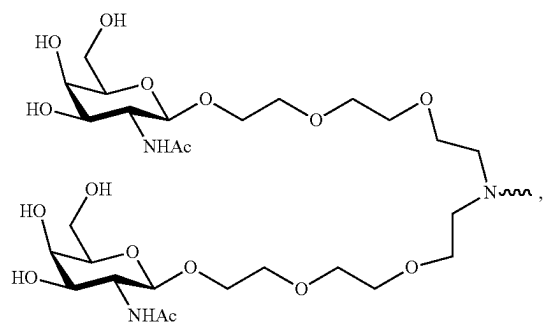
Ligand 1



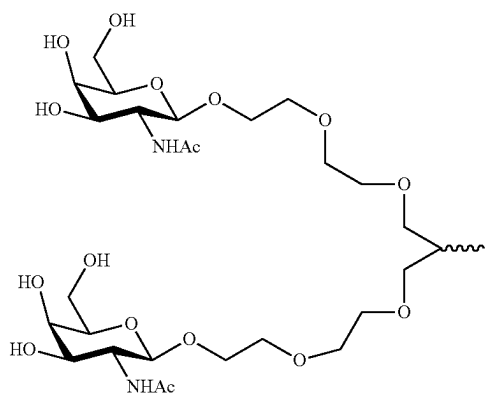
Ligand 2



Ligand 3

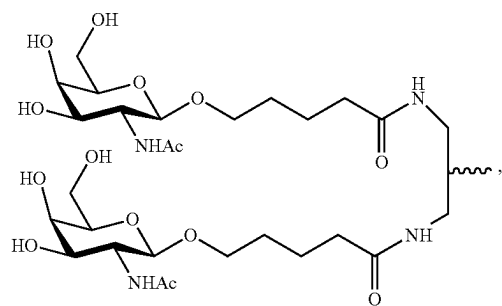


Ligand 4

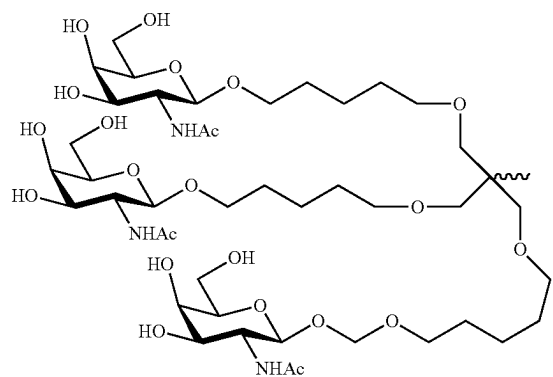


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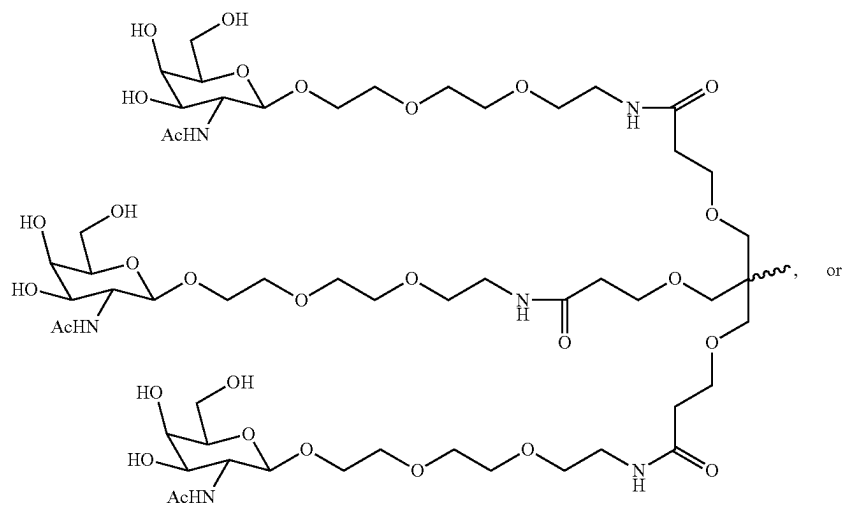
Ligand 5



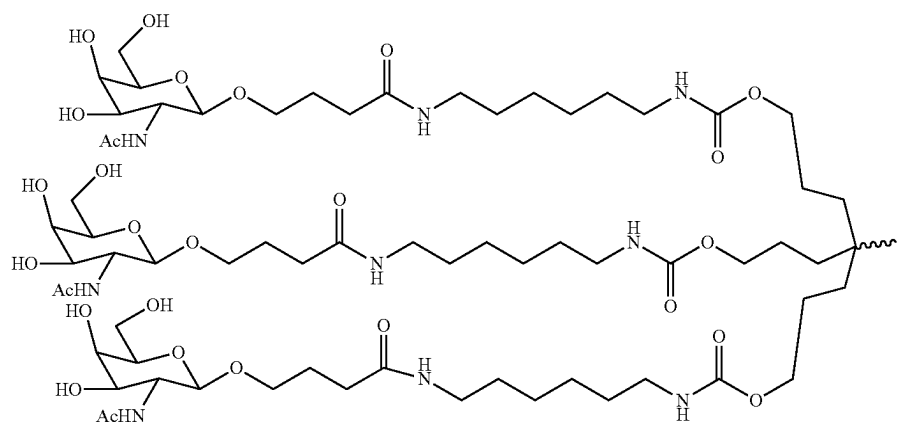
Ligand 6

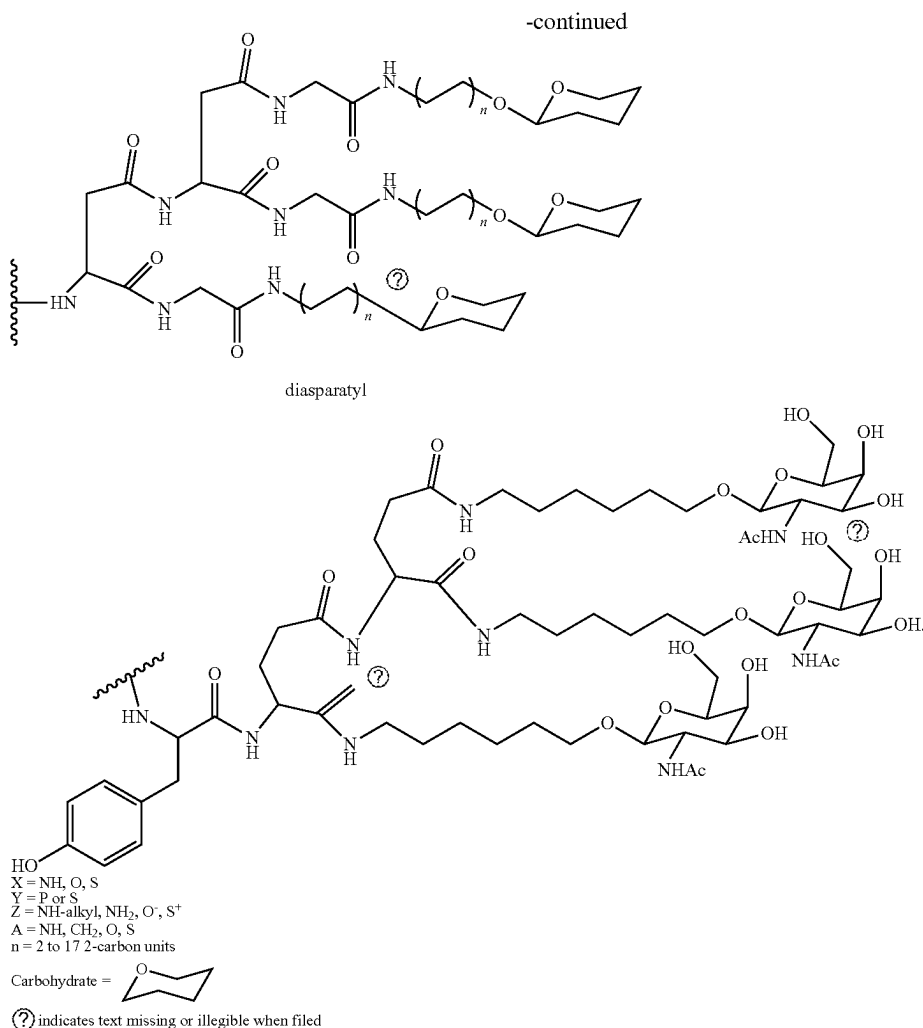


Ligand 7



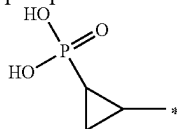
Ligand 8





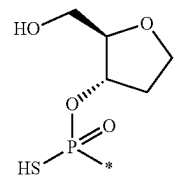
[0165] In some embodiments of any one of the aspects, the antisense strand comprises a phosphoryl analog or phosphate mimic at the 5'-terminus. In some embodiments, the antisense strand comprises an alkenylphosphonates, i.e., a vinyl phosphonate at the 5'-terminus. For example, the antisense strand comprises a 5'-E-vinyl phosphonate.

[0166] In some embodiments, the antisense strand comprises a cyclopropylphosphonate at the

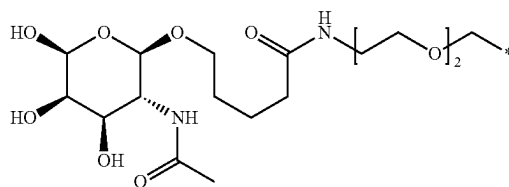


5'-terminus. For example, the antisense comprises at the 5'-terminus, where * is a bond to C5 position of the nucleotide at the 5'-terminus.

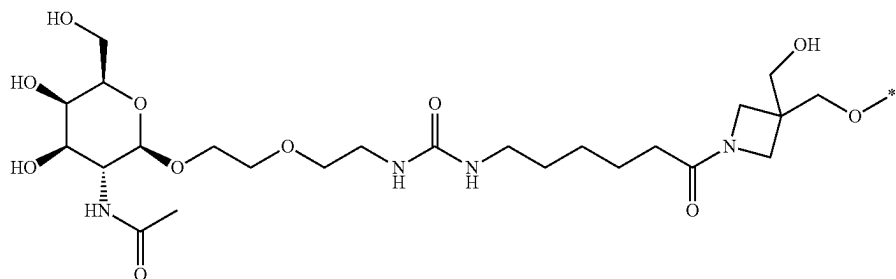
[0167] In some embodiments of any one of the aspects, at least one of the strands, e.g., the sense and/or the antisense strand of the double-stranded RNA comprises a monomer or ligand selected from the following:



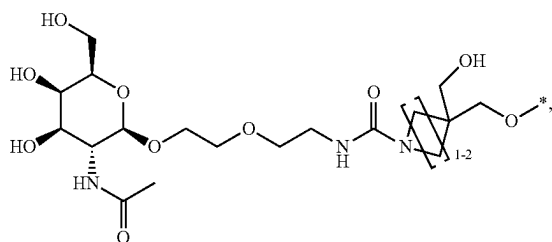
where * is a bond to a 5', 3'-terminal hydroxyl group of the strand;



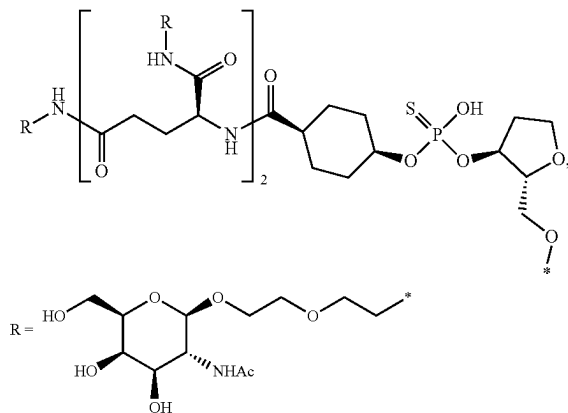
where * is a bond to 2'-hydroxyl of a nucleotide in the strand;



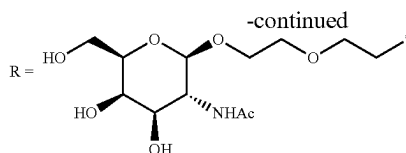
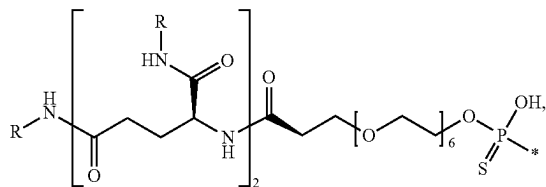
where * is a bond to a 5' or 3'-terminus of the strand;



where * is a bond to a 5' or 3'-terminus of the strand;



where —O—* is a connection to a 5' or 3'-terminus of the strand; and/or



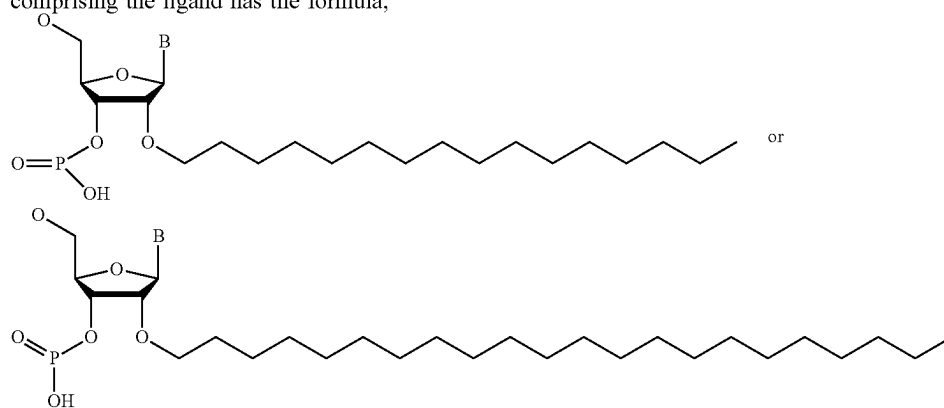
where —P—* is a connection to a 5' or 3'-hydroxyl group of the strand.

[0168] In some embodiments, the ligand comprises a lipophilic group. For example, the ligand can be a C_{6-30} aliphatic group or a C_{10-30} aliphatic group. "Aliphatic" as used herein means a saturated or unsaturated and straight, branched, and/or cyclic hydrocarbon having the defined number of carbon atoms; examples include alkyl, alkenyl, alkynyl, cycloalkyl, cycloalkylalkyl, cycloalkylalkenyl, and cycloalkylalkynyl, having the defined number of carbon atoms. In some embodiments, the ligand is a C_{10-30} alkyl group. For example, the ligand is a straight-chain or branched hexyl, octyl, decyl, dodecyl, tetradecyl, hexadecyl, octadecyl, icosyl, docosyl, or tetracosyl group. For example, the ligand is a straight-chain hexyl, octyl, decyl, dodecyl, tetradecyl, hexadecyl, octadecyl, icosyl, docosyl, or tetracosyl group. For example, the ligand is a straight-chain hexyl, octyl, decyl, dodecyl, hexadecyl, octadecyl, icosyl, or docosyl group. For example, the ligand is a straight-chain hexadecyl group. For example, the ligand is a straight-chain docosyl group.

[0169] In certain embodiments, the ligand is conjugated at the 2'-position of a nucleotide at an internal, i.e., non-terminal position of the sense strand and is a straight-chain or branched tetradecyl, hexadecyl, octadecyl, icosyl, docosyl, or tetracosyl group. For example, the ligand is conjugated at the 2'-position of a nucleotide at an internal, i.e., non-terminal position of the sense strand and is a straight-chain or branched hexyl, octyl, decyl, dodecyl, tetradecyl, hexadecyl, octadecyl, icosyl, docosyl, or tetracosyl group. For example, the ligand is conjugated at the 2'-position of a nucleotide at an internal, i.e., non-terminal position of the sense strand and is a straight-chain hexadecyl, octadecyl, icosyl, or docosyl group.

[0170] The internal sense strand nucleotide position can be all positions except the three terminal positions from each end of the sense strand. In some embodiments, the internal positions exclude a cleavage site region of the sense strand. In some embodiments, the internal positions exclude positions 9-12 or positions 11-13, counting from the 5'-end of the

sense strand. For example, the internal nucleotide position can be one or more of positions 4-8 and 13-18 on the sense strand, such as one or more of positions 5, 6, 7, 15, and 17 on the sense strand counting from the 5'-end of the sense strand. In one embodiment, the internal nucleotide position can be one of positions 5, 6, 7, or 8 of the sense strand, counting from the 5'-end. For example, each of these embodiments, the internal nucleotide position is position 6 or 7 of the sense strand, counting from the 5'-end. For example, each of these embodiments, the internal nucleotide position is position 6 of the sense strand, counting from the 5'-end. For example, each of these embodiments, the internal nucleotide position is position 7 of the sense strand, counting from the 5'-end. In certain embodiments, the internal nucleotide comprising the ligand has the formula,

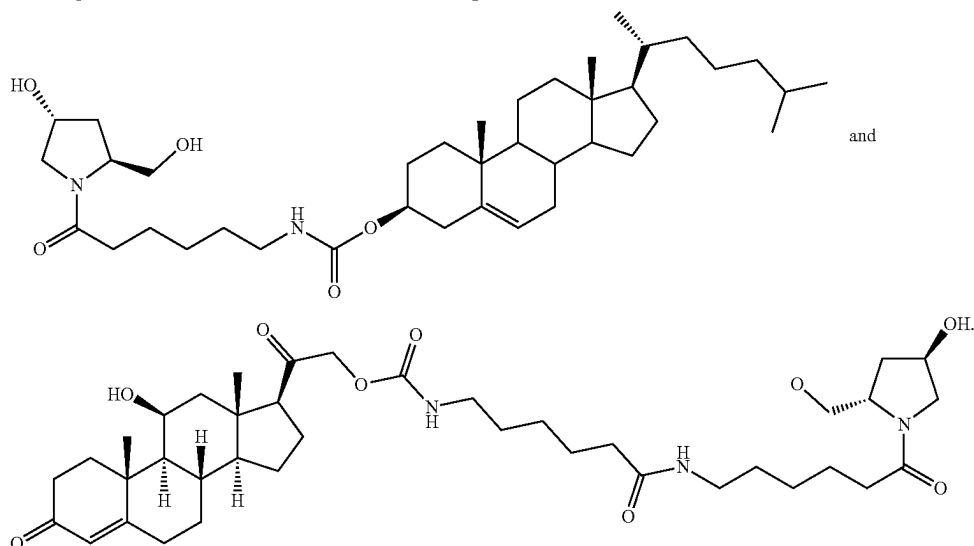


where B is a nucleotide base or a nucleotide base analog, optionally where B is adenine, guanine, cytosine, thymine or uracil.

[0171] In some embodiments, the ligand comprises an inverted nucleotide or an inverted abasic nucleotide. For example, the ligand comprises an abasic nucleotide linked via a 5'->5' or 3'->3' linkage to a strand of the dsRNA

molecule. In some embodiments, the ligand comprises an abasic nucleotide linked via a 3'->3' linkage to the 3'-end of the sense strand.

[0172] In other embodiments, the ligand comprises a lipophilic group that comprises a steroidal fused ring system. For example, the ligand can comprise cholesterol or corticosterone. Examples of such ligands, include, for example,



For example, such ligands may be attached to the 5' and/or 3' ends of a strand of the dsRNA molecule. In some embodiments, the ligand may be attached to the 5' or 3'-end of the sense strand. In some embodiments, the ligand may be attached to the 5'-end of the sense strand. In some embodiments, the ligand may be attached to the 3'-end of the sense strand. Attachment to the dsRNA molecule may be via a bond to the oxygen atom illustrated above having an open valence, or a bond formed with the 4-hydroxyl group of the pyrrolidine ring.

[0173] The ligand may be attached to the polynucleotide via a carrier. The carriers include (i) at least one "backbone attachment point," preferably two "backbone attachment points" and (ii) at least one "tethering attachment point." A "backbone attachment point" as used herein refers to a functional group, e.g., a hydroxyl group, or generally, a bond available for, and that is suitable for incorporation of the carrier into the backbone, e.g., the phosphate, or modified phosphate, e.g., sulfur containing, backbone, of a ribonucleic acid. A "tethering attachment point" (TAP) in some embodiments refers to a constituent ring atom of the cyclic carrier, e.g., a carbon atom or a heteroatom (distinct from an atom which provides a backbone attachment point), that connects a selected moiety. The moiety can be, e.g., a carbohydrate, e.g., monosaccharide, disaccharide, trisaccharide, tetrasaccharide, oligosaccharide and polysaccharide. Optionally, the selected moiety is connected by an intervening tether to the cyclic carrier. Thus, the cyclic carrier will often include a functional group, e.g., an amino group, or generally, provide a bond, that is suitable for incorporation or tethering of another chemical entity, e.g., a ligand to the constituent ring.

[0174] In one embodiment the dsRNA molecule of the invention is conjugated to a ligand via a carrier, wherein the carrier can be cyclic group or acyclic group; preferably, the cyclic group is selected from pyrrolidinyl, pyrazolinyl, pyrazolidinyl, imidazolinyl, imidazolidinyl, piperidinyl, piperazinyl, [1,3]dioxolane, oxazolidinyl, isoxazolidinyl, morpholinyl, thiazolidinyl, isothiazolidinyl, quinoxalinyl, pyridazinonyl, tetrahydrofuryl and decalin; preferably, the acyclic group is selected from serinol backbone or diethanolamine backbone.

[0175] The ligand can be attached to the sense strand, antisense strand or both strands, at the 3'-end, 5'-end or both ends. For instance, the ligand can be conjugated to the sense strand, in particular, the 3'-end of the sense strand.

[0176] The ligand can be conjugated to the sense strand or the antisense strand via a linker comprising a cleavable group. A cleavable linking group is one which is sufficiently stable outside the cell, but which upon entry into a target cell is cleaved to release the two parts the linker is holding together. In a preferred embodiment of the dsRNA molecule according to the present invention, the cleavable linking group is cleaved at least 10 times or more, preferably at least 100 times faster in the target cell or under a first reference condition (which can, e.g., be selected to mimic or represent intracellular conditions) than in the blood of a subject, or under a second reference condition (which can, e.g., be selected to mimic or represent conditions found in the blood or serum).

[0177] Cleavable linking groups are susceptible to cleavage agents, e.g., pH, redox potential or the presence of degradative molecules. Generally, cleavage agents are more prevalent or found at higher levels or activities inside cells

than in serum or blood. Examples of such degradative agents include: redox agents which are selected for particular substrates or which have no substrate specificity, including, e.g., oxidative or reductive enzymes or reductive agents such as mercaptans, present in cells, that can degrade a redox cleavable linking group by reduction; esterases; endosomes or agents that can create an acidic environment, e.g., those that result in a pH of five or lower; enzymes that can hydrolyze or degrade an acid cleavable linking group by acting as a general acid, peptidases (which can be substrate specific), and phosphatases.

[0178] A cleavable linkage group, such as a disulfide bond can be susceptible to pH. The pH of human serum is 7.4, while the average intracellular pH is slightly lower, ranging from about 7.1-7.3. Endosomes have a more acidic pH, in the range of 5.5-6.0, and lysosomes have an even more acidic pH at around 5.0. Some linkers will have a cleavable linking group that is cleaved at a preferred pH, thereby releasing the cationic lipid from the ligand inside the cell, or into the desired compartment of the cell.

[0179] A linker can include a cleavable linking group that is cleavable by a particular enzyme. The type of cleavable linking group incorporated into a linker can depend on the cell to be targeted.

[0180] For example, liver targeting ligands can be linked to the cationic lipids through a linker that includes an ester group. Liver cells are rich in esterases, and therefore the linker will be cleaved more efficiently in liver cells than in cell types that are not esterase-rich. Other cell-types rich in esterases include cells of the lung, renal cortex, and testis.

[0181] Linkers that contain peptide bonds can be used when targeting cell types rich in peptidases, such as liver cells and synoviocytes.

[0182] In general, the suitability of a candidate cleavable linking group can be evaluated by testing the ability of a degradative agent (or condition) to cleave the candidate linking group. It will also be desirable to also test the candidate cleavable linking group for the ability to resist cleavage in the blood or when in contact with other non-target tissue. Thus one can determine the relative susceptibility to cleavage between a first and a second condition, where the first is selected to be indicative of cleavage in a target cell and the second is selected to be indicative of cleavage in other tissues or biological fluids, e.g., blood or serum. The evaluations can be carried out in cell free systems, in cells, in cell culture, in organ or tissue culture, or in whole animals. It may be useful to make initial evaluations in cell-free or culture conditions and to confirm by further evaluations in whole animals. In preferred embodiments, useful candidate compounds are cleaved at least 2, 4, 10 or 100 times faster in the cell (or under in vitro conditions selected to mimic intracellular conditions) as compared to blood or serum (or under in vitro conditions selected to mimic extracellular conditions).

[0183] One class of cleavable linking groups is redox cleavable linking groups, which may be used in the dsRNA molecule according to the present invention that are cleaved upon reduction or oxidation. An example of reductively cleavable linking group is a disulfide linking group ($-S-S-$). To determine if a candidate cleavable linking group is a suitable "reductively cleavable linking group," or for example is suitable for use with a particular iRNA moiety and particular targeting agent one can look to methods described herein. For example, a candidate can be evaluated

by incubation with dithiothreitol (DTT), or other reducing agent using reagents known in the art, which mimic the rate of cleavage which would be observed in a cell, e.g., a target cell. The candidates can also be evaluated under conditions which are selected to mimic blood or serum conditions. In a preferred embodiment, candidate compounds are cleaved by at most 10% in the blood. In preferred embodiments, useful candidate compounds are degraded at least 2, 4, 10 or 100 times faster in the cell (or under in vitro conditions selected to mimic intracellular conditions) as compared to blood (or under in vitro conditions selected to mimic extracellular conditions). The rate of cleavage of candidate compounds can be determined using standard enzyme kinetics assays under conditions chosen to mimic intracellular media and compared to conditions chosen to mimic extracellular media.

[0184] Phosphate-based cleavable linking groups, which may be used in the dsRNA molecule according to the present invention, are cleaved by agents that degrade or hydrolyze the phosphate group. An example of an agent that cleaves phosphate groups in cells are enzymes such as phosphatases in cells. Examples of phosphate-based linking groups are —O—P(O)(ORk)—O— , —O—P(S)(ORk)—O— , —O—P(S)(SRk)—O— , —S—P(O)(ORk)—O— , —O—P(O)(ORk)—S— , —S—P(O)(ORk)—S— , —O—P(S)(ORk)—S— , —S—P(S)(ORk)—O— , —O—P(O)(Rk)—O— , —O—P(S)(Rk)—O— , —S—P(O)(Rk)—O— , —S—P(S)(Rk)—O— , —S—P(O)(Rk)—S— , —O—P(S)(Rk)—S— , wherein Rk at each occurrence can be, independently, hydrogen, C1-C20 alkyl, C1-C20 haloalkyl, C6-C10 aryl, C7-C12 aralkyl. Preferred embodiments are —O—P(O)(OH)—O— , —O—P(S)(OH)—O— , —O—P(S)(SH)—O— , —S—P(O)(OH)—O— , —O—P(O)(OH)—S— , —S—P(O)(OH)—S— , —O—P(S)(OH)—S— , —S—P(S)(OH)—O— , —O—P(O)(H)—O— , —O—P(S)(H)—O— , —S—P(O)(H)—O— , —S—P(S)(H)—O— , —S—P(O)(H)—S— , —O—P(S)(H)—S— . A preferred embodiment is —O—P(O)(OH)—O— . These candidates can be evaluated using methods analogous to those described above.

[0185] Acid cleavable linking groups, which may be used in the dsRNA molecule according to the present invention, are linking groups that are cleaved under acidic conditions. In preferred embodiments acid cleavable linking groups are cleaved in an acidic environment with a pH of about 6.5 or lower (e.g., about 6.0, 5.5, 5.0, or lower), or by agents such as enzymes that can act as a general acid. In a cell, specific low pH organelles, such as endosomes and lysosomes can provide a cleaving environment for acid cleavable linking groups. Examples of acid cleavable linking groups include but are not limited to hydrazones, esters, and esters of amino acids. Acid cleavable groups can have the general formula —C=NN— , C(O)O— , or —OC(O)— . A preferred embodiment is when the carbon attached to the oxygen of the ester (the alkoxy group) is an aryl group, substituted alkyl group, or tertiary alkyl group such as dimethyl pentyl or t-butyl. These candidates can be evaluated using methods analogous to those described above.

[0186] Ester-based cleavable linking groups, which may be used in the dsRNA molecule according to the present invention, are cleaved by enzymes such as esterases and amidases in cells. Examples of ester-based cleavable linking groups include but are not limited to esters of alkylene, alkenylene and alkynylene groups. Ester cleavable linking groups have the general formula —C(O)O— , or —OC— .

(O)—. These candidates can be evaluated using methods analogous to those described above.

[0187] Peptide-based cleavable linking groups, which may be used in the dsRNA molecule according to the present invention, are cleaved by enzymes such as peptidases and proteases in cells. Peptide-based cleavable linking groups are peptide bonds formed between amino acids to yield oligopeptides (e.g., dipeptides, tripeptides etc.) and polypeptides. Peptide-based cleavable groups do not include the amide group (—C(O)NH—). The amide group can be formed between any alkylene, alkenylene or alkynylene. A peptide bond is a special type of amide bond formed between amino acids to yield peptides and proteins. The peptide based cleavage group is generally limited to the peptide bond (i.e., the amide bond) formed between amino acids yielding peptides and proteins and does not include the entire amide functional group. Peptide-based cleavable linking groups have the general formula $\text{—NHCHR}^A\text{C(O)NHCHR}^B\text{C(O)—}$, where R^A and R^B are the R groups of the two adjacent amino acids. These candidates can be evaluated using methods analogous to those described above. As used herein, “carbohydrate” refers to a compound which is either a carbohydrate per se made up of one or more monosaccharide units having at least 6 carbon atoms (which may be linear, branched or cyclic) with an oxygen, nitrogen or sulfur atom bonded to each carbon atom; or a compound having as a part thereof a carbohydrate moiety made up of one or more monosaccharide units each having at least six carbon atoms (which may be linear, branched or cyclic), with an oxygen, nitrogen or sulfur atom bonded to each carbon atom. Representative carbohydrates include the sugars (mono-, di-, tri- and oligosaccharides containing from about 4-9 monosaccharide units), and polysaccharides such as starches, glycogen, cellulose and polysaccharide gums. Specific monosaccharides include C_5 and above (preferably C_5 - C_8) sugars; di- and trisaccharides include sugars having two or three monosaccharide units (preferably C_5 - C_8).

[0188] In some embodiments, the dsRNA molecule of the invention comprises one or more overhang regions and/or capping groups of dsRNA molecule at the 3'-end, or 5'-end or both ends of a strand. The overhang can be 1-10 nucleotides in length. For example, the overhang can be 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 nucleotides in length. In some embodiments, the overhang is 1-6 nucleotides in length, for instance 2-6 nucleotides in length, 1-5 nucleotides in length, 2-5 nucleotides in length, 1-4 nucleotides in length, 2-4 nucleotides in length, 1-3 nucleotides in length, 2-3 nucleotides in length, or 1-2 nucleotides in length. The overhangs can be the result of one strand being longer than the other, or the result of two strands of the same length being staggered. The overhang can form a mismatch with the target sequence or it can be complementary to the gene sequences being targeted or it can be the other sequence. The first and second strands can also be joined, e.g., by additional bases to form a hairpin, or by other non-base linkers.

[0189] In some embodiments, the nucleotides in the overhang region of the dsRNA molecule of the invention can each independently be a modified or unmodified nucleotide including, but not limited to 2'-sugar modified, such as, 2'-Fluoro 2'-O-methyl, thymidine (T), 2'-O-methoxyethyl-5-methyluridine, 2'-O-methoxyethyladenosine, 2'-O-methoxyethyl-5-methylcytidine, GNA, SNA, hGNA, hhGNA, mGNA, TNA, h'GNA, and any combinations thereof. For example, dTdT can be an overhang sequence for

either end on either strand. The overhang can form a mismatch with the target mRNA or it can be complementary to the gene sequences being targeted or can be other sequence.

[0190] The 5'- or 3'-overhangs at the sense strand, antisense strand or both strands of the dsRNA molecule of the invention may be phosphorylated. In some embodiments, the overhang region contains two nucleotides having a phosphorothioate between the two nucleotides, where the two nucleotides can be the same or different. In some embodiments, the overhang is present at the 3'-end of the sense strand, antisense strand or both strands. In some embodiments, this 3'-overhang is present in the antisense strand. In some embodiments, this 3'-overhang is present in the sense strand.

[0191] The dsRNA molecule of the invention may comprise only a single overhang, which can strengthen the interference activity of the dsRNA, without affecting its overall stability. For example, the single-stranded overhang is located at the 3'-terminal end of the sense strand or, alternatively, at the 3'-terminal end of the antisense strand. The dsRNA can also have a blunt end, located at the 5'-end of the antisense strand (or the 3'-end of the sense strand) or vice versa.

[0192] Generally, the antisense strand of the dsRNA has a nucleotide overhang at the 3'-end, and the 5'-end is blunt. While not bound by theory, the asymmetric blunt end at the 5'-end of the antisense strand and 3'-end overhang of the antisense strand favor the guide strand loading into RISC process. For example, the single overhang is at least one, two, three, four, five, six, seven, eight, nine, or ten nucleotides in length. In some embodiments, the dsRNA has a 2 nucleotide overhang on the 3'-end of the antisense strand and a blunt end at the 5'-end of the antisense strand.

[0193] The dsRNA of the invention can comprise one or more modified nucleotides. For example, every nucleotide in the sense strand and antisense strand of the dsRNA molecule can be modified. Each nucleotide can be modified with the same or different modification which can include one or more alteration of one or both of the non-linking phosphate oxygens and/or of one or more of the linking phosphate oxygens; alteration of a constituent of the ribose sugar; replacement of the ribose sugar; wholesale replacement of the phosphate moiety with "dephospho" linkers; modification or replacement of a naturally occurring base; and replacement or modification of the ribose-phosphate backbone.

[0194] As nucleic acids are polymers of subunits, many of the modifications occur at a position which is repeated within a nucleic acid, e.g., a modification of a base, or a phosphate moiety, or a non-linking O of a phosphate moiety. In some cases, the modification will occur at all of the subject positions in the nucleic acid but in many cases it will not. By way of example, a modification may only occur at a 3' or 5' terminal position, may only occur in a central region, may only occur at a non-terminal region, or may only occur in a terminal region, e.g., at a position on a terminal nucleotide or in the last 2, 3, 4, 5, or 10 nucleotides of a strand. A modification may occur in a double strand region, a single strand region, or in both. A modification may occur only in the double strand region of a RNA or may only occur in a single strand region of a RNA. For example, a phosphorothioate modification at a non-linking O position may only occur at one or both termini, may only occur in a

terminal region, e.g., at a position on a terminal nucleotide or in the last 2, 3, 4, 5, or 10 nucleotides of a strand, or may occur in double strand and single strand regions, particularly at termini. The 5' end or ends can be phosphorylated.

[0195] It may be possible, e.g., to enhance stability, to include particular bases in overhangs, or to include modified nucleotides or nucleotide surrogates, in single strand overhangs, e.g., in a 5' or 3' overhang, or in both. For example, it can be desirable to include purine nucleotides in overhangs. In some embodiments all or some of the bases in a 3' or 5' overhang may be modified, e.g., with a modification described herein. Modifications can include, e.g., the use of modifications at the 2' position of the ribose sugar with modifications that are known in the art, e.g., the use of deoxyribonucleotides, 2'-deoxy-2'-fluoro (2'-F) or 2'-O-methyl modified instead of the ribosugar of the nucleobase, and modifications in the phosphate group, e.g., phosphorothioate modifications. Overhangs need not be homologous with the target sequence.

[0196] In some embodiments, the dsRNA molecule of the invention comprises modifications of an alternating pattern, particular in the B1, B2, B3, B1', B2', B3', B4' regions. The term "alternating motif" or "alternative pattern" as used herein refers to a motif having one or more modifications, each modification occurring on alternating nucleotides of one strand. The alternating nucleotide may refer to one per every other nucleotide or one per every three nucleotides, or a similar pattern. For example, if A, B and C each represent one type of modification to the nucleotide, the alternating motif can be "ABABABABABAB . . ." "AABBAAB-BAABB . . ." "AABAABAABAAB . . ." "AAABAAA-BAAAB . . ." "AAABBBAAABBB . . .," or "ABCABCAB-CABC . . .," etc.

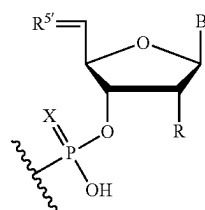
[0197] The type of modifications contained in the alternating motif may be the same or different. For example, if A, B, C, D each represent one type of modification on the nucleotide, the alternating pattern, i.e., modifications on every other nucleotide, may be the same, but each of the sense strand or antisense strand can be selected from several possibilities of modifications within the alternating motif such as "ABABAB . . .", "ACACAC . . ." "BDBDBD . . ." or "CDCDCD . . .," etc.

[0198] In some embodiments, the dsRNA molecule of the invention comprises the modification pattern for the alternating motif on the sense strand relative to the modification pattern for the alternating motif on the antisense strand is shifted. The shift may be such that the modified group of nucleotides of the sense strand corresponds to a differently modified group of nucleotides of the antisense strand and vice versa. For example, the sense strand when paired with the antisense strand in the dsRNA duplex, the alternating motif in the sense strand may start with "ABABAB" from 5'-3' of the strand and the alternating motif in the antisense strand may start with "BABABA" from 3'-5' of the strand within the duplex region. As another example, the alternating motif in the sense strand may start with "AABBAABB" from 5'-3' of the strand and the alternating motif in the antisense strand may start with "BBAABBAA" from 3'-5' of the strand within the duplex region, so that there is a complete or partial shift of the modification patterns between the sense strand and the antisense strand.

5'-Modifications

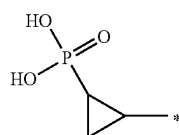
[0199] In some embodiments dsRNA molecules of the invention are 5' phosphorylated or include a phosphoryl analog or phosphate mimic at the 5' terminus. 5'-phosphate modifications include those which are compatible with RISC mediated gene silencing. Suitable modifications include: 5'-monophosphate ((HO)₂(O)P—O-5'); 5'-diphosphate ((HO)₂(O)P—O—P(HO)(O)—O-5'); 5'-triphosphate ((HO)₂(O)P—O—(HO)(O)P—O—P(HO)(O)—O-5'); 5'-guanosine cap (7-methylated or non-methylated)(7m-G-O-5'-(HO)(O)P—O—(HO)(O)P—O—P(HO)(O)—O-5'); 5'-adenosine cap (Appp), and any modified or unmodified nucleotide cap structure (N—O-5'-(HO)(O)P—O—(HO)(O)P—O—P(HO)(O)—O-5'); 5'-monothiophosphate (phosphorothioate; (HO)₂(S)P—O-5'); 5'-monodithiophosphate (phosphorodithioate; (HO)(HS)(S)P—O-5'); 5'-phosphorothiolate ((HO)₂(O)P—S-5'); any additional combination of oxygen/sulfur replaced monophosphate, diphosphate and triphosphates (e.g. 5'-alpha-thiotriphosphate, 5'-gamma-thiotriphosphate, etc.), 5'-phosphoramidates ((HO)₂(O)P—NH-5', (HO)(NH₂)(O)P—O-5'), 5'-alkylphosphonates (e.g. RP(OH)(O)—O-5', R=alkyl, such as methyl, ethyl, isopropyl, propyl, etc.), 5'-alkenylphosphonates (i.e. vinyl, substituted vinyl), (OH)₂(O)P—5'-CH₂-), cyclopropylphosphonates, 5'-alkyletherphosphonates (R=alkylether-methoxymethyl(MeOCH₂—), ethoxymethyl, etc., e.g. RP(OH)(O)—O-5'-). In one example, the modification can be placed in the antisense strand of a dsRNA molecule.

[0200] In some embodiments of any one of the aspects, the antisense strand comprises a phosphoryl analog or phosphate mimic at the 5'-terminus. In some embodiments, the antisense strand comprises an alkenylphosphonates, i.e., a vinyl phosphonate at the 5'-terminus. For example, the antisense strand comprises a 5'-E-vinyl phosphonate. In exemplary embodiments, a 5' vinyl phosphonate modified nucleotide at the 5'-terminus may have the structure:



wherein X is O or S; R is hydrogen, hydroxy, fluoro, or C₁₋₂₀alkoxy (e.g., methoxy); R^{5'} is =C(H)—P(O)(OH)₂ and the double bond between the C_{5'} carbon and R^{5'} is in the E or Z orientation (e.g., E orientation); and B is a nucleobase or a modified nucleobase, optionally where B is adenine, guanine, cytosine, thymine, or uracil.

[0201] In some embodiments, the antisense strand comprises a cyclopropylphosphonate at the 5'-terminus. For example, the antisense comprises



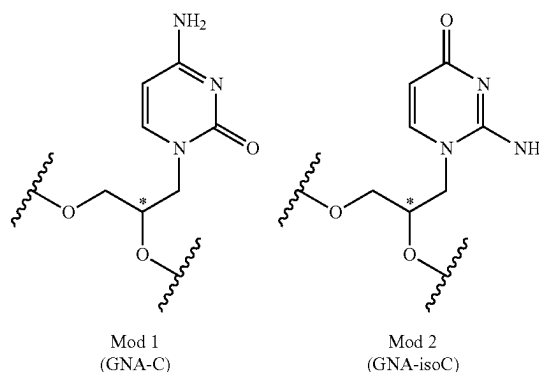
at the 5'-terminus (which, for example, may replace the 4'-group in immediately preceding structure).

[0202] The dsRNA agents of the invention can comprise thermally destabilizing modifications in the seed region of the antisense strand (i.e., at positions 2-9 of the 5'-end of the antisense strand) to reduce or inhibit off-target gene silencing. Without wishing to be bound by a theory, dsRNAs with an antisense strand comprising at least one thermally destabilizing modification of the duplex within the first 9 nucleotide positions, counting from the 5' end, of the antisense strand have reduced off-target gene silencing activity. Accordingly, in some embodiments, the antisense strand comprises at least one (e.g., one, two, three, four, five or more) thermally destabilizing modification of the duplex within the first 9 nucleotide positions of the 5' region of the antisense strand. In some embodiments, thermally destabilizing modification of the duplex is located in positions 2-9, or preferably positions 4-8, from the 5'-end of the antisense strand. In some further embodiments, the thermally destabilizing modification of the duplex is located at position 5, 6, 7 or 8 from the 5'-end of the antisense strand.

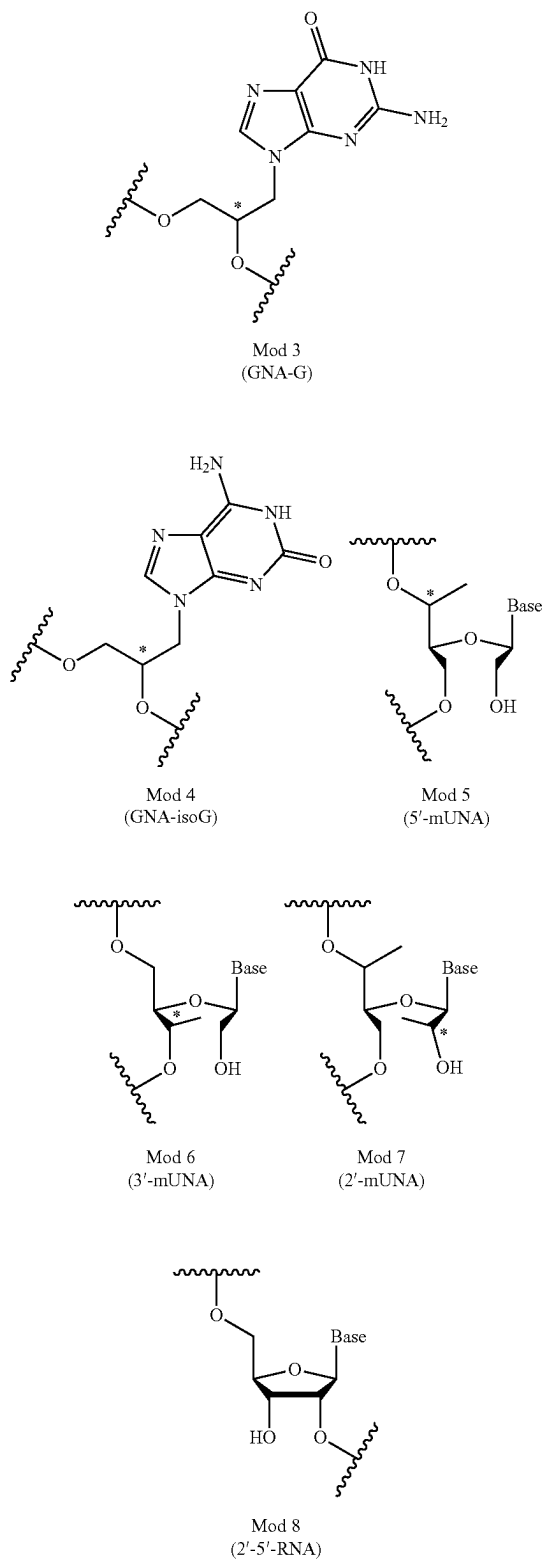
[0203] In still some further embodiments, the thermally destabilizing modification of the duplex is located at position 7 from the 5'-end of the antisense strand.

[0204] The term “thermally destabilizing modification(s)” includes modification(s) that would result with a dsRNA with a lower overall melting temperature (T_m)(preferably a T_m with one, two, three or four degrees lower than the T_m of the dsRNA without having such modification(s). In some embodiments, the thermally destabilizing modification of the duplex is located at position 2, 3, 4, 5, 6, 7, 8 or 9 from the 5'-end of the antisense strand.

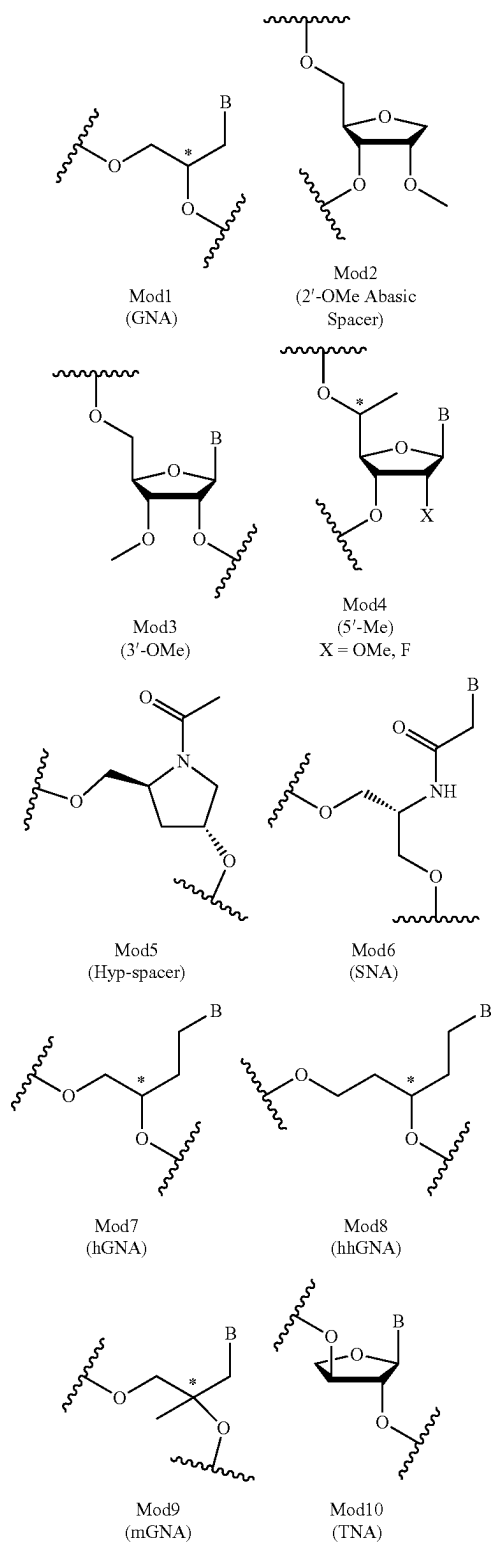
[0205] The thermally destabilizing modifications can include, but are not limited to, abasic modification; mismatch with the opposing nucleotide in the opposing strand; and sugar modification such as 2'-deoxy modification or acyclic nucleotide, e.g., unlocked nucleic acids (UNA) or glycol nucleic acid (GNA), or a 5'-2'-linked nucleotide (e.g., having 3'-OMe, 3'-F, 3'-H or 3'-OH, herein a “3'-RNA”). For example, the thermally destabilizing modifications can include, but are not limited to, mUNA and GNA building blocks as follows:



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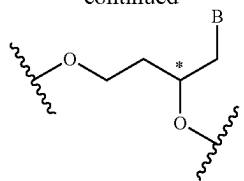


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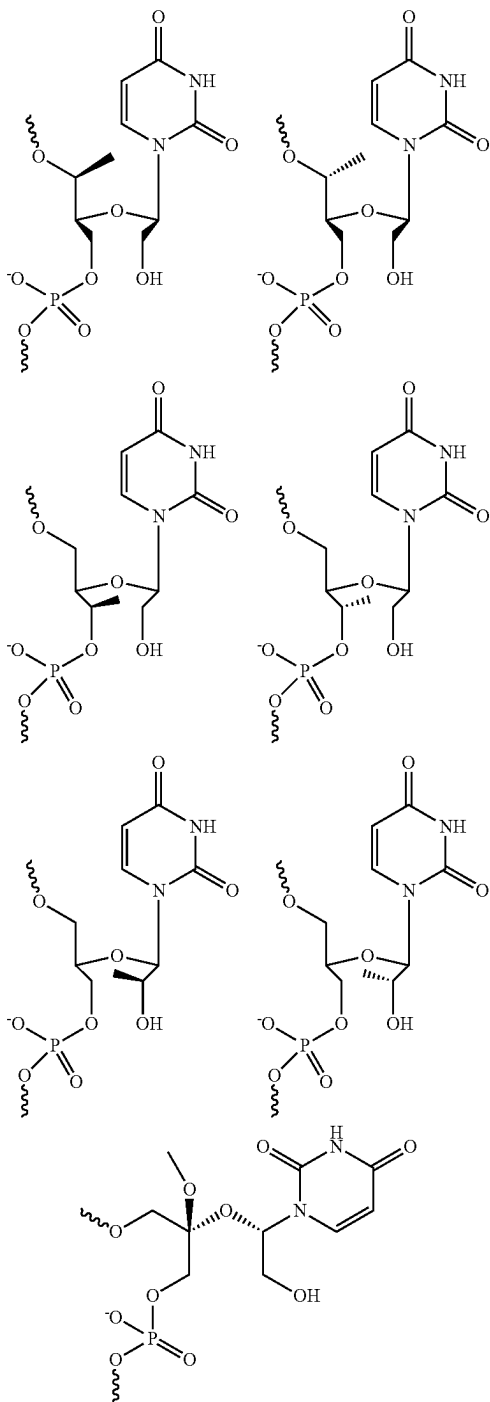


*Both stereoisomers tested

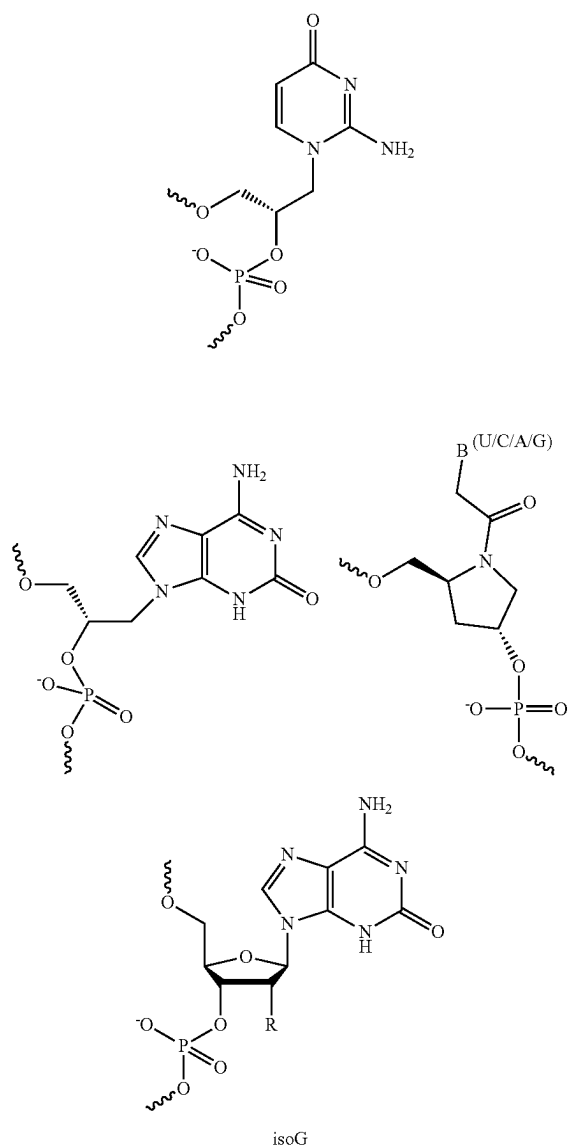
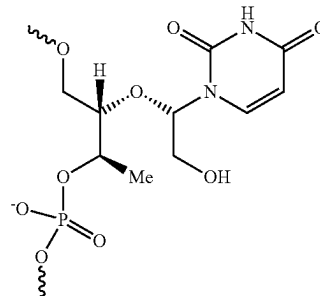
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h⁺GNA

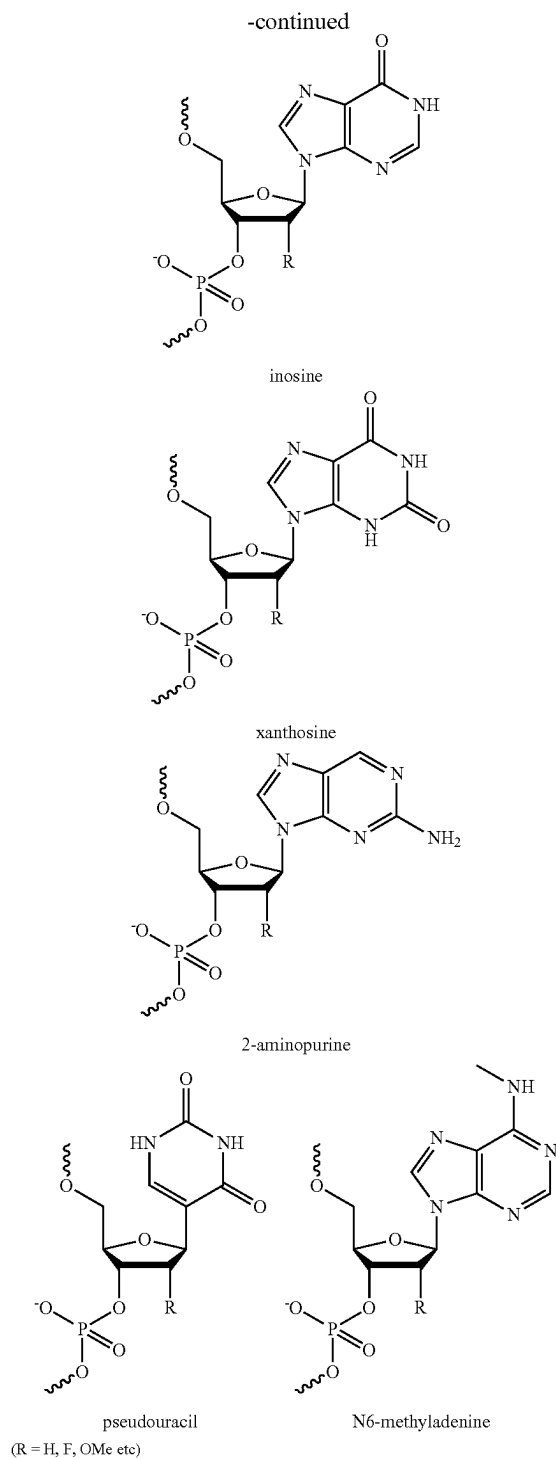
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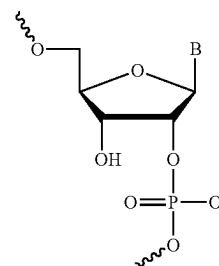


isoG



[0206] In some embodiments, the destabilizing modification is a 5'-2'-linked nucleotide (e.g., having 3'-OMe, 3'-F, 3'-H or 3'-OH. In some embodiments, the destabilizing modification is a 5'-2'-linked nucleotide having 3'-OMe, 3'-F, 3'-H or 3'-OH. In some embodiments, the destabilizing modification is a 5'-2'-linked nucleotide having 3'-OMe. In some embodiments, the destabilizing modification is a 5'-2'-

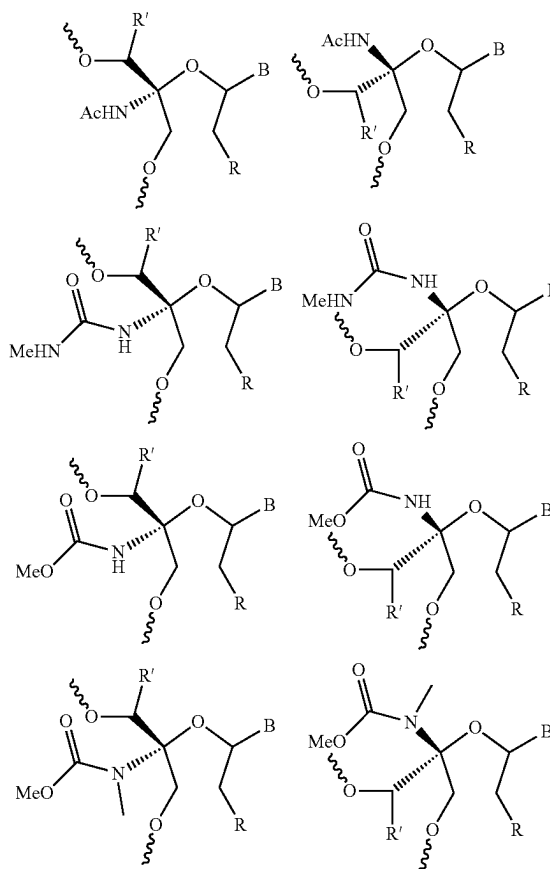
linked nucleotide having 3'-F. In some embodiments, the destabilizing modification is a 5'-2'-linked nucleotide having 3'-H. In some embodiments, the destabilizing modification is a 5'-2'-linked nucleotide having 3'-OH, e.g., having the formula

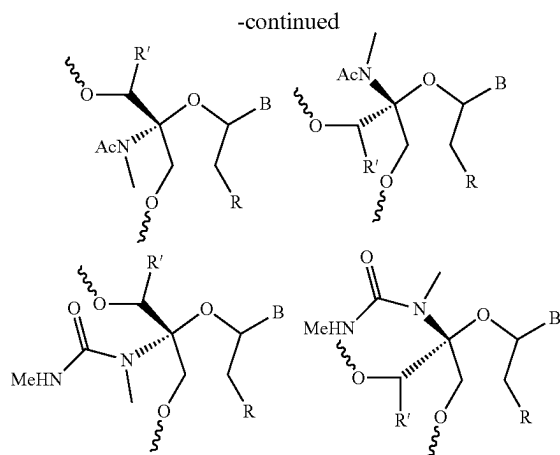


wherein B is a nucleotide base or a nucleotide base analog, optionally where B is adenine, guanine, cytosine, thymine or uracil.

[0207] In some embodiments, the destabilizing modification is selected from the group consisting of GNA-isoC, GNA-isoG, 5'-mUNA, 4'-mUNA, 3'-mUNA, and 2'-mUNA.

[0208] In some embodiments, the destabilizing modification mUNA is selected from the group consisting of





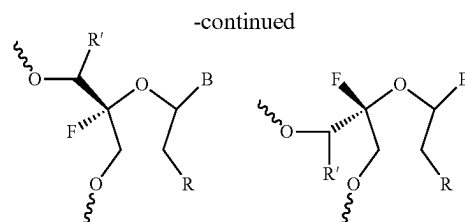
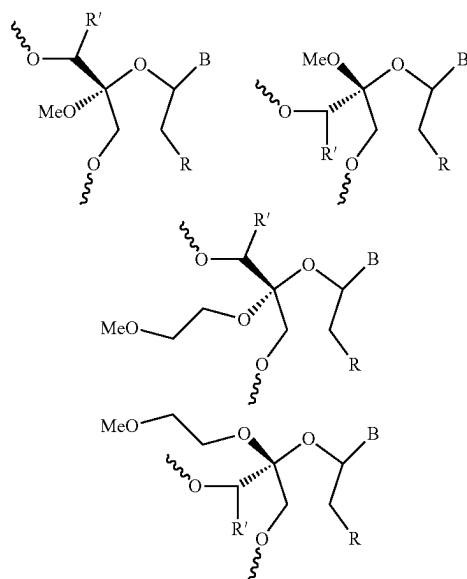
[0209] R=H, OH; OMe; Cl, F; OH; O—(CH₂)₂OMe; SMe, NMe₂; NH₂; Me; CCH(alkyne), O-nPr; O-alkyl; O-alkylamino;

[0210] R'=H, Me;

[0211] B=A; C; 5-Me-C; G; I; U; T; Y; 2-thiouridine; 4-thiouridine; C₅-modified pyrimidines; C₂-modified purines; N₈-modified purines; phenoxazine; G-clamp; non-canonical mono, bi and tricyclic heterocycles; pseudouracil; isoC; isoG; 2,6-diamminopurine; pseudocytosine; 2-aminopurine; xanthosine; N⁶-alkyl-A; O⁶-alkyl-G; 2-thiouridine; 4-thiouridine; C₅-modified pyrimidines; C₂-modified purines; N₈-modified purines; 7-deazapurines, phenoxazine; G-clamp; non-canonical mono, bi and tricyclic heterocycles; and

[0212] Stereochemistry is R or S and combination of R and S for the unspecified chiral centers.

[0213] In some embodiments, the destabilizing modification mUNA is selected from the group consisting of



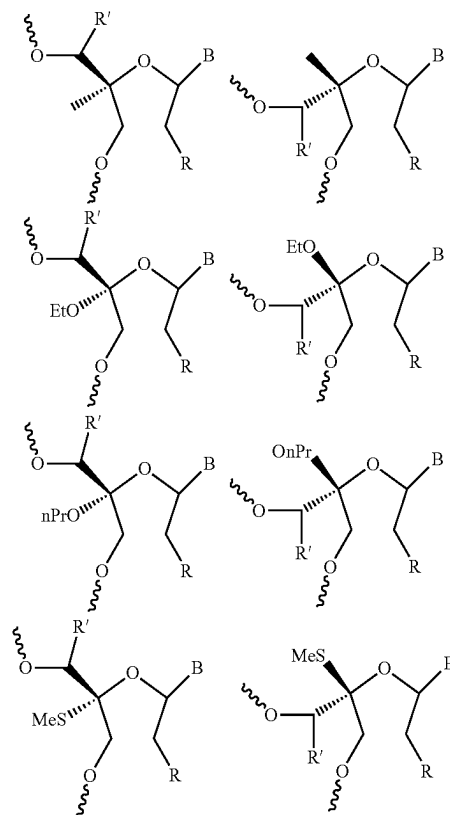
[0214] R=H, OH; OMe; Cl, F; OH; O—(CH₂)₂OMe; SMe, NMe₂; NH₂; Me; CCH(alkyne), O-nPr; O-alkyl; O-alkylamino;

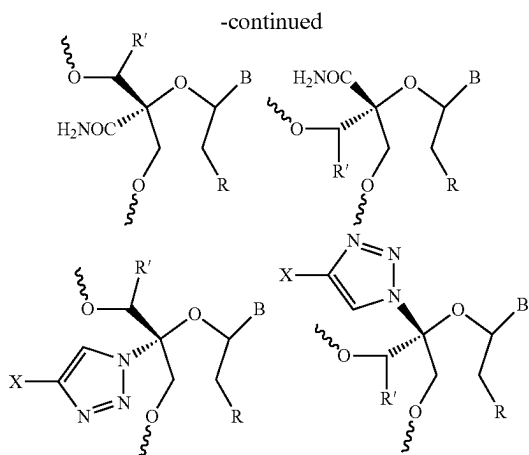
[0215] R'=H, Me;

[0216] B=A; C; 5-Me-C; G; I; U; T; Y; 2-thiouridine; 4-thiouridine; C₅-modified pyrimidines; C₂-modified purines; N₈-modified purines; phenoxazine; G-clamp; non-canonical mono, bi and tricyclic heterocycles; pseudouracil; isoC; isoG; 2,6-diamminopurine; pseudocytosine; 2-aminopurine; xanthosine; N⁶-alkyl-A; O⁶-alkyl-G; 2-thiouridine; 4-thiouridine; C₅-modified pyrimidines; C₂-modified purines; N₈-modified purines; 7-deazapurines, phenoxazine; G-clamp; non-canonical mono, bi and tricyclic heterocycles; and

[0217] Stereochemistry is R or S and combination of R and S for the unspecified chiral centers.

[0218] In some embodiments, the destabilizing modification mUNA is selected from the group consisting of

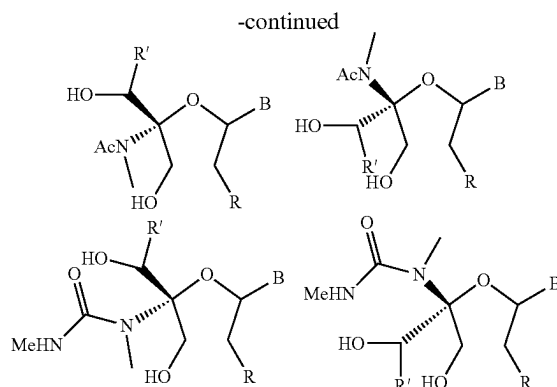




[0219] R=H, OMe; F; OH; O—(CH₂)₂OMe; SMe, NMe₂; NH₂; Me; O-nPr; O-alkyl; O-alkylamino; R'=H, Me;

[0220] B=A; C; 5-Me-C; G; I; U; T; Y; 2-thiouridine; 4-thiouridine; C₅-modified pyrimidines; C2-modified purines; N8-modified purines; phenoxazine; G-clamp; non-canonical mono, bi and tricyclic heterocycles; pseudouracil; isoC; isoG; 2,6-diamminopurine; pseudocytosine; 2-aminopurine; xanthosine; N⁶-alkyl-A; O⁶-alkyl-G; 7-deazapurines; and Stereochemistry is R or S and combination of R and S for the unspecified chiral centers.

[0221] In some embodiments, the destabilizing modification mUNA is selected from the group consisting of



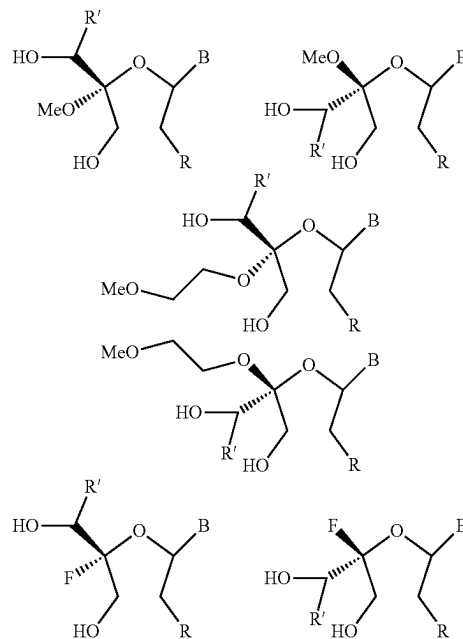
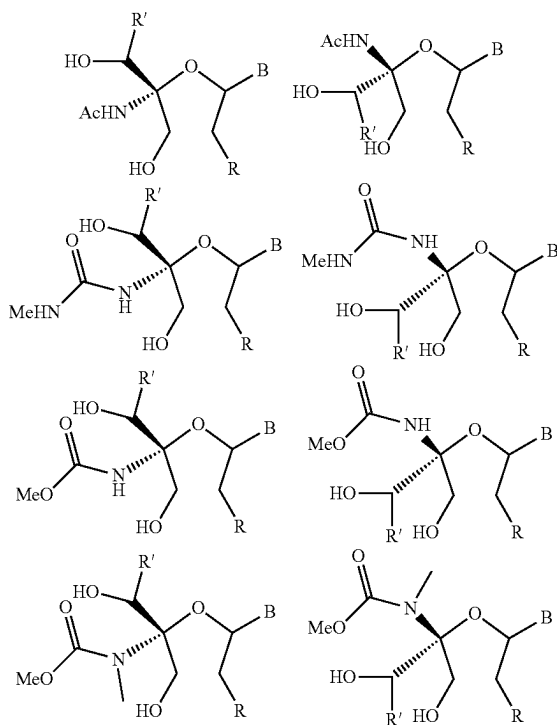
[0222] R=H, OH; OMe; Cl, F; OH; O—(CH₂)₂OMe; SMe, NMe₂; NH₂; Me; CCH(alkyne), O-nPr; O-alkyl; O-alkylamino;

[0223] R'=H, Me;

[0224] B=A; C; 5-Me-C; G; I; U; T; Y; 2-thiouridine; 4-thiouridine; C₅-modified pyrimidines; C2-modified purines; N8-modified purines; phenoxazine; G-clamp; non-canonical mono, bi and tricyclic heterocycles; pseudouracil; isoC; isoG; 2,6-diamminopurine; pseudocytosine; 2-aminopurine; xanthosine; N⁶-alkyl-A; O⁶-alkyl-G; 2-thiouridine; 4-thiouridine; C₅-modified pyrimidines; C2-modified purines; N8-modified purines; 7-deazapurines, phenoxazine; G-clamp; non-canonical mono, bi and tricyclic heterocycles; and

[0225] Stereochemistry is R or S and combination of R and S for the unspecified chiral centers

[0226] In some embodiments, the destabilizing modification mUNA is selected from the group consisting of



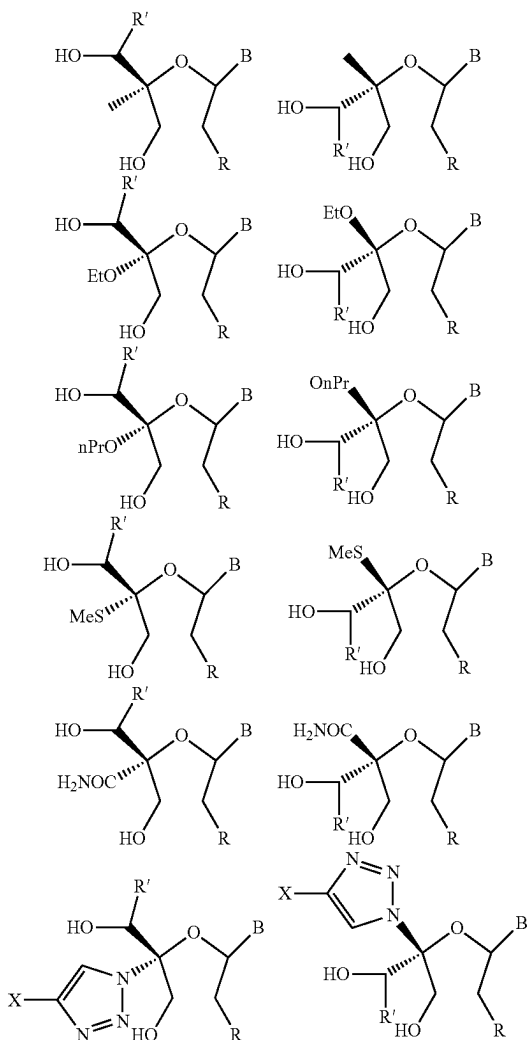
[0227] R=H, OH; OMe; Cl, F; OH; O—(CH₂)₂OMe; SMe, NMe₂; NH₂; Me; CCH(alkyne), O-nPr; O-alkyl; O-alkylamino;

[0228] R'=H, Me;

[0229] B=A; C; 5-Me-C; G; I; U; T; Y; 2-thiouridine; 4-thiouridine; C₅-modified pyrimidines; C2-modified purines; N8-modified purines; phenoxazine; G-clamp; non-canonical mono, bi and tricyclic heterocycles; pseudouracil; isoC; isoG; 2,6-diamminopurine; pseudocytosine; 2-aminopurine; xanthosine; N⁶-alkyl-A; O⁶-alkyl-G; 2-thiouridine; 4-thiouridine; C₅-modified pyrimidines; C2-modified purines; N8-modified purines; 7-deazapurines, phenoxazine; G-clamp; non-canonical mono, bi and tricyclic heterocycles; and

[0230] Stereochemistry is R or S and combination of R and S for the unspecified chiral centers

[0231] In some embodiments, the modification mUNA is selected from the group consisting of



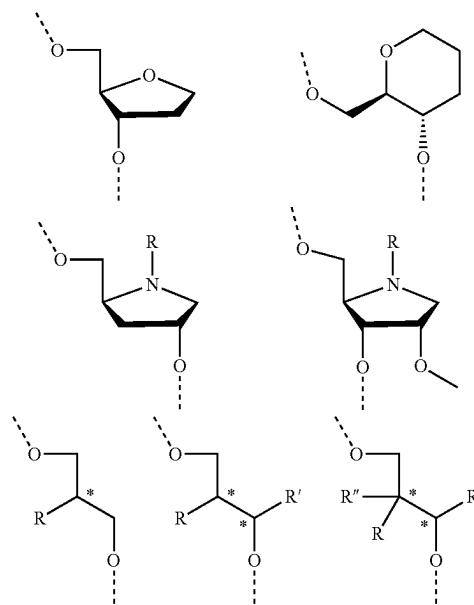
[0232] R=H, OMe; F; OH; O-(CH₂)₂OMe; SMe, NMe₂; NH₂; Me; O-nPr; O-alkyl; O-alkylamino;

[0233] R'=H, Me;

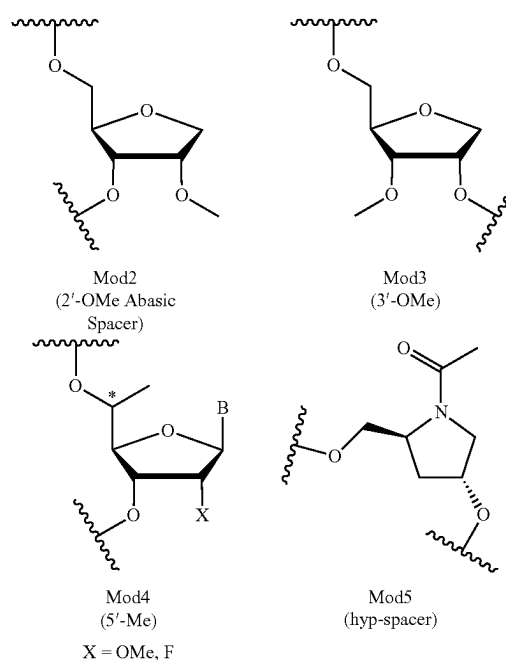
[0234] B=A; C; 5-Me-C; G; I; U; T; Y; 2-thiouridine; 4-thiouridine; C₅-modified pyrimidines; C2-modified purines; N8-modified purines; phenoxazine; G-clamp; non-canonical mono, bi and tricyclic heterocycles;

pseudouracil; isoC; isoG; 2,6-diamminopurine; pseudocytosine; 2-aminopurine; xanthosine; N⁶-alkyl-A; O⁶-alkyl-G; 7-deazapurines; and Stereochemistry is R or S and combination of R and S for the unspecified chiral centers

[0235] Exemplary abasic modifications include, but are not limited to the following:

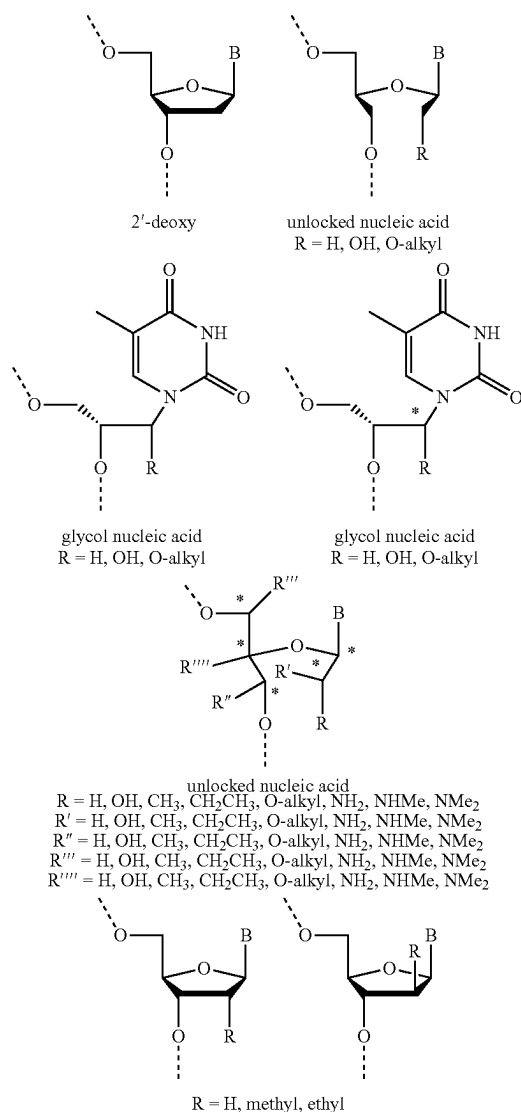


[0236] Wherein R=H, Me, Et or OMe; R'=H, Me, Et or OMe; R''=H, Me, Et or OMe



wherein B is a modified or unmodified nucleobase and the asterisk on each structure represents either R, S or racemic.

[0237] Exemplified sugar modifications include, but are not limited to the following:

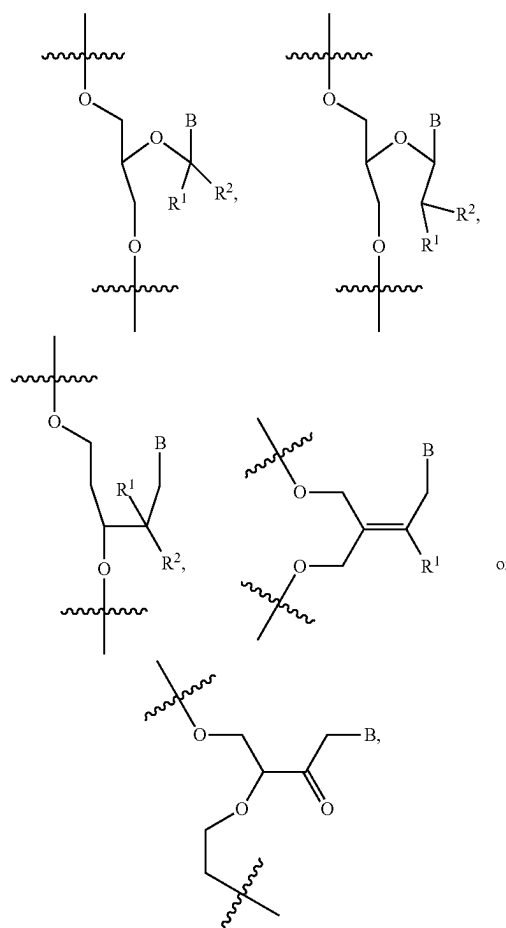


wherein B is a modified or unmodified nucleobase and the asterisk on each structure represents either R, S or racemic.

[0238] In some embodiments the thermally destabilizing modification of the duplex is selected from the mUNA and GNA building blocks described in Examples 1-3 herein. In some embodiments, the destabilizing modification is selected from the group consisting of GNA-isoC, GNA-isoG, 5'-mUNA, 4'-mUNA, 3'-mUNA, and 2'-mUNA. In some further embodiments of this, the dsRNA molecule further comprises at least one thermally destabilizing modification selected from the group consisting of GNA, 2'-OMe, 3'-OMe, 5'-Me, Hy p-spacer, SNA, hGNA, hhGNA, mGNA, TNA and h'GNA (Mod A-Mod K).

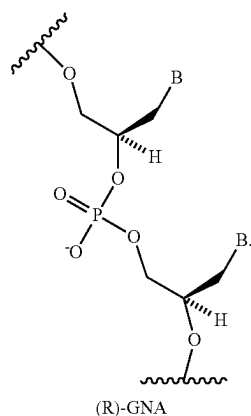
[0239] The term "acyclic nucleotide" refers to any nucleotide having an acyclic ribose sugar, for example, where any

of bonds between the ribose carbons (e.g., C1'-C2', C2'-C3', C3'-C4', C4'-O4', or C1'-O4') is absent and/or at least one of ribose carbons or oxygen (e.g., C1', C2', C3', C4' or O4') are independently or in combination absent from the nucleotide. In some embodiments, acyclic nucleotide is



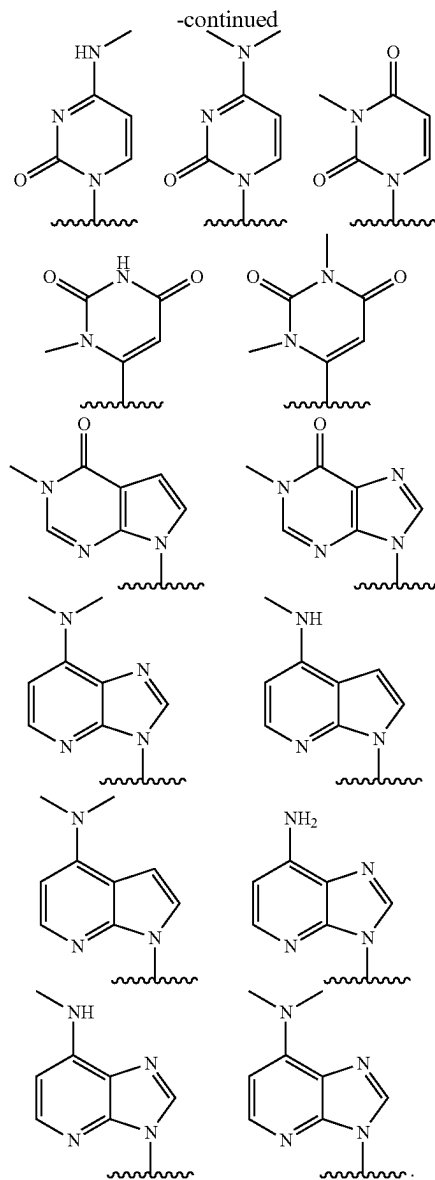
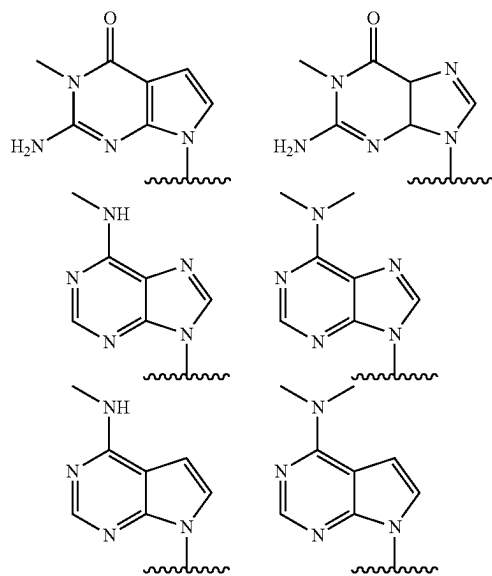
wherein B is a modified or unmodified nucleobase, R1 and R2 independently are H, halogen, OR3, or alkyl; and R3 is H, alkyl, cycloalkyl, aryl, aralkyl, heteroaryl or sugar). The term "UNA" refers to unlocked acyclic nucleic acid, wherein any of the bonds of the sugar has been removed, forming an unlocked "sugar" residue. In one example, UNA also encompasses monomers with bonds between C1'-C4' being removed (i.e. the covalent carbon-oxygen-carbon bond between the C1' and C4' carbons). In another example, the C2'-C3' bond (i.e. the covalent carbon-carbon bond between the C2' and C3' carbons) of the sugar is removed (see Mikhailov et. al., Tetrahedron Letters, 26(17): 2059 (1985); and Fluiter et al., Mol. Biosyst., 10:1039 (2009), which are hereby incorporated by reference in their entirety). The acyclic derivative provides greater backbone flexibility without affecting the Watson-Crick pairings. The acyclic nucleotide can be linked via 2'-5' or 3'-5' linkage.

[0240] The term 'GNA' refers to glycol nucleic acid which is a polymer similar to DNA or RNA but differing in the composition of its "backbone" in that is composed of repeating glycerol units linked by phosphodiester bonds:



[0241] The thermally destabilizing modification of the duplex can be mismatches (i.e., noncomplementary base pairs) between the thermally destabilizing nucleotide and the opposing nucleotide in the opposite strand within the dsRNA duplex. Exemplary mismatch base pairs include G: G, G: A, G: U, G: T, A: A, A: C, C: C, C: U, C: T, U: U, T: T, U: T, or a combination thereof. Other mismatch base pairings known in the art are also amenable to the present invention. A mismatch can occur between nucleotides that are either naturally occurring nucleotides or modified nucleotides, i.e., the mismatch base pairing can occur between the nucleobases from respective nucleotides independent of the modifications on the ribose sugars of the nucleotides. In certain embodiments, the dsRNA molecule contains at least one nucleobase in the mismatch pairing that is a 2'-deoxy nucleobase; e.g., the 2'-deoxy nucleobase is in the sense strand.

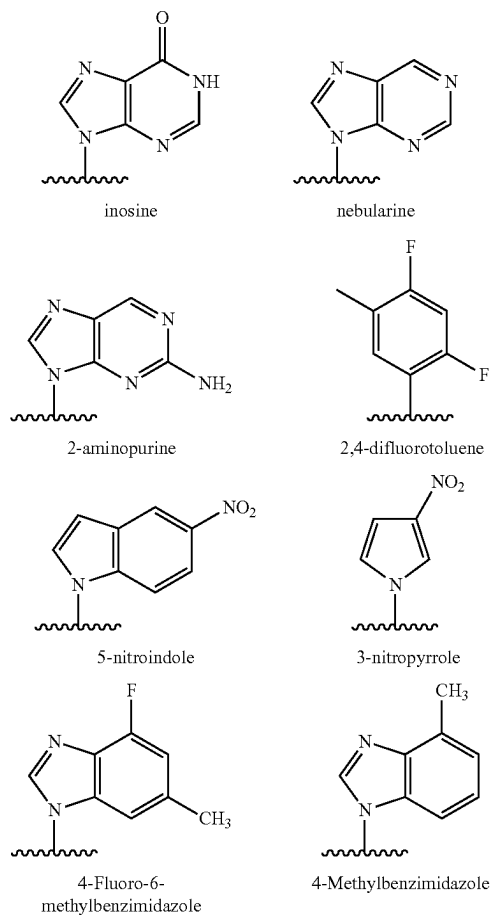
[0242] In some embodiments, the thermally destabilizing modification of the duplex in the seed region of the antisense strand includes nucleotides with impaired W-C H-bonding to complementary base on the target mRNA, such as:



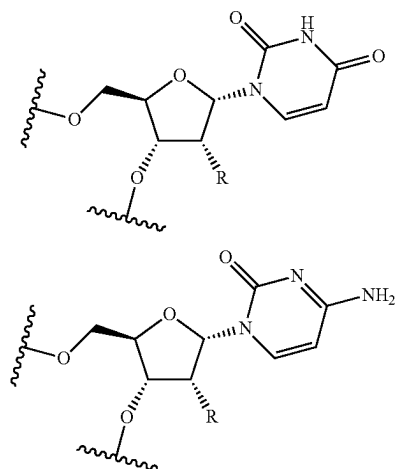
[0243] More examples of abasic nucleotide, acyclic nucleotide modifications (including UNA and GNA), and mismatch modifications have been described in detail in WO 2011/133876, which is herein incorporated by reference in its entirety.

[0244] The thermally destabilizing modifications may also include universal base with reduced or abolished capability to form hydrogen bonds with the opposing bases, and phosphate modifications.

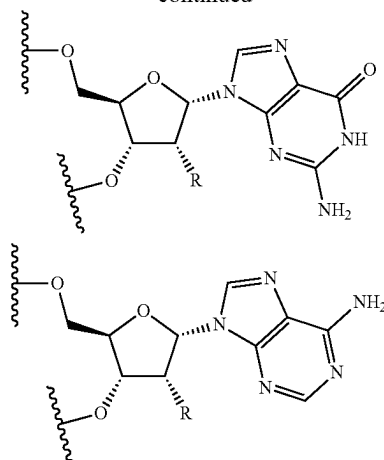
[0245] In some embodiments, the thermally destabilizing modification of the duplex includes nucleotides with non-canonical bases such as, but not limited to, nucleobase modifications with impaired or completely abolished capability to form hydrogen bonds with bases in the opposite strand. These nucleobase modifications have been evaluated for destabilization of the central region of the dsRNA duplex as described in WO 2010/0011895, which is herein incorporated by reference in its entirety. Exemplary nucleobase modifications are:



[0246] In some embodiments, the thermally destabilizing modification of the duplex in the seed region of the antisense strand includes one or more α -nucleotide complementary to the base on the target mRNA, such as:

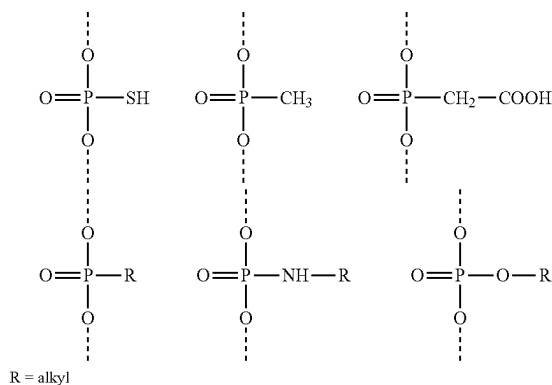


-continued



Wherein R is H, OH, OCH₃, F, NH₂, NHMe, NMe₂ or O-alkyl

[0247] Exemplary phosphate modifications known to decrease the thermal stability of dsRNA duplexes compared to natural phosphodiester linkages are:



[0248] The alkyl for the R group can be a C₁-C₆alkyl. Specific alkyls for the R group include, but are not limited to methyl, ethyl, propyl, isopropyl, butyl, pentyl and hexyl.

[0249] It is noted a thermally destabilizing modification can replace a 2'-deoxy nucleotide in the antisense strand. For example, a 2'-deoxy nucleotide at positions 2, 5, 7, 12, 14 and/or 16, counting from 5'-end, of the antisense strand can be replaced with a thermally destabilizing modification described herein. Thus, in some embodiments, the antisense strand comprises a thermally destabilizing modification at 1, 2, 3, 4, 5 and/or 6 of positions 2, 5, 7, 12, 14 and/or 16, counting from 5'-end of the antisense strand. For example, the antisense strand comprises a thermally destabilizing modification at positions 5 and 7, counting from 5'-end of the antisense strand.

[0250] In addition to the antisense strand comprising a thermally destabilizing modification, the dsRNA can also comprise one or more stabilizing modifications. For example, the dsRNA can comprise at least two (e.g., two, three, four, five, six, seven, eight, nine, ten or more) stabilizing modifications. Without limitations, the stabilizing

modifications all can be present in one strand. In some embodiments, both the sense and the antisense strands comprise at least two stabilizing modifications. The stabilizing modification can occur on any nucleotide of the sense strand or antisense strand. For instance, the stabilizing modification can occur on every nucleotide on the sense strand and/or antisense strand; each stabilizing modification can occur in an alternating pattern on the sense strand or antisense strand; or the sense strand or antisense strand comprises both stabilizing modification in an alternating pattern. The alternating pattern of the stabilizing modifications on the sense strand may be the same or different from the antisense strand, and the alternating pattern of the stabilizing modifications on the sense strand can have a shift relative to the alternating pattern of the stabilizing modifications on the antisense strand.

[0251] In some embodiments, the antisense strand comprises at least two (e.g., two, three, four, five, six, seven, eight, nine, ten or more) stabilizing modifications. Without limitations, a stabilizing modification in the antisense strand can be present at any positions. In some embodiments, the antisense comprises stabilizing modifications at positions 2, 6, 8, 9, 14 and 16 from the 5'-end. In some other embodiments, the antisense comprises stabilizing modifications at positions 2, 6, 14 and 16 from the 5'-end. In still some other embodiments, the antisense comprises stabilizing modifications at positions 2, 14 and 16 from the 5'-end.

[0252] In some embodiments, the antisense strand comprises at least one stabilizing modification adjacent to the destabilizing modification. For example, the stabilizing modification can be the nucleotide at the 5'-end or the 3'-end of the destabilizing modification, i.e., at position-1 or +1 from the position of the destabilizing modification. In some embodiments, the antisense strand comprises a stabilizing modification at each of the 5'-end and the 3'-end of the destabilizing modification, i.e., positions-1 and +1 from the position of the destabilizing modification.

[0253] In some embodiments, the antisense strand comprises at least two stabilizing modifications at the 3'-end of the destabilizing modification, i.e., at positions +1 and +2 from the position of the destabilizing modification. In some embodiments, the sense strand comprises at least two (e.g., two, three, four, five, six, seven, eight, nine, ten or more) stabilizing modifications. Without limitations, a stabilizing modification in the sense strand can be present at any positions. In some embodiments, the sense strand comprises stabilizing modifications at positions 7, 10 and 11 from the 5'-end. In some other embodiments, the sense strand comprises stabilizing modifications at positions 7, 9, 10 and 11 from the 5'-end. In some embodiments, the sense strand comprises stabilizing modifications at positions opposite or complementary to positions 11, 12 and 15 of the antisense strand, counting from the 5'-end of the antisense strand. In some other embodiments, the sense strand comprises stabilizing modifications at positions opposite or complementary to positions 11, 12, 13 and 15 of the antisense strand, counting from the 5'-end of the antisense strand. In some embodiments, the sense strand comprises a block of two, three or four stabilizing modifications.

[0254] In some embodiments, the sense strand does not comprise a stabilizing modification in position opposite or complementary to the thermally destabilizing modification of the duplex in the antisense strand.

[0255] Exemplary thermally stabilizing modifications include, but are not limited to 2'-fluoro modifications. Other thermally stabilizing modifications include, but are not limited to LNA.

[0256] It is noted a thermally stabilizing modification can replace a 2'-fluoro nucleotide in the sense and/or antisense strand. For example, a 2'-fluoro nucleotide at positions 8, 9, 10, 11 and/or 12, counting from 5'-end, of the sense strand, can be replaced with a thermally stabilizing modification. Similarly, a 2'-fluoro nucleotide at position 14, counting from 5'-end, of the antisense strand, can be replaced with a thermally stabilizing modification.

[0257] For the dsRNA molecules to be more effective in vivo, the antisense strand must have some metabolic stability. In other words, for the dsRNA molecules to be more effective in vivo, some amount of the antisense strand may need to be present in vivo after a period time after administration. Accordingly, in some embodiments, at least 40%, for example at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, or at least 80% of the antisense strand of the dsRNA is present in vivo, for example in mouse liver, at day 5 after in vivo administration. In some embodiments, at least 40%, for example at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, or at least 80% of the antisense strand of the dsRNA is present in vivo, for example in mouse liver, at day 6 after in vivo administration. In some embodiments, at least 40%, for example at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, or at least 80% of the antisense strand of the dsRNA is present in vivo, for example in mouse liver, at day 7 after in vivo administration. In some embodiments, at least 40%, for example at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, or at least 80% of the antisense strand of the dsRNA is present in vivo, for example in mouse liver, at day 8 after in vivo administration. In some embodiments, at least 40%, for example at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, or at least 80% of the antisense strand of the dsRNA is present in vivo, for example in mouse liver, at day 9 after in vivo administration. In some embodiments, at least 40%, for example at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, or at least 80% of the antisense strand of the dsRNA is present in vivo, for example in mouse liver, at day 10 after in vivo administration. In some embodiments, at least 40%, for example at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, or at least 80% of the antisense strand of the dsRNA is present in vivo, for example in mouse liver, at day 11 after in vivo administration. In some embodiments, at least 40%, for example at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, or at least 80% of the antisense strand of the dsRNA is present in vivo, for example in mouse liver, at day 12 after in vivo administration. In some embodiments, at least 40%, for example at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, or at least 80% of the antisense strand of the dsRNA is present in vivo, for example in mouse liver, at day 13 after in vivo administration. In some embodiments, at least 40%, for example at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, or at least 80% of the

antisense strand of the dsRNA is present in vivo, for example in mouse liver, at day 14 after in vivo administration. In some embodiments, at least 40%, for example at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, or at least 80% of the antisense strand of the dsRNA is present in vivo, for example in mouse liver, at day 15 after in vivo administration.

Uses of dsRNA

[0258] The present invention further relates to a use of a dsRNA molecule as defined herein for inhibiting expression of a target gene. In some embodiments, the present invention further relates to a use of a dsRNA molecule for inhibiting expression of a target gene in vitro.

[0259] The present invention further relates to a dsRNA molecule as defined herein for use in inhibiting expression of a target gene in a subject. The subject may be any animal, such as a mammal, e.g., a mouse, a rat, a sheep, a cattle, a dog, a cat, or a human

[0260] In some embodiments, the dsRNA molecule of the invention is administered in buffer. In some embodiments, siRNA compounds described herein can be formulated for administration to a subject. A formulated siRNA composition can assume a variety of states. In some examples, the composition is at least partially crystalline, uniformly crystalline, and/or anhydrous (e.g., less than 80, 50, 30, 20, or 10% water). In another example, the siRNA is in an aqueous phase, e.g., in a solution that includes water.

[0261] The aqueous phase or the crystalline compositions can, e.g., be incorporated into a delivery vehicle, e.g., a liposome (particularly for the aqueous phase) or a particle (e.g., a microparticle as can be appropriate for a crystalline composition). Generally, the siRNA composition is formulated in a manner that is compatible with the intended method of administration, as described herein. For example, in particular embodiments the composition is prepared by at least one of the following methods: spray drying, lyophilization, vacuum drying, evaporation, fluid bed drying, or a combination of these techniques; or sonication with a lipid, freeze-drying, condensation and other self-assembly.

[0262] A dsRNA preparation can be formulated in combination with another agent, e.g., another therapeutic agent or an agent that stabilizes a dsRNA, e.g., a protein that complexes with dsRNA to form an iRNP. Still other agents include chelating agents, e.g., EDTA (e.g., to remove divalent cations such as Mg^{2+}), salts, RNase inhibitors (e.g., a broad specificity RNase inhibitor such as RNasin) and so forth.

[0263] In some embodiments, the dsRNA preparation includes another dsRNA compound, e.g., a second dsRNA that can mediate RNAi with respect to a second gene, or with respect to the same gene. Still other preparation can include at least 3, 5, ten, twenty, fifty, or a hundred or more different siRNA species. Such dsRNAs can mediate RNAi with respect to a similar number of different genes.

[0264] In some embodiments, the dsRNA preparation includes at least a second therapeutic agent (e.g., an agent other than a RNA or a DNA). For example, a dsRNA composition for the treatment of a viral disease, e.g., HIV, might include a known antiviral agent (e.g., a protease inhibitor or reverse transcriptase inhibitor). In another example, a dsRNA composition for the treatment of a cancer might further comprise a chemotherapeutic agent.

[0265] Exemplary formulations which can be used for administering the dsRNA molecule according to the present invention are discussed below.

[0266] Liposomes. A dsRNA preparation can be formulated for delivery in a membranous molecular assembly, e.g., a liposome or a micelle. As used herein, the term “liposome” refers to a vesicle composed of amphiphilic lipids arranged in at least one bilayer, e.g., one bilayer or a plurality of bilayers. Liposomes include unilamellar and multilamellar vesicles that have a membrane formed from a lipophilic material and an aqueous interior. The aqueous portion contains the siRNA composition. The lipophilic material isolates the aqueous interior from an aqueous exterior, which typically does not include the siRNA composition, although in some examples, it may. Liposomes are useful for the transfer and delivery of active ingredients to the site of action. Because the liposomal membrane is structurally similar to biological membranes, when liposomes are applied to a tissue, the liposomal bilayer fuses with bilayer of the cellular membranes. As the merging of the liposome and cell progresses, the internal aqueous contents that include the dsRNA are delivered into the cell where the dsRNA can specifically bind to a target RNA and can mediate RNAi. In some cases, the liposomes are also specifically targeted, e.g., to direct the dsRNA to particular cell types.

[0267] A liposome containing a dsRNA can be prepared by a variety of methods. In one example, the lipid component of a liposome is dissolved in a detergent so that micelles are formed with the lipid component. For example, the lipid component can be an amphipathic cationic lipid or lipid conjugate. The detergent can have a high critical micelle concentration and may be nonionic. Exemplary detergents include cholate, CHAPS, octylglucoside, deoxycholate, and lauroyl sarcosine. The dsRNA preparation is then added to the micelles that include the lipid component. The cationic groups on the lipid interact with the siRNA and condense around the dsRNA to form a liposome. After condensation, the detergent is removed, e.g., by dialysis, to yield a liposomal preparation of dsRNA.

[0268] If necessary a carrier compound that assists in condensation can be added during the condensation reaction, e.g., by controlled addition. For example, the carrier compound can be a polymer other than a nucleic acid (e.g., spermine or spermidine). pH can also be adjusted to favor condensation.

[0269] Further description of methods for producing stable polynucleotide delivery vehicles, which incorporate a polynucleotide/cationic lipid complex as structural components of the delivery vehicle, are described in, e.g., WO 96/37194. Liposome formation can also include one or more aspects of exemplary methods described in Felgner, P. L. et al., *Proc. Natl. Acad. Sci., USA* 8:7413-7417, 1987; U.S. Pat. Nos. 4,897,355; 5,171,678; Bangham, et al. *M. Mol. Biol.* 23:238, 1965; Olson, et al. *Biochim. Biophys. Acta* 557:9, 1979; Szoka, et al. *Proc. Natl. Acad. Sci.* 75:4194, 1978; Mayhew, et al. *Biochim. Biophys. Acta* 775:169, 1984; Kim, et al. *Biochim. Biophys. Acta* 728:339, 1983; and Fukunaga, et al. *Endocrinol.* 115:757, 1984, which are incorporated by reference in their entirety. Commonly used techniques for preparing lipid aggregates of appropriate size for use as delivery vehicles include sonication and freeze-thaw plus extrusion (see, e.g., Mayer, et al. *Biochim. Biophys. Acta* 858:161, 1986, which is incorporated by reference in its

entirety). Microfluidization can be used when consistently small (50 to 200 nm) and relatively uniform aggregates are desired (Mayhew, et al. *Biochim. Biophys. Acta* 775:169, 1984, which is incorporated by reference in its entirety). These methods are readily adapted to packaging siRNA preparations into liposomes.

[0270] Liposomes that are pH-sensitive or negatively-charged entrap nucleic acid molecules rather than complex with them. Since both the nucleic acid molecules and the lipid are similarly charged, repulsion rather than complex formation occurs. Nevertheless, some nucleic acid molecules are entrapped within the aqueous interior of these liposomes. pH-sensitive liposomes have been used to deliver DNA encoding the thymidine kinase gene to cell monolayers in culture. Expression of the exogenous gene was detected in the target cells (Zhou et al., *Journal of Controlled Release*, 19, (1992) 269-274, which is incorporated by reference in its entirety).

[0271] One major type of liposomal composition includes phospholipids other than naturally-derived phosphatidylcholine. Neutral liposome compositions, for example, can be formed from dimyristoyl phosphatidylcholine (DMPC) or dipalmitoyl phosphatidylcholine (DPPC). Anionic liposome compositions generally are formed from dimyristoyl phosphatidylglycerol, while anionic fusogenic liposomes are formed primarily from dioleoyl phosphatidylethanolamine (DOPE). Another type of liposomal composition is formed from phosphatidylcholine (PC) such as, for example, soybean PC, and egg PC. Another type is formed from mixtures of phospholipid and/or phosphatidylcholine and/or cholesterol.

[0272] Examples of other methods to introduce liposomes into cells in vitro and include U.S. Pat. Nos. 5,283,185; 5,171,678; WO 94/00569; WO 93/24640; WO 91/16024; Felgner, *J. Biol. Chem.* 269:2550, 1994; Nabel, *Proc. Natl. Acad. Sci.* 90:11307, 1993; Nabel, *Human Gene Ther.* 3:649, 1992; Gershon, *Biochem.* 32:7143, 1993; and Strauss *EMBO J.* 11:417, 1992.

[0273] In some embodiments, cationic liposomes are used. Cationic liposomes possess the advantage of being able to fuse to the cell membrane. Non-cationic liposomes, although not able to fuse as efficiently with the plasma membrane, are taken up by macrophages in vivo and can be used to deliver siRNAs to macrophages.

[0274] Further advantages of liposomes include: liposomes obtained from natural phospholipids are biocompatible and biodegradable; liposomes can incorporate a wide range of water and lipid soluble drugs; liposomes can protect encapsulated siRNAs in their internal compartments from metabolism and degradation (Rosoff, in "Pharmaceutical Dosage Forms," Lieberman, Rieger and Banker (Eds.), 1988, volume 1, p. 245). Important considerations in the preparation of liposome formulations are the lipid surface charge, vesicle size and the aqueous volume of the liposomes.

[0275] A positively charged synthetic cationic lipid, N—[1-(2,3-dioleoyloxy) propyl]-N,N,N-trimethylammonium chloride (DOTMA) can be used to form small liposomes that interact spontaneously with nucleic acid to form lipid-nucleic acid complexes which are capable of fusing with the negatively charged lipids of the cell membranes of tissue culture cells, resulting in delivery of siRNA (see, e.g., Felgner, P. L. et al., *Proc. Natl. Acad. Sci., USA* 8:7413-

7417, 1987 and U.S. Pat. No. 4,897,355 for a description of DOTMA and its use with DNA, which are incorporated by reference in their entirety).

[0276] A DOTMA analogue, 1,2-bis(oleoyloxy)-3-(trimethylammonia) propane (DOTAP) can be used in combination with a phospholipid to form DNA-complexing vesicles. Lipofectin™ (Bethesda Research Laboratories, Gaithersburg, Md.) is an effective agent for the delivery of highly anionic nucleic acids into living tissue culture cells that comprise positively charged DOTMA liposomes which interact spontaneously with negatively charged polynucleotides to form complexes. When enough positively charged liposomes are used, the net charge on the resulting complexes is also positive. Positively charged complexes prepared in this way spontaneously attach to negatively charged cell surfaces, fuse with the plasma membrane, and efficiently deliver functional nucleic acids into, for example, tissue culture cells. Another commercially available cationic lipid, 1,2-bis(oleoyloxy)-3,3-(trimethylammonia) propane ("DOTAP") (Boehringer Mannheim, Indianapolis, Indiana) differs from DOTMA in that the oleoyl moieties are linked by ester, rather than ether linkages.

[0277] Other reported cationic lipid compounds include those that have been conjugated to a variety of moieties including, for example, carboxyspermine which has been conjugated to one of two types of lipids and includes compounds such as 5-carboxyspermylglycine dioctaoyleylamide ("DOGS") (Transfectam™, Promega, Madison, Wisconsin) and dipalmitoylphosphatidylethanolamine 5-carboxyspermyl-amide ("DPPES") (see, e.g., U.S. Pat. No. 5,171,678).

[0278] Another cationic lipid conjugate includes derivatization of the lipid with cholesterol ("DC-Chol") which has been formulated into liposomes in combination with DOPE (See, Gao, X. and Huang, L., *Biochim. Biophys. Res. Commun.* 179:280, 1991). Lipopolylysine, made by conjugating polylysine to DOPE, has been reported to be effective for transfection in the presence of serum (Zhou, X. et al., *Biochim. Biophys. Acta* 1065:8, 1991, which is incorporated by reference in its entirety). For certain cell lines, these liposomes containing conjugated cationic lipids, are said to exhibit lower toxicity and provide more efficient transfection than the DOTMA-containing compositions. Other commercially available cationic lipid products include DMRIE and DMRIE-HP (Vical, La Jolla, California) and Lipofectamine (DOSPA) (Life Technology, Inc., Gaithersburg, Maryland). Other cationic lipids suitable for the delivery of oligonucleotides are described in WO 98/39359 and WO 96/37194.

[0279] Liposomal formulations are particularly suited for topical administration. Liposomes present several advantages over other formulations. Such advantages include reduced side effects related to high systemic absorption of the administered drug, increased accumulation of the administered drug at the desired target, and the ability to administer siRNA, into the skin. In some implementations, liposomes are used for delivering siRNA to epidermal cells and also to enhance the penetration of siRNA into dermal tissues, e.g., into skin. For example, the liposomes can be applied topically. Topical delivery of drugs formulated as liposomes to the skin has been documented (see, e.g., Weiner et al., *Journal of Drug Targeting*, 1992, vol. 2, 405-410 and du Plessis et al., *Antiviral Research*, 18, 1992, 259-265; Mannino, R. J. and Fould-Fogerite, S., *Biotechniques* 6:682-690, 1988; Itani, T. et al. *Gene* 56:267-276, 1987; Nicolau, C. et

al. *Meth. Enz.* 149:157-176, 1987; Straubinger, R. M. and Papahadjopoulos, D. *Meth. Enz.* 101:512-527, 1983; Wang, C. Y. and Huang, L., *Proc. Natl. Acad. Sci. USA* 84:7851-7855, 1987, which are incorporated by reference in their entirety).

[0280] Non-ionic liposomal systems have also been examined to determine their utility in the delivery of drugs to the skin, in particular systems comprising non-ionic surfactant and cholesterol. Non-ionic liposomal formulations comprising Novasome I (glyceryl dilaurate/cholesterol/polyoxyethylene-10-stearyl ether) and Novasome II (glyceryl distearate/cholesterol/polyoxyethylene-10-stearyl ether) were used to deliver a drug into the dermis of mouse skin. Such formulations with dsRNA described herein are useful for treating a dermatological disorder.

[0281] Liposomes that include dsRNA described herein can be made highly deformable. Such deformability can enable the liposomes to penetrate through pore that are smaller than the average radius of the liposome. For example, transfersomes are a type of deformable liposomes. Transfersomes can be made by adding surface edge activators, usually surfactants, to a standard liposomal composition. Transfersomes that include dsRNA described herein can be delivered, for example, subcutaneously by infection in order to deliver dsRNA to keratinocytes in the skin. In order to cross intact mammalian skin, lipid vesicles must pass through a series of fine pores, each with a diameter less than 50 nm, under the influence of a suitable transdermal gradient. In addition, due to the lipid properties, these transfersomes can be self-optimizing (adaptive to the shape of pores, e.g., in the skin), self-repairing, and can frequently reach their targets without fragmenting, and often self-loading.

[0282] Other formulations amenable to the present invention are described in U.S. provisional application Ser. No. 61/018,616, filed Jan. 2, 2008; 61/018,611, filed Jan. 2, 2008; 61/039,748, filed Mar. 26, 2008; 61/047,087, filed Apr. 22, 2008 and 61/051,528, filed May 8, 2008. PCT application no PCT/US2007/080331, filed Oct. 3, 2007 also describes formulations that are amenable to the present invention.

[0283] Surfactants. The dsRNA compositions can include a surfactant. In some embodiments, the dsRNA is formulated as an emulsion that includes a surfactant. The most common way of classifying and ranking the properties of the many different types of surfactants, both natural and synthetic, is by the use of the hydrophile/lipophile balance (HLB). The nature of the hydrophilic group provides the most useful means for categorizing the different surfactants used in formulations (Rieger, in "Pharmaceutical Dosage Forms," Marcel Dekker, Inc., New York, NY, 1988, p. 285).

[0284] If the surfactant molecule is not ionized, it is classified as a nonionic surfactant. Nonionic surfactants find wide application in pharmaceutical products and are usable over a wide range of pH values. In general, their HLB values range from 2 to about 18 depending on their structure. Nonionic surfactants include nonionic esters such as ethylene glycol esters, propylene glycol esters, glyceryl esters, polyglyceryl esters, sorbitan esters, sucrose esters, and ethoxylated esters. Nonionic alkanolamides and ethers such as fatty alcohol ethoxylates, propoxylated alcohols, and ethoxylated/propoxylated block polymers are also included in this class. The polyoxyethylene surfactants are the most popular members of the nonionic surfactant class.

[0285] If the surfactant molecule carries a negative charge when it is dissolved or dispersed in water, the surfactant is classified as anionic. Anionic surfactants include carboxylates such as soaps, acyl lactylates, acyl amides of amino acids, esters of sulfuric acid such as alkyl sulfates and ethoxylated alkyl sulfates, sulfonates such as alkyl benzene sulfonates, acyl isethionates, acyl taurates and sulfosuccinates, and phosphates. The most important members of the anionic surfactant class are the alkyl sulfates and the soaps.

[0286] If the surfactant molecule carries a positive charge when it is dissolved or dispersed in water, the surfactant is classified as cationic. Cationic surfactants include quaternary ammonium salts and ethoxylated amines. The quaternary ammonium salts are the most used members of this class.

[0287] If the surfactant molecule has the ability to carry either a positive or negative charge, the surfactant is classified as amphoteric. Amphoteric surfactants include acrylic acid derivatives, substituted alkylamides, N-alkylbetaines and phosphatides.

[0288] The use of surfactants in drug products, formulations and in emulsions has been reviewed (Rieger, in "Pharmaceutical Dosage Forms," Marcel Dekker, Inc., New York, NY, 1988, p. 285).

[0289] Micelles and other Membranous Formulations. For ease of exposition the micelles and other formulations, compositions and methods in this section are discussed largely with regard to unmodified siRNA compounds. It may be understood, however, that these micelles and other formulations, compositions and methods can be practiced with other siRNA compounds, e.g., modified siRNA compounds, and such practice is within the invention. The siRNA compound, e.g., a double-stranded siRNA compound, or ssiRNA compound, (e.g., a precursor, e.g., a larger siRNA compound which can be processed into a ssiRNA compound, or a DNA which encodes an siRNA compound, e.g., a double-stranded siRNA compound, or ssiRNA compound, or precursor thereof) composition can be provided as a micellar formulation. "Micelles" are defined herein as a particular type of molecular assembly in which amphipathic molecules are arranged in a spherical structure such that all the hydrophobic portions of the molecules are directed inward, leaving the hydrophilic portions in contact with the surrounding aqueous phase. The converse arrangement exists if the environment is hydrophobic.

[0290] A mixed micellar formulation suitable for delivery through transdermal membranes may be prepared by mixing an aqueous solution of the dsRNA composition, an alkali metal C₈ to C₂₂ alkyl sulphate, and a micelle forming compounds. Exemplary micelle forming compounds include lecithin, hyaluronic acid, pharmaceutically acceptable salts of hyaluronic acid, glycolic acid, lactic acid, chamomile extract, cucumber extract, oleic acid, linoleic acid, linolenic acid, monoolein, monooleates, monolaurates, borage oil, evening of primrose oil, menthol, trihydroxy oxo cholanyl glycine and pharmaceutically acceptable salts thereof, glycerin, polyglycerin, lysine, polylysine, triolein, polyoxyethylene ethers and analogues thereof, polidocanol alkyl ethers and analogues thereof, chenodeoxycholate, deoxycholate, and mixtures thereof. The micelle forming compounds may be added at the same time or after addition of the alkali metal alkyl sulphate. Mixed micelles will form with substantially any kind of mixing of the ingredients but vigorous mixing in order to provide smaller size micelles.

[0291] In one method, a first micellar composition is prepared which contains the dsRNA composition and at least the alkali metal alkyl sulphate. The first micellar composition is then mixed with at least three micelle forming compounds to form a mixed micellar composition. In another method, the micellar composition is prepared by mixing the dsRNA composition, the alkali metal alkyl sulphate and at least one of the micelle forming compounds, followed by addition of the remaining micelle forming compounds, with vigorous mixing.

[0292] Phenol and/or m-cresol may be added to the mixed micellar composition to stabilize the formulation and protect against bacterial growth. Alternatively, phenol and/or m-cresol may be added with the micelle forming ingredients. An isotonic agent such as glycerin may also be added after formation of the mixed micellar composition.

[0293] For delivery of the micellar formulation as a spray, the formulation can be put into an aerosol dispenser and the dispenser is charged with a propellant. The propellant, which is under pressure, is in liquid form in the dispenser. The ratios of the ingredients are adjusted so that the aqueous and propellant phases become one, i.e., there is one phase. If there are two phases, it is necessary to shake the dispenser prior to dispensing a portion of the contents, e.g., through a metered valve. The dispensed dose of pharmaceutical agent is propelled from the metered valve in a fine spray.

[0294] Propellants may include hydrogen-containing chlorofluorocarbons, hydrogen-containing fluorocarbons, dimethyl ether and diethyl ether. In certain embodiments, HFA 134a (1,1,1,2 tetrafluoroethane) may be used.

[0295] The specific concentrations of the essential ingredients can be determined by relatively straightforward experimentation. For absorption through the oral cavities, it is often desirable to increase, e.g., at least double or triple, the dosage for through injection or administration through the gastrointestinal tract.

[0296] Particles. In some embodiments, dsRNA preparations can be incorporated into a particle, e.g., a microparticle. Microparticles can be produced by spray-drying, but may also be produced by other methods including lyophilization, evaporation, fluid bed drying, vacuum drying, or a combination of these techniques.

Pharmaceutical Compositions

[0297] The dsRNA agents of the invention can be formulated for pharmaceutical use. The present invention further relates to a pharmaceutical composition comprising the dsRNA molecule as defined herein. Pharmaceutically acceptable compositions comprise a therapeutically-effective amount of one or more of the dsRNA molecules in any of the preceding embodiments, taken alone or formulated together with one or more pharmaceutically acceptable carriers (additives), excipient and/or diluents.

[0298] The pharmaceutical compositions may be specially formulated for administration in solid or liquid form, including those adapted for the following: (1) oral administration, for example, drenches (aqueous or non-aqueous solutions or suspensions), tablets, e.g., those targeted for buccal, sublingual, and systemic absorption, boluses, powders, granules, pastes for application to the tongue; (2) parenteral administration, for example, by subcutaneous, intramuscular, intravenous or epidural injection as, for example, a sterile solution or suspension, or sustained-release formulation; (3) topical application, for example, as a cream, ointment, or a

controlled-release patch or spray applied to the skin; (4) intravaginally or intrarectally, for example, as a pessary, cream or foam; (5) sublingually; (6) ocularly; (7) transdermally; or (8) nasally. Delivery using subcutaneous or intravenous methods can be particularly advantageous.

[0299] The phrase “therapeutically-effective amount” as used herein means that amount of a compound, material, or composition comprising a compound of the invention which is effective for producing some desired therapeutic effect in at least a sub-population of cells in an animal at a reasonable benefit/risk ratio applicable to any medical treatment.

[0300] The phrase “pharmaceutically acceptable” is employed herein to refer to those compounds, materials, compositions, and/or dosage forms which are, within the scope of sound medical judgment, suitable for use in contact with the tissues of human beings and animals without excessive toxicity, irritation, allergic response, or other problem or complication, commensurate with a reasonable benefit/risk ratio.

[0301] The phrase “pharmaceutically-acceptable carrier” as used herein means a pharmaceutically-acceptable material, composition or vehicle, such as a liquid or solid filler, diluent, excipient, manufacturing aid (e.g., lubricant, talc magnesium, calcium or zinc stearate, or steric acid), or solvent encapsulating material, involved in carrying or transporting the subject compound from one organ, or portion of the body, to another organ, or portion of the body. Each carrier must be “acceptable” in the sense of being compatible with the other ingredients of the formulation and not injurious to the patient. Some examples of materials which can serve as pharmaceutically-acceptable carriers include: (1) sugars, such as lactose, glucose and sucrose; (2) starches, such as corn starch and potato starch; (3) cellulose, and its derivatives, such as sodium carboxymethyl cellulose, ethyl cellulose and cellulose acetate; (4) powdered tragacanth; (5) malt; (6) gelatin; (7) lubricating agents, such as magnesium stearate, sodium lauryl sulfate and talc; (8) excipients, such as cocoa butter and suppository waxes; (9) oils, such as peanut oil, cottonseed oil, safflower oil, sesame oil, olive oil, corn oil and soybean oil; (10) glycols, such as propylene glycol; (11) polyols, such as glycerin, sorbitol, mannitol and polyethylene glycol; (12) esters, such as ethyl oleate and ethyl laurate; (13) agar; (14) buffering agents, such as magnesium hydroxide and aluminum hydroxide; (15) alginic acid; (16) pyrogen-free water; (17) isotonic saline; (18) Ringer’s solution; (19) ethyl alcohol; (20) pH buffered solutions; (21) polyesters, polycarbonates and/or polyanhydrides; (22) bulking agents, such as polypeptides and amino acids (23) serum component, such as serum albumin, HDL and LDL; and (22) other non-toxic compatible substances employed in pharmaceutical formulations.

[0302] The formulations may conveniently be presented in unit dosage form and may be prepared by any methods well known in the art of pharmacy. The amount of active ingredient which can be combined with a carrier material to produce a single dosage form will vary depending upon the host being treated, the particular mode of administration. The amount of active ingredient which can be combined with a carrier material to produce a single dosage form will generally be that amount of the compound which produces a therapeutic effect. Generally, out of one hundred percent, this amount will range from about 0.1 percent to about ninety-nine percent of active ingredient, preferably from

about 5 percent to about 70 percent, most preferably from about 10 percent to about 30 percent.

[0303] In certain embodiments, a formulation of the present invention comprises an excipient selected from the group consisting of cyclodextrins, celluloses, liposomes, micelle forming agents, e.g., bile acids, and polymeric carriers, e.g., polyesters and polyanhydrides; and a compound of the present invention. In certain embodiments, an aforementioned formulation renders orally bioavailable a compound of the present invention.

[0304] The dsRNA agent preparation can be formulated in combination with another agent, e.g., another therapeutic agent or an agent that stabilizes a dsRNA, e.g., a protein that complexes with the dsRNA to form an iRNP. Still other agents include chelating agents, e.g., EDTA (e.g., to remove divalent cations such as Mg^{2+}), salts, RNase inhibitors (e.g., a broad specificity RNase inhibitor such as RNasin) and so forth.

[0305] Methods of preparing these formulations or compositions include the step of bringing into association a compound of the present invention with the carrier and, optionally, one or more accessory ingredients. In general, the formulations are prepared by uniformly and intimately bringing into association a compound of the present invention with liquid carriers, or finely divided solid carriers, or both, and then, if necessary, shaping the product.

[0306] In some cases, in order to prolong the effect of a drug, it is desirable to slow the absorption of the drug from subcutaneous or intramuscular injection. This may be accomplished by the use of a liquid suspension of crystalline or amorphous material having poor water solubility.

[0307] The rate of absorption of the drug then depends upon its rate of dissolution which, in turn, may depend upon crystal size and crystalline form. Alternatively, delayed absorption of a parenterally-administered drug form is accomplished by dissolving or suspending the drug in an oil vehicle.

[0308] The compounds according to the invention may be formulated for administration in any convenient way for use in human or veterinary medicine, by analogy with other pharmaceuticals.

[0309] The term "treatment" is intended to encompass therapy and cure. The patient receiving this treatment is any animal in need, including primates, in particular humans, and other mammals such as equines, cattle, swine and sheep; and poultry and pets in general.

[0310] Double-stranded RNA agents are produced in a cell in vivo, e.g., from exogenous DNA templates that are delivered into the cell. For example, the DNA templates can be inserted into vectors and used as gene therapy vectors. Gene therapy vectors can be delivered to a subject by, for example, intravenous injection, local administration (U.S. Pat. No. 5,328,470, which is incorporated by reference in its entirety), or by stereotactic injection (see, e.g., Chen et al. (1994) *Proc. Natl. Acad. Sci. USA* 91:3054-3057, which is incorporated by reference in its entirety). The pharmaceutical preparation of the gene therapy vector can include the gene therapy vector in an acceptable diluent, or can comprise a slow release matrix in which the gene delivery vehicle is imbedded. The DNA templates, for example, can include two transcription units, one that produces a transcript that includes the top strand of a dsRNA molecule and one that produces a transcript that includes the bottom strand of a dsRNA molecule. When the templates are transcribed,

the dsRNA molecule is produced, and processed into siRNA agent fragments that mediate gene silencing.

[0311] The dsRNA molecule as defined herein or a pharmaceutical composition comprising a dsRNA molecule as defined herein can be administered to a subject using different routes of delivery. A composition that includes a dsRNA described herein can be delivered to a subject by a variety of routes. Exemplary routes include: intravenous, subcutaneous, topical, rectal, anal, vaginal, nasal, pulmonary, ocular.

[0312] The dsRNA molecule of the invention can be incorporated into pharmaceutical compositions suitable for administration. Such compositions typically include one or more species of dsRNAs and a pharmaceutically acceptable carrier. As used herein the language "pharmaceutically acceptable carrier" is intended to include any and all solvents, dispersion media, coatings, antibacterial and antifungal agents, isotonic and absorption delaying agents, and the like, compatible with pharmaceutical administration. The use of such media and agents for pharmaceutically active substances is well known in the art. Except insofar as any conventional media or agent is incompatible with the active compound, use thereof in the compositions is contemplated. Supplementary active compounds can also be incorporated into the compositions.

[0313] The compositions of the present invention may be administered in a number of ways depending upon whether local or systemic treatment is desired and upon the area to be treated. Administration may be topical (including ophthalmic, vaginal, rectal, intranasal, transdermal), oral or parenteral. Parenteral administration includes intravenous drip, subcutaneous, intraperitoneal or intramuscular injection, or intrathecal or intraventricular administration.

[0314] The route and site of administration may be chosen to enhance targeting. For example, to target muscle cells, intramuscular injection into the muscles of interest would be a logical choice. Lung cells might be targeted by administering the dsRNA in aerosol form. The vascular endothelial cells could be targeted by coating a balloon catheter with the dsRNA and mechanically introducing the dsRNA.

[0315] In one aspect, the invention features a method of administering a dsRNA molecule, e.g., a dsRNA agent described herein, to a subject (e.g., a human subject). In another aspect, the present invention relates to a dsRNA molecule as defined herein for use in inhibiting expression of a target gene in a subject. The method or the medical use includes administering a unit dose of the dsRNA molecule, e.g., a dsRNA agent described herein. In some embodiments, the unit dose is less than 10 mg per kg of bodyweight, or less than 10, 5, 2, 1, 0.5, 0.1, 0.05, 0.01, 0.005, 0.001, 0.0005, 0.0001, 0.00005 or 0.00001 mg per kg of bodyweight, and less than 200 nmole of RNA agent (e.g., about 4.4×10^{16} copies) per kg of bodyweight, or less than 1500, 750, 300, 150, 75, 15, 7.5, 1.5, 0.75, 0.15, 0.075, 0.015, 0.0075, 0.0015, 0.00075, 0.00015 nmole of RNA agent per kg of bodyweight.

[0316] The defined amount can be an amount effective to treat or prevent a disease or disorder, e.g., a disease or disorder associated with the target gene. The unit dose, for example, can be administered by injection (e.g., intravenous, subcutaneous or intramuscular), an inhaled dose, or a topical application. In some embodiments dosages may be less than 10, 5, 2, 1, or 0.1 mg/kg of body weight.

[0317] In some embodiments, the unit dose is administered less frequently than once a day, e.g., less than every 2, 4, 8 or 30 days. In another embodiment, the unit dose is not administered with a frequency (e.g., not a regular frequency). For example, the unit dose may be administered a single time.

[0318] In some embodiments, the effective dose is administered with other traditional therapeutic modalities. In some embodiments, the subject has a viral infection and the modality is an antiviral agent other than a dsRNA molecule, e.g., other than a siRNA agent. In another embodiment, the subject has atherosclerosis and the effective dose of a dsRNA molecule, e.g., a siRNA agent, is administered in combination with, e.g., after surgical intervention, e.g., angioplasty.

[0319] In some embodiments, a subject is administered an initial dose and one or more maintenance doses of a dsRNA molecule, e.g., a siRNA agent, (e.g., a precursor, e.g., a larger dsRNA molecule which can be processed into a siRNA agent, or a DNA which encodes a dsRNA molecule, e.g., a siRNA agent, or precursor thereof). The maintenance dose or doses can be the same or lower than the initial dose, e.g., one-half less of the initial dose. A maintenance regimen can include treating the subject with a dose or doses ranging from 0.01 μ g to 15 mg/kg of body weight per day, e.g., 10, 1, 0.1, 0.01, 0.001, or 0.00001 mg per kg of bodyweight per day. The maintenance doses are, for example, administered no more than once every 2, 5, 10, or 30 days. Further, the treatment regimen may last for a period of time which will vary depending upon the nature of the particular disease, its severity and the overall condition of the patient. In certain embodiments the dosage may be delivered no more than once per day, e.g., no more than once per 24, 36, 48, or more hours, e.g., no more than once for every 5 or 8 days. Following treatment, the patient can be monitored for changes in his condition and for alleviation of the symptoms of the disease state. The dosage of the compound may either be increased in the event the patient does not respond significantly to current dosage levels, or the dose may be decreased if an alleviation of the symptoms of the disease state is observed, if the disease state has been ablated, or if undesired side-effects are observed.

[0320] The effective dose can be administered in a single dose or in two or more doses, as desired or considered appropriate under the specific circumstances. If desired to facilitate repeated or frequent infusions, implantation of a delivery device, e.g., a pump, semi-permanent stent (e.g., intravenous, intraperitoneal, intracisternal or intracapsular), or reservoir may be advisable.

[0321] In some embodiments, the composition includes a plurality of dsRNA molecule species. In another embodiment, the dsRNA molecule species has sequences that are non-overlapping and non-adjacent to another species with respect to a naturally occurring target sequence. In another embodiment, the plurality of dsRNA molecule species is specific for different naturally occurring target genes. In another embodiment, the dsRNA molecule is allele specific.

[0322] The dsRNA molecules of the invention described herein can be administered to mammals, particularly large mammals such as nonhuman primates or humans in a number of ways.

[0323] In some embodiments, the administration of the dsRNA molecule, e.g., a siRNA agent, composition is parenteral, e.g., intravenous (e.g., as a bolus or as a diffusible

infusion), intradermal, intraperitoneal, intramuscular, intrathecal, intraventricular, intracranial, subcutaneous, transmucosal, buccal, sublingual, endoscopic, rectal, oral, vaginal, topical, pulmonary, intranasal, urethral or ocular. Administration can be provided by the subject or by another person, e.g., a health care provider. The medication can be provided in measured doses or in a dispenser which delivers a metered dose. Selected modes of delivery are discussed in more detail below.

[0324] The invention provides methods, compositions, and kits, for rectal administration or delivery of dsRNA molecules described herein

[0325] In particular embodiments, the present invention relates to the dsRNA molecules of the present invention for use in the methods described above.

[0326] Methods of inhibiting expression of the target gene

[0327] Embodiments of the invention also relate to methods for inhibiting the expression of a target gene. The method comprises the step of administering the dsRNA molecules in any of the preceding embodiments, in an amount sufficient to inhibit expression of the target gene. The present invention further relates to a use of a dsRNA molecule as defined herein for inhibiting expression of a target gene in a target cell. In a preferred embodiment, the present invention further relates to a use of a dsRNA molecule for inhibiting expression of a target gene in a target cell in vitro.

[0328] Another aspect the invention relates to a method of modulating the expression of a target gene in a cell, comprising providing to said cell a dsRNA molecule of this invention. In some embodiments, the target gene is selected from the group consisting of Factor VII, Eg5, PCSK9, TPX2, apoB, SAA, TTR, RSV, PDGF beta gene, Erb-B gene, Src gene, CRK gene, GRB2 gene, RAS gene, MEKK gene, JNK gene, RAF gene, Erk1/2 gene, PCNA (p21) gene, MYB gene, JUN gene, FOS gene, BCL-2 gene, hepcidin, Activated Protein C, Cyclin D gene, VEGF gene, EGFR gene, Cyclin A gene, Cyclin E gene, WNT-1 gene, beta-catenin gene, c-MET gene, PKC gene, NFkB gene, STAT3 gene, survivin gene, Her2/Neu gene, topoisomerase I gene, topoisomerase II alpha gene, mutations in the p73 gene, mutations in the p21 (WAF1/CIP1) gene, mutations in the p27 (KIP1) gene, mutations in the PPMID gene, mutations in the RAS gene, mutations in the caveolin I gene, mutations in the MIB I gene, mutations in the MTAI gene, mutations in the M68 gene, mutations in tumor suppressor genes, and mutations in the p53 tumor suppressor gene.

[0329] In particular embodiments, the present invention relates to the dsRNA molecules of the present invention for use in the methods described above.

[0330] Exemplary embodiments of the various aspects can be described by the following lettered embodiments:

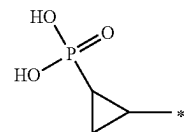
[0331] Embodiment A: A dsRNA agent comprising a sense strand and antisense, each strand independently having a length of 15-35 nucleotides, wherein the sense strand comprises a 2'-fluoro nucleotide at position 10, counting from 5'-end of the sense strand, and wherein the antisense strand comprises a 2'-deoxy nucleotide at positions 5 and 7, counting from 5'-end of the antisense strand.

[0332] Embodiment B: The dsRNA agent of Embodiment A, wherein the sense strand further comprises a 2'-fluoro nucleotide at position 11, counting from 5'-end of the sense strand.

- [0333] Embodiment C: The dsRNA agent of Embodiment A or B, wherein the sense strand further comprises a 2'-fluoro nucleotide at position 9, counting from 5'-end of the sense strand.
- [0334] Embodiment D: The dsRNA of any one of Embodiments A to C, wherein the sense strand further comprises a 2'-fluoro nucleotide at positions 9 and 11, counting from 5'-end of the sense strand.
- [0335] Embodiment E: The dsRNA agent of any one of Embodiments A to D, wherein the sense strand comprises a 2'-fluoro nucleotide at positions 8 and 9, counting from 5'-end of the sense strand.
- [0336] Embodiment F: The dsRNA agent of any one of Embodiments A to E, wherein the sense strand comprises a 2'-fluoro nucleotide at positions 11 and 12, counting from 5'-end of the sense strand.
- [0337] Embodiment G: The dsRNA agent of any one of Embodiments A to F, wherein the sense strand comprises at least one 2'-OMe nucleotide.
- [0338] Embodiment H: The dsRNA agent of any one of Embodiments A to G, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 2, counting from 5'-end of the antisense strand.
- [0339] Embodiment I: The dsRNA agent of any one of Embodiments A to H, wherein the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5 and 7, counting from 5'-end of the antisense strand.
- [0340] Embodiment J: The dsRNA agent of any one of Embodiments A to I, wherein the antisense strand comprises at least one 2'-fluoro nucleotide.
- [0341] Embodiment K: The dsRNA agent of any one of Embodiments A to J, wherein the antisense strand comprises a 2'-fluoro nucleotide at position 14 of the antisense strand, counting from 5'-end of the antisense strand.
- [0342] Embodiment L: The dsRNA agent of any one of Embodiments A to K, wherein the dsRNA agent comprises a ligand.
- [0343] Embodiment M: The dsRNA agent of any one of Embodiments A to L, wherein the sense strand comprises a ligand.
- [0344] Embodiment N: The dsRNA agent of Embodiment L or M, wherein the ligand is an ASGPR ligand.
- [0345] Embodiment O: The dsRNA agent of any one of Embodiments A to N, wherein the dsRNA agent comprises at least two phosphorothioate internucleotide linkages.
- [0346] Embodiment P: The dsRNA agent of any one of Embodiments A to O, wherein the sense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 5' end of the sense strand.
- [0347] Embodiment Q: The dsRNA agent of any one of Embodiments A to P, wherein the antisense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 5' end of the antisense strand and at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 3' end of the antisense strand.
- [0348] Embodiment R: The dsRNA agent of any one of Embodiments A to Q, wherein the dsRNA has a duplex region of from 18 to about 25 basepairs.
- [0349] Embodiment S: The dsRNA agent of any one of Embodiments A to R, wherein the sense strand is 18-23 nucleotides in length.
- [0350] Embodiment T: The dsRNA agent of any one of Embodiments A to S, wherein the antisense strand is 18-25 nucleotides in length.
- [0351] Embodiment U: A dsRNA agent comprising a sense strand and an antisense strand, wherein the sense strand is 18-23 nucleotides in length and comprises a 2'-fluoro nucleotide at position 10, counting from 5'-end of the sense strand and a 2'-fluoro nucleotide at position 9 or 11, counting from 5'-end of the sense strand, and wherein the antisense strand is 18-25 nucleotide in length and comprises a 2'-deoxy nucleotide at positions 5 and 7, counting from 5'-end of the antisense strand.
- [0352] Embodiment V: A dsRNA agent comprising a sense strand and an antisense strand, wherein the sense strand is 18-23 nucleotides in length and comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, counting from 5'-end of the sense strand, and wherein the antisense is 18-25 nucleotide in length and comprises a 2'-deoxy nucleotide at position 5 and 7, counting from 5'-end of the antisense strand.
- [0353] Additional exemplary embodiments can be described by one or more of the following numbered embodiments:
- [0354] Embodiment 1: dsRNA agent comprising a sense strand and antisense, each strand independently having a length of 15-35 nucleotides wherein each nucleotide is independently modified or unmodified, wherein: the sense strand comprises a 2'-fluoro nucleotide at position 10, counting from 5'-end of the sense strand, and the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7 and 12, counting from 5'-end of the antisense strand, and wherein: (i) the antisense strand comprises a 2'-fluoro nucleotide at position 14 and a nucleotide other than a 2'-deoxy or 2'-fluoro nucleotide at position 16, counting from the 5'-end of the antisense strand; or (ii) the antisense strand comprises a 2'-deoxy nucleotide at position 14 or 16, counting from the 5'-end of the antisense strand, and the sense strand comprises a nucleotide other than a 2'-fluoro nucleotide at position 7, counting from the 5'-end of the sense strand.
- [0355] Embodiment 2: The dsRNA agent of Embodiment 1, wherein the sense strand further comprises a 2'-fluoro nucleotide at position 11, counting from 5'-end of the sense strand.
- [0356] Embodiment 3: The dsRNA agent of Embodiment 1 or 2, wherein the sense strand further comprises a 2'-fluoro nucleotide at position 9, counting from 5'-end of the sense strand.
- [0357] Embodiment 4: The dsRNA of any one of Embodiments 1-3, wherein the sense strand further comprises a 2'-fluoro nucleotide at positions 9 and 11, counting from 5'-end of the sense strand.
- [0358] Embodiment 5: The dsRNA agent of any one of Embodiments 1-4, wherein the sense strand comprises a 2'-fluoro nucleotide at positions 8 and 9, counting from 5'-end of the sense strand.
- [0359] Embodiment 6: The dsRNA agent of any one of Embodiments 1-5, wherein the sense strand comprises

- a 2'-fluoro nucleotide at positions 11 and 12, counting from 5'-end of the sense strand.
- [0360] Embodiment 7: The dsRNA agent of any one of Embodiments 1-6, wherein the sense strand comprises a nucleotide other than a 2'-fluoro at position 7, counting from the 5'-end of the sense strand.
- [0361] Embodiment 8: The dsRNA of any one of Embodiments 1-7, wherein the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, and a nucleotide other than a 2'-fluoro at position 7, counting from the 5'-end of the sense strand.
- [0362] Embodiment 9: The dsRNA agent of any one of Embodiments 1-8, wherein the sense strand comprises at least one 2'-OMe nucleotide.
- [0363] Embodiment 10: The dsRNA agent of any one of Embodiments 1-9, wherein the sense strand comprises a 2'-OMe nucleotide at position 7, counting from the 5'-end of the sense strand.
- [0364] Embodiment 11: The dsRNA agent of any one of Embodiments 1-10, wherein the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, and a 2'-OMe nucleotide at position 7, counting from the 5'-end of the sense strand.
- [0365] Embodiment 12: The dsRNA agent of any one of Embodiments 1-11, wherein the antisense strand comprises a 2'-fluoro nucleotide at position 14 of the antisense strand, and a nucleotide other than a 2'-deoxy or 2'-fluoro nucleotide at position 16, counting from 5'-end of the antisense strand.
- [0366] Embodiment 13: The dsRNA agent of any one of Embodiments 1-12, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 2, 5, 7, and 12, a 2'-fluoro nucleotide at position 14 and a nucleotide other than a 2'-deoxy or 2'-fluoro nucleotide at position 16, counting from the 5'-end of the antisense strand.
- [0367] Embodiment 14: The dsRNA agent of any one of Embodiments 1-13, wherein the antisense strand comprises a 2'-OMe nucleotide at position 16, counting from the 5'-end of the antisense strand.
- [0368] Embodiment 15: The dsRNA agent of any one of Embodiments 1-14, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 2, 5, 7, and 12, a 2'-fluoro nucleotide at position 14, and a 2'-OMe nucleotide at position 16, counting from the 5'-end of the antisense strand.
- [0369] Embodiment 16: The dsRNA agent of any one of Embodiments 1-11, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 14 of the antisense strand, counting from 5'-end of the antisense strand, and the sense strand comprises a nucleotide other than a 2'-fluoro nucleotide at position 7, counting from 5'-end of the sense strand.
- [0370] Embodiment 17: The dsRNA agent of any one of Embodiments 1-11 or 16, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 14 of the antisense strand, counting from 5'-end of the antisense strand, and the sense strand comprises a 2'-OMe nucleotide at position 7, counting from 5'-end of the sense strand.
- [0371] Embodiment 18: The dsRNA agent of any one of Embodiments 1-11 or 16-17, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 2, 5, 7, 12 and 14 of the antisense strand, counting from 5'-end of the antisense strand, and the sense strand comprises a nucleotide other than 2'-fluoro nucleotide at position 7, counting from 5'-end of the sense strand.
- [0372] Embodiment 19: The dsRNA agent of any one of Embodiments 1-18, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 16 of the antisense strand, counting from 5'-end of the antisense strand, and the sense strand comprises a nucleotide other than a 2'-fluoro nucleotide at position 7, counting from 5'-end of the sense strand.
- [0373] Embodiment 20: The dsRNA agent of any one of Embodiments 1-19, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 16 of the antisense strand, counting from 5'-end of the antisense strand, and the sense strand comprises 2'-OMe nucleotide at position 7, counting from 5'-end of the sense strand.
- [0374] Embodiment 21: The dsRNA agent of any one of Embodiments 1-20, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 2, 5, 7, 12 and 16 of the antisense strand, counting from 5'-end of the antisense strand, and the sense strand comprises a nucleotide other than a 2'-fluoro at position 7, counting from 5'-end of the sense strand.
- [0375] Embodiment 22: The dsRNA agent of any one of Embodiments 1-11 or 16-21, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 2, 5, 7, 12, 14 and 16 of the antisense strand, counting from 5'-end of the antisense strand, and the sense strand comprises a nucleotide other than a 2'-fluoro at position 7, counting from 5'-end of the sense strand.
- [0376] Embodiment 23: The dsRNA agent of any one of Embodiments 1-11 or 16-22, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 2, 5, 7, 12, 14 and 16, and the sense strand comprises a nucleotide other than a 2'-fluoro at position 7, counting from 5'-end of the sense strand.
- [0377] Embodiment 24: The dsRNA agent of any one of Embodiments 1-15 or 19-20, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 2, 5, 7, 12, and 16 and a 2'-fluoro at position 14 of the antisense strand, counting from 5'-end of the antisense strand, and the sense strand comprises a nucleotide other than a 2'-fluoro at position 7, counting from 5'-end of the sense strand.
- [0378] Embodiment 25: The dsRNA agent of any one of Embodiments 1-15, 19-20 or 24, wherein the antisense strand comprises a 2'-deoxy nucleotide at position 2, 5, 7, 12, and 16 and a 2'-fluoro at position 14 of the antisense strand, counting from 5'-end of the antisense strand, and the sense strand comprises a 2'-OMe nucleotide at position 7, counting from 5'-end of the sense strand.
- [0379] Embodiment 26: The dsRNA agent of any one of Embodiments 1-25, wherein the dsRNA agent comprises a ligand.
- [0380] Embodiment 27: The dsRNA agent of any one of Embodiments 1-26, wherein the sense strand comprises a ligand.
- [0381] Embodiment 28: The dsRNA agent of Embodiment 27, wherein the ligand is at 3'-end of the sense strand.
- [0382] Embodiment 29: The dsRNA agent of Embodiment 27, wherein the ligand is at 5'-end of the sense strand.

- [0383] Embodiment 30: The dsRNA agent of any one of Embodiments 26-29, wherein the ligand comprises an ASGPR ligand.
- [0384] Embodiment 31: The dsRNA agent of any one of Embodiments 26-29, wherein the ligand is lipophilic group.
- [0385] Embodiment 32: The dsRNA agent of Embodiment 31, wherein the ligand is a C₁₀₋₃₀aliphatic group.
- [0386] Embodiment 33: The dsRNA agent of Embodiment 32, wherein the C₁₀₋₃₀aliphatic group is a C₁₀₋₃₀alkyl group.
- [0387] Embodiment 34: The dsRNA agent of Embodiment 33, wherein the C₁₀₋₃₀alkyl group is a straight-chain or branched tetradecyl, hexadecyl, octadecyl, icosyl, docosyl, or tetracosyl group.
- [0389] Embodiment 35: The dsRNA agent of any one of Embodiments 27, wherein the ligand is conjugated to a non-terminal nucleotide of the sense strand.
- [0390] Embodiment 36: The dsRNA agent of Embodiment 35, wherein the ligand is conjugated to the 2'-position of a non-terminal nucleotide of the sense strand, optionally conjugated to one of positions 5, 6, 7, or 8 of the sense strand, counting from the 5' end).
- [0391] Embodiment 37: The dsRNA agent of Embodiment 26-36, wherein the ligand comprises an abasic nucleotide, optionally the abasic nucleotide is an inverted nucleotide and linked via a 5'->5' or a 3'->3' linkage to a strand of the dsRNA agent.
- [0392] Embodiment 38: The dsRNA agent of any one of Embodiments 26-37, wherein the ligand is attached at the 3'-end of the sense strand.
- [0393] Embodiment 39: The dsRNA agent of Embodiment 38, wherein the ligand is attached at the 3'-end of the sense strand via a 3'->3' linkage.
- [0394] Embodiment 40: The dsRNA agent of any one of Embodiments 1-39, wherein the dsRNA comprises two ligands.
- [0395] Embodiment 41: The dsRNA of Embodiment 40, wherein the sense strand comprises a first ligand attached at the 3'-end of the sense strand and a second ligand attached at the 5'-end of the sense strand.
- [0396] Embodiment 42: The dsRNA of Embodiment 41, wherein the first ligand comprises an abasic nucleotide and the second ligand comprises an ASGPR ligand, optionally the abasic nucleotide is an inverted nucleotide and linked via a 3'->3' linkage to the sense strand.
- [0397] Embodiment 43: The dsRNA agent of any one of Embodiments 1-42, wherein the dsRNA agent comprises at least two phosphorothioate internucleotide linkages.
- [0398] Embodiment 44: The dsRNA agent of any one of Embodiments 1-43, wherein the sense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 5' end of the sense strand.
- [0399] Embodiment 45: The dsRNA agent of any one of Embodiments 1-44, wherein the antisense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 5' end of the antisense strand and at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 3' end of the antisense strand.
- [0400] Embodiment 46: The dsRNA agent of any one of Embodiments 1-45, wherein the dsRNA has a duplex region of from 18 to about 25 basepairs.
- [0401] Embodiment 47: The dsRNA agent of any one of Embodiments 1-46, wherein the sense strand is 18-23 nucleotides in length.
- [0402] Embodiment 48: The dsRNA agent of any one of Embodiments 1-47, wherein the antisense strand is 18-25 nucleotides in length.
- [0403] Embodiment 49: A dsRNA agent comprising a sense strand and an antisense strand, wherein the sense strand is 18-23 nucleotides in length and comprises a 2'-fluoro nucleotide at position 10, counting from 5'-end of the sense strand and a 2'-fluoro nucleotide at position 9 or 11, counting from 5'-end of the sense strand, and the antisense strand is 18-25 nucleotide in length and comprises a 2'-deoxy nucleotide at positions 2, 5, 7, and 12, counting from 5'-end of the antisense strand, wherein: (i) the antisense strand comprises a 2'-fluoro nucleotide at position 14 and a nucleotide other than a 2'-deoxy or 2'-fluoro nucleotide at position 16, counting from the 5'-end of the antisense strand; or (ii) the antisense strand comprises a 2'-deoxy nucleotide at position 14 or 16, counting from the 5'-end of the antisense strand, and the sense strand comprises a nucleotide other than a 2'-fluoro nucleotide at position 7, counting from the 5'-end of the sense strand.
- [0404] Embodiment 50: A dsRNA agent comprising a sense strand and an antisense strand, wherein the sense strand is 18-23 nucleotides in length and comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, counting from 5'-end of the sense strand, and the antisense is 18-25 nucleotide in length and comprises a 2'-deoxy nucleotide at position 2, 5, 7, and 12, counting from 5'-end of the antisense strand, wherein: (i) the antisense strand comprises a 2'-fluoro nucleotide at position 14 and a nucleotide other than a 2'-deoxy or 2'-fluoro nucleotide at position 16, counting from the 5'-end of the antisense strand; or (ii) the antisense strand comprises a 2'-deoxy nucleotide at position 14 or 16, counting from the 5'-end of the antisense strand, and the sense strand comprises a nucleotide other than a 2'-fluoro nucleotide at position 7, counting from the 5'-end of the sense strand.
- [0405] Embodiment 51: The dsRNA agent of any one of Embodiments 1-50, comprising a phosphate mimic at the 5'-end of the antisense strand.
- [0406] Embodiment 52: The dsRNA agent of Embodiment 51, wherein the phosphate mimic is a 5'-E-vinyl phosphonate.
- [0407] Embodiment 53: The dsRNA agent of Embodiment 52, wherein the phosphate mimic is a 5'-cyclopropylphosphonate having the structure:



where * is a bond to C₅ position of the nucleotide at the 5'-terminus.

[0408] Embodiment 54: The dsRNA agent of any one of Embodiments 1-53, wherein remaining nucleotides (i.e., nucleotides at positions not otherwise defined) in the sense strand are unmodified nucleotides or modified nucleotides, optionally selected from the groups consisting of 2'-OMe, 2'-F, 2'-H, and an 2'-O—C10-30aliphatic group, provided no more than one modified nucleotide is an 2'-O—C10-30aliphatic group.

[0409] Embodiment 55: The dsRNA agent of any one of Embodiments 1-54, wherein remaining nucleotides (i.e., nucleotides at positions not otherwise defined) in the sense strand are modified nucleotides selected from the group consisting of 2'-OMe, 2'-F, 2'-H, and an 2'-O—C10-30aliphatic group, provided no more than one modified nucleotide is an 2'-O—C10-30aliphatic group.

[0410] Embodiment 56: The dsRNA agent of any one of Embodiments 1-55, wherein remaining nucleotides (i.e., nucleotides at positions not otherwise defined) in the antisense strand are unmodified nucleotides or modified nucleotides, optionally selected from the group consisting of 2'-OMe, 2'-F, 2'-H, GNA and 3'-RNA, the 3'-RNA being optionally 3'-OH, provided no more than one modified nucleotide is GNA or 3'-RNA.

[0411] Embodiment 57: The dsRNA agent of any one of Embodiments 1-56, wherein remaining nucleotides (i.e., nucleotides at positions not otherwise defined) in the antisense strand are modified nucleotides selected from the group consisting of 2'-OMe, 2'-F, 2'-H, GNA, and 3'-RNA, the 3'-RNA being optionally 3'-OH, provided no more than one modified nucleotide is GNA or 3'-RNA.

Some Selected Definitions

[0412] For convenience, certain terms employed herein, in the specification, examples and appended claims are collected herein. Unless stated otherwise, or implicit from context, the following terms and phrases include the meanings provided below. Unless explicitly stated otherwise, or apparent from context, the terms and phrases below do not exclude the meaning that the term or phrase has acquired in the art to which it pertains. The definitions are provided to aid in describing particular embodiments, and are not intended to limit the claimed invention, because the scope of the invention is limited only by the claims. Further, unless otherwise required by context, singular terms shall include pluralities and plural terms shall include the singular.

[0413] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as those commonly understood to one of ordinary skill in the art to which this invention pertains. Although any known methods, devices, and materials may be used in the practice or testing of the invention, the methods, devices, and materials in this regard are described herein.

[0414] Further, the practice of the present invention can employ, unless otherwise indicated, conventional techniques of molecular biology (including recombinant techniques), microbiology, cell biology, biochemistry, and immunology, which are within the skill of the art. Such techniques are explained fully in the literature, such as, *"Molecular Cloning: A Laboratory Manual"*, second edition (Sambrook et al., 1989); *"Oligonucleotide Synthesis"* (M. J. Gait, ed., 1984); *"Animal Cell Culture"* (R. I. Freshney, ed., 1987); *"Methods in Enzymology"* (Academic Press, Inc.); *"Current*

Protocols in Molecular Biology" (F. M. Ausubel et al., eds., 1987, and periodic updates); *"PCR: The Polymerase Chain Reaction"*, (Mullis et al., ed., 1994); *"A Practical Guide to Molecular Cloning"* (Perbal Bernard V., 1988); *"Phage Display: A Laboratory Manual"* (Barbas et al., 2001).

[0415] Where a range of values is provided, it is understood that each intervening value, to the tenth of the unit of the lower limit unless the context clearly dictates otherwise, between the upper and lower limit of that range and any other stated or intervening value in that stated range, is encompassed within the invention. The upper and lower limits of these smaller ranges may independently be included in the smaller ranges and are also encompassed within the invention, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either or both of those included limits are also included in the invention.

[0416] Certain ranges are presented herein with numerical values being preceded by the term "about." The term "about" is used herein to provide literal support for the exact number that it precedes, as well as a number that is near to or approximately the number that the term precedes. In determining whether a number is near to or approximately a specifically recited number, the near or approximating unrecited number may be a number which, in the context in which it is presented, provides the substantial equivalent of the specifically recited number.

[0417] As used herein the term "comprising" or "comprises" is used in reference to compositions, methods, and respective component(s) thereof, that are essential to the invention, yet open to the inclusion of unspecified elements, whether essential or not.

[0418] The singular terms "a," "an," and "the" include plural referents unless context clearly indicates otherwise. Similarly, the word "or" is intended to include "and" unless the context clearly indicates otherwise. It is further noted that the claims can be drafted to exclude any optional element. As such, this statement is intended to serve as antecedent basis for use of such exclusive terminology as "solely," "only" and the like in connection with the recitation of claim elements, or use of a "negative" limitation.

[0419] As used herein, the terms "dsRNA", "siRNA", and "iRNA agent" are used interchangeably to refer to agents that can mediate silencing of a target RNA, e.g., mRNA, e.g., a transcript of a gene that encodes a protein. For convenience, such mRNA is also referred to herein as mRNA to be silenced. Such a gene is also referred to as a target gene. In general, the RNA to be silenced is an endogenous gene, exogenous gene or a pathogen gene. In addition, RNAs other than mRNA, e.g., tRNAs, and viral RNAs, can also be targeted.

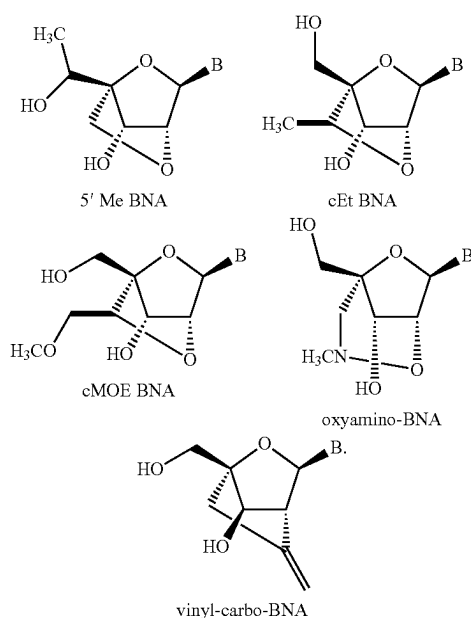
[0420] As used herein, the phrase "mediates RNAi" refers to the ability to silence, in a sequence specific manner, a target gene, e.g., mRNA. While not wishing to be bound by theory, it is believed that silencing uses the RNAi machinery or process and a guide RNA, e.g., antisense strand of a dsRNA, where the antisense strand is 21 to 23 nucleotides in length.

[0421] As used herein, "specifically hybridizable" and "complementary" are terms which are used to indicate a sufficient degree of complementarity such that stable and specific binding occurs between a compound of the invention and a target RNA molecule. Specific binding requires a sufficient degree of complementarity to avoid non-specific

binding of the oligomeric compound to non-target sequences under conditions in which specific binding is desired, i.e., under physiological conditions in the case of assays or therapeutic treatment, or in the case of in vitro assays, under conditions in which the assays are performed. The non-target sequences typically differ by at least 5 nucleotides.

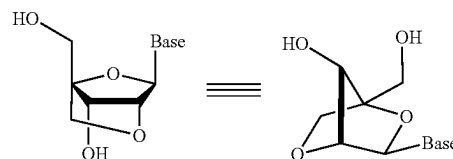
[0422] In some embodiments, a dsRNA molecule of the invention is “sufficiently complementary” to a target RNA, e.g., a target mRNA, such that the dsRNA molecule silences production of protein encoded by the target mRNA. In another embodiment, the dsRNA molecule of the invention is “exactly complementary” to a target RNA, e.g., the target RNA and the dsRNA duplex agent anneal, for example to form a hybrid made exclusively of Watson-Crick base pairs in the region of exact complementarity. A “sufficiently complementary” target RNA can include an internal region (e.g., of at least 10 nucleotides) that is exactly complementary to a target RNA. Moreover, in some embodiments, the dsRNA molecule of the invention specifically discriminates a single-nucleotide difference. In this case, the dsRNA molecule only mediates RNAi if exact complementarity is found in the region (e.g., within 7 nucleotides of) the single-nucleotide difference.

[0423] The term ‘BNA’ refers to bridged nucleic acid, and is often referred as constrained or inaccessible RNA. BNA can contain a 5-, 6-membered, or even a 7-membered bridged structure with a “fixed” C3'-endo sugar pucker. The bridge is typically incorporated at the 2', 4'-position of the ribose to afford a 2', 4'-BNA nucleotide (e.g., LNA, or ENA). Examples of BNA nucleotides include the following nucleosides:



[0424] The term ‘LNA’ refers to locked nucleic acid, and is often referred as constrained or inaccessible RNA. LNA is a modified RNA nucleotide. The ribose moiety of an LNA nucleotide is modified with an extra bridge (e.g., a methylene bridge or an ethylene bridge) connecting the 2' hydroxyl

to the 4' carbon of the same ribose sugar. For instance, the bridge can “lock” the ribose in the 3'-endo North) conformation:



[0425] The term ‘ENA’ refers to ethylene-bridged nucleic acid, and is often referred as constrained or inaccessible RNA.

[0426] The “cleavage site” herein means the backbone linkage in the target gene or the sense strand that is cleaved by the RISC mechanism by utilizing the iRNA agent. And the target cleavage site region comprises at least one or at least two nucleotides on both side of the cleavage site. For the sense strand, the cleavage site is the backbone linkage in the sense strand that would get cleaved if the sense strand itself was the target to be cleaved by the RNAi mechanism. The cleavage site can be determined using methods known in the art, for example the 5'-RACE assay as detailed in Soutschek et al., *Nature* (2004) 432, 173-178, which is incorporated by reference in its entirety. As is well understood in the art, the cleavage site region for a conical double stranded RNAi agent comprising two 21-nucleotides long strands (wherein the strands form a double stranded region of 19 consecutive base pairs having 2-nucleotide single stranded overhangs at the 3'-ends), the cleavage site region corresponds to positions 9-12 from the 5'-end of the sense strand.

[0427] The terms “decrease”, “reduced”, “reduction”, or “inhibit” are all used herein to mean a decrease by a statistically significant amount. In some embodiments, “reduce,” “reduction” or “decrease” or “inhibit” typically means a decrease by at least 10% as compared to a reference level (e.g. the absence of a given treatment) and can include, for example, a decrease by at least about 10%, at least about 20%, at least about 25%, at least about 30%, at least about 35%, at least about 40%, at least about 45%, at least about 50%, at least about 55%, at least about 60%, at least about 65%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 95%, at least about 98%, at least about 99%, or more. As used herein, “reduction” or “inhibition” does not encompass a complete inhibition or reduction as compared to a reference level. “Complete inhibition” is a 100% inhibition as compared to a reference level. A decrease can be preferably down to a level accepted as within the range of normal for an individual without a given disorder.

[0428] As used herein, a “central region” of a strand refers to positions 5-17, e.g., positions 6-16, positions 6-15, positions 6-14, positions 6-13, positions 6-12, positions 7-15, positions 7-14, positions 7-13, positions 7-12, positions 8-16, positions 8-15, positions 8-14, positions 8-13, positions 8-12, positions 9-16, positions 9-15, positions 9-14, positions 9-13, positions 9-12, positions 10-16, positions 10-15, positions 10-14, positions 10-13 or positions 10-12, counting from the 5'-end of the strand. For example, the central region of a strand means positions 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 or 17 of the strand. A preferred central

region for the sense strand is positions 6, 7, 8, 9, 10, 11, 12, 13, and 14, counting from the 5'-end of the sense strand. A more preferred central region for the sense strand is positions 7, 8, 9, 10, 11, 12 and 13, counting from the 5'-end of the sense strand. A preferred central region for the antisense strand is positions 9, 10, 11, 12, 13, 14, 15 16 and 17, counting from 5'-end of the antisense strand. A more preferred central region for the antisense strand is positions 10, 11, 12, 13, 14, 15 and 16, counting from 5'-end of the antisense strand.

[0429] As will be apparent to those of skill in the art upon reading this disclosure, each of the individual aspects described and illustrated herein has discrete components and features which can be readily separated from or combined with the features of any of the other several aspects without departing from the scope or spirit of the present invention. Any recited method can be carried out in the order of events recited or in any other order which is logically possible.

[0430] The invention is further illustrated by the following examples, which should not be construed as further limiting. The contents of all references, pending patent applications and published patents, cited throughout this application are hereby expressly incorporated by reference.

Examples

Oligonucleotide Synthesis and Purification

[0431] All oligonucleotides were prepared on a MerMade 192 synthesizer on a 1 μ mole scale using universal or custom supports. All phosphoramidites were used at a concentration 100 mM in 100% Acetonitrile or 9:1 Acetonitrile: DMF with a standard protocol for 2-cyanoethyl phosphoramidites, except that the coupling time was extended to 400 seconds. Oxidation of the newly formed linkages was achieved using a solution of 50 mM 12 in 9:1 Acetonitrile: Water to create phosphate linkages and 100 mM DDTT in 9:1 Pyridine:

Acetonitrile to create phosphorothioate linkages. After the trityl-off synthesis, columns were incubated with 150 μ L of 40% aqueous Methylamine for 45 minutes and the solution drained via vacuum into a 96-well plate. After repeating the incubation and draining with a fresh portion of aqueous Methylamine, the plate containing crude oligonucleotide solution was sealed and shaken at room temperature for an additional 60 minutes to completely remove all protecting groups. Precipitation of the crude oligonucleotides was accomplished via the addition of 1.2 mL of 9:1 Acetonitrile: EtOH to each well followed by incubation at -20° C. overnight. The plate was then centrifuged at 3000 RPM for 45 minutes, the supernatant removed from each well, and the pellets resuspended in 950 μ L of 20 mM aqueous NaOAc. Each crude solution was finally desalted over a GE Hi-Trap Desalting Column (Sephadex G25 Superfine) using water to elute the final oligonucleotide products. All identities and purities were confirmed using ESI-MS and IEX HPLC, respectively.

Cell Culture and Transfections

[0432] Primary Mouse or Cyno Hepatocytes (Thermo Fisher Scientific/Gibco) were transfected by adding 4.9 μ L of Opti-MEM plus 0.1 μ L of Lipofectamine RNAiMax per well (Invitrogen, cat #13778-150) to 5 μ L of siRNA duplexes per well into a 384-well plate and incubated at room temperature for 15 minutes. 40 μ L of Dulbecco's Modified Eagle Medium (PCH) or William's Medium (PMH) containing $\sim 5 \times 10^3$ cells were then added to the siRNA mixture. Cells were incubated for 24 hours at 37° C. and then processed for RNA purification. Experiments were performed at 10 nM, 1 nM, and 0.1 nM doses of siRNA.

[0433] Sequences of parent dsRNA molecules are shown in Tables 1-3 and abbreviations used in the sequences are summarized in Table 4.

TABLE 1

Sequences of parent dsRNA molecules					
target	Duplex ID	Sense Sequence (5'→3')	SEQ ID NO:	Antisense Sequence (5'→3')	SEQ ID NO:
AGT	AD-67327	uscsuccAfcCfUfUfuccuucuaaL96	5	asUfsuagAfagaaaGfUfGfgagascsu	37
AGT	AD-85435	csascaauGfaGfAfGfuaccugugaaL96	6	usUfscacagguacucUfcAfuugugsgsa	38
AGT	AD-85438	cscsucaaCfuGfGfAfugaagaaacuL96	7	asGfsuuucuaucAfgUfugagsgsa	39
AGT	AD-85446	gscsugagAfaGfAfUfugacagguaaL96	8	usAfsaccuguaucUfuCfucagcsasg	40
F9	AD-674282	gsasagacCfcUfGfAfacaccgucuuL96	9	asUfsgacGfgUfGfuucaGfgUfgcuucsasu	41
F9	AD-674283	asasugugCfaGfUfUfuccuugauuL96	10	asAfsucaAfgGfAfaaacUfgCfacauuscsg	42
F9	AD-674287	uscscuugUfuUfAfAfaugguggaaL96	11	asUfsuccAfcCfAfuuaaAfaCfaaggasusu	43
F9	AD-674289	asasaaugGfcAfGfGfugcaagcaguL96	12	asCfsugcUfuGfCfaccuGfcCfauuuususa	44
F9	AD-674293	uscсаuucAfaGfAfUfgacaucauuL96	13	asAfsugcUfuGfUfcaucUfuGfaaugasasu	45
F9	AD-674299	gsasacugGfaUfAfAfaaccuuuaaL96	14	usAfsuaaAfaGfGfuuaUfcCfaguucscsa	46
F9	AD-674303	usgsaaggAfgGfCfAfaagauucguL96	15	asAfsugaUfuCfUfuugcCfuCfuucascsg	47
F9	AD-674305	asasuuaUfaCfUfAfaaguuuccuL96	16	asGfsggaAfaCfCfuuaUfaUfauuuuscsc	48
F9	AD-674307	ascсagacUfaGfUfCfuuuuacuuL96	17	asAfsugguAfgAfAfaagacUfaGfucugusasa	49

TABLE 1-continued

Sequences of parent dsRNA molecules					
target	Duplex ID	Sense Sequence (5'→3')	SEQ ID NO:	Antisense Sequence (5'→3')	SEQ ID NO:
F9	AD-674310	csasuaagGfcGfUfGfaugguucuuuL96	18	asAfsagaAfcCfAfucacGfcCfuuaugsgsg	50
F9	AD-674312	ususuaugGfcCfAfGfaagaauacaaL96	19	usUfsguaUfuCfUfucugGfcCfauaasusg	51
F9	AD-674314	csuscuccGfuCfAfUfucucaccuuuL96	20	asAfsaggUfgAfGfaaugAfcGfgagagsgsu	52
F9	AD-674323	uscscuauCfuGfAfGfacuuggugaaL96	21	usUfscacCfaAfGfucucAfGafugagasasa	53
F9	AD-674325	gsascucuGfcUfAfAfcguuccacuaL96	22	usAfsugugGfaAfCfguuaGfcAfagagcsasg	54
F9	AD-68435	usgsugcaAfuGfAfAfaaggcaauauL96	23	asUfsauuUfgCfCfuuucAfuUfgcacascsu	55
mTTR	AD-1181392	csasguguUfcUfUfGfcucuauaaaaL96	24	usUfsuuaUfagagcaaGfaAfcacugsusu	56
mTTR	AD-1181401 (Sequence 1)	usc suugcUfcUfAfUfaaaccguguuL96	25	asAfsacGfguuuauaGfaGfcaagasasc	57
mTTR	AD-1181410 (Sequence 2)	asgsuguuCfuUfGfCfucuaauaaacaL96	26	usGfsuuuAfuagagcaAfgAfacacugsu	58
mTTR	AD-1181417	asgsaacuGfgAfCfAfcacaaucguaL96	27	usAfsccaUfuugguguCfcAfguucuasasc	59
mTTR	AD-1181426 (Sequence 3)	ascsagugUfuCfUfUfgcucuauaaaL96	28	usUfsuauAfgagcaagAfaCfacugususu	60
mTTR	AD-1181443	ususcuugCfuCfUfAfuaaaccguguL96	29	asCfsacgGfuuuauagAfgCfaagaascsa	61
mTTR	AD-1181451 (Sequence 4)	gsusguucUfuGfCfUfcuauaaaccaL96	30	usGfsguuUfauagagcAfaGfaacacsusg	62
mTTR	AD-1181460	asascuggAfcAfCfCfaaaucguacuL96	31	asGfsuacGfauuugguGfuCfcaguuscsu	63
mTTR	AD-1181469	csasggagGfaCfCfAfggaucuuugcaL96	32	usGfscaaGfauccuggUfcCfuccugsgsg	64
TTR	AD-157448	csasauaaAfaCfAfUfuccugugaaaL96	33	usUfsucacaggaugUfuUfuauugsusc	65
TTR	AD-157464	csusaaagCfaGfUfGfuuuucaccuaL96	34	usAfsggugaaaaacacUfgCfuuuagsusa	66
TTR	AD-157468	csasgagaCfaAfUfAfaaacauuccuL96	35	asGfsgaauuuuuauUfgUfcucugscsc	67
none	AD-64972	uscscucuGfaUfGfGfucaaaaguccuL96	36	asGfsgacUfuugaccaUfcAfgaggascsa	68

TABLE 2

Additional exemplary sequences of parent dsRNA molecules.					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1531684	Marc1	asasaaaaGfuUfCfUfcaaaaugaaL96	69	usUfscauUfuUfUfgagaAfcAfuuuusasa	237

TABLE 2-continued

Additional exemplary sequences of parent dsRNA molecules.					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1531719	Marc1	asasaaucAfcCfAfCfucuuugggcaL96	70	usGfseccAfaAfGfagugGfuGfauuuuscsc	238
AD-1531682	Marc1	asasagugGfgAfGfAfccucuguaaL96	71	usUfsacaCfaGfGfgucuCfcCfacuuusgsa	239
AD-75247	PNPLA3	asasaugaAfaGfAfCfaagguggauL96	72	asUfsccaCfcUfUfugucUfuUfcauuuscscu	240
AD-1531655	C3	asasaugaGfgGfUfUfucacagucuaL96	73	usUfsgacUfgUfGfaaacCfcUfcauuususc	241
AD-67589	PNPLA3	asascuugCfuAfCfCfcauuaggauaL96	74	usAfsuccUfaAfUfggguAfgCfaaguusgsc	242
AD-1531703	Marc1	asasucacCfaCfUfCfuugggcagucL96	75	asCfsugcCfcAfAfagagUfgGfugauususu	243
AD-1531665	C3	ascsauggGfcCfAfGfuggaagaucaL96	76	usGfsaucUfuCfCfacugGfcCfcaugususc	244
AD-1531660	C3	ascscaggAfaCfUfGfaaccuugauaL96	77	usAfsucaAfgGfUfucagUfuCfcuggusgsg	245
AD-1531672	C3	ascscaggAfuGfCfCfacuauugucuaL96	78	usAfsagacAfuAfGfuggcAfuCfcugguscsu	246
AD-1010735	PNPLA3	ascscuaaCfuAfAfAfauaauuuuaL96	79	usAfsaacAfuUfAfuuuuAfgUfuaggusgsa	247
AD-1531721	Marc1	ascscucgCfcUfGfGfuccugauuuuL96	80	usAfsaauCfaGfGfaccagfgCfaggususc	248
AD-67605	PNPLA3	ascscuguUfgAfAfUfuuuuauuuuL96	81	asUfsaaUAfcAfAfauuUfaAfcaggusasa	249
AD-571552	C3	ascsuacaUfgAfAfCfcuacagagauL96	82	asUfsucUfgUfAfgguuCfaUfaguususc	250
AD-1531716	Marc1	ascsugauUfaUfGfGfaauaguucuuL96	83	asAfsagaaCfuAfUfuccaUfaAfcagusus	251
AD-569269	C3	asgsaaaUfcUfAfCfuacaucauuL96	84	asAfsuagAfuGfUfaguaGfaAfuuuuscscu	252
AD-1531718	Marc1	asgsacagGfaUfUfCfugaaaacucaL96	85	usGfsaguUfuUfCfagaaUfcCfugucususc	253
AD-1531663	C3	asgsagcgGfgUfAfCfcucuucuaL96	86	usGfsaugAfaGfAfgguaCfcCfgcucusgsc	254
AD-1531765	SCN9A	asgscaaaGfgUfCfAfcaauuuccuaL96	87	usAfsaggaAfaUfUfgugaCfcUfuugcuscsa	255
AD-1531735	SCN9A	asgscauaAfaUfGfUfuucgaaauaL96	88	usAfsuuUcfcgAfAfaacaUfuUfaugcususc	256
AD-519933	PNPLA3	asgscaugAfgGfUfUfcuagaauuL96	89	asCfsauuCfuAfAfgaacCfuCfaugcusgsg	257
AD-519780	PNPLA3	asgsaagCfaAfCfCfuucgccucuaL96	90	asCfsaggCfcgAfAfagguUfgCfuuccusag	258
AD-1531679	Marc1	asgsagaccAfgAfUfUfgcuuacuaaL96	91	usUfsagagUfaAfGfcaauCfuGfguccususc	259
AD-1531692	Marc1	asgsagagaAfgAfAfAfagugauucaL96	92	usUfsagaaUfcAfCfuuuuUfuUfcuccuscsc	260
AD-1531741	SCN9A	asgsucuuCfaAfGfUfuggcaaaaL96	93	usAfsuuUfgCfCfaacuUfgAfagacuscsg	261

TABLE 2-continued

Additional exemplary sequences of parent dsRNA molecules.				
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO: Antisense sequence (5'→3')	SEQ ID NO:
AD-572575	C3	asgsuggaCfuAfUfGfuguacaagauL96	94 asUfscuuGfuAfCfacauAfgUfccacusasc	262
AD-1531744	SCN9A	asgsuuccUfaUfCfUfccuuucagaaL96	95 usUfscugAfaAfGfgagaUfaGfgaacusasc	263
AD-519354	PNPLA3	asusaaugUfcUfUfAfuguaaugcuuL96	96 asAfgcaUfuAfCfauaaGfaCfauuauasc	264
AD-67551	PNPLA3	asusacauGfaGfCfAfagauuugcaaL96	97 usUfgcaAfaUfCfuugcUfcAfugauasc	265
AD-1531687	Marc1	asuscaacCfaGfGfAfgggaaacauaL96	98 usAfsuguUfuCfCfcuccUfgGfuugauscsa	266
AD-1531736	SCN9A	asuscaucUfuUfGfGfgucauucuuuL96	99 asAfsagaAfuGfAfcccaAfaGfauausasa	267
AD-1531688	Marc1	asuscugaUfgAfAfGfuauuuuuuuL96	100 asAfsaaaAfuAfUfacuuCfaUfcagausc	268
AD-1531747	SCN9A	asuscucUfuUfGfUfcguagauuL96	101 asAfsucaCfuAfCfagacaAfaGfaagauscsa	269
AD-1531756	SCN9A	asusgcugAfgAfAfAfugucgaaaaL96	102 usUfsuucGfaCfAfauuuCfuCfagcausc	270
AD-1531696	Marc1	asusggcuUfgUfUfCfcagaucauL96	103 asAfsugcAfuCfUfggaaCfaAfgccausc	271
AD-1531658	C3	asusguacCfaUfGfCfuaaggccaaaL96	104 usUfsuggCfcUfUfagcaUfgGfuacausg	272
AD-1531757	SCN9A	asusgucgAfgUfAfCfacuuiuacuuL96	105 asAfsguaAfaAfGfuguaCfuCfagacausu	273
AD-67565	PNPLA3	asusguuaGfuAfGfAfauaagccuuL96	106 usAfsaggCfuUfAfuuuAfcUfaacausc	274
AD-1010714	PNPLA3	asusuaggAfuAfAfUfgucuuauguaL96	107 usAfscauAfaGfAfcauuAfuCfcauagsg	275
AD-1531728	SCN9A	asusucucUfuCfGfUfucacagaugL96	108 usCfsaucUfgUfGfaacgAfaGfagausc	276
AD-1531686	Marc1	csasacacUfuGfAfAfgcaugguguuL96	109 asAfsacCfaUfGfcuucAfaGfuguugsusc	277
AD-67583	PNPLA3	csasagauUfuGfCfAfacuugcuacaL96	110 usGfsuagCfaAfGfuugcAfaAfucuugsc	278
AD-572388	C3	csasagguCfuAfCfGfccuauuacauL96	111 asUfsguaAfuAfGfgcguAfgAfcuugsasc	279
AD-1531662	C3	csascccuCfaUfCfAfucuaaccuggaL96	112 usCfscagGfuAfGfaugaUfgAfgggugsusu	280
AD-571901	C3	csasccguAfuCfCfAfcugggaaucuL96	113 asGfsauuCfcCfAfguggAfuAfcggugsgsg	281
AD-1010732	PNPLA3	csasccuuUfuUfCfAfcuaacuaaaL96	114 usUfsuagUfuAfGfgugaAfaAfggugsusu	282
AD-520061	PNPLA3	csasccuuUfuUfCfAfcuaacuaaaL96	115 asUfsuagUfuAfGfgugaAfaAfggugsusu	283
AD-1531770	SCN9A	csascuccUfuCfCfUfgauuguguuuL96	116 asAfsacaCfaAfUfcaggAfaGfgagugsgsa	284
AD-1531705	Marc1	csasgaacGfaAfAfGfuuauggaaL96	117 usUfsccaUfaUfAfacuuUfcGfuucugsasa	285

TABLE 2-continued

Additional exemplary sequences of parent dsRNA molecules.					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-569266	C3	csasgagaAfaUfUfCfuacuacauL96	118	asGfsaugUfaGfUfagaaUfuUfcucugsusa	286
AD-1531740	SCN9A	csasugaaCfGfCfUfucuccacuL96	119	asAfsugGfAfaGfaaguCfGfUfucugsusg	287
AD-75265	PNPLA3	csasugagCfaAfGfAfuugcaacuL96	120	asAfsuguGfCfAfaucUfGfCfucugsusa	288
AD-1531709	Marc1	csasugguGfuUfUfCfagaacugagaL96	121	usCfsucaGfuUfCfugaaAfcAfcuagscsu	289
AD-569048	C3	csasuugaGfaAfCfCfcggaaggcauL96	122	asUfsgccUfuCfCfggguUfcUfcaugsusu	290
AD-571633	C3	cscsacagCfcAfAfAfgauaagaacuL96	123	asGfsuucUfuAfUfcuuGfGfCfugugsusc	291
AD-569516	C3	cscsagauCfcAfCfUfucaccaagauL96	124	asUfscuuGfgUfGfaaguGfgAfucugsusa	292
AD-519346	PNPLA3	cscsauuaGfgAfUfAfaugcuuauL96	125	asAfsuaaGfaCfAfuauCfcUfaaugsgsu	293
AD-571748	C3	cscscgucGfuGfCfGfuuggcucauL96	126	asUfsugaGfcCfAfacgcAfcGfacgggsasg	294
AD-1531708	Marc1	cscsgaccCfaAfGfGfaccagauuguL96	127	asCfsaaUfGfGfuccUfGfGfucggsasa	295
AD-570712	C3	cscsgagcCfGfUfUfCfucuaauuuL96	128	asAfsauUfGfUfGfagaaCfGfCfucggsasu	296
AD-1531706	Marc1	cscsguauGfuCfCfUfggaauuuauL96	129	asUfsaaUfUfUfcaggAfcAfuacggsusu	297
AD-67584	PNPLA3	cscsuAACfAfaAfAfaauuuuuL96	130	usUfsaaaCfaUfUfauuUfaGfuuaggsusg	298
AD-1531668	C3	cscsuuguCfuUfCfUfcagaaccagaL96	131	usCfsuggUfuCfUfgagaAfgAfaaggsasg	299
AD-67577	PNPLA3	csgsacauCfuGfCfCfuaaagucuaL96	132	usUfsgacUfuUfAfgggcAfgAfuugcgsusa	300
AD-571753	C3	csgsugcgUfuGfGfCfucuaugaacuL96	133	asGfsuucAfuUfGfagccAfaCfGcacgsasc	301
AD-1531657	C3	csusacccUfaCfUfCfuguuguucaL96	134	usCfsgaaCfaAfCfagagUfaGfGguagscsc	302
AD-571715	C3	csusacugCfaGfCfUfaaaagacuL96	135	asAfsaguCfuUfUfuagcUfgCfaguagsgsg	303
AD-1531659	C3	csusagugCfuGfUfCfcagugagaaaL96	136	usUfsucuCfaCfUfggacAfgCfacuagsusu	304
AD-67560	PNPLA3	csusauuaAfuGfGfUfcagacuguaL96	137	usAfsacaGfuCfUfgaccAfuUfaauagsgsg	305
AD-1531674	PNPLA3	csusccauGfgCfGfGfggguaacaaaL96	138	usUfsuguUfaCfCfcccGfCfAfuaggagsasc	306
AD-519757	PNPLA3	csusgaguUfgGfUfUfuuaugaaaaL96	139	asUfsuuUfaUfAfaaacCfaAfcucagscsu	307
AD-1531732	SCN9A	csusgcccAfaAfAfUfacugauaaL96	140	usAfsuaUfcAfGfuuuUfuGfggcagscsa	308
AD-1531656	C3	csusgggaGfgAfCfCfcugguaagcaL96	141	usGfscuuAfcCfAfgggUfcUfcccagsgsg	309
AD-1531678	Marc1	csusguggAfgGfAfaagaaaaguaL96	142	usAfsuuUfuCfUfucucCfuCfcacagsasa	310

TABLE 2-continued

Additional exemplary sequences of parent dsRNA molecules.					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1531701	Marc1	csusucuuAfuUfGfGfugacguggaaL96	143	usUfsccaCfGfUfCfaccaAfuAfagaagscsu	311
AD-1531680	Marc1	csusuguuCfcAfGfAfugcauuuuuaL96	144	usUfsaaaAfuGfCfaucUgGfAfacaagscsc	312
AD-1531734	SCN9A	csusuuguCfGfUfAfGfugauuuuccL96	145	asGfsgaaAfaUfCfacuacGfAfcaaagsasa	313
AD-568977	C3	gsascagaCfaAfGfAfccaucuacauL96	146	asUfsguaGfaUfGfgucuUfgUfcugucsusg	314
AD-569268	C3	gsasgaaaUfuCfUfAfcauacauL96	147	asUfsagaUfgUfAfguagAfaUfuucucsusg	315
AD-572818	C3	gsasgaacCfaGfAfAfacaauGCCauL96	148	asUfsggcAfuUfGfuuucUfgGfuucucsusu	316
AD-1531729	SCN9A	gsasggucAfaGfAfCfaucuuuauL96	149	usCfsauaAfaGfAfugucUfuGfaccucscsa	317
AD-1531689	Marc1	gsasgugcUfcCfUfUfcuccaguucL96	150	asGfsaacUfgGfAfgaagGfaGfcacucscsg	318
AD-75275	PNPLA3	gsasuuugCfaAfCfUfugcuacccauL96	151	asUfsgggUfaGfCfaaguUfgCfaaaucsusu	319
AD-67568	PNPLA3	gscsacagGfgAfAfCfcucuaccuuaL96	152	usAfsaggUfaGfAfgguuCfcCfugucscasg	320
AD-569494	C3	gscsaggcAfgAfGfCfGcagcggaL96	153	asUfscccGfcUfGfcgcuCfuGfccucscasc	321
AD-1531722	Marc1	gscscuuCfcCfCfUfcagcuauaL96	154	usCfsauuAfgCfUfgaggGfgAfauggcscsa	322
AD-571610	C3	gscscucuUfcUfUfAfacaaauuucL96	155	asGfsaaaUfuUfGfuuaaGfaAfgaggcscsc	323
AD-1531753	SCN9A	gscsguugUfaGfUfUfccuauccuL96	156	asGfsgagAfuAfGfgaacUfaCfaacgcscsu	324
AD-1531759	SCN9A	gscsucuCfaUfGfUfgcacuauucL96	157	asGfsauuAfgUfGfcacaUfgAfugagcsasu	325
AD-572495	C3	gscsugagGfaGfAfAfuuucuauL96	158	asAfsugaAfgCfAfauucUfcCfucagcsasc	326
AD-1010719	PNPLA3	gscsugagUfuGfGfUfuuuauaaaaL96	159	usUfsuucAfuAfAfaaccAfaCfucagcsusc	327
AD-1531683	Marc1	gscsuucCfaGfAfCfagcauuggauL96	160	asUfsccaAfuGfCfugucUfgAfgaagscsag	328
AD-1531711	Marc1	gsgsaccaGfaUfUfGfcuuacucagaL96	161	usCfsugaGfuAfAfgcaaUfcUfgguccsusu	329
AD-1531664	C3	gsgsagguUfgUfGfCfugagccggaaL96	162	usUfsccgGfcUfCfagcaCfaAfccuccscsc	330
AD-1531762	SCN9A	gsgscacaUfgAfAfCfagcuucuaL96	163	usGfsaagAfaGfUfcguuCfaUfgugccsasc	331
AD-67526	PNPLA3	gsgscuuAfuCfCfCfuccuuccuuaL96	164	usAfsaggAfaGfGfagggAfuAfaggccsasc	332
AD-1531725	SCN9A	gsgscuggAfuUfUfCfcauuuguuuL96	165	asAfsacaAfuUfAfggaaAfuCfcagccsasa	333
AD-1531737	SCN9A	gsgsgaaaAfcAfAfUfcuuccguuuL96	166	usAfsaacGfgAfAfgauuGfuUfuucccsusu	334

TABLE 2-continued

Additional exemplary sequences of parent dsRNA molecules.					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1531673	PNPLA3	gsgsgguaAfcAfAfGfaugauaauL96	167	asGfsauuAfuCfAfucuuGfuUfaccscscsg	335
AD-1531717	Marc1	gsgsugucUfcAfAfUfgcuucaauL96	168	asCfsauuGfaAfGfcauuGfaGfacacsasg	336
AD-570644	C3	gsusaaugCfaGfGfAfcuucuauL96	169	asAfsugaAfgAfAfguccUfgCfauuacsusg	337
AD-568962	C3	gsusaccuCfuUfCfAfuccagacaguL96	170	asCfsuguCfuGfGfaugaAfgAfgguacscsc	338
AD-1531690	Marc1	gsusauaaCfuCfUfAfagaucugauL96	171	asAfsucaGfaUfCfuuaGfGfauuacsasa	339
AD-1531750	SCN9A	gsusccucUfaAfGfAfagaauaucuaL96	172	usAfsgauAfuUfCfuucuUfaGfaggacsusg	340
AD-1531710	Marc1	gsusgaccCfuUfCfAfgaacgaaaguL96	173	asCfsuuuCfgUfUfcugaAfgGfgucacsasc	341
AD-67578	PNPLA3	gsusgaguGfaCfAfAfcguaccuuaL96	174	usAfsaggGfuAfCfguugUfcAfcucacsusc	342
AD-67582	PNPLA3	gsusgcuaAfaGfUfUfuccaucuuL96	175	asAfsagaUfgGfGfaaacUfuUfagcacsusu	343
AD-1531666	C3	gsusgggaGfaAfGfUfucggccuagaL96	176	usCfsuagGfcCfGfaacuUfcUfcccacsusg	344
AD-1531695	Marc1	usasacucUfaAfGfAfucugaugaauL96	177	asUfsucaUfcAfGfaucuUfaGfaguasusa	345
AD-1531723	SCN9A	usascaugAfuCfUfUfcuuugucguaL96	178	usAfsccaCfaAfAfgaagAfuCfauguasgsg	346
AD-75270	PNPLA3	usasccugUfuGfAfAfuuuuguauuaL96	179	usAfsauaCfaAfAfauucAfaCfagguasasc	347
AD-1531671	C3	usascgugCfuGfCfCfcaguucgaaL96	180	usUfscgaAfaCfUfgggcAfgCfacguascsu	348
AD-519350	PNPLA3	usasggauAfaUfGfUfcuuauaauL96	181	asUfsuacAfuAfAfgacaUfuAfuccuasasu	349
AD-1531764	SCN9A	usasugccAfaAfAfUfccuuuuuauaL96	182	usAfsuaaAfaAfGfgauuUfuGfgcauasgsa	350
AD-1531739	SCN9A	usasugugAfaAfCfAfaaccuuacgaL96	183	usCfsguaAfgGfUfuuguUfuCfacuasasu	351
AD-1531681	Marc1	usasuuuAfaUfUfUfcaggauccgaL96	184	usCfsgcaUfcCfUfgaaaUfuAfcauasusu	352
AD-1531704	Marc1	uscsaaugCfuUfCfAfauguccaguL96	185	asCfsuggGfaCfAfuuugaAfgCfauugasgsa	353
AD-67564	PNPLA3	uscsacuuGfaGfGfAfggcgagucuaL96	186	usAfsagacUfcGfCfcuccUfcAfaugagascu	354
AD-1531676	Marc1	uscsagaaCfgAfAfAfguuauauggaL96	187	usCfscuuAfuAfAfcuuuCfgUfucugasasg	355
AD-1531700	Marc1	uscsaggaUfgCfGfAfugucuaugcaL96	188	usGfscuuAfgAfCfaucgCfaUfccugasasa	356
AD-1531702	Marc1	uscscauaGfaUfCfUfggaucuggcaL96	189	usGfscgaGfaUfCfcagaUfcUfauggasasa	357
AD-1531730	SCN9A	uscscauuGfuCfUfUfgacauuuauL96	190	asUfsaagAfuGfUfcaagAfcAfauggasusc	358
AD-1531738	SCN9A	uscsccucAfaGfAfAfgaaauaucuaL96	191	asUfsagaUfaUfUfcuucUfuAfgaggascsu	359

TABLE 2-continued

Additional exemplary sequences of parent dsRNA molecules.					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1531733	SCN9A	uscscugcAfaGfUfCfaaguuccaaaL96	192	usUfsuggAfaCfUfugacUfuGfcaggasasa	360
AD-67554	PNPLA3	uscugagCfuGfAfGfuuguuuuauL96	193	asUfsaaaAfcCfAfacucAfgCfucagassg	361
AD-1531685	Marc1	usgsacccUfuCfAfGfaacgaaguL96	194	asAfsuuUfcGfUfucugAfaGfggucascsa	362
AD-75269	PNPLA3	usgsagugAfaGfAfAfaugaagacaL96	195	usGfsucuUfuCfAfuuucUfuCfacucasgu	363
AD-1531752	SCN9A	usgsauagUfuAfCfCfuaguugcaaL96	196	usUfsgcaAfaCfUfagguAfaCfuauacasasa	364
AD-1531754	SCN9A	usgscagaCfaAfGfAfucuuacuuL96	197	usAfsaguGfaAfGfaucuUfgUfcugcasusa	365
AD-518942	PNPLA3	usgsccaaAfaCfAfAfccaucaccguL96	198	asCfsgguGfaUfGfguugUfuUfuggcasusc	366
AD-571932	C3	usgscgauCfaGfAfAfgagaccaaguL96	199	asCfsuugGfuCfUfcuucUfgAfucgcassg	367
AD-572022	C3	usgscuaaGfgCfCfAfaagaucaacuL96	200	asGfsuugAfuCfUfuuggCfcUfuagcasusg	368
AD-1531760	SCN9A	usgscucuCfcAfUfAfuggauaaaaL96	201	usUfsuaaUfcCfAfauauGfgAfgagcasasu	369
AD-1531766	SCN9A	usgscucuCfcUfUfUfgugguuuauL96	202	asUfsgaaAfcCfAfcaaaGfgAfgagcasusc	370
AD-572577	C3	usgsgacuAfuGfUfGfuacaagaccuL96	203	asGfsgucUfuGfUfacacAfuAfguccascsu	371
AD-1531697	Marc1	usgsgaggAfgAfAfGfaaaagugauL96	204	asAfsucaCfuUfUfucuuCfuCfuccascsa	372
AD-67561	PNPLA3	usgsgauaCfaUfGfAfgcaagauuuL96	205	usAfsaauCfuUfGfcucaUfgUfauccascsc	373
AD-67586	PNPLA3	usgsggagAfgAfUfAfugccuucgaaL96	206	usUfscgaAfgGfCfaauCfuCfucccasgsc	374
AD-569763	C3	usgsggcaAfcUfCfCfaacaaauacuL96	207	asGfsuaaUfuGfUfuggaGfuUfgcccascs	375
AD-1531726	SCN9A	usgsggucAfuUfCfUfucacuuugaaL96	208	usUfscaaAfgUfGfaagaAfuGfaccasasa	376
AD-1531742	SCN9A	usgsgucuUfuAfCfUfggaauuuuL96	209	asAfsaagAfuUfCfcaguAfaAfgaccasasa	377
AD-67573	PNPLA3	usgsgugaCfaUfGfGfcuuccagauL96	210	usAfsucuGfgAfAfgccaUfgUfcaccasgu	378
AD-1531761	SCN9A	usgsguguCfaUfCfAfuagauaaauuL96	211	asAfsauuAfuCfUfaugaUfgAfcaccasasu	379
AD-1531712	Marc1	usgsguguCfuCfAfAfugcuucaauL96	212	usAfsuugAfaGfCfaaugAfgAfcaccasgsa	380
AD-1531669	C3	usgsuacaAfgAfCfCfcgacuggucaL96	213	usGfsaccAfgUfCfggguCfuUfguacascsa	381
AD-1531745	SCN9A	usgsuaggAfgAfAfUfucacuuuucL96	214	usGfsaaaAfgUfGfaauuCfuCfcuacascsa	382
AD-1531724	SCN9A	usgsuaggAfgAfAfUfucacuuuucL96	215	asGfsaaaAfgUfGfaauuCfuCfcuacascsa	383
AD-1531731	SCN9A	usgsucgaGfuAfCfAfcuuuuacugaL96	216	usCfsaguAfaAfAfguguAfcUfcgacacusu	384

TABLE 2-continued

Additional exemplary sequences of parent dsRNA molecules.					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-570132	C3	usgsuucgUfgCfUfGfaauaagaaguL96	217	asCfsuucUfuAfUfucagCfaCfgaacascsg	385
AD-1531769	SCN9A	usgsuucuGfuCfUfGfaguguguuuuL96	218	usAfsaacAfcAfCfucagAfcAfgaacascsa	386
AD-1531694	Marc1	usgsuuuaAfaAfCfCfcaauaucuauL96	219	asUfsagaUfaUfUfggggUfuUfaaacasasc	387
AD-1531677	Marc1	ususaaaaCfuGfUfGfaauaaauggaL96	220	usCfscauUfuUfUfucacAfgUfuuaasasa	388
AD-1531727	SCN9A	ususaccuAfuCfUfCfugcuucaaguL96	221	asCfsuugAfaGfCfagagAfuAfgguaascsc	389
AD-67575	PNPLA3	ususaccuGfuUfGfAfauuuuguauuL96	222	asAfsuacAfaAfAfuucaAfcAfgguaascsa	390
AD-75274	PNPLA3	ususauguAfaUfGfCfugcccuguaaL96	223	usUfsacaGfgGfCfagcaUfuAfcuaaasgsa	391
AD-1531768	SCN9A	ususcucAfaGfGfAfaaaagauaaaL96	224	usUfsuauCfuUfUfuuccUfuGfaggaasasu	392
AD-570134	C3	ususcgugCfuGfAfAfaagaagaauL96	225	asUfsucuUfcUfUfaauCfagCfaggaascsa	393
AD-1531707	Marc1	ususgccuUfuUfUfGfuccuuugauuL96	226	asAfsucaAfaGfGfacaAfaUfggcaasusa	394
AD-75272	PNPLA3	usugcuaCfcCfAfUfuaggauaauaL96	227	usAfsuuaUfcCfUfaaugGfgUfagcaasgsu	395
AD-67567	PNPLA3	usugguuUfuAfUfGfaaaagcuagaL96	228	usCfsuagCfuUfUfucauAfaAfaccaascsu	396
AD-1531714	Marc1	usuguaaUfuUfCfAfggaugcgauaL96	229	usAfsucgCfaUfCfcugaAfaUfuacaasusa	397
AD-1531720	Marc1	usugnuuCfaGfAfUfgcauuuuauL96	230	asUfsuaaAfaUfGfcaucUfgGfaacaasgsc	398
AD-1531746	SCN9A	ususuaguAfcAfCfUfccuuauucauL96	231	asUfsgaaUfaAfGfgaguGfuAfcuaaasasu	399
AD-1531755	SCN9A	ususuauCfuCfUfUfgggucuuuL96	232	usAfsaugAfcCfCfaagAfuGfauaaasgsa	400
AD-1531675	Marc1	ususuccaUfaGfAfUfcuggaucugaL96	233	usCfsagaUfcCfAfgaucUfaUfggaaasasu	401
AD-1531767	SCN9A	ususuguaGfaUfCfUfugcauuuacaL96	234	usGfsuaaUfuGfCfaagaUfcUfacaasasg	402
AD-520053	PNPLA3	ususuagAfaCfAfCfcuuuuucacuL96	235	asGfsugaAfaAfAfggugUfuCfuaaaaasusu	403
AD-1010734	PNPLA3	ususuucAfcCfUfAfacuaaaaauaaL96	236	usUfsauuUfuAfGfuugGfuGfaaaaasgsg	404

TABLE 3

More additional exemplary sequences of parent dsRNA molecules.					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1632799	AGT	asgscugAfgGfGfCfcaccauccuuL96	405	asAfsghaUfgGfUfgggccCfuCfagggcuscua	752

TABLE 3-continued

More additional exemplary sequences of parent dsRNA molecules.					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1632801	AGT	cscsugagGfgCfCfAfccauccucuL96	406	asAfsagagGfaUfGfguggCfcCfucaggscsu	753
AD-1632805	AGT	asgsggccAfcCfAfUfccucugccuuL96	407	asAfsggcAfgAfGfgaugGfuGfgccuscscsa	754
AD-1632838	AGT	gsusgaccGfgGfUfGfuacauacacuL96	408	asGfsuguAfuGfUfacacCfcGfgucacscsu	755
AD-1684490	AGT	cscsggggGfuAfCfAfuaacccccuuL96	409	asAfsgggGfuGfUfauguAfcAfcccggsusc	756
AD-1684492	AGT	csgsggggUfaCfAfUfacacccccuuL96	410	asAfsaggGfgUfGfuaugUfaCfacccgsgsu	757
AD-1684494	AGT	gsgsguguAfcAfUfAfcaacccuucuL96	411	asGfsaagGfgGfUfguauGfuAfcacccsgsg	758
AD-1684496	AGT	gsgsguaCfaUfAfCfaccuuccuL96	412	asGfsgaaGfgGfGfuguaUfgUfacaccscsg	759
AD-1684498	AGT	gsusguacAfuAfCfAfcccuuccauL96	413	asUfsggaAfgGfGfguguAfuGfuacacscsc	760
AD-1684500	AGT	usgsuacaUfaCfAfCfccccuuccacuL96	414	asGfsgggAfaGfGfgggugUfaUfguacascsc	761
AD-1684502	AGT	gsusacauAfcAfCfCfccuuccaccuL96	415	asGfsgugGfaAfGfggggGfuAfuuguacsaasc	762
AD-1684504	AGT	usascauaCfaCfCfCfcuuccaccuuL96	416	asAfsgggGfgAfAfggggUfgUfauguascsa	763
AD-1684506	AGT	ascsauacAfcCfCfCfcuuccaccucuL96	417	asGfsaggUfgGfAfagggGfuGfuaugusaasc	764
AD-1684508	AGT	csasuacaCfcCfCfUfuccaccucguL96	418	asCfsgagGfuGfGfaaggGfgUfguauugsusa	765
AD-1684510	AGT	asusacacCfcCfUfUfccaccucguuL96	419	asAfsccaGfgUfGfgaagGfgGfuguauugsu	766
AD-1684512	AGT	cscscuucCfaCfCfUfcgucauccauL96	420	asUfsggaUfgAfCfagggUfgGfaagggsgsu	767
AD-1684514	AGT	cscsuuccAfcCfUfCfgucauccacuL96	421	asGfsgggAfuGfAfagagGfuGfgaagggsgsg	768
AD-1632840	AGT	csusuccaCfcUfCfGfucauccacauL96	422	asUfsgugGfaUfGfacgaGfgUfggaagsgsg	769
AD-1632841	AGT	ususccacCfuCfGfUfcauccacaauL96	423	asUfsuguGfgAfUfgacgAfgGfuggaasgsg	770
AD-1632842	AGT	uscscaccUfcGfUfCfauccacaauuL96	424	asAfsuugUfgGfAfugacGfaGfguggasasg	771
AD-1632843	AGT	cscsaccuCfGfUfCfAfuccacaauL96	425	asCfsauuGfuGfGfaugaCfGfagguggsasa	772
AD-1632844	AGT	csascuccGfuCfAfUfccacaauL96	426	asUfscuUfgUfGfgaugAfcGfagguggsa	773
AD-1632846	AGT	cscsucguCfaUfCfCfacaauagauL96	427	asUfscucAfuUfGfuggaUfgAfcgaggsusg	774
AD-1632847	AGT	csuscgucAfuCfCfAfcaauagagauL96	428	asCfsucuCfaUfUfguggAfuGfacgaggsu	775

TABLE 3-continued

More additional exemplary sequences of parent dsRNA molecules.					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1632848	AGT	uscsgucaUfcCfAfCfaaugagaguul96	429	asAfsucUfcAfUfugugGfaUfgacgasgsg	776
AD-1632849	AGT	csgsucauCfcAfCfAfaugagaguauL96	430	asUfsacuCfuCfAfuuguGfgAfugacgsasg	777
AD-1632836	AGT	gsuscauCfaCfAfAfugagaguacuL96	431	asGfsuacUfcUfCfauugUfgGfugacsgsa	778
AD-1632850	AGT	uscсаucAfcAfAfUfgagaguaccuL96	432	asGfsguaCfuCfUfcuauGfuGfgaugascsg	779
AD-1632851	AGT	csasuccaCfaAfUfGfagaguaccuL96	433	asAfsagguAfcUfCfucauUfgUfggaugsasc	780
AD-1632852	AGT	asusccacAfaUfGfAfgaguaccuguL96	434	asCfsagguUfaCfUfcucaUfuGfuggausgsa	781
AD-1632853	AGT	uscscacaAfuGfAfGfaguaccuguL96	435	asAfsacagGfuAfcUfcucaUfuUfguggasusg	782
AD-1632854	AGT	cscsacaaUfgAfGfAfguaccuguguL96	436	asCfsacaGfgUfAfcucaUfuUfguggsasus	783
AD-1632855	AGT	csascaauGfaGfAfGfuaccugugauL96	437	asUfsacacAfgGfUfacucUfcAfuugugsgsa	784
AD-1632856	AGT	ascsaaugAfgAfGfUfaccugugaguL96	438	asCfsucaCfaGfGfuacuUfuCfauugusgsg	785
AD-1632857	AGT	csasaugaGfaGfUfAfcugugagcuL96	439	asGfsucUfcAfAfGfguacUfcUfcuugusug	786
AD-1632858	AGT	asasugagAfgUfAfcUfcugugagcuL96	440	asUfsgcuCfaCfAfgguacUfuCfcauugsgsu	787
AD-1632859	AGT	asusgagaGfuAfcUfcUfgugagcaguL96	441	asCfsugcUfcAfCfagguAfcUfcucausug	788
AD-1632860	AGT	usgsagagUfaCfCfUfgugagcagcuL96	442	asGfsucugCfuCfAfcaggUfaCfucucasusu	789
AD-1632861	AGT	gsasgaguAfcCfUfGfugagcagcuL96	443	asAfsagcuGfcUfCfagagGfuAfcucucsasus	790
AD-1632862	AGT	asgsaguaCfcUfGfUfgagcagcuguL96	444	asCfsagcUfgCfUfcacaGfgUfacucuscasa	791
AD-1632863	AGT	gsasguacCfuGfUfGfagcagcugguL96	445	asCfsacagCfuGfCfucacAfgGfuacucusc	792
AD-1632864	AGT	asgsuaccUfgUfGfAfgcagcugguL96	446	asGfsccaGfcUfGfcucaCfaGfguacuscusu	793
AD-1632865	AGT	gsusaccuGfuGfAfGfagcagcugguL96	447	asUfsgccAfgCfUfgcucAfcAfgguacusc	794
AD-1632866	AGT	usascugUfgAfGfGfagcuggccauL96	448	asUfsugcCfaGfCfugcuCfaCfagguascusu	795
AD-1632991	AGT	asasugguCfGfGfAfgcuggccauL96	449	asUfsggcCfaGfCfauccCfGfAfcuugsgsc	796
AD-1632992	AGT	asusggucGfgGfAfUfgcuggccauL96	450	asUfsuggCfcAfGfcaucCfcGfaccausug	797
AD-1632993	AGT	usgsugcgGfgAfUfGfugcuggccacuL96	451	asGfsuugGfcCfAfgcauUfcCfagaccasusu	798

TABLE 3-continued

More additional exemplary sequences of parent dsRNA molecules.					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1632994	AGT	gsgsucggGfaUfGfCfuggccaacuL96	452	asAfsuuGfgCfCfagcaUfcCfcgaccsas	799
AD-1632995	AGT	gsuscgggAfuGfCfUfggccaacuuL96	453	asAfsaguUfgGfCfcagcAfuCfccgacscsa	800
AD-1632996	AGT	uscsgggaUfgCfUfGfgccaacuucL96	454	asGfsaagUfuGfGfccagCfaUfccgascsc	801
AD-1632997	AGT	csgsggauGfcUfGfGfccacuucuuL96	455	asAfsгааGfuUfGfgccaGfcAfucccgasc	802
AD-1632998	AGT	gsgsgaugCfuGfGfCfcaacuucuuL96	456	asAfsagaAfgUfUfgggcAfgCfaucccgsga	803
AD-1632999	AGT	gsgsaugcUfgGfCfCfaacuucuuL96	457	asCfsaagAfaGfUfuggcCfaGfcaucccgsg	804
AD-1633000	AGT	gsasugcuGfgCfCfAfacuucugguL96	458	asCfscaaGfaAfGfuuggCfcAfgcauccsc	805
AD-1633003	AGT	gscsuggcCfaAfCfUfucuggggcuL96	459	asAfsgccCfaAfGfaaguUfgGfccagcsasu	806
AD-1633004	AGT	csusggccAfaCfUfUfucuggggcuL96	460	asAfsagcCfcAfAfgaagUfuGfgccagscsa	807
AD-1633007	AGT	gscscaacUfuCfUfUfgggcuuccguL96	461	asCfsaggAfgCfCfcaagAfaGfuuggcscsa	808
AD-1633008	AGT	cscsaacuUfcUfUfGfgggcuuccguL96	462	asAfsaggAfaGfCfccaaGfaAfguuggscsc	809
AD-1633009	AGT	csasacuUfuUfGfGfgcuuccguauL96	463	asUfsacgGfaAfGfccccAfgAfguugsgsc	810
AD-1633010	AGT	asascuucUfuGfGfGfcuuccguauL96	464	asAfsuacGfgAfAfgcccAfaGfaaguusgsg	811
AD-1633011	AGT	ascsuucUfgGfGfCfuuccguauauL96	465	asUfsauaCfgGfAfgccCfaAfgaagusug	812
AD-1633012	AGT	csusucuuGfgGfCfUfuccguauauL96	466	asAfsuauAfcGfGfaagcCfcAfgaagsusu	813
AD-1633013	AGT	ususcuugGfgCfUfUfccguauauauL96	467	asUfsauaUfaCfGfgaagCfcCfaagaasgu	814
AD-1633014	AGT	uscscuugGfcUfUfCfcguauauauL96	468	asAfsuauAfuAfcfggaaGfcCfcaagasag	815
AD-1633015	AGT	csusuggCfuUfCfCfguauauauguL96	469	asCfsauaUfaUfAfcggaAfgCfccaagsasa	816
AD-1633016	AGT	ususgggCfuCfCfGfuauauauguL96	470	asCfscauAfuAfuacggAfaGfcccaasgsa	817
AD-1633018	AGT	gsgsgcuuCfcGfUfAfuauauggcauL96	471	asUfsgccAfuAfuauacGfgAfgcccsasa	818
AD-1633019	AGT	gsgscuucCfGfUfAfuauauggcauL96	472	asAfsugcCfaUfAfuauaCfGfgaagccscsa	819
AD-1633020	AGT	gscsuuccGfuAfuAfuauauggcauL96	473	asCfsaugCfcAfUfauauAfcGfgaagccsc	820
AD-1633027	AGT	usasaauaUfgGfCfAfgcacaguguL96	474	asCfsacuGfuGfCfaugcCfaUfauauascsg	821

TABLE 3-continued

More additional exemplary sequences of parent dsRNA molecules.					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1633028	AGT	asusauauGfgCfAfUfgcacagugauL96	475	asUfscacUfgUfGfcaugCfcAfuaauasasc	822
AD-1633029	AGT	usasuaugGfcAfUfGfcacagugaguL96	476	asCfsucaCfuGfUfgcaUGfcCfauauasusa	823
AD-1633030	AGT	asusauggCfaUfGfCfacagugagcuL96	477	asGfscucAfcUfGfugcaUfgCfcauauasasu	824
AD-1633031	AGT	usasuggcAfuGfCfAfcagugagcuL96	478	asAfsagcuCfaCfUfgugcAfuGfccauasusa	825
AD-84731	AGT	asusggcaUfgCfAfCfagugagcuauL96	479	asUfsagcUfcAfCfugugCfaUfgccauasasu	826
AD-1633032	AGT	usgsgcaUGfcAfCfAfgugagcuauL96	480	asAfsuagCfuCfAfcuguGfcAfugccasusa	827
AD-1633033	AGT	gsgscaugCfaCfAfgfugagcuauL96	481	asCfsauaGfcUfCfagugUfgCfaugccsasau	828
AD-1633034	AGT	gscsaugcAfcAfGfUfgagcuauL96	482	asCfscuAfgCfUfcacUGfuGfcaugcscsa	829
AD-1684516	AGT	csasugcaCfaGfUfGfagcuauL96	483	asCfscuAfaGfCfucUGfuGfcaugcscsc	830
AD-1684518	AGT	asusgcacAfgUfGfAfgcuauL96	484	asCfscuAfuAfgGfcuAfuGfugcaugsgsc	831
AD-1684520	AGT	cscsugcCfcCfAfcgugcuauL96	485	asAfsagcAfgCfCfugugGfgGfagaggsasc	832
AD-1633048	AGT	usgsgcAcCfCfUfGfGfcucucucuuL96	486	asAfsagagAfgAfgGfccaGfgGfugccasasa	833
AD-1633049	AGT	gsgscaccCfuGfGfCfcucucucuuL96	487	asUfsagaGfaGfAfgggcAfgGfugccasasa	834
AD-1633094	AGT	gsascaggCfuAfcAfggcauL96	488	asAfsaggaUfuGfCfcuguAfgCfcugucsasg	835
AD-1633095	AGT	ascsaggcUfaCfAfgfgcauL96	489	asCfsaggAfuUfGfGfcugUfaGfGfcugscsa	836
AD-1633119	AGT	ususcuuGfgAfgGfGfagacaauL96	490	asGfsuucUfuGfUfGfcuAfgaggaascsa	837
AD-1633121	AGT	cscsuuggAfaGfGfAfaagaacugL96	491	asCfsaguUfcUfUfguccUfuCfcaaggsasa	838
AD-1633122	AGT	csusuggaAfgGfAfaagaacugcuL96	492	asGfscagUfuCfUfugucCfuUfccaagsgsa	839
AD-1633254	AGT	csascugAfaGfCfAfgccuuuL96	493	asAfscaaAfcGfGfcugCfuCfagguagscsa	840
AD-1633257	AGT	csusgaagCfaGfCfCfuuugugcuL96	494	asUfsgcaCfaAfaCfGgCfUfgCfuucagsgsu	841
AD-1633269	AGT	usugugCfaGfGfGfcuggcucuuL96	495	asAfsagagCfcAfgGgcccUfgCfacaascsa	842
AD-1633270	AGT	usugugCfaGfGfCfcuggcucucuuL96	496	asGfsagaGfcCfAfgggcCfuGfcaasasc	843
AD-1633271	AGT	usgsgcaGfgGfCfcuggcucucuuL96	497	asAfsagagAfgCfCfagggCfcUfgcacasasa	844
AD-1633272	AGT	gsusgcagGfgCfCfUfggcucucuuL96	498	asUfsagaGfaGfCfcaggCfcCfugcacsasa	845

TABLE 3-continued

More additional exemplary sequences of parent dsRNA molecules.					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1633273	AGT	usgscaggGfcCfUfGfgcucucuaauL96	499	asAfsuagAfgAfGfccagGfcCfcugcascsa	846
AD-1633290	AGT	ascsgcucUfcUfGfGfacuucacaguL96	500	asCfsuguGfaAfGfuccaGfaGfagcgusgsg	847
AD-1633291	AGT	csgscucUfcUfGfGfAfcuucacagauL96	501	asUfscugUfgAfAfguccAfgAfgagcgusug	848
AD-1633324	AGT	csusgagaAfgAfUfUfgacagguucL96	502	asGfsaacCfuGfUfcauCfuUfcucagscsa	849
AD-84739	AGT	usgsagaaGfaUfUfGfacaggucauL96	503	asUfsgaaCfcUfGfucaaUfcUfucucasgsc	850
AD-1633325	AGT	gsasgaagAfuUfGfAfcagguucauL96	504	asAfsugaAfcCfUfgucaAfuCfuucucsasg	851
AD-1633326	AGT	asgsaagaUfuGfAfCfagguucauguL96	505	asCfsaugAfaCfCfugucAfaUfcuucscsa	852
AD-1633327	AGT	gsasagauUfgAfCfAfgguucaugcuL96	506	asGfscauGfaAfCfcuguCfaAfucucusc	853
AD-1633328	AGT	asasgauuGfaCfAfGfguucaugcauL96	507	asUfsgcaUfgAfAfccugUfcAfauucscsu	854
AD-1633329	AGT	asgsauugAfcAfGfGfuucaugcaguL96	508	asCfsugcAfuGfAfaccuGfuCfaucusc	855
AD-1633330	AGT	gsasuugaCfaGfGfUfcaugcagguL96	509	asCfscugCfaUfGfaaccUfgUfcaucscsu	856
AD-1633331	AGT	asusugacAfgGfUfUfcaugcagguL96	510	asGfscuGfcAfUfgaacCfuGfucaucscsu	857
AD-1633332	AGT	usugacaGfgUfUfCfaugcagguL96	511	asAfsgccUfgCfAfugaaCfcUfgucausc	858
AD-1633333	AGT	usgsacagGfuUfCfAfcagcagguL96	512	asCfsagcCfuGfCfaugaAfcCfugucasasu	859
AD-1633334	AGT	gsascaggUfuCfAfUfGfcagcuguuL96	513	asAfsacagCfcUfGfcaugAfaCfcugucsa	860
AD-1633335	AGT	ascscagguUfcAfUfGfGfcagcuguuL96	514	asCfsacaGfcCfUfGcauGfaAfccugucsa	861
AD-1633343	AGT	asusgcagGfcUfGfUfGacaggauguL96	515	asCfsaucCfuGfUfcacaGfcCfugcausgsa	862
AD-1633345	AGT	gscsagguUfgUfGfAfcaggaugauL96	516	asUfscuUfcCfUfGcauGfaGfccugcsasu	863
AD-1633346	AGT	csasggcuGfuGfAfCfaggaugaaL96	517	asUfsuccAfuCfCfugucAfcAfccugscsa	864
AD-1633409	AGT	gscsuuucAfaCfAfCfcuacguccauL96	518	asUfsggaCfgUfAfggugUfuGfaaagcscsa	865
AD-1633453	AGT	gsasguucUfgGfGfUfggacaacaguL96	519	asCfsuguUfgUfCfcaccCfaGfaacucscsu	866
AD-1633464	AGT	gsgsacaaCfaGfCfAfcucaguguL96	520	asAfsacUfgAfGfgugUfgUfugucscsa	867
AD-1633465	AGT	gsascaacAfgCfAfCfcucagugucL96	521	asGfsacaCfuGfAfggugCfuGfuugucscsa	868
AD-1633466	AGT	ascsaacaGfcAfCfCfucagugucL96	522	asAfsagacAfcUfGfagguGfcUfguugucsc	869

TABLE 3-continued

More additional exemplary sequences of parent dsRNA molecules.					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1633467	AGT	csasacagCfaCfCfUfcagugucugul96	523	asCfsagaCfaCfUfgaggUfgCfuguugsusc	870
AD-1633468	AGT	asascagcAfcCfUfCfagugucuguul96	524	asAfsacagAfcAfcUfagagGfuGfcuguusgsu	871
AD-1633604	AGT	asusgccuCfuGfAfCfCfuggacaagul96	525	asCfsuugUfcCfAfCfaggucAfgAfggcausasg	872
AD-1633621	AGT	asasggugGfaGfGfGfucucacuuul96	526	asAfsaagUfgAfGfaccCfCfaccuusgsu	873
AD-1633622	AGT	asgsguggAfgGfGfUfCfucacuuucul96	527	asGfsaaaGfuGfAfgaccCfCfaccususg	874
AD-1633623	AGT	gsgsuggaGfgGfUfCfucacuuuccul96	528	asGfsgaaAfgUfGfagacCfCfaccususu	875
AD-1633627	AGT	gsasggguCfuCfAfCfuuccagcaul96	529	asUfsgcuGfgAfAfagugAfgAfcuccscsa	876
AD-1633628	AGT	asgsggucUfcAfCfUfuuccagcaul96	530	asUfsugcUfgGfAfaaguGfaGfaccuscsc	877
AD-1633630	AGT	gsgsucucAfcUfUfUfccagcaaaul96	531	asUfsuuuGfcUfGfgaaaGfuGfagaccscsu	878
AD-1633631	AGT	gsuscucaCfuUfUfCfCfagcaaaacul96	532	asGfsuuuUfgCfUfggaaAfgUfgagaccscsc	879
AD-1633632	AGT	uscscacUfuUfCfCfagcaaaacul96	533	asAfsuuUfuGfCfCfuggaAfaGfugagascsc	880
AD-1633633	AGT	csuscacuUfuCfCfAfgcaaaacucul96	534	asGfsaguUfuUfGfcuggAfaAfgugagsasc	881
AD-1633634	AGT	uscscacuUfcCfAfGfcaaaacuccul96	535	asGfsgagUfuUfUfgcugGfaAfgugagsga	882
AD-1633635	AGT	csascuuUfcAfGfCfaaaacuccul96	536	asGfsggaGfuUfUfugcuGfgAfaagugsasg	883
AD-1633636	AGT	ascsuuucCfaGfCfAfaaaacuccul96	537	asAfsaggAfgUfUfuugcUfgGfaaagugsaa	884
AD-1633637	AGT	csusuuccAfgCfAfAfaaacuccucul96	538	asGfsaggGfaGfUfuugCfuGfgaaagsusg	885
AD-1633638	AGT	ususuccaGfcAfAfAfacuccucaul96	539	asUfsgagGfgAfGfuuuuGfcUfggaaasgsu	886
AD-1633639	AGT	ususcagCfaAfAfAfcuccucaul96	540	asUfsugaGfgGfAfguuUfgCfuggaasasg	887
AD-1633640	AGT	uscscagcAfaAfAfCfuccucaacuL96	541	asGfsuugAfgGfGfaguUfuGfcuggasasa	888
AD-1633641	AGT	cscsagcaAfaAfCfUfccucaacuL96	542	asAfsuuGfaGfGfgaguUfuUfgcuggasasa	889
AD-1633642	AGT	csasgcaaAfaCfUfCfuccucaacugul96	543	asCfsaguUfgAfGfggagUfuUfgcuggsgsa	890
AD-1633643	AGT	asgscaaaAfcUfCfCfucuaacuggul96	544	asCfscagUfuGfAfgggaGfuUfuugcusgsg	891
AD-1633644	AGT	gscsaaaaCfuCfCfCfucaacuggauL96	545	asUfscgaGfuUfGfaggGfgUfuugcususg	892

TABLE 3-continued

More additional exemplary sequences of parent dsRNA molecules.					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1633645	AGT	csasaaacUfcCfCfUfcaacuggauuL96	546	asAfsuccAfgUfUfgaggGfaGfuuuugscsu	893
AD-1633646	AGT	asasaacuCfcCfUfCfaacuggauguL96	547	asCfsaucCfaGfUfugagGfgAfguuuusgsc	894
AD-1633647	AGT	asasacucCfcUfCfAfacuggaugauL96	548	asUfscuacCfcAfGfuugaGfgGfaguuuusug	895
AD-1633648	AGT	asascuccCfuCfAfAfcuggaugaauL96	549	asUfsucaUfcCfAfguugAfgGfaguususu	896
AD-1633649	AGT	ascsuccUfcAfAfCfuggaugaaguL96	550	asCfsuucAfuCfCfaguUGfaGfggagususu	897
AD-1633650	AGT	csuscccuCfaAfCfUfggaugaagauL96	551	asUfscuuCfaUfCfcaguUfgAfgggagsusu	898
AD-1633651	AGT	uscscuccAfaCfUfGfgaugaagaauL96	552	asUfsucuUfcAfUfccagUfuGfagggasgsu	899
AD-1633652	AGT	cscscucaAfcUfGfGfaugaagaaauL96	553	asUfsuucUfuCfAfuccaGfuUfgagggasag	900
AD-84707	AGT	cscsucaaCfuGfGfAfugaagaaacuL96	7	asGfsuuuCfuUfCfauccAfgUfugagsgsa	901
AD-1633653	AGT	csuscaacUfgGfAfUfgaagaacuuL96	554	asAfsguuUfcUfUfcaucCfaGfuugagsgsg	902
AD-1633678	AGT	asgsgaucUfuAfUfGfaccugcagguL96	555	asCfscugCfaGfGfucauAfaGfauccusug	903
AD-1633683	AGT	csusuagAfcCfUfGfCaggaccuguL96	556	asCfsaggUfcCfUfGfcagGfuCfaaagsasu	904
AD-1633732	AGT	cgsagcuGfaAfCfCfugcaaaaauuL96	557	asAfsuuuUfuGfCfagguUfcAfgcucgsgsu	905
AD-1633733	AGT	gsasgcugAfaCfCfUfgcaaaaauuL96	558	asAfsauuUfuUfGfcaggUfuCfagcucsgsg	906
AD-1633734	AGT	asgscugaAfcCfUfGfcaaaaauuguL96	559	asCfsauuUfuUfUfgcagGfuUfcagcuscsg	907
AD-1633735	AGT	gscsugaaCfcUfGfCfaaaaauugauL96	560	asUfscuuUfuUfUfugcaGfgUfucagcsusc	908
AD-1633736	AGT	csusgaacCfuGfCfAfaaaauagaguL96	561	asCfsucaAfuUfUfuugcAfgGfuucagscsu	909
AD-1633737	AGT	usgsaaccUfgCfAfAfaaaugagcuL96	562	asGfscucAfaUfUfuuuUGfaGfguucagsgc	910
AD-1633738	AGT	gsasaccuGfcAfAfAfaauugagcauL96	563	asUfsgcuCfaAfUfuuuUGfcAfgguucsasg	911
AD-1633739	AGT	asascugCfaAfAfAfaauaggaauL96	564	asUfsugcUfcAfAfuuuUGfcAfgguucscsa	912
AD-1633740	AGT	ascscugCfaAfAfAfaauagcauuL96	565	asAfsuugCfuCfAfauuUfuGfcaggususuc	913
AD-1633741	AGT	cscsugcaAfaAfAfUfugagcaauguL96	566	asCfsauuGfcUfCfaauUfuUfgcaggususu	914
AD-1633742	AGT	csusgcaaAfaAfUfUfgagcaaugauL96	567	asUfscuuUfgCfUfcauuUfuUfugcagsgsu	915
AD-1633743	AGT	usgscaaaAfaUfUfGfagcaaugacuL96	568	asGfsucaUfuGfCfucaaUfuUfuugcagsgg	916

TABLE 3-continued

More additional exemplary sequences of parent dsRNA molecules.					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1633759	AGT	gsasggugCfuGfAfAfCagcauuuuuL96	569	asAfsaaaUfgCfUfguucAfgCfaccucscsc	917
AD-1684522	AGT	asgsugucUfgAfAfCfagcauuuuuL96	570	asAfsaaaAfuGfCfuguuCfaGfcaccucscsc	918
AD-1684523	AGT	gsgsugcuGfaAfCfAfCagcauuuuuL96	571	asAfsaaaAfaUfGfcuguUfcAfgcaccusuc	919
AD-1684524	AGT	gsusgcugAfaCfAfGfcauuuuuuuL96	572	asAfsaaaAfaAfUfgcugUfuCfagcacscsu	920
AD-1684525	AGT	usgscugaAfcAfGfCfauuuuuuuuL96	573	asAfsaaaAfaAfAfugcuGfuUfcagcascsc	921
AD-1684526	AGT	gscsugaaCfaGfCfAfuuuuuuuuL96	574	asCfsaaaAfaAfAfaugUfgUfcagcsasc	922
AD-1684527	AGT	csusgaacAfgCfAfUfuuuuuuL96	575	asUfscAAfaAfAfaugCfuGfuucagscsa	923
AD-1633777	AGT	usgsagagAfgAfGfCfaccagaguL96	576	asAfsCucUfgUfGfggcuCfuCfucucasusc	924
AD-1633779	AGT	asgsagagAfgCfCfCfacagagucuL96	577	asAfsagacUfcUfGfugggCfuCfucucscsa	925
AD-1633780	AGT	gsasgagaGfcCfCfAfCagagucuauL96	578	asUfsagaCfuCfUfguggGfcUfcucucsusuc	926
AD-1633840	AGT	gsasaccgCfcCfAfUfuccuguuL96	579	asCfsaaaCfaGfGfaaugGfgCfggguccsag	927
AD-1633841	AGT	asascgcgCfcAfUfUfuccuguuL96	580	asGfscaaAfcAfGfgaaUfgGfcgguuscsa	928
AD-1633842	AGT	ascscgcCfAfUfUfCfcuguuL96	581	asAfsagcaAfaCfAfaggaaUfgGfcgguusuc	929
AD-1633843	AGT	cscsgcccAfuUfCfCfuguuL96	582	asCfsagcAfaAfCfaggAfuGfggcggsusu	930
AD-1633844	AGT	csgscccaUfuCfCfUfguuL96	583	asAfsagCfaAfAfCaggAfaUfgggcgsgsu	931
AD-1633845	AGT	gscsccauUfcCfUfGfuuL96	584	asCfsacaGfcAfAfacagGfaAfugggcsag	932
AD-1633846	AGT	cscscauUfcUfGfUfuL96	585	asAfsacAfCfAfaacaGfgAfaugggscag	933
AD-84712	AGT	cscsauucCfuGfUfUfugcuguL96	586	asUfsacaCfaGfCfaaacAfgGfauggsgsc	934
AD-1633847	AGT	csasuuccUfgUfUfUfgcuguL96	587	asAfsuacAfcAfGfcaaaCfaGfgaaugsgsg	935
AD-1633848	AGT	asusuccuGfuUfUfGfugcuguL96	588	asCfsauaCfaCfAfgcaaAfcAfggaugsgsg	936
AD-1633849	AGT	ususcugUfuUfGfCfuguguL96	589	asUfscAUfcAfCfagcaAfaCfaggasusg	937
AD-1633850	AGT	uscscuguUfuGfCfUfguguL96	590	asAfsucaUfaCfAfcagcAfaAfcaggasasu	938
AD-1633851	AGT	cscsuguUfgCfUfGfugaugauL96	591	asGfsaucAfuAfCfagcaAfaAfcaggasasa	939
AD-1633852	AGT	csusguuuGfcUfGfUfguauL96	592	asUfsgauCfaUfAfcacaGfcAfaacagsgsa	940

TABLE 3-continued

More additional exemplary sequences of parent dsRNA molecules.					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1633853	AGT	usgsuuugCfuGfUfGfuaugaucaauL96	593	asUfsugaUfcAfUfacacAfgCfaaacasgsg	941
AD-1633854	AGT	gsusuugcUfgUfGfUfaugaucaauL96	594	asUfsuugAfuCfAfuacaCfaGfcaaacasg	942
AD-1633855	AGT	ususugcuGfuGfUfAfugaucaaaagL96	595	asCfsuuuGfaUfCfauacAfcAfgcaaaasca	943
AD-1684528	AGT	cscscagUfcUfCfCfcaccuuuucL96	596	asGfsaaaAfgGfUfgggaGfaCfugggsgsu	944
AD-1684530	AGT	cscscaguCfuCfCfCfaccuuuucL96	597	asAfsagaAfaGfGfugggAfgAfcugggsgsg	945
AD-1684532	AGT	cscsagucUfcCfCfAfcuuuucL96	598	asAfsagaAfaAfgGfuggGfaGfacuggsgsg	946
AD-1684534	AGT	asgsucucCfcAfCfCfuuuucL96	599	asAfsagaGfaAfAfaagguGfgGfagacugsg	947
AD-1633946	AGT	gsuscuccCfaCfCfUfuucuaauL96	600	asUfsagaAfgAfAfaaggUfgGfgagacsug	948
AD-67328	AGT	uscscuccAfcCfUfUfuucuaauL96	5	asUfsuagAfaGfAfaaagGfuGfggagascsu	949
AD-1633947	AGT	csuscccaCfcUfUfUfucuaauL96	601	asAfsuuaGfaAfGfaaaGfgUfgggagsasc	950
AD-1633948	AGT	uscscacCfuUfUfUfucuaaagL96	602	asCfsauuAfgAfAfgaaaAfgGfgggagsa	951
AD-1633949	AGT	cscscaccUfuUfUfCfuucuaagL96	603	asUfscuUfaGfAfaagaaAfaGfguggsgsg	952
AD-84700	AGT	cscsaccUfuUfCfUfucuaagL96	604	asCfsucaUfuAfGfaagaAfaAfgguggsgsa	953
AD-1633950	AGT	csascuuUfuCfUfUfucuaagL96	605	asAfsucUfuUfAfgaagAfaAfgguggsgsg	954
AD-1633951	AGT	ascscuuUfcUfUfCfuaagagucL96	606	asGfsacuCfaUfUfagaaGfaAfaaggusgsg	955
AD-1633991	AGT	cscsguuCfuCfCfUfuggucuaagL96	607	asCfsuuaGfaCfCfaaggAfgAfaacggscsu	956
AD-1633992	AGT	csgsuuucUfcCfUfUfuggucuaagL96	608	asAfsuuAfgAfcfaagGfaGfaaacgsgsc	957
AD-1633993	AGT	gsusuucCfcUfUfGfgucuaagL96	609	asCfsacuUfaGfAfcfaaGfgAfgaaacsgsg	958
AD-1634065	AGT	gsusuugcUfgGfUfUfaauuuuagL96	610	asCfsuaaAfaUfAfaaccCfaGfcaaacusg	959
AD-1634066	AGT	ususugcuGfgGfUfUfaauuuuagL96	611	asUfscuaAfaAfUfaaacCfcAfgcaaaascu	960
AD-1634067	AGT	ususgcugGfgUfUfUfaauuuuagL96	612	asCfsucUfaAfaAfuaaaCfcCfagcaasasc	961
AD-1634068	AGT	usgscuggGfuUfUfAfuuuuagL96	613	asUfscucUfaAfAfaaaAfcCfcagcasasa	962
AD-84730	AGT	gscsuggUfuUfAfUfaauuagL96	614	asUfsucUfuAfAfaaaAfaCfccagcsasa	963
AD-1634069	AGT	csusggguUfuUfUfuaagaaauL96	615	asAfsuucUfcUfAfaaaAfaAfccagcsa	964

TABLE 3-continued

More additional exemplary sequences of parent dsRNA molecules.					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1634070	AGT	usgsgguuUfaUfUfUfuagagaauL96	616	asCfsauuCfuCfUfaaaaUfaAfaccasgsc	965
AD-1634071	AGT	gsgsguuuAfuUfUfUfagagaauL96	617	asCfscuUfcUfCfuaaaAfuAfaaccsasg	966
AD-1684536	AGT	gsgsuuuuUfuUfUfAfgagaauL96	618	asCfscuUfuCfUfcuaaAfaUfaaacscsa	967
AD-1684538	AGT	gsusuuuUfuUfAfgfagaauL96	619	asCfscuUfuUfCfucuaAfaUfaaacscsc	968
AD-1684540	AGT	ususuauUfuUfAfgfagaauL96	620	asCfscuUfuUfCfucuaAfaUfaaacscsc	969
AD-1684542	AGT	asgsaaugGfgGfGfUfgggaggcauL96	621	asUfsgccUfcCfCfcaccCfcCfauucscsu	970
AD-1684544	AGT	gsasaugGfgGfUfGfgggaggcauL96	622	asUfsugcCfuCfCfccacCfcCfauucscsu	971
AD-1684546	AGT	asasugggGfgGfGfGfgaggcaauL96	623	asCfsuugCfcUfCfccccCfcCfcauucscsu	972
AD-1684548	AGT	asusggggGfuGfGfGfgaggcaauL96	624	asUfscuUfcCfUfccccCfcCfcauucscsu	973
AD-1684550	AGT	usgsgggGfgGfGfGfaggcaauL96	625	asUfsucuUfgCfCfucccCfaCfccccasusu	974
AD-1684552	AGT	gsgsgguuGfgGfGfAfggcaagaauL96	626	asGfsuucUfuGfCfcuccCfcAfccccsasusu	975
AD-1684554	AGT	gsusggggAfgGfCfAfgaaccaguuL96	627	asAfsucgGfuUfCfuugcCfuCfccccscsc	976
AD-1684556	AGT	usgsgggaGfgCfAfAfgaaccaguuL96	628	asCfsacuGfgUfUfCuugCfcUfccccscsc	977
AD-1684558	AGT	gsgsggagGfcAfAfGfaaccaguuL96	629	asAfsacUfgGfUfucuuGfcCfccccscsc	978
AD-1634072	AGT	gsgsgaggCfaAfGfAfacaguuuL96	630	asAfsacaCfuGfGfuucuUfgCfccccscsa	979
AD-1634073	AGT	gsgsaggCfaGfAfAfcaguuuuL96	631	asAfsaacAfcUfGfguuUfuGfccccscsc	980
AD-1634074	AGT	gsasggcaAfgAfAfCfcaguuuuL96	632	asUfsaaaCfaCfUfgguuCfuUfgccucscsc	981
AD-1634075	AGT	asgsgcaaGfaAfCfCfaguuuuL96	633	asCfsuaaAfcAfCfugguUfcUfgccucscsc	982
AD-1634076	AGT	gsgscaagAfaCfCfAfguuuagcuL96	634	asGfscuaAfaCfAfcuggUfuCfuugccscsu	983
AD-1634077	AGT	gscsaagaAfcCfAfguuuagcgcuL96	635	asCfsgcuAfaAfCfacugGfuUfcuugccscsu	984
AD-1634078	AGT	csasagaaCfcAfGfUfguuuagcgcuL96	636	asGfscgcUfaAfAfcacuGfgUfucuuugscsc	985
AD-1634079	AGT	asasgaacCfaGfUfGfuuuagcgcuL96	637	asCfsgcgCfuAfaAfacacUfgGfuucuuugsc	986
AD-1634080	AGT	asgsaaccAfgUfGfUfuuuagcgcuL96	638	asCfscgcGfcUfAfaacaCfuGfguucuuugsc	987

TABLE 3-continued

More additional exemplary sequences of parent dsRNA molecules.					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1634081	AGT	gsasaccaGfuGfUfUfuagcgcgguL96	639	asCfscgcCfGfUfaaacAfcUfgguucsusu	988
AD-1634082	AGT	asascagUfgUfUfUfagcgcggaL96	640	asUfscgcGfcGfCfuaaaCfaCfugguuscsu	989
AD-1634105	AGT	csusguuCfaAfAfAfagaauuccauL96	641	asUfsggaAfuUfCfuuuUfgGfaacagsusa	990
AD-1634107	AGT	gsusuccaAfaAfAfGfaauccaacuL96	642	asGfsuugGfaAfUfucuuUfuUfggaacsasg	991
AD-1634109	AGT	uscscaaaAfaGfAfAfauccaaccguL96	643	asCfsgguUfgGfAfauuUfuUfuggasasc	992
AD-1634110	AGT	cscsaaaaAfgAfAfUfuccaaccgaL96	644	asUfscggUfuGfGfaauCfuUfuuuugsasa	993
AD-1634111	AGT	csasaaaaGfaAfUfUfccaaccgacuL96	645	asGfsucgGfuUfGfgaaUfcUfuuuugsasa	994
AD-1634112	AGT	asasaaagAfaUfUfCfcaaccgaccuL96	646	asGfsgucGfgUfUfggaaUfuCfuuuusgsg	995
AD-1634113	AGT	asasaagaAfuUfCfCfcaaccgaccuL96	647	asUfsgguCfGfUfuggaAfuUfcuuuusug	996
AD-1634114	AGT	asasagaaUfuCfCfAfaccgaccaguL96	648	asCfsuggUfcGfGfuuggAfaUfcuuuususu	997
AD-1634115	AGT	asasgaauUfcCfAfAfccgaccagcuL96	649	asGfscugGfuCfGfguugGfaAfuucuuusu	998
AD-84724	AGT	asgsaaauCfcAfAfCfcgaccagcuL96	650	asAfsGcuGfgUfCfGguGfgAfaucuuusu	999
AD-1634116	AGT	gsasauucCfaAfCfCfGaccagcuuL96	651	asAfsagcUfgGfUfcgguUfgGfaauucsusu	1000
AD-1634117	AGT	asasuuccAfaCfCfGfaccagcuuguL96	652	asCfsaagCfuGfGfucggUfuGfgaauscsu	1001
AD-1634118	AGT	asusuccaAfcCfGfAfccagcuuguL96	653	asAfscaaGfcUfGfgucgGfuUfggausus	1002
AD-68579	AGT	ususcacCfcGfAfCfcagcuuguuL96	654	asAfsacaAfgCfUfggucGfgUfggausus	1003
AD-1634119	AGT	uscscacCfGfAfCfCfagcuuguuL96	655	asAfsaacAfaGfCfugguCfGfuggasasu	1004
AD-1634120	AGT	cscsaaccGfaCfCfAfGcuuguuguL96	656	asCfsaaaCfaAfGfcuggUfcGfguugsasa	1005
AD-1634121	AGT	csasaccGfCfAfGfCfuuguuguuL96	657	asAfscaaAfcAfAfGcugGfuCfGguugsasa	1006
AD-1634122	AGT	asascgaCfCfAfGfCfuuguuguuL96	658	asCfsacaAfaCfAfagcuGfgUfcgguusgsg	1007
AD-1634123	AGT	ascscgacCfaGfCfUfuguuguuL96	659	asUfscacAfaAfCfaagcUfgGfucgguusg	1008
AD-1634124	AGT	cscsgaccAfgCfUfUfuguugaaL96	660	asUfsucaCfaAfAfcaagCfuGfgucggsusu	1009
AD-1634125	AGT	csgsaccaGfcUfUfGfuuguugaaL96	661	asUfsuucAfcAfAfacaaGfcUfggucgsgsu	1010
AD-1634126	AGT	gsasccagCfuUfGfUfuguugaaacuL96	662	asGfsuuuCfaCfAfaacaAfgCfugguccsgsg	1011

TABLE 3-continued

More additional exemplary sequences of parent dsRNA molecules.					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1634127	AGT	ascscagcUfuGfUfUfugugaaacauL96	663	asUfsguuUfcAfCfaaacAfaGfcugguscsg	1012
AD-1634128	AGT	cscsagcuUfgUfUfUfgugaaacauL96	664	asUfsuguUfuCfAfcaaaCfaAfgcuggsusuc	1013
AD-1634129	AGT	csasgcuUfuUfUfGfugaaacaaauL96	665	asUfsuugUfuUfCfacaAfaAfgcugsgsu	1014
AD-1634130	AGT	asgscuugUfuUfGfUfgaaacaaauL96	666	asUfsuuUfuUfUfcacaAfaCfaagcugsg	1015
AD-1684562	AGT	gscsuuguUfuGfUfGfaaaacaaauL96	667	asUfsuuUfgUfUfucacAfaAfaagcuscsg	1016
AD-1684563	AGT	csusuguUfgUfGfAfaacaaauL96	668	asUfsuuUfuGfUfuucaCfaAfaagcscsu	1017
AD-1684564	AGT	usugsuuUfuGfAfaacaaauL96	669	asCfsuuUfuUfGfuucAfaAfaagcsc	1018
AD-1634135	AGT	usgsuuccCfuUfUfCfaaguugaguL96	670	asCfsucaAfcUfUfgaaaAfgGfgaacscsu	1019
AD-1634136	AGT	gsusuccCfuUfUfCfaaguugaguL96	671	asUfscucAfaCfUfugaaAfaGfggaacsasc	1020
AD-1634137	AGT	ususccuUfuUfCfAfaaguugaguL96	672	asUfsucuCfaAfCfuugaAfaAfgggaacsca	1021
AD-1634146	AGT	csasaguUfaGfAfaacaaauL96	673	asCfsaaUfuUfUfguucUfcAfacuugsasa	1022
AD-1634147	AGT	asasguugAfgAfaCfaaaauL96	674	asCfscaaUfuUfUfuguUcfUfaacuugsasa	1023
AD-68585	AGT	asgsuugaGfaAfCfAfaaaauL96	675	asCfsccaAfuUfUfuuguUfcUfcaacususc	1024
AD-1634148	AGT	gsusugagAfaCfAfaaaauL96	676	asAfscccAfaUfUfuuuUfuCfucacuusu	1025
AD-1634149	AGT	usugagaAfaCfAfaaaauL96	677	asAfsaccCfaAfUfuuuUfuUfcuacuusu	1026
AD-1634150	AGT	usgsagaaCfaAfAfaauL96	678	asAfsaacCfcAfAfuuuUfgUfucacasasc	1027
AD-1634151	AGT	gsasgaacAfaAfAfaauL96	679	asAfsaaaCfcCfAfauuUfuGfuucacsasa	1028
AD-1634152	AGT	asgsaacaAfaAfAfuL96	680	asUfsaaaAfcCfCfauuUfuUfguucscsa	1029
AD-1634153	AGT	gsasacaaAfaAfUfUfgguuuL96	681	asUfsuaaAfaCfCfcauuUfuUfguucscsa	1030
AD-1634162	AGT	asgsuauaCfaUfUfUfuugcauugcuL96	682	asGfscaaUfgCfAfaaaUfgUfauacususu	1031
AD-1634163	AGT	gsusauacAfuUfUfUfgcauugccuL96	683	asGfsgcaAfuGfCfaaaAfuGfuauacususu	1032
AD-1634164	AGT	usasuacaUfuUfUfUfgcauugccuL96	684	asAfsaggAfaUfGfcaaaAfaUfguauascsu	1033
AD-1634165	AGT	asusacauUfuUfUfGfcauugccuL96	685	asAfsaggCfaAfUfgcaaAfaUfguauasc	1034
AD-1634169	AGT	asusuuuGfcAfUfUfgccuucgguL96	686	asAfsccgAfaGfGfcauuGfcAfaaausgsu	1035

TABLE 3-continued

More additional exemplary sequences of parent dsRNA molecules.					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1634170	AGT	ususuugCfaUfUfGfccuucgguuuL96	687	asAfsaccGfaAfGfgcaaUfgCfaaaaausg	1036
AD-1634171	AGT	ususuugcAfuUfGfCfcuucgguuuuL96	688	asAfsaacCfGfAfGfgcaAfuGfcaaaaasasu	1037
AD-1634172	AGT	ususugcaUfuGfCfCfuucgguuuguL96	689	asCfsaaaCfcGfAfaggcAfaUfgcaaaasasa	1038
AD-1634173	AGT	ususgcauUfgCfCfUfucgguuuguuL96	690	asAfscaaAfcCfGfaaggCfaAfugcaasasa	1039
AD-1634174	AGT	usgscauuGfcCfUfUfGcgguuguauL96	691	asUfsacaAfaCfCfGaagGfcAfaugcasasa	1040
AD-1634175	AGT	gscsauugCfcUfUfCfGguuguauuL96	692	asAfsuacAfaAfCfcgaGfgCfaaugcsasa	1041
AD-1634176	AGT	csasuugcCfuUfCfGfguuuguauuuL96	693	asAfsauaCfaAfAfccgaAfgGfcaaugscsa	1042
AD-1634177	AGT	asusugccUfuCfGfGfuuguauuuuL96	694	asAfsaaUfCfAfAfaccgAfaGfgcaausgsc	1043
AD-1634178	AGT	ususgccuUfcGfGfUfuuguauuuauL96	695	asUfsaaaUfaCfAfaaccGfaAfggcaausg	1044
AD-1634179	AGT	usgsccuuCfGfUfUfuguauuuaguL96	696	asCfsuaaAfuAfCfaaacCfGfaggcasasu	1045
AD-1634180	AGT	gscscuucGfgUfUfUfuguauuuaguL96	697	asAfscaAfaUfAfcaaaCfcGfaaggcsasa	1046
AD-1634181	AGT	cscsuucgGfuUfUfGfuauuuaguguL96	698	asCfsacuAfaAfUfacaAfcCfGaaggscsa	1047
AD-1634182	AGT	csusucggUfuUfGfUfauuuaguguuL96	699	asAfsacUfaAfAfuaaAfaCfcaagsgsc	1048
AD-1634183	AGT	ususcgguUfuGfUfAfuuuagugucuL96	700	asGfsacaCfuAfAfauaAfaAfccgaasgsg	1049
AD-1634184	AGT	uscsgguuUfgUfAfUfuuuagugucuL96	701	asAfsagAcAfUfAfaaAfaAfaccgasasg	1050
AD-1634185	AGT	csgsguuuGfuAfUfUfuagugucuL96	702	asAfsagaCfaCfUfaaaAfaAaacgsasa	1051
AD-1634186	AGT	gsgsuuugUfaUfUfUfagugucuuguL96	703	asCfsaagAfcAfCfuaaaUfaCfaaacsgsa	1052
AD-1634187	AGT	gsusuuguAfuUfUfAfgugucuuguL96	704	asUfscaaGfaCfAfcaaaAfuAfaaacscsg	1053
AD-1634188	AGT	ususuguaUfuUfAfGfugucuugaauL96	705	asUfsucaAfgAfCfacaAfaUfacaascsc	1054
AD-1634189	AGT	ususguauUfuAfGfUfgucugaauuL96	706	asAfsuucAfaGfAfcaAfaAfaaacsasc	1055
AD-1634190	AGT	usgsauuUfaGfUfGfucugaauuL96	707	asCfsauuCfaAfGfacacUfaAfaucasasa	1056
AD-1634191	AGT	gsusauuuAfgUfGfUfcuugaauuL96	708	asAfscaUfCfAfGgacaCfuAfaauacsasa	1057
AD-1634192	AGT	usasuuaGfuGfUfCfuugaauuL96	709	asUfsacaUfuCfAfagacAfcUfaaauscsa	1058
AD-1634193	AGT	asusuagUfgUfCfUfugaauuL96	710	asUfsuacAfuUfCfaagaCfaCfuaaaasasc	1059

TABLE 3-continued

More additional exemplary sequences of parent dsRNA molecules.					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1634194	AGT	ususuguGfuCfUfUfgaaguaagL96	711	asCfsuuaCfaUfUfcaagAfcAfcuaasasa	1060
AD-1634195	AGT	ususagugUfcUfUfGfaaaguaagauL96	712	asUfscuuAfcAfUfucaaGfaCfacuaasasu	1061
AD-1634196	AGT	usasguguCfuUfGfAfauguaagaauL96	713	asUfscuuUfaCfAfuucaAfgAfcacuasasa	1062
AD-1634197	AGT	asgsugucUfuGfAfAfuguaagaacuL96	714	asGfsuucUfuAfcFauucAfaGfacacusasa	1063
AD-1634199	AGT	usgsucuuGfaAfUfGfuaagaacaauL96	715	asAfsuguUfcUfUfacauUfcAfagacascsu	1064
AD-1634200	AGT	gsuscuugAfaUfGfUfaagaacaugL96	716	asCfsaugUfuCfUfuacaUfuCfaagacsasc	1065
AD-1634203	AGT	ususgaauGfuAfAfGfaacaugaccuL96	717	asGfsgucAfuGfUfucuuAfcAfuucaasgsa	1066
AD-1634209	AGT	gsusaagaAfcAfUfGfaccuccguguL96	718	asCfsacgGfaGfGfucuuGfuUfcuacsasu	1067
AD-1634210	AGT	usasagaaCfaUfGfAfcucccguguL96	719	asAfsacGfgAfGfgucaUfgUfucuuascsa	1068
AD-1634211	AGT	asasgaacAfuGfAfcucccguguauL96	720	asUfsacaCfGfAfggucAfuGfuucuasasc	1069
AD-1634212	AGT	asgsaacaUfgAfcCfucccguguaguL96	721	asCfsuacAfcGfGfagguCfaUfguucususa	1070
AD-1634213	AGT	gsasacauGfaCfCfUfccguguaguL96	722	asAfsucaCfaCfGfgaggUfcAfuguucsusu	1071
AD-1634214	AGT	asascaugAfcCfUfcCfuguguaguL96	723	asCfsacuAfcAfcCfaggGfuCfauguuscscu	1072
AD-1634215	AGT	ascsaugaCfcUfcCfCfuguguaguL96	724	asAfsacUfaCfAfcggaGfgUfcaugususc	1073
AD-1634216	AGT	csasugacCfuCfCfGfuguguagucuL96	725	asGfsacaCfuAfcCfaggAfgGfucugsusu	1074
AD-1634217	AGT	asusgaccUfcCfGfUfuguguagucuL96	726	asAfsagacAfcUfAfcacgGfaGfgucausgsu	1075
AD-1684565	AGT	csusuaguUfuUfUfUfccacagaugL96	727	asCfsaucUfgUfGfgaaaAfaAfcuaagsgsu	1076
AD-1684566	AGT	ususaguUfuUfUfCfcacagaugcuL96	728	asGfscacuCfuGfUfggaaAfaAfacuaasgsg	1077
AD-1684567	AGT	usasguuuUfuUfCfCfcaagaugcuL96	729	asAfsagcaUfcUfGfuggaAfaAfaacuasag	1078
AD-1684568	AGT	gsusuuuuUfcCfAfcfagaugcuuguL96	730	asCfsaagCfaUfCfugugGfaAfaaaacsusa	1079
AD-1684569	AGT	ususuuucCfaCfAfcfagaugcuuguL96	731	asCfsacaAfgCfAfucugUfgGfaaaaaasasc	1080
AD-1684570	AGT	ususuuccAfcAfcAfcfagaugcuuguL96	732	asUfscacAfaGfCfaucuuGfuGfgaaaaasasa	1081
AD-1634234	AGT	ususuccaCfaGfAfUfgcuugugauL96	733	asAfsucaCfaAfcGfcaucUfgUfggaaaasasa	1082

TABLE 3-continued

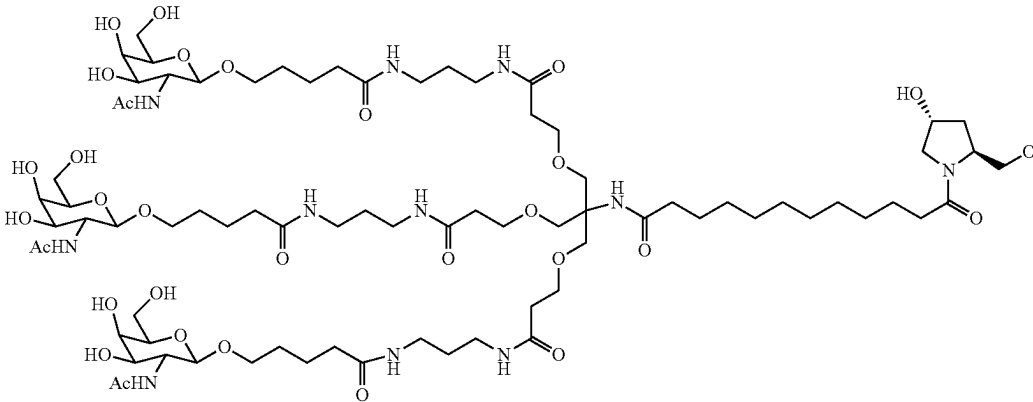
More additional exemplary sequences of parent dsRNA molecules.					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1634235	AGT	ususcacAfgAfUfGfCuugauuuL96	734	asAfsaucAfcAfAfgcauCfuGfuggaasasa	1083
AD-1634236	AGT	uscscacaGfaUfGfCfCuugauuuL96	735	asAfsaauCfaCfAfagcaUfcUfuggasasa	1084
AD-1634237	AGT	cscscacagAfuGfCfUfugauuuuuL96	736	asAfsaaaUfcAfCfaagcAfuCfuguggsasa	1085
AD-1634238	AGT	csascagaUfgCfUfUfgugauuuuuL96	737	asAfsaaaAfuCfAfcaagCfaUfcugugsgsa	1086
AD-1634282	AGT	ascscugaAfuUfUfCfuguuugaauL96	738	asAfsuucAfaAfCfagaaAfuUfcaggusgsc	1087
AD-1634283	AGT	cscsugaaUfuUfCfUfguuugaauL96	739	asCfsauuCfaAfAfcagaAfaUfcaggusg	1088
AD-1634304	AGT	gsgsaaccAfuAfGfCfugguuuuuL96	740	asAfsaauAfaCfCfagcuAfuGfguuccsgsc	1089
AD-1634305	AGT	gsasaccaUfaGfCfUfgguuuuuL96	741	asGfsaaaUfaAfCfcagcUfaUfgguuccsg	1090
AD-1634306	AGT	asasccauAfgCfUfGfGuuuuuL96	742	asAfsagaAfuAfAfcagCfuUfugguuscsc	1091
AD-1634307	AGT	ascscuaGfCfUfGfGuuuuuL96	743	asGfsagaAfaUfAfaccaGfcUfauggususc	1092
AD-1634308	AGT	cscsauagCfuGfGfUfuuuuL96	744	asGfsagAfaAfUfaaccAfgCfuauugsusu	1093
AD-1634327	AGT	cscsuuguGfuUfAfGfuaaaacguL96	745	asCfsguuUfaUfUfacuaAfcAfcaaggsgsa	1094
AD-1634328	AGT	csusugugUfuAfGfUfaaaacguuL96	746	asAfsaguUfuAfUfuacuAfaCfacaaggsg	1095
AD-1634329	AGT	usuguguUfaGfUfAfaaaacgucuL96	747	asGfsacgUfuUfAfuuaUfaAfcacasgsg	1096
AD-1634330	AGT	usgsuguuAfgUfAfAfaaaacgucuL96	748	asAfsagacGfuUfUfaauaCfuAfacacasg	1097
AD-1634331	AGT	gsusguuaGfuAfAfUfaaacgucuL96	749	asAfsagaCfGfUfUfuuuAfcUfaacacsasa	1098
AD-1634332	AGT	usgsuuagUfaAfUfAfaaacgucuL96	750	asCfsaagAfcGfUfuuuUfaCfuaacacsasa	1099
AD-1634333	AGT	gsusuaguAfaUfAfAfaaacgucuL96	751	asGfscaaGfaCfGfuuuUfuAfcuaacsasc	1100

TABLE 4

Abbreviations of nucleotide monomers used in nucleic acid sequence representation*	
Abbreviation	Nucleotide(s)
A	Adenosine-3'-phosphate
Ab	beta-L-adenosine-3'-phosphate
Abs	beta-L-adenosine-3'-phosphorothioate
Af	2'-fluoroadenosine-3'-phosphate
Afs	2'-fluoroadenosine-3'-phosphorothioate
As	adenosine-3'-phosphorothioate
C	cytidine-3'-phosphate

TABLE 4-continued

Abbreviations of nucleotide monomers used in nucleic acid sequence representation*	
Abbreviation	Nucleotide(s)
Cb	beta-L-cytidine-3'-phosphate
Cbs	beta-L-cytidine-3'-phosphorothioate
Cf	2'-fluorocytidine-3'-phosphate
Cfs	2'-fluorocytidine-3'-phosphorothioate
Cs	cytidine-3'-phosphorothioate
G	guanosine-3'-phosphate
Gb	beta-L-guanosine-3'-phosphate
Gbs	beta-L-guanosine-3'-phosphorothioate
Gf	2'-fluoroguanosine-3'-phosphate
Gfs	2'-fluoroguanosine-3'-phosphorothioate
Gs	guanosine-3'-phosphorothioate
T	5'-methyluridine-3'-phosphate
Tf	2'-fluoro-5-methyluridine-3'-phosphate
Tfs	2'-fluoro-5-methyluridine-3'-phosphorothioate
Ts	5-methyluridine-3'-phosphorothioate
U	Uridine-3'-phosphate
Uf	2'-fluorouridine-3'-phosphate
Ufs	2'-fluorouridine-3'-phosphorothioate
Us	uridine-3'-phosphorothioate
N	any nucleotide, modified or unmodified
a	2'-O-methyladenosine-3'-phosphate
as	2'-O-methyladenosine-3'-phosphorothioate
c	2'-O-methylcytidine-3'-phosphate
CS	2'-O-methylcytidine-3'-phosphorothioate
g	2'-O-methylguanosine-3'-phosphate
gs	2'-O-methylguanosine-3'-phosphorothioate
t	2'-O-methyl-5-methyluridine-3'-phosphate
ts	2'-O-methyl-5-methyluridine-3'-phosphorothioate
u	2'-O-methyluridine-3'-phosphate
us	2'-O-methyluridine-3'-phosphorothioate
s	phosphorothioate linkage
L96	N-[tris(GalNAc-alkyl)-amido-dodecanoyl]-4-hydroxyprolinol [Hyp-(GalNAc-alkyl) ₃]



(Agn)	Adenosine-glycol nucleic acid (GNA)
(Cgn)	Cytidine-glycol nucleic acid (GNA)
(Ggn)	Guanosine-glycol nucleic acid (GNA)
(Tgn)	Thymidine-glycol nucleic acid (GNA) S-Isomer
P	Phosphate
VP	Vinyl-phosphonate (e.g., 5'-E-vinylphosphonate)
dA	2'-deoxyadenosine-3'-phosphate
dAs	2'-deoxyadenosine-3'-phosphorothioate
dC	2'-deoxycytidine-3-phosphate
dCs	2'-deoxycytidine-3'-phosphorothioate
dG	2'-deoxyguanosine-3'-phosphate
dGs	2'-deoxyguanosine-3'-phosphorothioate
dT	2'-deoxythymidine-3'-phosphate
dTs	2'-deoxythymidine-3'-phosphorothioate
dU	2'-deoxyuridine
dUs	2'-deoxyuridine-3-phosphorothioate
(Ahd)	2'-O-hexadecyl-adenosine-3'-phosphate
(AhdS)	2'-O-hexadecyl-adenosine-3'-phosphorothioate
(Chd)	2'-O-hexadecyl-cytidine-3'-phosphate

TABLE 4-continued

Abbreviations of nucleotide monomers used in nucleic acid sequence representation*	
Abbreviation	Nucleotide(s)
(Chds)	2'-O-hexadecyl-cytidine-3'-phosphorothioate
(Ghd)	2'-O-hexadecyl-guanosine-3'-phosphate
(Ghds)	2'-O-hexadecyl-guanosine-3'-phosphorothioate
(Uhd)	2'-O-hexadecyl-uridine-3'-phosphate
(Uhds)	2'-O-hexadecyl-uridine-3'-phosphorothioate

*It will be understood that these monomers, when present in an oligonucleotide, are mutually linked by 5'-3'-phosphodiester bonds; and it is understood that when the nucleotide contains a 2'-fluoro modification, then the fluoro replaces the hydroxy at that position of the parent nucleotide (i.e., it is a 2'-deoxy-2'-fluoronucleotide).

[0434] Some exemplary dsRNA molecule designs according to embodiments of the disclosure are shown schematically in FIGS. 5A and 5B. Exemplary dsRNA molecules according to some embodiments of the disclosure are listed in Table 5 and 6.

TABLE 5

Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO	Antisense sequence (5'→3')	SEQ ID NO:
AD-1636769	Marc1	asasaaaugu UfCfUfcaaaaugaaL96	1101	usdTscadTudTuugadGaAfcauuuuusasa	1269
AD-1636805	Marc1	asasaaucacCfAfCfucuuugggcaL96	1102	usdGscddCadAagagdTgGfugauuuuscsc	1270
AD-1636782	Marc1	asasagugggAfGfAfccuguguaaL96	1103	usdTsacdAcddAgggudCuCfccacuuusgsa	1271
AD-1636754	PNPLA3	asasaugaaaGfAfCfaaagguggauL96	1104	asdTsccdAcddCuugdTcUfuucauuuscscu	1272
AD-1636695	C3	asasaugaggGfUfUfucacagucuaL96	1105	usdTsgadCudGugaadAcCfcucauuususc	1273
AD-1636733	PNPLA3	asascuugcuAfCfCfcauaggauaL96	1106	usdAsucdCudAauggdGuAfgaaguusgsc	1274
AD-1636798	Marc1	asasucaccaCfUfCfuugggagucuaL96	1107	asdCsugdCcdCaaagdAgUfggugauususu	1275
AD-1636720	C3	ascsaugggcCfAfGfuggaagaucuaL96	1108	usdGsaudCudTccacdTgGfcccagugusug	1276
AD-1636701	C3	ascscaggaaCfUfGfaaccuuguaL96	1109	usdAsucdAadGguucdAgUfuccuggugsg	1277
AD-1636716	C3	ascscaggauGfCfCfcauugucuaL96	1110	usdAsgadCadTagugdGcAfuccugguscscu	1278
AD-1636729	PNPLA3	ascscuaacuAfAfAfauaauuuuaL96	1111	usdAsaadCadTuauudTuAfguuaggusgsa	1279
AD-1636806	Marc1	ascscucgccUfGfGfuccugauuuuaL96	1112	usdAsaadTcdAggacdCaGfgcgaggususc	1280
AD-1636735	PNPLA3	ascscugugAfAfUfuuuuauuuuaL96	1113	asdTsaaadTadCaaaadTuCfaacaggusasa	1281
AD-1636714	C3	ascsuacaugAfAfCfcuacagagauL96	1114	asdTscudCudGuaggdTuCfauguagusug	1282
AD-1636778	Marc1	ascsugauuaUfGfGfaauaguucuuL96	1115	asdAsgadAcddAuucdCaUfaaucagususa	1283
AD-1479350	C3	asgsaaaucUfAfCfuacauuuuuL96	1116	asdAsuadGadTguagdTaGfaauuuucscscu	1284

TABLE 5-continued

Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO	Antisense sequence (5'→3')	SEQ ID NO:
AD-1636785	Marc1	asgsacaggaUfUfCfugaaaaacucaL96	1117	usdGsagdTudTucagdAaUfccugucususg	1285
AD-1636763	C3	asgsagcgggUfAfCfcucucaucaL96	1118	usdGsaudGadAgaggdTacfcgcgucucsgsc	1286
AD-1636834	SCN9A	asgscaaaggUfCfAfcaauuuccuaL96	1119	usdAsggdAadAuugudGacfcuuugcuscsc	1287
AD-1636811	SCN9A	asgscauaaaaUfGfUfuuucgaaaL96	1120	usdAsuudTcdGaaaadCaUfuuaugcuscsc	1288
AD-1636757	PNPLA3	asgscaugagGfUfUfcuuagaauL96	1121	asdCsaudTcdTaagadAcCfucaugcucsgsg	1289
AD-1636758	PNPLA3	asgsgaagcaAfCfCfuucgccugL96	1122	asdCsagdGcdGaaagdGuUfgcuuccuscs	1290
AD-1636791	Marc1	asgsgaccagAfUfUfgcuuacuaL96	1123	usdTsgadGudAagcadAuCfugguccususg	1291
AD-1636777	Marc1	asgsgagaagAfAfAfagugauucaaL96	1124	usdTsgadAudCacuudTuCfuuccuccuscsc	1292
AD-1636838	SCN9A	asgsucucaAfGfUfuggcaaaaL96	1125	usdAsuudTudGccaadCuUfgaagacuscsg	1293
AD-1636703	C3	asgsuggacuAfUfGfuguacaagauL96	1126	asdTscudTgdTcacadAuAfguccacuscsc	1294
AD-1636831	SCN9A	asgsuuccuaUfCfUfccuuucagaaL96	1127	usdTscudGadAaggadGaUfaggaacuscsc	1295
AD-1636732	PNPLA3	asusaaugucUfUfAfuguaaugcuL96	1128	asdAsgcdAudTacaadAaGfacauuauscsc	1296
AD-1636737	PNPLA3	asusacaugaGfCfAfagauuugcaaL96	1129	usdTsgcdAadAucuudGcUfcauguauscsc	1297
AD-1636779	Marc1	asuscaaccaGfGfAfgggaaacuaL96	1130	usdAsugdTudTccudCcUfggguugauscsc	1298
AD-1636843	SCN9A	asuscaucuuUfGfGfgucauucuuL96	1131	asdAsagdAadTgaccdCaAfagaugausasa	1299
AD-1636772	Marc1	asuscugaugAfAfGfuaauuuuuuL96	1132	asdAsaadAadTauacdTuCfaucagauscsc	1300
AD-1636836	SCN9A	asuscuucuuUfGfUfcguagugauL96	1133	asdAsucdAcdTaccgadCaAfagaugauscsc	1301
AD-1636813	SCN9A	asusgcugagAfAfAfugucgaaaaL96	1134	usdTsuudCgdAcaudTuCfucagcauscsc	1302
AD-1636783	Marc1	asusggcuugUfUfCfcagaucauL96	1135	asdAsugdCadTcuggdAaCfaagccauscsc	1303
AD-1636692	C3	asusguaccaUfGfCfuaaggccaaaL96	1136	usdTsgdGcdCuagdCaUfggguacauscsc	1304
AD-1636844	SCN9A	asusgucgagUfAfCfauuuuacuuL96	1137	asdAsgudAadAagudTaCfucgacaususu	1305
AD-1636726	PNPLA3	asusguaguAfGfAfauaagccuaL96	1138	usdAsagdGcdTuauudCuAfcuaacauscsc	1306
AD-1636760	PNPLA3	asusuaggauAfAfUfgucuuauuaL96	1139	usdAscadTadAgacadTuAfuccuaucsgsg	1307
AD-1636841	SCN9A	asusucucuuCfGfUfucacagaugaL96	1140	usdCsaudCudGugaadCgAfagagaauscsc	1308

TABLE 5-continued

Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO	Antisense sequence (5'→3')	SEQ ID NO:
AD-1636789	Marc1	csasacacuuGfAfAfgcaugguguuL96	1141	asdAscadCcdAugcudTcAfaguguugsusc	1309
AD-1636762	PNPLA3	csasagauuuGfCfAfacuugcuacaL96	1142	usdGsuadGcdAaguudGcAfaaucuugscsu	1310
AD-1636766	C3	csasaggucuAfCfGfccuauuacauL96	1143	asdTsgudAadTaggcdGuAfgaccuugsasc	1311
AD-1636718	C3	csascccuaUfCfAfucuaccuggaL96	1144	usdCscadGgdTagaudGaUfgagggugsusu	1312
AD-1636707	C3	csasccguauCfCfAfcugggaaucL96	1145	asdGsaudTcdCcagudGgAfuacggugsgsg	1313
AD-1636727	PNPLA3	csasccuuuuUfCfAfccuaacuaaaL96	1146	usdTsuadGudTaggudGaAfaaaggugsusu	1314
AD-1636741	PNPLA3	csasccuuuuUfCfAfccuaacuaauL96	1147	asdTsuadGudTaggudGaAfaaaggugsusu	1315
AD-1636830	SCN9A	csascuccuuCfCfUfgauuguguuuL96	1148	asdAsacdAcdAaucadGgAfgagagugsgsa	1316
AD-1636780	Marc1	csasgaacgaAfAfGfuuaauaggaaL96	1149	usdTscddAudAuaacdTuUfcguucugsasa	1317
AD-1636689	C3	csasgagaaaUfUfCfuacuacauL96	1150	asdGsaudGudAguagdAaUfuucucugsusa	1318
AD-1636839	SCN9A	csasugaacgAfCfUfucuccacuL96	1151	asdAsgudGgdAagaadGuCfguucaugsusg	1319
AD-1636753	PNPLA3	csasugagcaAfGfAfuugcaacuL96	1152	asdAsgudTgdCaaudCuUfgcucaugsusa	1320
AD-1636792	Marc1	csasugguguUfUfCfagaacugagaL96	1153	usdCsucdAgdTucugdAaAfcaccaugscsu	1321
AD-1636709	C3	csasuugagaAfCfCfcggaaggcauL96	1154	asdTsgcdCudTccggdGuUfcucaauggsusu	1322
AD-1636721	C3	cscsacagccAfAfAfgauaagaacuL96	1155	asdGsuudCudTaucudTuGfgcuguggsusc	1323
AD-1636719	C3	cscsagauccAfCfUfucaccaagauL96	1156	asdTscudTgdGugaadGuGfgaucuggsusa	1324
AD-1636765	PNPLA3	cscsauuaggAfUfAfaugucuuaauL96	1157	asdAsuadAgdAcauudAuCfcuaaugggsusu	1325
AD-1636693	C3	cscscgucguGfCfGfuuggcucaauL96	1158	asdTsugdAgdCcaacdGcAfcgacgggsasg	1326
AD-1636807	Marc1	cscsgacccaAfGfGfaccagauuguL96	1159	asdCsaadTcdTggucdCuUfgggucggsasa	1327
AD-1479283	C3	cscsgagccgUfUfCfucuaaauuL96	1160	asdAsaudTgdTagagdAaCfggcucggsasu	1328
AD-1636774	Marc1	cscsguauguCfCfUfggaauauuauL96	1161	asdTsaadTadTuccadGgAfcuacggsusu	1329
AD-1636725	PNPLA3	cscsuaacuaAfAfAfaaauuuuuL96	1162	usdTsaadAcdAuaudTuUfaguuauggsusg	1330
AD-1636715	C3	cscsuugucuUfCfUfcagaaccagaL96	1163	usdCsugdGudTcugadGaAfgacaaggsasg	1331

TABLE 5-continued

Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO	Antisense sequence (5'→3')	SEQ ID NO:
AD-1636750	PNPLA3	csgsacaucuGfCfCfcaaaagucuaL96	1164	usdTsgadCudTuaggdGcAfgaugucgsusa	1332
AD-1636702	C3	csgsugcgguGfGfCfcaugaacuL96	1165	asdGsuudCadTugagdCcAfacgcacgsasc	1333
AD-1636705	C3	csusacccuaCfUfCfuguuguucgaL96	1166	usdCsgadAcdAacagdAgUfaggguaagscsc	1334
AD-1636690	C3	csusacugcaGfCfUfaaaagacuuuL96	1167	asdAsagdTcdTuuaadGcUfgcaguagsgsg	1335
AD-1636694	C3	csusagugcuGfUfCfcagugagaaaL96	1168	usdTsuudTcdAcuggdAcAfgcacuagsusu	1336
AD-1636743	PNPLA3	csusauuaauGfGfUfcagacuguaaL96	1169	usdAsacdAgdTcugadCcAfuaauagsgsg	1337
AD-1636756	PNPLA3	csusccauggCfGfGfgguuaaaaaL96	1170	usdTsuudTcdAcuccdCgCfcauggagsasc	1338
AD-1636734	PNPLA3	csusgaguugGfUfUfuaugaaaaL96	1171	asdTsuudTcdAuaaadAcCfaacucagscsu	1339
AD-1636840	SCN9A	csusgccccaaAfAfUfacugauaauL96	1172	usdAsuudAudCaguadTuUfugggcagscsa	1340
AD-1636712	C3	csusgggaggAfCfCfcugguaagcaL96	1173	usdGscudTadCcaggdGuCfcucccagscsg	1341
AD-1636800	Marc1	csusguggagGfAfGfaagaaaaguaL96	1174	usdAscudTudTcuudTcCfuccacagsasa	1342
AD-1636790	Marc1	csusucuauUfGfGfugacguggaaL96	1175	usdTscddAcdGucacdCaAfuaagaagscsu	1343
AD-1636770	Marc1	csusuguuccAfGfAfugcauuuuL96	1176	usdTsaadAadTgcaudCuGfgaacaagscsc	1344
AD-1636837	SCN9A	csusuugucGfAfGfugauuuuccuL96	1177	asdGsgadAadAucacdTaCfagacaaagsasa	1345
AD-1636699	C3	gsascagacaAfGfAfccaucacauL96	1178	asdTsgudAgdAuggdCuUfgucugucsusg	1346
AD-1479341	C3	gsasgaaauCfUfAfcaucaucuuL96	1179	asdTsagdAudGuagudAgAfauuucucsusg	1347
AD-1636691	C3	gsasgaaccaGfAfAfacaauccauL96	1180	asdTsggdCadTuguudTcUfgguuucsusu	1348
AD-1636815	SCN9A	gsasggucaaGfAfCfaucuuuauL96	1181	usdCsaudAadAgaugdTcUfugaccucscsa	1349
AD-1636795	Marc1	gsasgugcucCfUfUfcuccaguucL96	1182	asdGsaadCudGgagadAgGfagcacucscsg	1350
AD-1636731	PNPLA3	gsasuugcaAfCfUfugcuaccuL96	1183	asdTsggdGudAgcaadGuUfgcaauucsusu	1351
AD-1636744	PNPLA3	gscsacagggAfAfCfcucuaaccuL96	1184	usdAsagdGudAgaggdTuCfccugucgsasg	1352
AD-1636717	C3	gscsaggcagAfGfCfagcagggauL96	1185	asdTscddCgdCugcgdCuCfugccucgsasc	1353
AD-1636796	Marc1	gscscuuuccCfCfUfcagcuaaL96	1186	usdCsaudTadGcugadGgGfgaauggcsasa	1354

TABLE 5-continued

Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO	Antisense sequence (5'→3')	SEQ ID NO:
AD-1636710	C3	gscscucucUfUfAfacaaauucU96	1187	asdGsaadAudTuguudAaGfaagaggcscsc	1355
AD-1636822	SCN9A	gscsguuguaGfUfUfccuauccuU96	1188	asdGsgadGadTaggadAcUfacaacgcscsu	1356
AD-1636828	SCN9A	gscsucaucaUfGfUfgcacuauucU96	1189	asdGsaadTadGugcadCaUfgaugagcsasu	1357
AD-1636767	C3	gscsugaggaGfAfAfuugcucauU96	1190	asdAsugdAadGcaudTcUfccucagcsasc	1358
AD-1636739	PNPLA3	gscsugaguUfGfUfuuuugaaaaL96	1191	usdTsudCadTaaaadCcAfacucagcsusc	1359
AD-1636803	Marc1	gscsuucucaGfAfCfagcauuggauL96	1192	asdTscddAadTgcugdTcUfgagaagcsasg	1360
AD-1636801	Marc1	gsgsaccagaUfUfGfcuucacagaL96	1193	usdCsugdAgdTaagcdAaUfcugguccsusu	1361
AD-1636713	C3	gsgsagguugUfGfCfugagccggaaL96	1194	usdTsccdGgdCucagdCaCfaaccuccscsc	1362
AD-1636833	SCN9A	gsgscacaugAfAfCfagcuucuaL96	1195	usdGsaadGadAgucgdTuCfaugugccsasc	1363
AD-1636740	PNPLA3	gsgsccuuaUfCfCfuccuuccuuaL96	1196	usdAsagdGadAggagdGgAfuaaggccsasc	1364
AD-1636823	SCN9A	gsgscuggauUfUfCfcuaauuguuuL96	1197	asdAsacdAadTuaggdAaAfuccagccsasa	1365
AD-1636827	SCN9A	gsgsgaaaacAfAfUfcuuccguuaL96	1198	usdAsaadCgdGaagadTuGfuuuucccsusu	1366
AD-1636749	PNPLA3	gsgsgguaacAfAfGfauaauaauL96	1199	asdGsaudTadTcaudTuGfuuaacccscsg	1367
AD-1636799	Marc1	gsgsugucucAfAfUfgcuucaauguL96	1200	asdCsaudTgdAagcadTuGfagacaccsag	1368
AD-1636700	C3	gsusaaugcaGfGfAfcuucuaauL96	1201	asdAsugdAadGaagudCcUfgcauuacsusg	1369
AD-1636706	C3	gsusaccucuUfCfAfuccagacaguL96	1202	asdCsugdTcdTggadGaAfgagguacscsc	1370
AD-1636771	Marc1	gsusauaacUfUfAfagaucugauL96	1203	asdAsucdAgdAucuudAgAfguuauacsasa	1371
AD-1636826	SCN9A	gsusccuuaAfGfAfagaauaucuaL96	1204	usdAsgadTadTucuudCuUfagaggacsusg	1372
AD-1636848	Marc1	gsusgaccuUfCfAfgaacgaaguL96	1205	asdCsudTcdGuucudGaAfgggucacsasc	1373
AD-1636746	PNPLA3	gsusgagugaCfAfAfcguaccuuaL96	1206	usdAsagdGgdTacgudTgUfcacucacsusc	1374
AD-1636747	PNPLA3	gsusgcuaaaGfUfUfucccaucuuL96	1207	asdAsagdAudGggaadAcUfuuagcacscsu	1375
AD-1636723	C3	gsusgggagaAfGfUfucggccuagaL96	1208	usdCsudGgdCgaadCuUfcuccccacsusg	1376
AD-1636847	Marc1	usasacucuaAfGfAfucugaugaauL96	1209	asdTsucdAudCagaudCuUfagaguuasusa	1377

TABLE 5-continued

Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO	Antisense sequence (5'→3')	SEQ ID NO:
AD-1636824	SCN9A	usascaugauCfUfUfcuuugucguaL96	1210	usdAscgdAcdAaagadAgAfucauguasgsg	1378
AD-1636748	PNPLA3	usascuguuGfAfAfuuuuguauuaL96	1211	usdAsaudAcdAaaudTcAfacagguasasc	1379
AD-1636704	C3	usascgugcuGfCfCfcaguuucgaaL96	1212	usdTscgdAadAcuggdGcAfgcacguascsu	1380
AD-1636724	PNPLA3	usasggauaaUfGfUfcuuauguaauL96	1213	asdTsuaCdCadTaagadCaUfuaucuasasu	1381
AD-1636832	SCN9A	usasugccaaAfAfUfccuuuuuauaL96	1214	usdAsuaCdAadAaggadTuUfuggcauasgsa	1382
AD-1636835	SCN9A	usasugugaaAfCfAfaaccuuacgaL96	1215	usdCsgudAadGguudGuUfucacauasasu	1383
AD-1636788	Marc1	usasuuuguaaUfUfUfcaggaugcgcaL96	1216	usdCsgcdAudCcugadAaUfuacaauasusu	1384
AD-1636787	Marc1	uscscsaugcuUfCfAfaugucccaguL96	1217	asdCsugdGgdAcaudGaAfgcaugagasgsa	1385
AD-1636761	PNPLA3	uscscacuugaGfGfAfggcgagucuaL96	1218	usdAsgadCudCgccudCcUfcaagugascsu	1386
AD-1636794	Marc1	uscscagaacgAfAfAfguuauauggaL96	1219	usdCscadTadTaacudTuCfguucugasasg	1387
AD-1636804	Marc1	uscscaggauGfGfAfgucuaugcaL96	1220	usdGscadTadGacaudCgCfaucugasasasa	1388
AD-1636802	Marc1	uscscauagaUfCfUfggaucuggcaL96	1221	usdGscddAgdAuccadGaUfcauaggasasa	1389
AD-1636814	SCN9A	uscscsuugcuCfUfUfgacauuuauL96	1222	asdTsaadGadTgucadAgAfcauaggasusc	1390
AD-1636812	SCN9A	uscscucuaaGfAfAfgaauaucuaL96	1223	asdTsagdAudAuucudTcUfuagaggascsu	1391
AD-1636850	SCN9A	uscscugcaaGfUfCfaaguuccaaaL96	1224	usdTsgdGadAcuugdAcUfugcaggasasa	1392
AD-1636738	PNPLA3	uscscugagcuGfAfGfuuuguuuuauL96	1225	asdTsaadAadCcaacdTcAfgcucagasgsg	1393
AD-1636797	Marc1	usgsacccuuCfAfGfaacgaaaguL96	1226	asdAscudTudCguucdTgAfgggucascsa	1394
AD-1636728	PNPLA3	usgsagugaaGfAfAfaugaaagacaL96	1227	usdGsucdTudTcauudTcUfucacucasgsu	1395
AD-1636842	SCN9A	usgsauaguAfCfCfuaguuugcaaL96	1228	usdTsgcdAadAcuagdGuAfacuauacasasa	1396
AD-1636852	SCN9A	usgscagacaAfGfAfucuuacuaL96	1229	usdAsagdTgdAagadCuUfgucugcasusa	1397
AD-1636764	PNPLA3	usgscaaaaCfAfAfccaucaccguL96	1230	asdCsggdTgdAuggudTgUfuuuggcasusc	1398
AD-1636698	C3	usgscgaucaGfAfAfgagaccaaguL96	1231	asdCsudGgdTcucudTcUfgaucgcasgsg	1399
AD-1636697	C3	usgscuaaggCfCfAfaagaucaacuL96	1232	asdGsuudGadTcuudGgCfcuuagcasusg	1400

TABLE 5-continued

Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO	Antisense sequence (5'→3')	SEQ ID NO:
AD-1636816	SCN9A	usgscucuccAfUfAfuuggauaaaaL96	1233	usdTsudAudCcaudAuGfgagagcasasu	1401
AD-1636846	SCN9A	usgscucuccUfUfUfgugguuucauL96	1234	asdTsgadAadCcacadAaGfgagagcasusc	1402
AD-1636711	C3	usgsgacuauGfUfGfuacaagaccuL96	1235	asdGsgudCudTguacdAcAfuaguccascsu	1403
AD-1636793	Marc1	usgsgaggagAfAfGfaaaagugauuL96	1236	asdAsucdAcdTuucdTuCfuccuccascsa	1404
AD-1636742	PNPLA3	usgsgauacaUfGfAfgcaagauuuL96	1237	usdAsaadTcdTugcudCaUfguauccascsc	1405
AD-1636755	PNPLA3	usgsggagagAfUfAfugccuucgaaL96	1238	usdTscgdAadGgcaudAuCfucuccascgsc	1406
AD-1636696	C3	usgsggcaacUfCfCfaacaauuacuL96	1239	asdGsuadAudTguugdGaGfuugcccascsg	1407
AD-1636821	SCN9A	usgsggucauUfCfUfucacuugaaL96	1240	usdTscadAadGugaadGaAfugaccasasa	1408
AD-1636851	SCN9A	usgsgucuuuAfCfUfpgaauuuuL96	1241	asdAsaadGadTuccadGuAfaagaccasasa	1409
AD-1636751	PNPLA3	usgsgugacaUfGfGfcuuccagauaL96	1242	usdAsucdTgdGaagcdCaUfguaccascgsu	1410
AD-1636818	SCN9A	usgsgugucaUfCfAfuagauaaauuL96	1243	asdAsaudTadTcaudGaUfgacaccasasu	1411
AD-1636781	Marc1	usgsgugucuCfAfAfugcuucaauaL96	1244	usdAsuudGadAgcaudTgAfgacaccascgsa	1412
AD-1636722	C3	usgsuacaagAfCfCfcgacuggucaL96	1245	usdGsacdCadGucggdGuCfuuguacascsa	1413
AD-1636820	SCN9A	usgsuaggagAfAfUfucacuuuuaL96	1246	usdGsaadAadGugaadTuCfuccuacascsa	1414
AD-1636817	SCN9A	usgsuaggagAfAfUfucacuuuucuL96	1247	asdGsaadAadGugaadTuCfuccuacascsa	1415
AD-1636825	SCN9A	usgsucgaguAfCfAfcuuuuacugaL96	1248	usdCsagdTadAaagudGuAfcucgacacusu	1416
AD-1636708	C3	usgsuucgugCfUfGfaauaagaaguL96	1249	asdCsuudCudTauucdAgCfacgaacascsg	1417
AD-1636819	SCN9A	usgsuucuguCfUfGfaguguguuuL96	1250	usdAsaadCadCacucdAgAfcagaacascsa	1418
AD-1636775	Marc1	usgsuuuuuAfCfCfcaauaucuauL96	1251	asdTsagdAudAuuggdGuUfuuaaacasasc	1419
AD-1636784	Marc1	ususaaaacuGfUfGfaauaaauggaL96	1252	usdCscadTudTauucdAcAfguuuuasasa	1420
AD-1636829	SCN9A	ususaccuauCfUfCfugcuucaaguL96	1253	asdCsuudGadAgcagdAgAfuagguaascsc	1421
AD-1636768	PNPLA3	ususaccuguUfGfAfauuuuguuuuL96	1254	asdAsuadCadAaaudCaAfcagguaascsa	1422
AD-1636745	PNPLA3	ususauguaaUfGfCfugcccuguaaL96	1255	usdTscadAgdGgcagdCaUfuacauaasgsa	1423

TABLE 5-continued

Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO	Antisense sequence (5'→3')	SEQ ID NO:
AD-1636809	SCN9A	usussccucaaGfGfAfaaaagauaaaL96	1256	usdTsuadTcdTuuuudCcUfugaggaasasu	1424
AD-1636688	C3	ususcgugcuGfAfAfaaagaagaauL96	1257	asdTsucdTudCuuaudTcAfgcacgaascsa	1425
AD-1636773	Marc1	usugccauUfUfGfuccuuugauuL96	1258	asdAsucdAadAggacdAaAfauggcaasusa	1426
AD-1636752	PNPLA3	usugcuaccCfAfUfuaggauaauL96	1259	usdAsuudAudCcuaadTgGfguagcaasgsu	1427
AD-1636759	PNPLA3	usugguuuUfUfGfaaaagcuagaL96	1260	usdCsuaDGcdTuuucdAuAfaaccaascsu	1428
AD-1636849	Marc1	usuguaauUfCfAfggaugcgauaL96	1261	usdAsucdGcdAuccudGaAfaauacaasusa	1429
AD-1636776	Marc1	usugguccaGfAfUfgcauuuuauL96	1262	asdTsuaDAadAugcadTcUfggacaasgsc	1430
AD-1636845	SCN9A	ususuaguacAfCfUfccuuauuauL96	1263	asdTsgadAudAaggadGuGfuacuaaasasu	1431
AD-1636810	SCN9A	ususuaucaUfUfUfugggucauuL96	1264	usdAsaudGadCccaadAgAfugauaaasgsa	1432
AD-1636786	Marc1	ususucauaGfAfUfcuggaucugaL96	1265	usdCsagdAudCcagadTcUfauggaaasasu	1433
AD-1636808	SCN9A	ususuuguagaUfCfUfugcauuuacaL96	1266	usdGsuaDAudTgcaadGaUfcuacaaasasg	1434
AD-1636736	PNPLA3	ususuagaaCfAfCfcuuuuucacuL96	1267	asdGsugdAadAaaggdTgUfucuaaaasusu	1435
AD-1636730	PNPLA3	ususuucacCfUfAfacuaaaauaaL96	1268	usdTsaudTudTaguudAgGfugaaaaasgsg	1436

TABLE 6

More Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1657992	AGT	asgscugagGfGfCfcaccauccuuL96	1437	asdAsggdAudGguggdCcCfucagguuscsa	1786
AD-1657994	AGT	cscsugagggCfCfAfccauccucuuL96	1438	asdAsgadGgdAuggudGgCfccucaggsusu	1787
AD-1657998	AGT	asgsggccacCfAfUfccucugccuuL96	1439	asdAsggdCadGaggadTgGfuggccuucsa	1788
AD-1658030	AGT	gsusgaccggGfUfGfuacauacacuL96	1440	asdGsugdTadTguacdAcCfcggucacscsu	1789
AD-1684491	AGT	cscsggguguAfCfAfuaacccccuuL96	1441	asdAsggdGgdTguaudGuAfcacccggususc	1790
AD-1684493	AGT	csgsggguguaCfAfUfacacccccuuL96	1442	asdAsagdGgdGuguadTgUfacacccggsusu	1791
AD-1684495	AGT	gsgsguguacAfUfAfcacccccuucL96	1443	asdGsaadGgdGgugudAuGfuacacccsgsg	1792

TABLE 6-continued

More Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1684497	AGT	gsgsuguacaUfAfCfaccuccuL96	1444	asdGsgadAgdGggugdTaUfguacaccscsg	1793
AD-1684499	AGT	gsusguacauAfCfAfccuccuL96	1445	asdTsggdAadGgggudGuAfuguacacscsc	1794
AD-1684501	AGT	usgsuacauAfCfAfccuccuL96	1446	asdGsugdGadAggggdTgUfauguacacscsc	1795
AD-1684503	AGT	gsusacauAcAfCfCfuccuL96	1447	asdGsgudGgdAagggdGuGfuauguacsasc	1796
AD-1684505	AGT	usascauacaCfCfCfuccuL96	1448	asdAsggdTgdGaaggdGgUfguauguascsa	1797
AD-1684507	AGT	ascsauacacCfCfCfuccuL96	1449	asdGsagdGudGgaagdGgGfuguauguasasc	1798
AD-1684509	AGT	csasucaccCfCfUfuccuL96	1450	asdCsgadGgdTggaadGgGfguguugsusa	1799
AD-1684511	AGT	asusacaccCfUfUfccuL96	1451	asdAscgdAgdGuggadAgGfgguguauugsu	1800
AD-1684513	AGT	cscscuuccaCfCfUfcguL96	1452	asdTsggdAudGacgadGgUfggaagggsgsu	1801
AD-1684515	AGT	cscsuuccacCfUfCfguL96	1453	asdGsugdGadTgacgdAgGfuggaagsgsg	1802
AD-1658032	AGT	csusuccaccUfCfGfucL96	1454	asdTsgudGgdAugacdGaGfuggaagsgsg	1803
AD-1658033	AGT	ususccaccUfGfUfcaL96	1455	asdTsugdTgdGaugadCgAfgguggaagsgsg	1804
AD-1658034	AGT	uscscaccucGfUfCfauL96	1456	asdAsuudGudGgaugdAcGfagguggasasg	1805
AD-1658035	AGT	cscsaccucGfUfAfuccaL96	1457	asdCsaudTgdTggaudGaCfagguggasasa	1806
AD-1658036	AGT	csasccucguCfAfUfccL96	1458	asdTscadTudGuggadTgAfcgagguggsgsa	1807
AD-1658038	AGT	cscsucgucaUfCfCfacaL96	1459	asdTscudCadTugugdGaUfgacgaggsusg	1808
AD-1658039	AGT	csuscgucaUfCfAfcaL96	1460	asdCsucdTcdAuugudGgAfugacgaggsu	1809
AD-1658040	AGT	uscsgucaucCfAfCfaL96	1461	asdAscudCudCauugdTgGfaugacgagsg	1810
AD-1658041	AGT	csgsucauccAfCfAfaL96	1462	asdTsacdTcdTcauudGuGfugaugacgsasg	1811
AD-1658042	AGT	gsuscauccaCfAfAfugL96	1463	asdGsuadCudCucaudTgUfggaugacsgsa	1812
AD-1658043	AGT	uscсаuccacAfAfUfL96	1464	asdGsgudAcdTcucadTuGfuggaugacscsg	1813
AD-1658044	AGT	csasuccacaAfUfGfagL96	1465	asdAsggdTadCucudAuUfguggaugasasc	1814
AD-1658045	AGT	asusccacaaUfGfAfgL96	1466	asdCsagdGudAcucudCaUfuguggauugsu	1815

TABLE 6-continued

More Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1658046	AGT	uscscacaauGfAfGfaguaccuguuL96	1467	asdAscadGgdTacudTcAfuuguggasusg	1816
AD-1658047	AGT	cscsacaauGfAfGfaguaccuguguL96	1468	asdCsacdAgdGuacudCuCfauguggsasus	1817
AD-1658048	AGT	csascaaugaGfAfGfuaccugugauL96	1469	asdTscadCadGguacdTcUfcauugugsgsa	1818
AD-1658049	AGT	ascsaaugagAfGfUfaccugugaguL96	1470	asdCsucdAc dAgguadCuCfcauugugsgsg	1819
AD-1658050	AGT	csasaugagaGfUfAfccugugagcuL96	1471	asdGscudCadCaggudAcUfcucauugsusg	1820
AD-1658051	AGT	asasugagagUfAfCfcugugagcauL96	1472	asdTsgcdTcdAcaggdTcCfucucauugsu	1821
AD-1658052	AGT	asusgagaguAfCfCfugugagcaguL96	1473	asdCsugdCudCacagdGuAfcucucausug	1822
AD-1658053	AGT	usgsagaguaCfCfUfgugagcagcuL96	1474	asdGscudGcdTcacadGgUfacucucasusu	1823
AD-1658054	AGT	gsasgaguacCfUfGfugagcagcuL96	1475	asdAsgcdTgdCucacdAgGfuacucucasu	1824
AD-1658055	AGT	asgsaguaccUfGfUfgagcagcuguL96	1476	asdCsagdCudGcucadCaGfguacucscsa	1825
AD-1658056	AGT	gsasguaccuGfUfGfagcagcugguL96	1477	asdCscadGcdTgcudAcAfgguaucusc	1826
AD-1658057	AGT	asgsuaccugUfGfAfagcagcugguL96	1478	asdGscddAgdCugcudCaCfagguacusc	1827
AD-1658058	AGT	gsusaccuguGfAfGfagcagcugcauL96	1479	asdTsgcdCadGcugcdTcAfcagguacsusc	1828
AD-1658059	AGT	usasccugugAfGfCfagcugccaauL96	1480	asdTsugdCcdAgcugdCuCfacagguasc	1829
AD-1658184	AGT	asasuggucgGfGfAfugcuggccauL96	1481	asdTsggdCcdAgcaudCcCfagcauugsg	1830
AD-1658185	AGT	asusggucggGfAfUfgcuggccauL96	1482	asdTsugdGcdCagcadTcCfcgaccausug	1831
AD-1658186	AGT	usgsugcgggAfUfGfcuggccaaucL96	1483	asdGsuudGgdCcagcdAuCfcgaccasusu	1832
AD-1658187	AGT	gsgsucgggaUfGfCfuggccaaucL96	1484	asdAsgudTgdGccagdCaUfcccagccasu	1833
AD-1658188	AGT	gsuscgggauGfCfUfggccaaucuuL96	1485	asdAsagdTudGgcccadGcAfucccgacscsa	1834
AD-1658189	AGT	uscsgggauGfUfGfgccaacuucL96	1486	asdGsaadGudTggccdAgCfaucccgascsc	1835
AD-1658190	AGT	csgsggaugcUfGfGfccaacuucuuL96	1487	asdAsgadAgdTuggcdCaGfcaucccgasc	1836
AD-1658191	AGT	gsgsgaugcuGfGfCfcaacuucuuL96	1488	asdAsagdAadGuuggdCcAfgcaucccgsgsa	1837
AD-1658192	AGT	gsgsaugcugGfCfCfcaacuucuuL96	1489	asdCsaadGadAguugdGcCfagcaucccgscg	1838

TABLE 6-continued

More Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1658193	AGT	gsasugcuggCfCfAfacuucugguL96	1490	asdCscadAgdAaguudGgCfcagcaucscsc	1839
AD-1658196	AGT	gscsuggccaAfCfUfucugggcuuL96	1491	asdAsgcdCcdAgaadGuUfgggcagcsasu	1840
AD-1658197	AGT	csusggccaaCfUfUfucugggcuuL96	1492	asdAsagdCcdCaagadAgUfuggccagcsa	1841
AD-1658200	AGT	gscscaacuUCfUfUfgggcuuccguL96	1493	asdCsggdAadGcccadAgAfauguugcscsa	1842
AD-1658201	AGT	cscsaacuucUfUfGfggcuuccguuL96	1494	asdAscgdGadAgcccdAaGfauguugscsc	1843
AD-1658202	AGT	csasacuucUfGfGfgcuuccguauL96	1495	asdTsacdGgdAagccdCaAfauguugsgsc	1844
AD-1658203	AGT	asascuucUfGfGfgcuuccguauuL96	1496	asdAsuadCgdGaagcdCcAfauguugsg	1845
AD-1658204	AGT	ascsuucUfGfGfCuuccguauuL96	1497	asdTsaudAcGgaagdCcCfaagaagusg	1846
AD-1658205	AGT	csusucUfGfGfCuuccguauuL96	1498	asdAsuadTadCggaadGcCfaagaagsusu	1847
AD-1658206	AGT	ususcUfGfGfCuuccguauuL96	1499	asdTsaudAudAcggadAgCfccaagaagsu	1848
AD-1658207	AGT	uscscUfGfGfCuuccguauuL96	1500	asdAsuadTadTacggdAaGfccccagasa	1849
AD-1658208	AGT	csusuggcuUfCfCfGuauauauguL96	1501	asdCsaudAudAuacgdGaAfgcccaagsasa	1850
AD-1658209	AGT	usugggcuUfCfGfGuauauauguL96	1502	asdCscadTadTauacdGgAfgcccaagsa	1851
AD-1658211	AGT	ggsggcuUfUfAfuaauauggcauL96	1503	asdTsgcdCadTauaudAcGfgaagcccsasa	1852
AD-1658212	AGT	ggsgcuUfUfAfuaauauggcauL96	1504	asdAsugdCcdAuauadTaCfpgaagccscsa	1853
AD-1658213	AGT	gscsuUfUfAfuaauauggcauL96	1505	asdCsaudGcdCauaudAuAfcggaagcscsc	1854
AD-1658220	AGT	usasuaugGfCfAfugcacaguguL96	1506	asdCsacdTgdTgcaudGcCfauauuascs	1855
AD-1658221	AGT	asusauugGfCfAfugcacaguguL96	1507	asdTscadCudGugcadTgCfcauauuas	1856
AD-1658222	AGT	usasuaugGfUfGfCfagagagauL96	1508	asdCsucdAcdTgugcdAuGfccaauasusa	1857
AD-1658223	AGT	asusaugGfUfGfCfagagagauL96	1509	asdGscudCadCugudCaUfgccauasasu	1858
AD-1658224	AGT	usasugGfUfGfCfagagagauL96	1510	asdAsgcdTcdAcugudGcAfgccauasusa	1859
AD-1658225	AGT	asusggGfUfGfCfagagagauL96	1511	asdTsagdCudCacugdTgCfagccausasu	1860
AD-1658226	AGT	usggGfUfGfCfagagagauL96	1512	asdAsuadGcdTcacudGuGfcaugccasusa	1861

TABLE 6-continued

More Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1658227	AGT	ggscaugcaCfAfGfugagcuauL96	1513	asdCsaudAgdCucacdTgUfgcaugccsas	1862
AD-1658228	AGT	gscsaugcacAfGfUfgagcuauL96	1514	asdCscadTadGcucadCuGfugcaugcscsa	1863
AD-1684517	AGT	csasugcacaGfUfGfagcuauL96	1515	asdCsccdAudAgcudAcUfgugcaugscsc	1864
AD-1684519	AGT	asusgcacagUfGfAfGfagcuauL96	1516	asdCsccdCadTagcudCaCfugugcausgsc	1865
AD-1684521	AGT	cscsucuccCfAfAfGfGgcuauL96	1517	asdAsgadCadGccgudTgGfGgagaggsasc	1866
AD-1658242	AGT	usgsgcaccUfGfGfGccucucuuL96	1518	asdAsgadGadGaggcdCaGfGgugccasasa	1867
AD-1658243	AGT	gsgscaccUfGfGfGccucucuuL96	1519	asdTsagdAgdAgaggdCcAfGgugccsasa	1868
AD-1658288	AGT	gsascaggcuAfCfAfGgcaauccuuL96	1520	asdAsggdAudTgccudGuAfGccugucsasg	1869
AD-1658289	AGT	ascasaggcuAfCfAfGfGcaauccuuL96	1521	asdCsagdGadTugccdTgUfagccugucscsa	1870
AD-1658313	AGT	ususccuuggAfAfGfGacaagaacuL96	1522	asdGsuudCudTguccdTuCfcaaggaascsa	1871
AD-1658315	AGT	cscsuuggaaGfGfAfcaagaacugL96	1523	asdCsagdTudCuugudCcUfuccaaggsasa	1872
AD-1658316	AGT	csusuggaagGfAfCfaagaacugcuL96	1524	asdGscadGudTcuugdTcCfuuccaagsgsa	1873
AD-1658448	AGT	csascugaaGfCfAfGccguuuguL96	1525	asdAscadAadCggcudGcUfucaggugscsa	1874
AD-1658451	AGT	csusgaagcaGfCfCfGuuugucauL96	1526	asdTsgcdAcdAaacgdGcUfGcuucagsgsu	1875
AD-1658463	AGT	ususugugcaGfGfGfGccugcucuuL96	1527	asdAsgadGcdCaggcdCcUfgcacaascs	1876
AD-1658464	AGT	ususugugcagGfGfGfGccugcucuuL96	1528	asdGsagdAgdCcaggdCcCfugcacaasasc	1877
AD-1658465	AGT	usgsugcaggGfCfCfGgucucuuL96	1529	asdAsgadGadGccagdGcCfugcacasasa	1878
AD-1658466	AGT	gsusgcaggGfCfUfGgucucuuL96	1530	asdTsagdAgdAgccadGgCfGgucacsa	1879
AD-1658467	AGT	usgscaggGfCfUfGfGcucuuL96	1531	asdAsuadGadGagcdAgGfGccugcascsa	1880
AD-1658484	AGT	ascsgcucucUfGfGfGfGfGcucuuL96	1532	asdCsugdTgdAagudCaGfagagcgusg	1881
AD-1658485	AGT	csgscucucUfGfGfGfGfGfGcucuuL96	1533	asdTscudGudGaagudCcAfGagagcgusg	1882
AD-1658519	AGT	csusgagaagAfUfUfGfGfGfGfGcucuuL96	1534	asdGsaadCcdTgucadAuCfuucucagscsa	1883
AD-1658520	AGT	usgsagaagaUfUfGfGfGfGfGfGcucuuL96	1535	asdTsgadAcuCugudAaUfGfGfGfGfGcucuuL96	1884

TABLE 6-continued

More Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1658521	AGT	gsasgaagauUfGfAfcagguucauuL96	1536	asdAsugdAadCcugudCaAfucuucucsasg	1885
AD-1658522	AGT	asgsaagauUfGfAfcagguucauguL96	1537	asdCsaudGadAccugdTcAfaucuucuscscsa	1886
AD-1658523	AGT	gsasagauugAfCfAfgguucaugcuL96	1538	asdGscadTgdAaccudGuCfaaucuucusc	1887
AD-1658524	AGT	asasgauugaCfAfGfguucaugcauL96	1539	asdTsgcdAudGaaccdTgUfcaaucuucscsu	1888
AD-1658525	AGT	asgsauugacAfGfGfuucaugcaguL96	1540	asdCsugdCadTgaacdCuGfucaaucuscusc	1889
AD-1658526	AGT	gsasuugacaGfGfUfcaugcagguL96	1541	asdCscudGcdAugaadCcUfgucaaucsusu	1890
AD-1658527	AGT	asusugacagGfUfUfcaugcaggcuL96	1542	asdGscddTgdCaugadAcCfugucaaucscsu	1891
AD-1658528	AGT	ususgacaggUfUfCfaugcaggcuuL96	1543	asdAsgcdCudGcaugdAaCfcugucaasusc	1892
AD-1658529	AGT	usgsacagguUfCfAfugcaggcuguL96	1544	asdCsagdCcdTgcaudGaAfcugucacasasu	1893
AD-1658530	AGT	gsascagguUcAfUfGfcaggcuguuL96	1545	asdAscadGcdCugcadTgAfacugucscasa	1894
AD-1658531	AGT	ascsagguUcAfUfGfcaggcuguguL96	1546	asdCsacdAgdCcugcdAuGfaaccuguscscsa	1895
AD-1658539	AGT	asusgcaggcUfGfUfgacaggauL96	1547	asdCsaudCcdTgucadCaGfccugcausgsa	1896
AD-1658541	AGT	gscsaggcugUfGfAfcaggauggauL96	1548	asdTscddAudCcugudCaCfagccugcsasu	1897
AD-1658542	AGT	csasggcuguGfAfCfaggauggaaL96	1549	asdTsucdCadTccugdTcAfcagccugscscsa	1898
AD-1658605	AGT	gscsuuucAaCfAfCfcuacguccauL96	1550	asdTsggdAcddGuaggdTgUfugaagcscscsa	1899
AD-1658650	AGT	gsasguucugGfGfUfggacaacaguL96	1551	asdCsugdTudGuccadCcCfagaacucscscsu	1900
AD-1658661	AGT	gsgsacaacaGfCfAfcucaguguuL96	1552	asdAscadCudGaggudGcUfguuguccsasc	1901
AD-1658662	AGT	gsascaacagCfAfCfcucagugucuL96	1553	asdGsacdAcddTgaggdTgCfuguugucscscsa	1902
AD-1658663	AGT	ascsaacagCfAfCfcucagugucuL96	1554	asdAsgadCadCugagdGuGfcuguuguscsc	1903
AD-1658664	AGT	csasacagcaCfCfUfcagugucuguL96	1555	asdCsagdAcddAcugadGgUfgcuguugsusc	1904
AD-1658665	AGT	asascagcacCfUfCfagugucuguuL96	1556	asdAscadGadCacugdAgGfugcuguugsusu	1905
AD-1658801	AGT	asusgccucuGfAfCfcuggacaaguL96	1557	asdCsudGudCcaggdTcAfgaggcausasg	1906
AD-1658818	AGT	asasgguggaGfGfGfucacacuuuL96	1558	asdAsaadGudGagacdCcUfccacuuugsusu	1907

TABLE 6-continued

More Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1658819	AGT	asgsguggagGfGfUfcucacuuucuL96	1559	asdGsaadAgdTgagadCcCfuccaccusug	1908
AD-1658820	AGT	gsgsuggaggGfUfCfucacuuuccuL96	1560	asdGsgadAadGugagadAcCfuccaccsusu	1909
AD-1658824	AGT	gsasgggucuCfAfCfuuccagcauL96	1561	asdTsgcdTgdGaaagdTgAfgaccucscsa	1910
AD-1658825	AGT	asgsggucucAfCfUfuuccagcaauL96	1562	asdTsugdCudGgaaadGuGfagaccuscsc	1911
AD-1658827	AGT	gsgsucucacUfUfUfccagcaaaauL96	1563	asdTsuudTgdCuggadAaGfugagaccscsu	1912
AD-1658828	AGT	gsuscucacuUfUfCfcagcaaaacuL96	1564	asdGsuudTudGcuggdAaAfgugagacscsc	1913
AD-1658829	AGT	uscscacuuUfCfCfagcaaaacuL96	1565	asdAsgudTudTgcugdGaAfgugagascsc	1914
AD-1658830	AGT	csuscacuuUfCfAfgcaaaacucuL96	1566	asdGsagdTudTugcudGgAfaagugagsasc	1915
AD-1658831	AGT	uscscacuuUfCfAfgcaaaacuccuL96	1567	asdGsgadGudTuugcdTgGfaaagugasgsa	1916
AD-1658832	AGT	csascuuuccAfGfCfaaaacucccuL96	1568	asdGsggdAgdTuuugdCuGfgaaagugasg	1917
AD-1658833	AGT	ascsuuuccaGfCfAfaaacucccuL96	1569	asdAsggdGadGuuudGcUfggaaagugasg	1918
AD-1658834	AGT	csusuuccagCfAfAfaaacuccucuL96	1570	asdGsagdGgdAguuudTgCfuggaaagsug	1919
AD-1658835	AGT	ususuccagcAfAfAfacuccucauL96	1571	asdTsgadGgdGaguudTuGfcuggaaagsu	1920
AD-1658836	AGT	ususccagcaAfAfAfcuccucaauL96	1572	asdTsugdAgdGgagudTuUfgcuggaasag	1921
AD-1658837	AGT	uscscagcaaAfAfCfuccucaacuL96	1573	asdGsuudGadGggagdTuUfugcuggasasa	1922
AD-1658838	AGT	cscsagcaaaAfCfUfccucaacuL96	1574	asdAsgudTgdAgggadGuUfuugcuggsasa	1923
AD-1658839	AGT	csasgcaaaaCfUfCfccucaacuguL96	1575	asdCsagdTudGagggdAgUfuugcuggsgsa	1924
AD-1658840	AGT	asgscaaaacUfCfCfcucaacugguL96	1576	asdCscadGudTgaggdGaGfuuuugcusgsg	1925
AD-1658841	AGT	gscsaaaacuCfCfCfucaacuggauL96	1577	asdTscddAgdTugagdGgAfguuuugcsusg	1926
AD-1658842	AGT	csasaaacucCfCfUfcaacuggauL96	1578	asdAsucdCadGuugadGgGfaguuuugscsu	1927
AD-1658843	AGT	asasaacuccCfUfCfaacuggauguL96	1579	asdCsaudCcdAguugdAgGfgaguuuugsc	1928
AD-1658844	AGT	asasacuccUfCfAfacuggaugauL96	1580	asdTscadTcdCaguudGaGfggaguuuusug	1929
AD-1658845	AGT	asascucccuCfAfAfcuggaauL96	1581	asdTsucdAudCcagudTgAfgggaguususu	1930

TABLE 6-continued

More Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1658846	AGT	ascsuccucAfAfCfuggaugaaguL96	1582	asdCsuudCadTccagdTuGfaggagagususu	1931
AD-1658847	AGT	csuscccucaAfCfUfggaugaagauL96	1583	asdTscudTcdAuccadGuUfgaggaggsusu	1932
AD-1658848	AGT	uscscucaaaCfUfGfgaugaagaauL96	1584	asdTsucdTudCauccdAgUfugaggaggsu	1933
AD-1658849	AGT	cscscucaacUfGfGfugaagaauL96	1585	asdTsudCudTcaucdCaGfuugagggsasg	1934
AD-1321390	AGT	cscsucaacuGfGfAfugaagaacuL96	1586	asdGsuudTcdTucaudCcAfguugagggsa	1935
AD-1658850	AGT	csuscaacugGfAfUfgaagaacuL96	1587	asdAsgudTudCuucadTcCfaguugaggsg	1936
AD-1658875	AGT	asgsgaucuuAfUfGfaccugcagguL96	1588	asdCscudGcdAgguudAuAfagauccusug	1937
AD-1658880	AGT	csusaugacCfUfGfcaggaccuguL96	1589	asdCsagdGudCcugcdAgGfucuaagsasu	1938
AD-1658929	AGT	csgsagcugaAfCfCfugcaaaaauL96	1590	asdAsuudTudTgcagdGuUfcagcucgsgsu	1939
AD-1658930	AGT	gsasgcugaaCfCfUfgcaaaaauuL96	1591	asdAsaudTudTugcadGgUfucagcucsgsg	1940
AD-1658931	AGT	asgscugaacCfUfGfcaaaaauuguL96	1592	asdCsaadTudTuugcdAgGfuucagcuscsg	1941
AD-1658932	AGT	gscsugaaccUfGfCfaaaaauugauL96	1593	asdTscadAudTuugcdCaGfguucagcsusc	1942
AD-1658933	AGT	csusgaaccuGfCfAfaaaaugaguL96	1594	asdCsucdAadTuuuudGcAfgguucagcsu	1943
AD-1658934	AGT	usgsaaccugCfAfAfaaaugagcuL96	1595	asdGscudCadAuuudTgCfagguucagsgc	1944
AD-1658935	AGT	gsasaccugcAfAfAfaauugagcauL96	1596	asdTsgcdTcdAuuudTuGfcagguucsasg	1945
AD-1658936	AGT	asasccugcaAfAfAfaugagcauL96	1597	asdTsugdCudCaauudTuUfgcagguucscsa	1946
AD-1658937	AGT	ascscugcaaAfAfAfuugagcauuL96	1598	asdAsuudGcdTcaaudTuUfugcaggususc	1947
AD-1658938	AGT	cscsugcaaaaAfAfUfugagcaauguL96	1599	asdCsaudTgdCucaadTuUfuugcagggsusu	1948
AD-1658939	AGT	csusgcaaaaAfUfUfgagcaaugauL96	1600	asdTscadTudGcucadAuUfuuugcagsgsu	1949
AD-1658940	AGT	usgscaaaaaUfUfGfagcaaugacuL96	1601	asdGsucdAudTgcucdAaUfuuuugcasgsg	1950
AD-1658954	AGT	gsasggugcuGfAfAfcagcauuuuuL96	1602	asdAsaadAudGcugudTcAfgcaccucscsc	1951
AD-1658955	AGT	asgsgugcugAfAfCfagcauuuuuuL96	1603	asdAsaadAadTgcugdTuCfagcaccucscsc	1952
AD-1658956	AGT	gsgsugcugaAfCfAfgcauuuuuuuL96	1604	asdAsaadAadAugcudGuUfcagcaccususc	1953

TABLE 6-continued

More Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1658957	AGT	gsusgcugaaCfAfGfcauuuuuuuL96	1605	asdAsaadAadAaugcdTgUfucagcacscsu	1954
AD-1658958	AGT	usgscugaacAfGfCfauuuuuuuuL96	1606	asdAsaadAadAaaugdCuGfuucagcascsc	1955
AD-1658959	AGT	gscsugaacaGfCfAfuuuuuuuuL96	1607	asdCsaadAadAaaadGcUfguucagcsasc	1956
AD-1658960	AGT	csusgaacagCfAfUfuuuuuuuuL96	1608	asdTscadAadAaaaadTgCfuguucagcsa	1957
AD-1658992	AGT	usgsagagagAfGfCfccacagaguL96	1609	asdAscudCudGugggdCuCfucucucasusc	1958
AD-1658994	AGT	asgsagagagCfCfCfacagagucuL96	1610	asdAsgadCudCugugdGgCfucucucusc	1959
AD-1658995	AGT	gsasgagagcCfCfAfcagagucuL96	1611	asdTsagdAcdTcugudGgGfcucucucusc	1960
AD-1659055	AGT	gsasaccgccCfAfUfuccuguuuL96	1612	asdCsaadAcdAggaadTgGfgcgguucsasg	1961
AD-1659056	AGT	asasccgcccAfUfUfccuguuugcuL96	1613	asdGscadAadCaggadAuGfggcgguusc	1962
AD-1659057	AGT	ascscgcccAfuUfCfcuguuugcuL96	1614	asdAsgcdAadAcaggdAaUfgggcgguusc	1963
AD-1659058	AGT	cscsgcccAfuUfCfcuguuugcuguL96	1615	asdCsagdCadAacagdGaAfugggcggsusu	1964
AD-1659059	AGT	csgscccAfuUfUfguguuugcuguL96	1616	asdAscadGcdAaacadGgAfaugggcggsu	1965
AD-1659060	AGT	gscsccAfuUfUfguguuugcuguL96	1617	asdCsacdAgdCaaacdAgGfaaugggcgsg	1966
AD-1659061	AGT	cscscauuccUfGfUfuugcuguguL96	1618	asdAscadCadGcaaadCaGfgaaugggcsg	1967
AD-1659062	AGT	cscscauuccUfUfUfugcuguguL96	1619	asdTsacdAcdAgcaadAcAfggaaugggcsg	1968
AD-1659063	AGT	csasuuccUfUfUfgcuguguauL96	1620	asdAsuadCadCagcadAaCfaggaaugggcsg	1969
AD-1659064	AGT	asusuccUfUfUfgcuguguauL96	1621	asdCsaudAcdAcagcdAaAfcaggaaugggcsg	1970
AD-1659065	AGT	ususccUfUfUfgcuguguauL96	1622	asdTscadTadCacagdCaAfacaggaasusg	1971
AD-1659066	AGT	uscscuguUfUfUfguguauL96	1623	asdAsucdAudAcacadGcAfaacaggasasu	1972
AD-1659067	AGT	cscsuguUfUfUfguguauL96	1624	asdGsaudCadTacacdAgCfaaacaggasasa	1973
AD-1659068	AGT	csusguUfUfUfguguauL96	1625	asdTsgadTcdAuacadCaGfcaaacagggsa	1974
AD-1659069	AGT	usgsuuUfUfUfguguauL96	1626	asdTsugdAudCauacdAcAfgcaaacagggsa	1975
AD-1659070	AGT	gsusuUfUfUfguguauL96	1627	asdTsuudGadTcauadCaCfagcaaacasag	1976

TABLE 6-continued

More Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1659071	AGT	ususugcuguGfUfAfugaucaaaaguL96	1628	asdCsuudTgdAucaudAcAfcagcaaascsa	1977
AD-1684529	AGT	cscscagucUfCfCfaccuuuucuL96	1629	asdGsaadAadGguggdGaGfacuggggsgsu	1978
AD-1684531	AGT	cscscagucUfCfCfaccuuuucuL96	1630	asdAsgadAadAggugdGgAfacuggggsgsg	1979
AD-1684533	AGT	cscsagucUfCfCfaccuuuucuL96	1631	asdAsagdAadAaggudGgGfacuggggsgsg	1980
AD-1684535	AGT	asgsucucccAfCfCfuuuucuucuL96	1632	asdAsgadAgdAaaagdGuGfggagacugsgsg	1981
AD-1659162	AGT	gsuscucccaCfCfUfuuuucuuaL96	1633	asdTsagdAadGaaaadGgUfgggagacugsg	1982
AD-1321384	hAGT	uscsucccacCfUfUfuuuucuuaL96	1634	asdTsuadGadAgaadAgGfugggagascsu	1983
AD-1659163	AGT	csuscccaccUfUfUfucuuuuaL96	1635	asdAsuudAgdAagaadAaGfuggggagsasc	1984
AD-1659164	AGT	uscscaccuUfUfUfucuuuaaguL96	1636	asdCsaudTadGaagadAaAfggugggagsa	1985
AD-1659165	AGT	cscscaccuuUfUfCfuuuuaagauL96	1637	asdTscadTudAgaagdAaAfggugggagsa	1986
AD-1659166	AGT	cscsaccuuUfCfUfucuaaagaguL96	1638	asdCsucdAudTagaadGaAfaagugggagsa	1987
AD-1659167	AGT	csasccuuUfUfUfucuaaagaguL96	1639	asdAscudCadTuagadAgAfaaaggugsgsg	1988
AD-1659168	AGT	ascscuuuUfUfCfuaaagagucuL96	1640	asdGsacdTcdAuagadAaGfaaaaggugsgsg	1989
AD-1659208	AGT	cscsguuuUfCfUfuggucuaaguL96	1641	asdCsuudAgdAccaadGgAfgaaacggsgsu	1990
AD-1659209	AGT	csgsuuucUfUfUfuggucuaaguL96	1642	asdAscudTadGaccadAgGfagaaacggsgsc	1991
AD-1659210	AGT	gsusuucuccUfUfGfgucuaaguguL96	1643	asdCsacdTudAgaccdAaGfgagaaacggsg	1992
AD-1659282	AGT	gsusuugcugGfGfUfuuuuuuaguL96	1644	asdCsuadAadAuaaadCcCfagcaaacsug	1993
AD-1659283	AGT	ususugcuggGfUfUfuuuuuuagauL96	1645	asdTscudAadAuaaadAcCfcagcaaacsu	1994
AD-1659284	AGT	ususgcugggUfUfUfauuuuagaguL96	1646	asdCsucdTadAaaudAaCfccagcaasasc	1995
AD-1659285	AGT	usgscuggguUfUfAfuuuuuagaguL96	1647	asdTscudCudAaaudAaAfcccagcasasa	1996
AD-1659286	AGT	gscsuggguUfAfUfuuuagagaauL96	1648	asdTsucdTcdTaaaadTaAfaccagcsasa	1997
AD-1659287	AGT	csusggguUfUfUfuuagagaauL96	1649	asdAsuudCudCuaaadAuAfaaaccagcsa	1998
AD-1659288	AGT	usgsgguuUfUfUfuagagaauL96	1650	asdCsaudTcdTcuaadAaUfaaaccagsgsc	1999

TABLE 6-continued

More Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1659289	AGT	gsgsguuuuUfUfUfagagaagguL96	1651	asdCscadTudCucudAaAfuaaaacccsasg	2000
AD-1684537	AGT	gsgsuuuuuUfUfAfgagaagggguL96	1652	asdCscddAudTcucudAaAfauaaacccscsa	2001
AD-1684539	AGT	gsusuuuuUfAfGfagaaugggguL96	1653	asdCscddCadTucudTaAfaauaaacscsc	2002
AD-1684541	AGT	ususuuuuUfGfAfgaaugggguL96	1654	asdCscddCcdAuucudCuAfaauaaascsc	2003
AD-1684543	AGT	asgsaaugggGfGfUfgggaggcauL96	1655	asdTsgcdCudCcccdCcCfccauucscsu	2004
AD-1684545	AGT	gsasaugggGfUfGfaggagcauL96	1656	asdTsugdCcdTcccdAcCfccauucscsu	2005
AD-1684547	AGT	asasuggggUfGfGfagggaaguL96	1657	asdCsuudGcdCucccdCaCfcccauucscsu	2006
AD-1684549	AGT	asusgggggUfGfGfagggaaguL96	1658	asdTscudTgdCcuccdCcAfccccaususc	2007
AD-1684551	AGT	usgsggggUfGfGfagggaaguL96	1659	asdTsucdTudGccudCcCfacccccassu	2008
AD-1684553	AGT	gsgsggggUfGfAfggcaagaacL96	1660	asdGsuudCudTgccudCcCfacccccassu	2009
AD-1684555	AGT	gsusggggGfGfAfagaaccaguuL96	1661	asdAscudGgdTucudGcCfuccccacscsc	2010
AD-1684557	AGT	usgsgggGfGfAfagaaccaguuL96	1662	asdCsacdTgdGuucdTgCfcuccccacscsc	2011
AD-1684559	AGT	gsgsgggGfGfAfagaaccaguuL96	1663	asdAscudCudGguudTuGfccuccccsasc	2012
AD-1684560	AGT	gsgsgaggGfGfAfaccaguuuL96	1664	asdAsacdAcdTgguudCuUfgccuccccscsa	2013
AD-1684561	AGT	gsgsaggGfGfAfaccaguuuuL96	1665	asdAsaadCadCuggudTcUfgccuccscsc	2014
AD-1659290	AGT	gsasggcaGfGfAfccaguuuuL96	1666	asdTsaadAcAcuggdTcUfugccuccscsc	2015
AD-1659291	AGT	asgsGcaGfGfAfccaguuuuL96	1667	asdCsuadAadCacugdTcUfugccuccscsc	2016
AD-1659292	AGT	gsgscaGfGfAfccaguuuuL96	1668	asdGscudAadAcacudGgUfucuccscsc	2017
AD-1659293	AGT	gscsaGfGfAfccaguuuuL96	1669	asdCsgcdTadAacacdTgGfucuccscsu	2018
AD-1659294	AGT	csasagaGfGfUfguuuagcgcuL96	1670	asdGscgdCudAaacadCuGfucuccscsc	2019
AD-1659295	AGT	asasgaGfGfUfguuuagcgcuL96	1671	asdCsgcdGcdTaaadAcUfgguucuccscsc	2020
AD-1659296	AGT	asgsaGfGfUfguuuagcgcuL96	1672	asdCsgcdGcdCuaadCaCfugguucuccscsc	2021
AD-1659297	AGT	gsasaccGfGfUfguuuagcgcuL96	1673	asdCscddGcdGcuadAcAfcugguucuccsu	2022

TABLE 6-continued

More Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1659298	AGT	asasccagugUfUfUfagcgggauL96	1674	asdTscddCgdCgcuadAaCfacugguuscsu	2023
AD-1659321	AGT	csusguuccaAfAfAfagaauuccauL96	1675	asdTsggdAadTucuudTuUfggaaacsusa	2024
AD-1659323	AGT	gsusuccaaaAfAfGfaauuccaacuL96	1676	asdGsuudGgdAauucdTuUfuuggaacsasg	2025
AD-1659325	AGT	uscscaaaaaGfAfAfauccaaccguL96	1677	asdCsggdTudGgaudTcUfuuuuggasasc	2026
AD-1659326	AGT	cscsaaaaagAfAfUfuccaaccgauL96	1678	asdTscgdGudTggaadTuCfuuuuuggsasa	2027
AD-1659327	AGT	csasaaaagaAfUfUfccaaaccgacuL96	1679	asdGsucdGgdTuggadAuUfcuuuuugsgsa	2028
AD-1659328	AGT	asasaagaaUfUfCfcaaccgaccuL96	1680	asdGsgudCgdGuuggdAaUfucuuuuugsg	2029
AD-1659329	AGT	asasaagaaUfCfCfaaccgaccuL96	1681	asdTsggdTcdGguugdGaAfuucuuuuusg	2030
AD-1659330	AGT	asasagaauCfCfAfaccgaccaguL96	1682	asdCsugdGudCggudGgAfaucuuuususu	2031
AD-1659331	AGT	asasgaauuCfAfAfccgaccagcuL96	1683	asdGscudGgdTcgudTgGfaauucuuususu	2032
AD-1659332	AGT	asgsaauccAfAfCfcgaccagcuL96	1684	asdAsgcdTgdGucggdTuGfgaauucuuususu	2033
AD-1659333	AGT	gsasauuccaAfCfCfaccagcuuL96	1685	asdAsagdCudGgucgdGuUfggaaucuuususu	2034
AD-1659334	AGT	asasuuccaaCfCfGfaccagcuuguL96	1686	asdCsaadGcdTggucdGgUfuggaaauuscsu	2035
AD-1659335	AGT	asusuccaacCfGfAfccagcuuguL96	1687	asdAscadAgdCuggudCgGfuuggaaususc	2036
AD-1659336	AGT	ususccaaccGfAfCfcagcuuguuL96	1688	asdAsacdAadGcuggdTcGfguuggaasusu	2037
AD-1659337	AGT	uscscaaaccGfCfCfagcuuguuuL96	1689	asdAsaadCadAgcugdGuCfggauuggasasu	2038
AD-1659338	AGT	cscsaaccgaCfCfAfGcuuguuguL96	1690	asdCsaadAcdAagcudGgUfcgguuggsasa	2039
AD-1659339	AGT	csasaccgacCfAfGfGcuuguuguL96	1691	asdAscadAadCaagdTgGfucgguugsgsa	2040
AD-1659340	AGT	asasccgaccAfGfCfuuguuguguL96	1692	asdCsacdAadAcaagdCuGfgucgguugsg	2041
AD-1659341	AGT	ascscgaccaGfCfUfuguuugugauL96	1693	asdTscadCadAacaadGcUfggucggusug	2042
AD-1659342	AGT	cscsgaccagCfUfUfuguuugugaaL96	1694	asdTsucdAcdAaacadAgCfuggucggsusu	2043
AD-1659343	AGT	csgsaccagcUfUfGfuugugaaaaL96	1695	asdTsuudCadCaaacdAaGfcuggucggsu	2044
AD-1659344	AGT	gsasccagcuUfGfUfuugugaaacuL96	1696	asdGsuudTcdAcaaadCaAfgcuggucsgsg	2045

TABLE 6-continued

More Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1659345	AGT	ascscagcuuGfUfUfugugaaacauL96	1697	asdTsgudTudCacaadAcAfacgugguscsg	2046
AD-1659346	AGT	cscsagcuugUfUfUfgugaaacauL96	1698	asdTsugdTudTcacadAaCfaagcuggsusc	2047
AD-1659347	AGT	csasgcuuguUfUfGfugaacaaaauL96	1699	asdTsuudGudTucacdAaAfcaagcugsgsu	2048
AD-1659348	AGT	asgscuuguuUfGfUfgaaacaaaauL96	1700	asdTsuudTgdTuucadCaAfacaagcugsg	2049
AD-1659349	AGT	gscsuuguuuGfUfGfaaacaaaauL96	1701	asdTsuudTudGuuucdAcAfaacaagcsusg	2050
AD-1659350	AGT	csusuguuugUfGfAfaacaaaauL96	1702	asdTsuudTudTguuudCaCfaacaagscsu	2051
AD-1659351	AGT	ususguuuguGfAfAfacaacaaaauL96	1703	asdCsuudTudTuguudTcAfcaacaagsgc	2052
AD-1659371	AGT	usgsuucccuUfUfUfcaaguagauL96	1704	asdCsucdAadCuugadAaAfgggaacascsu	2053
AD-1659372	AGT	gsusucccuUfUfCfaaguagauL96	1705	asdTscudCadAcuugdAaAfaaggaacsasc	2054
AD-1659373	AGT	ususcccuUfCfAfaguugagaauL96	1706	asdTsucdTcdAacuudGaAfaaggaacsca	2055
AD-1659382	AGT	csasaguugaGfAfAfaacaaaauL96	1707	asdCsaadTudTuugudTcUfcaacuugsasa	2056
AD-1659383	AGT	asasguugagAfAfCfaaaaauugguL96	1708	asdCscadAudTuugdTcUfcaacuugsga	2057
AD-1659384	AGT	asgsuugagaAfCfAfaaaauggguL96	1709	asdCscddAadTuuuudGuUfcucaacusug	2058
AD-1659385	AGT	gsusugagaaCfAfAfaaaauggguL96	1710	asdAsccdCadAuuuudTgUfucucaacsusu	2059
AD-1659386	AGT	ususgagaaCfAfAfaaaauggguL96	1711	asdAsacdCcdAuuuudTuGfuucucaacsu	2060
AD-1659387	AGT	usgsagaacaAfAfAfaauuggguuuL96	1712	asdAsaadCcdCaauudTuUfguucucasasc	2061
AD-1659388	AGT	gsasgaacaaAfAfAfaauuggguuuL96	1713	asdAsaadAcddCaauudTuUfuguucucsa	2062
AD-1659389	AGT	asgsaacaaaAfAfUfuggguuuauL96	1714	asdTsaadAadCcaadTuUfuguucucsa	2063
AD-1659390	AGT	gsasacaaaAfUfUfuggguuuauL96	1715	asdTsuaadAadAcccadAuUfuuguucusc	2064
AD-1659399	AGT	asgsuauacaUfUfUfugcauugcuL96	1716	asdGscadAudGcaaadAaUfuguauacsusu	2065
AD-1659400	AGT	gsusauacauUfUfUfugcauugccuL96	1717	asdGsgcdAadTgcaadAaUfuguauacsusu	2066
AD-1659401	AGT	usasuaacauUfUfUfugcauugccuL96	1718	asdAsggdCadAugcadAaAfauguauacsu	2067
AD-1659402	AGT	asusacauuuUfUfGfcauugccuuL96	1719	asdAsagdGcdAaugcdAaAfauguauasasc	2068

TABLE 6-continued

More Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1659406	AGT	asusuuuugcAfUfUfgccuucgguuL96	1720	asdAsccdGadAggcadAuGfcaaaaausgsu	2069
AD-1659407	AGT	ususuugcaUfUfGfccuucgguuL96	1721	asdAsacdCgdAaggcdAaUfgcaaaaasug	2070
AD-1659408	AGT	ususuugcauUfGfCfcuucgguuuL96	1722	asdAsaadCcdGaaggdCaAfugcaaaasasu	2071
AD-1659409	AGT	ususugcauUfGfCfcuucgguuL96	1723	asdCsaadAcCdgaagdGcAfaugcaasasa	2072
AD-1659410	AGT	usugcauUfGfCfcuucgguuL96	1724	asdAscadAadCcgaadGgCfaaugcaasasa	2073
AD-1659411	AGT	usgscauUfGfCfcuucgguuL96	1725	asdTsacdAadAccgadAgGfcaaugcasasa	2074
AD-1659412	AGT	gscsauUfGfCfcuucgguuL96	1726	asdAsuadCadAaccgdAaGfgcaaugcsasa	2075
AD-1659413	AGT	csasuUfGfCfcuucgguuL96	1727	asdAsaudAcCdAaacdGaAfggcaaugscsa	2076
AD-1659414	AGT	asusugccuUfGfCfcuucgguuL96	1728	asdAsaadTadCaaacdCgAfaaggcaaugsc	2077
AD-1659415	AGT	usugccuUfGfCfcuucgguuL96	1729	asdTsadAudAcaaadCcGfaaggcaasug	2078
AD-1659416	AGT	usgscuUfGfCfcuucgguuL96	1730	asdCsudAadTacaadAcCfgaaggcasasu	2079
AD-1659417	AGT	gscscuUfGfCfcuucgguuL96	1731	asdAscudAadAuacadAaCfcaaggcsasa	2080
AD-1659418	AGT	cscsuUfGfCfcuucgguuL96	1732	asdCsacdTadAauacdAaAfcgaaggscsa	2081
AD-1659419	AGT	csuscgguUfGfCfcuucgguuL96	1733	asdAscadCudAaaudCaAfaaccgaagsgsc	2082
AD-1659420	AGT	ususcgguUfGfCfcuucgguuL96	1734	asdGsacdAcdTaaudAcAfaaccgaagsg	2083
AD-1659421	AGT	uscsgguUfGfCfcuucgguuL96	1735	asdAsgadCadCuaaadTaCfaaacccgasag	2084
AD-1659422	AGT	csgsgguUfGfCfcuucgguuL96	1736	asdAsagdAcDacuaadAuAfaaacccgsasa	2085
AD-1659423	AGT	gsgsuUfGfCfcuucgguuL96	1737	asdCsaadGadCacuadAaUfaaacccsgsa	2086
AD-1659424	AGT	gsusuUfGfCfcuucgguuL96	1738	asdTscadAgdAcacudAaAfuacaacscsg	2087
AD-1659425	AGT	ususuUfGfCfcuucgguuL96	1739	asdTsucdAadGacacdTaAfaucacaascsc	2088
AD-1659426	AGT	usugsuUfGfCfcuucgguuL96	1740	asdAsuudCadAgacadCuAfaauacaasasc	2089
AD-1659427	AGT	usgsuUfGfCfcuucgguuL96	1741	asdCsaudTcdAagacdAcUfaaaucacasasa	2090
AD-1659428	AGT	gsusaUfGfCfcuucgguuL96	1742	asdAscadTudCaagadCaCfuuaauacsasa	2091

TABLE 6-continued

More Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1659429	AGT	usasuuuaguGfUfCfuugaauL96	1743	asdTsacdAudTcaagdAcAfcuaaaascsa	2092
AD-1659430	AGT	asusuagugUfCfUfugaauL96	1744	asdTsudCadTucaadGaCfacuaaaasasc	2093
AD-1659431	AGT	ususuaguguCfUfUfgaauL96	1745	asdCsudAcdAuucadAgAfcacuaaasusa	2094
AD-1659432	AGT	ususagugucUfUfGfaugaauL96	1746	asdTscudTadCauucdAaGfacacuaasasu	2095
AD-1659433	AGT	usasgugucUfGfAfaugaagaauL96	1747	asdTsucdTudAcaudCaAfgacacuasasa	2096
AD-1659434	AGT	asgsugucUfGfAfafugaagaacuL96	1748	asdGsudCudTacaudTcAfgacacuasasa	2097
AD-1659436	AGT	usgsucuugaAfUfGfuaagaacauL96	1749	asdAsugdTudCuucadAuUfcaagacascsu	2098
AD-1659437	AGT	gsuscuugaaUfGfUfaagaacauL96	1750	asdCsudGudTcuucadCaUfcaagacasc	2099
AD-1659440	AGT	ususgauguAfAfGfaacaugaccuL96	1751	asdGsgudCadTguucdTuaAfcuucasgsa	2100
AD-1659446	AGT	gsusaagaacAfUfGfaccuccguguL96	1752	asdCsacdGgdAggudAuGfuucuuacsasu	2101
AD-1659447	AGT	usasagaacaUfGfAfccuccguguL96	1753	asdAscadCgdGaggudCaUfguucuaascsa	2102
AD-1659448	AGT	asasgaacauGfAfCfcuccguguauL96	1754	asdTsacdAcdGgaggdTcAfuguucuuasasc	2103
AD-1659449	AGT	asgsaacaugAfCfCfuccguguaguL96	1755	asdCsudCadCggagdGuCfauguucusus	2104
AD-1659450	AGT	gsasacaugaCfCfUfccguguaguL96	1756	asdAscudAcdAcggadGgUfcauguucusus	2105
AD-1659451	AGT	asascaugacCfUfCfCfuguaguguL96	1757	asdCsacdTadCacggdAgGfucauguucsu	2106
AD-1659452	AGT	ascsaugaccUfCfCfuguaguguL96	1758	asdAscadCudAcacgdGaGfugucaugususc	2107
AD-1659453	AGT	csasugaccUfCfGfuguagugucL96	1759	asdGsacdAcdTacacdGgAfggucaugsusu	2108
AD-1659454	AGT	asusgaccUfCfGfuguagugucL96	1760	asdAsgadCadCuacadCgGfaggucaugsu	2109
AD-1659481	AGT	csusuaguuUfUfUfccacagauguL96	1761	asdCsudCudGuggadAaAfaacuaagsgu	2110
AD-1659482	AGT	ususaguuUfUfCfCfacaugaucL96	1762	asdGscadTcdTguggdAaAfaaacuaasgs	2111
AD-1659483	AGT	usasguuuUfCfCfacaugaucL96	1763	asdAsgcdAudCugudGaAfaaaacuasag	2112
AD-1659485	AGT	gsusuuuucCfAfCfagaugcuL96	1764	asdCsadGcdAucugdTgGfaaaaaacsusa	2113
AD-1659487	AGT	ususuuccaCfAfGfauugugucL96	1765	asdCsacdAadGcaudTgUfggaaaaasasc	2114

TABLE 6-continued

More Exemplary dsRNA molecule according to some embodiments of the disclosure					
Duplex ID	Target	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1659488	AGT	ususuccacAfGfAfugcuugugauL96	1766	asdTscadCadAgcaudCuGfuggaaaasasa	2115
AD-1659489	AGT	ususuccacaGfAfUfgcuugugauuL96	1767	asdAsucdAcdAagcadTcUfguggaaasasa	2116
AD-1659490	AGT	ususcacagAfUfGfugcuugauuuL96	1768	asdAsaudCadCaagcdAuCfuguggaasasa	2117
AD-1659491	AGT	uscscacagaUfGfCfugugauuuuL96	1769	asdAsaadTcdAcaagdCaUfcuguggasasa	2118
AD-1659492	AGT	cscscacagauGfCfUfugugauuuuL96	1770	asdAsaadAudCacaadGcAfucuguggsasa	2119
AD-1659493	AGT	csascagaugCfUfUfgugauuuuuL96	1771	asdAsaadAadTcacadAgCfaucugugsgsa	2120
AD-1659537	AGT	ascscugaauUfUfCfuguuugaauuL96	1772	asdAsuudCadAacagdAaAfuucaggugsgc	2121
AD-1659538	AGT	cscsugaauUfCfUfguuugaauuL96	1773	asdCsaudTcdAaacadGaAfaucaggsgusg	2122
AD-1659559	AGT	gsgsaaccauAfGfCfugguuuuuL96	1774	asdAsaadTadAccagdCuAfuugguuccsgsc	2123
AD-1659560	AGT	gsasaccauaGfCfUfgguuuuuuL96	1775	asdGsaadAudAaccadGcUfaugguucscsg	2124
AD-1659561	AGT	asascgauagCfUfGfuguuuuuuL96	1776	asdAsgadAadTaaccdAgCfuaugguuuscsc	2125
AD-1659562	AGT	ascscauagcUfGfGfuuuuuucucuL96	1777	asdGsagdAadAuaacdCaGfcauaggususc	2126
AD-1659563	AGT	cscsauagcuGfGfUfuuuuuccuL96	1778	asdGsgadGadAuaadCcAfgcuagggsusu	2127
AD-1659582	AGT	cscsuuguguUfAfGfuaaauaacguL96	1779	asdCsgudTudAuuaacdTaAfcacaaggsgsa	2128
AD-1659583	AGT	csusuguguUfAfGfuaaauaacguL96	1780	asdAscgdTudTauuadCuAfacacaaggsgsg	2129
AD-1659584	AGT	ususguguuaGfUfAfuaaaacgucuL96	1781	asdGsacdGudTuauudAcUfaacacaagsgsg	2130
AD-1659585	AGT	usgsuguuagUfAfAfuaaaacgucuuL96	1782	asdAsgadCgdTuauudTaCfuaacacasasg	2131
AD-1659586	AGT	gsusguuaguAfAfUfaaacgucuuuL96	1783	asdAsagdAcdGuuudTuAfcuaacacsasa	2132
AD-1659587	AGT	usgsuuaguaAfUfAfaaacgucuuuL96	1784	asdCsaadGadCguuudAuUfacuaacacsasa	2133
AD-1659588	AGT	gsusuaguuaUfAfAfacgucuuuL96	1785	asdGscadAgdAcguudTaUfuacuaacsasc	2134

[0435] In some embodiments, the dsRNA molecule is not a dsRNA molecule listed in Table 7.

TABLE 7

Additional dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1289862.1	cscsgcu (Chd) agGfUfUfcugcuuuuuaaL96	2135	VPusdTsaadAadGcagadAcCfudGagcggscsc	2136
AD-1193318.1	uscsugagCfuGfAfGfuugguuuuauL96	193	asdTsaadAadCcaacdTcdAgdCucagascgsg	2137
AD-1193320.1	uscsugagCfuGfAfGfuugguuuuauL96	193	asdTsaadAadCcaacdTcdAgdCucagascsc	2138
AD-1193410.1	ususuuagAfaCfAfCfcuuuuucacuL96	235	asdGsugdAadAaaggdTgUfuCfuaaaasusu	2139
AD-1193413.1	ususuuagAfaCfAfCfcuuuuucacuL96	235	asdGsugdAadAaaggdTgUfuCfuaaaascsc	2140

Total RNA isolation using DYNABEADS mRNA Isolation Kit (Invitrogen™, part #: 610-12)

[0436] Cells were lysed in 75 μ L of Lysis/Binding Buffer containing 3 μ L of beads per well and mixed for 10 minutes on an electrostatic shaker. The washing steps were automated on a Biotek EL406, using a magnetic plate support. Beads were washed (in 90 μ L) once in Buffer A, once in Buffer B, and twice in Buffer E, with aspiration steps in between. Following a final aspiration, complete 10 μ L RT mixture was added to each well, as described below.

Cdna Synthesis Using ABI High Capacity cDNA Reverse Transcription Kit (Applied Biosystems, Foster City, CA, Cat #4368813)

[0437] A master mix of 1 μ L 10 $\times$ Buffer, 0.4 μ L 25 $\times$ dNTPs, 1 μ L Random primers, 0.5 L Reverse Transcriptase, 0.5 μ L RNase inhibitor and 6.6 μ L of H₂O per reaction were added per well. Plates were sealed, agitated for 10 minutes on an electrostatic shaker, and then incubated at 37° C. for 2 hours. Following this, the plates were agitated at 80° C. for 8 minutes

Real Time PCR

[0438] Two μ L of cDNA were added to a master mix containing 0.5 μ L of human GAPDH TaqMan Probe (4326317E), 0.5 μ L human AGT (HsO0174854 ml), 2 μ L nuclease-free water and 5 μ L Lightcycler 480 probe master mix (Roche Cat #O4887301001) per well in a 384 well plates (Roche cat #O4887301001). Real time PCR was done in a LightCycler480 Real Time PCR system (Roche).

[0439] To calculate relative fold change, data were analyzed using the $\Delta\Delta C_t$ method and normalized to assays performed with cells transfected with 10 nM AD-1955, or mock transfected cells.

[0440] Results are shown in FIGS. 1A-1B and 3A-4C and summarized in Tables 8-14. The results show a consistently improved in vitro activity relative to the parent design across several exemplary designs and across a large set of exemplary sequences, targets, and cell lines.

TABLE 8

In vitro activity for various designs in primary mouse hepatocytes											
Parent Duplex ID	Dose	Parent	StDev	AS5-7	StDev	AS14	StDev	2015	StDev	D1	StDev
AD-1181392	10 nM	4.2	0.5	4.9	0.3	5.6	0.5	4.7	0.6	4.4	0.3
AD-1181401		7.4	1.0	5.9	1.2	10.9	3.2	6.1	0.5	4.6	0.2
AD-1181410		5.8	0.3	4.4	0.5	5.8	0.9	6.1	1.1	6.0	0.7
AD-1181417		5.8	0.8	6.1	1.8	8.1	1.9	10.1	2.3	6.3	1.4
AD-1181426		6.2	1.5	6.4	0.2	5.3	0.7	6.4	0.6	4.2	0.7
AD-64972		5.6	0.9	5.2	0.6	5.7	1.7	7.7	0.7	4.9	1.1
AD-1181443		5.3	0.4	4.7	1.0	6.6	0.4	3.8	0.7	4.6	0.6
AD-1181451		6.3	1.0	4.2	0.2	5.4	0.4	6.7	1.0	5.8	0.7
AD-1181460		5.5	0.7	5.4	0.5	7.5	0.4	7.5	0.5	5.6	0.8
AD-1181469		11.2	1.8	8.2	1.8	10.3	2.8	8.0	1.3	7.1	0.8
AD-674282		5.3	4.9	10.9	0.9					27.0	10.3
AD-674283		1.3	0.4	2.2	0.3					1.4	0.4
AD-674287		4.1	1.2	4.5	0.7					12.4	2.0
AD-674289		9.2	6.3	11.8	2.0					14.0	2.8
AD-674293		3.1	3.4	7.7	1.7					11.0	3.8
AD-674299		1.5	1.0	2.4	1.4					3.2	0.7
AD-674303		29.5	6.4	26.3	4.4					29.3	4.6
AD-674305		2.1	1.7	5.5	2.3					6.0	2.7

TABLE 8-continued

In vitro activity for various designs in primary mouse hepatocytes											
Parent Duplex ID	Dose	Parent	StDev	AS5-7	StDev	AS14	StDev	2015	StDev	D1	StDev
AD-674307	1 nM	2.9	3.0	2.3	1.0					3.1	0.2
AD-674310		4.8	0.9	2.9	1.0					5.5	2.6
AD-674312		6.3	3.7	4.0	1.4					5.0	2.3
AD-674314		16.8	2.9	19.8	6.0					21.1	9.3
AD-674323		21.4	5.9	16.0	1.9					30.8	6.8
AD-674325		14.4	2.6	13.8	0.7					17.9	2.8
AD-68435		4.2	2.0	1.9	0.5					4.9	1.4
AD-1181392		2.8	0.2	2.9	0.3	3.0	0.2	2.3	0.2	3.0	0.2
AD-1181401		8.9	1.6	3.9	0.3	18.3	3.8	6.5	0.6	3.6	0.1
AD-1181410		3.8	0.4	3.5	0.2	3.8	0.1	3.7	0.5	3.6	0.3
AD-1181417		11.5	2.0	7.7	0.8	17.1	1.9	10.5	1.4	4.2	1.0
AD-1181426		3.6	0.4	3.7	0.6	3.9	0.4	4.7	0.6	2.7	0.3
AD-64972		5.3	0.5	4.3	0.4	10.8	2.5	8.6	2.2	4.6	0.7
AD-1181443		4.5	0.5	4.7	1.1	6.5	1.3	3.4	0.5	3.3	0.4
AD-1181451		10.8	1.9	9.9	1.5	17.2	3.0	10.2	2.3	6.8	1.1
AD-1181460		18.7	2.0	4.1	0.5	35.5	1.9	19.9	2.0	4.5	0.6
AD-1181469		33.5	3.1	13.1	2.4	68.6	2.5	44.9	1.6	16.5	1.5
AD-674282		11.5	11.6	15.9	6.3					16.5	4.0
AD-674283		11.8	6.2	5.2	1.6					3.5	0.4
AD-674287		14.8	2.2	7.3	2.1					12.4	2.4
AD-674289		67.0	11.6	29.3	6.8					19.0	4.5
AD-674293		30.4	9.4	17.7	2.8					9.1	1.0
AD-674299		16.0	4.4	7.7	1.4					4.4	1.2
AD-674303	0.1 nM	87.2	4.5	42.6	5.9					39.1	4.4
AD-674305		28.8	8.2	8.5	1.5					6.7	1.4
AD-674307		15.9	2.0	3.5	1.4					2.4	1.3
AD-674310		18.7	4.5	5.9	1.9					4.8	2.4
AD-674312		29.1	5.0	15.1	5.2					6.3	2.0
AD-674314		36.5	9.1	20.1	6.3					15.9	3.2
AD-674323		63.6	7.5	18.7	4.2					14.5	4.2
AD-674325		36.9	6.4	16.1	2.5					11.4	1.7
AD-68435		14.9	1.7	4.5	0.8					4.0	0.9
AD-674282		3.5	2.0	17.6	3.4					31.5	8.3
AD-674283		5.9	2.9	9.7	3.5					7.5	1.5
AD-674287		13.4	7.8	14.7	4.8					34.3	15.5
AD-674289		53.5	6.6	42.0	7.3					52.0	16.3
AD-674293		22.9	1.1	30.2	4.6					42.9	5.5
AD-674299		13.5	4.3	9.3	1.3					17.9	7.1
AD-674303		57.4	5.9	64.5	11.7					67.7	11.8
AD-674305		25.7	4.9	15.2	7.6					32.2	11.3
AD-674307		14.8	1.4	5.9	0.4					17.2	2.0
AD-674310		23.3	11.8	13.7	3.8					33.6	7.0
AD-674312		35.9	12.0	27.0	4.5					29.8	9.4
AD-674314		34.9	2.5	41.3	10.2					60.8	9.8
AD-674323		70.3	6.8	46.4	13.8					77.4	1.8
AD-674325		28.8	2.4	32.1	11.7					51.6	10.9
AD-68435		15.8	2.4	9.2	4.8					46.3	16.4

TABLE 9

In vitro activity for various designs in primary mouse hepatocytes											
Parent Duplex ID	Dose	Parent	StDev	D2	StDev	D3	StDev	D4	StDev	v2	StDev
AD-1181392	10 nM	4.2	0.5	5.0	0.1	9.1	1.2	9.6	0.3	7.0	0.4
AD-1181401		7.4	1.0	6.2	1.8	8.2	0.8	7.8	1.2	7.4	0.5
AD-1181410		5.8	0.3	4.7	0.2	5.8	0.8	7.5	0.2	5.7	1.5
AD-1181417		5.8	0.8	5.0	1.4	11.8	2.1	12.1	1.5	8.6	0.6
AD-1181426		6.2	1.5	5.9	0.8	5.9	0.8	9.0	2.1	7.2	1.8
AD-64972		5.6	0.9	5.9	1.1	7.8	1.7	10.5	0.8	13.0	0.5
AD-1181443		5.3	0.4	6.0	1.4	6.8	0.5	4.0	0.5	5.2	0.3
AD-1181451		6.3	1.0	4.9	0.5	8.8	4.9	7.6	0.6	7.2	0.3
AD-1181460		5.5	0.7	6.0	0.7	8.9	1.1	12.4	3.3	10.7	2.0
AD-1181469		11.2	1.8	6.0	1.2	7.6	1.2	9.9	2.3	18.3	5.2
AD-674282		5.3	4.9								
AD-674283		1.3	0.4								
AD-674287		4.1	1.2								
AD-674289		9.2	6.3								

TABLE 9-continued

In vitro activity for various designs in primary mouse hepatocytes											
Parent Duplex ID	Dose	Parent	StDev	D2	StDev	D3	StDev	D4	StDev	v2	StDev
AD-674293	1 nM	3.1	3.4								
AD-674299		1.5	1.0								
AD-674303		29.5	6.4								
AD-674305		2.1	1.7								
AD-674307		2.9	3.0								
AD-674310		4.8	0.9								
AD-674312		6.3	3.7								
AD-674314		16.8	2.9								
AD-674323		21.4	5.9								
AD-674325		14.4	2.6								
AD-68435		4.2	2.0								
AD-1181392		2.8	0.2	2.9	0.1	5.7	0.7	9.7	0.6	4.1	0.4
AD-1181401		8.9	1.6	4.6	0.8	6.6	0.5	7.8	0.4	5.0	0.2
AD-1181410		3.8	0.4	3.0	0.2	3.6	0.7	3.4	0.9	4.2	0.4
AD-1181417		11.5	2.0	4.3	0.5	8.3	1.2	43.7	5.6	17.7	4.3
AD-1181426		3.6	0.4	2.8	0.5	5.3	0.6	14.5	3.2	5.3	0.4
AD-64972		5.3	0.5	5.4	2.3	8.3	0.8	17.7	1.1	19.3	1.5
AD-1181443		4.5	0.5	4.5	0.5	5.4	0.3	4.0	0.5	4.9	0.8
AD-1181451		10.8	1.9	18.0	5.2	11.1	0.2	11.0	1.0	34.9	3.8
AD-1181460		18.7	2.0	5.9	0.7	17.6	2.0	38.2	6.6	22.1	2.6
AD-1181469		33.5	3.1	21.2	0.9	33.8	0.8	56.1	2.2	74.3	5.8
AD-674282		11.5	11.6								
AD-674283		11.8	6.2								
AD-674287		14.8	2.2								
AD-674289		67.0	11.6								
AD-674293		30.4	9.4								
AD-674299		16.0	4.4								
AD-674303		87.2	4.5								
AD-674305		28.8	8.2								
AD-674307		15.9	2.0								
AD-674310		18.7	4.5								
AD-674312		29.1	5.0								
AD-674314		36.5	9.1								
AD-674323		63.6	7.5								
AD-674325		36.9	6.4								
AD-68435	0.1 nM	14.9	1.7								
AD-674282		3.5	2.0								
AD-674283		5.9	2.9								
AD-674287		13.4	7.8								
AD-674289		53.5	6.6								
AD-674293		22.9	1.1								
AD-674299		13.5	4.3								
AD-674303		57.4	5.9								
AD-674305		25.7	4.9								
AD-674307		14.8	1.4								
AD-674310		23.3	11.8								
AD-674312		35.9	12.0								
AD-674314		34.9	2.5								
AD-674323		70.3	6.8								
AD-674325		28.8	2.4								
AD-68435		15.8	2.4								

TABLE 10

In vitro activity for various designs in primary mouse hepatocytes											
Parent Duplex ID	Dose	Parent	StDev	D1	StDev	D5	StDev	D6	StDev	D7	StDev
AD-1181392	10 nM	1.2	0.1	0.9	0.1	1.8	0.3	8.6	1.2	2.8	0.3
AD-1181401		24.2	2.3	4.5	1.1	7.4	1.1	19.1	1.1	8.5	2.1
AD-1181410		5.3	1.9	1.7	0.4	2.2	0.5	3.5	1.2	2.5	0.3
AD-1181417		15.4	3.6	5.5	1.6	9.8	2.6	59.8	21.7	13.6	4.3
AD-1181426		2.6	0.6	1.4	0.3	4.5	0.7	9.7	1.5	4.7	1.1
AD-64972		12.1	2.9	5.0	1.4	9.4	2.5	15.0	5.1	11.8	1.8
AD-1181443		4.0	0.5	1.5	0.6	5.1	1.0	10.8	2.0	5.5	0.3
AD-1181451		13.1	0.7	6.3	1.4	13.9	1.8	17.2	2.0	14.8	1.9
AD-1181460		22.6	4.7	3.0	0.5	4.9	0.2	16.4	1.8	7.7	1.1
AD-1181392	1 nM	5.5	0.5	3.7	0.2	7.0	2.1	22.7	4.9	10.9	2.5

TABLE 10-continued

In vitro activity for various designs in primary mouse hepatocytes											
Parent Duplex ID	Dose	Parent	StDev	D1	StDev	D5	StDev	D6	StDev	D7	StDev
AD-1181401	0.1 nM	47.8	13.8	15.0	2.9	18.6	4.6	39.1	2.5	17.9	1.3
AD-1181410		16.7	2.9	9.2	1.3	8.9	0.9	12.0	3.2	14.0	2.3
AD-1181417		49.0	14.9	27.9	3.2	42.6	9.8	88.7	11.6	34.8	7.4
AD-1181426		15.8	3.8	8.1	0.8	8.6	3.5	20.5	2.6	12.0	2.9
AD-64972		39.9	6.4	17.9	2.7	32.0	8.5	36.2	8.5	31.2	6.0
AD-1181443		16.9	2.3	9.5	4.1	21.4	4.0	28.3	4.3	20.8	4.2
AD-1181451		44.9	3.3	31.9	9.0	49.0	6.0	53.4	6.9	43.3	4.4
AD-1181460		74.8	15.4	23.9	0.7	23.7	0.8	46.7	6.7	28.4	6.8
AD-1181392		23.5	1.4	18.7	1.4	26.1	7.9	56.7	7.6	27.7	3.2
AD-1181401		61.8	6.4	33.9	3.7	37.7	4.1	75.4	15.2	45.9	8.7
AD-1181410		48.7	4.0	28.3	3.9	20.9	1.8	32.0	2.3	35.5	7.8
AD-1181417		62.5	7.4	56.2	14.9	88.6	13.4	132.8	33.8	84.7	22.3
AD-1181426		71.0	16.7	53.9	4.2	46.1	17.4	136.9	20.8	69.9	16.6
AD-64972		153.9	12.2	94.2	14.1	108.8	22.1	56.3	2.6	54.8	13.3
AD-1181443		80.2	5.8	20.2	2.4	42.0	2.4	45.0	9.6	34.3	5.8
AD-1181451		69.6	6.4	58.7	7.8	71.2	9.0	80.8	25.3	65.2	10.0
AD-1181460		72.3	4.0	44.3	5.7	48.3	2.9	53.6	7.2	51.1	1.9

TABLE 11

In vitro activity for various designs in primary mouse hepatocytes													
Parent Duplex ID	Dose	Parent	StDev	D8	StDev	D9	StDev	D10	StDev	D11	StDev	D12	StDev
AD-1181392	10 nM	1.2	0.1	11.0	2.4	9.3	1.7	4.7	0.7	10.5	3.0	14.7	2.5
AD-1181401		24.2	2.3	21.5	3.2	13.6	2.6	11.4	2.0	24.1	3.0	23.4	3.6
AD-1181410		5.3	1.9	3.0	0.3	4.3	0.6	3.6	0.7	10.5	3.6	5.7	1.7
AD-1181417		15.4	3.6	48.9	10.6	48.8	13.3	23.4	11.8	54.9	16.4	53.8	6.7
AD-1181426		2.6	0.6	10.7	3.6	6.1	1.3	6.9	2.2	21.0	5.4	18.3	4.6
AD-64972		12.1	2.9	13.7	3.1	13.4	4.4	15.1	5.6	27.0	5.7	30.5	3.6
AD-1181443		4.0	0.5	11.4	3.1	9.2	1.8	6.9	1.0	16.8	3.9	14.9	3.3
AD-1181451		13.1	0.7	19.6	3.2	20.4	4.1	15.2	3.3	30.9	3.4	39.0	6.9
AD-1181460		22.6	4.7	24.3	5.2	23.0	2.8	15.0	1.8	35.8	3.9	47.1	2.6
AD-1181392	1 nM	5.5	0.5	29.8	1.0	27.1	3.3	12.6	0.9	28.1	8.1	32.2	3.7
AD-1181401		47.8	13.8	44.7	5.3	30.3	4.6	29.8	6.7	43.3	0.4	48.2	11.0
AD-1181410		16.7	2.9	13.2	1.4	15.2	4.2	12.4	3.0	24.2	6.3	16.2	2.3
AD-1181417		49.0	14.9	68.4	6.6	74.2	16.5	40.2	5.1	76.1	7.5	109.0	13.3
AD-1181426		15.8	3.8	30.7	9.4	20.1	2.9	19.6	6.0	41.7	3.8	30.4	7.2
AD-64972		39.9	6.4	33.8	5.9	41.3	5.1	34.7	8.0	62.8	3.6	62.5	7.1
AD-1181443		16.9	2.3	34.9	4.2	28.8	5.8	23.2	6.5	45.6	11.2	39.0	3.7
AD-1181451		44.9	3.3	66.8	4.3	62.9	15.2	55.0	13.0	76.2	10.5	82.0	19.4
AD-1181460		74.8	15.4	70.3	13.8	49.4	6.2	50.2	7.3	77.5	9.4	79.8	13.4
AD-1181392	0.1 nM	23.5	1.4	54.1	10.4	47.7	2.3	28.9	1.1	49.2	3.2	54.4	2.2
AD-1181401		61.8	6.4	60.0	5.6	47.9	5.6	51.8	7.7	66.5	6.4	68.6	2.0
AD-1181410		48.7	4.0	36.9	4.3	34.1	7.6	33.4	1.8	62.0	9.3	43.6	6.5
AD-1181417		62.5	7.4	160.5	30.4	98.3	27.8	118.4	35.9	76.4	7.7	110.2	13.7
AD-1181426		71.0	16.7	46.9	6.8	48.4	12.4	77.7	25.4	107.5	28.7	88.3	36.8
AD-64972		153.9	12.2	55.6	11.8	68.1	24.6	69.6	22.2	130.8	26.1	123.2	18.6
AD-1181443		80.2	5.8	60.7	9.7	63.6	8.4	46.4	5.7	56.6	7.5	58.8	1.6
AD-1181451		69.6	6.4	53.8	2.3	60.8	1.8	51.2	1.0	68.4	8.3	77.0	4.5
AD-1181460		72.3	4.0	64.1	9.0	71.2	3.6	64.4	3.5	68.4	3.2	68.6	3.3

TABLE 12

In vitro activity for various designs in primary cynomolgus hepatocytes							
Parent Duplex ID	Dose	Parent	StDev	D1	StDev	D2	StDev
AD-157464.8	10 nM	1.0	0.0	1.0	0.3	1.2	0.3
AD-157448.7		1.5	0.3	2.4	0.3	3.0	1.5
AD-157468.11		2.2	1.2	1.6	0.2	1.8	0.1
AD-67327.28		8.3	11.8	1.7	0.5	1.8	0.1
AD-85446.5		1.9	0.3	1.7	0.7	1.8	0.5
AD-85435.13		2.9	0.1	2.1	0.2	1.7	0.2
AD-85438.5		1.7	0.3	1.2	0.1	1.8	0.1
AD-157464.8	1 nM	3.0	0.6	1.4	0.4	1.6	0.4

TABLE 12-continued

In vitro activity for various designs in primary cynomolgus hepatocytes							
Parent Duplex ID	Dose	Parent	StDev	D1	StDev	D2	StDev
AD-157448.7	0.1 nM	3.7	0.7	3.5	0.6	2.8	0.8
AD-157468.11		5.7	0.6	4.4	0.2	3.6	0.4
AD-67327.28		1.6	0.0	1.9	0.1	2.2	0.2
AD-85446.5		4.5	0.6	2.6	0.7	2.7	0.7
AD-85435.13		6.7	0.5	2.8	0.3	2.8	0.2
AD-85438.5		3.5	0.1	3.2	0.5	5.9	1.0
AD-157464.8		10.2	0.1	1.9	0.5	2.9	0.9
AD-157448.7		9.0	1.0	8.9	1.7	6.9	1.0
AD-157468.11		16.9	3.9	13.1	1.5	10.0	1.3
AD-67327.28		4.6	0.7	3.9	0.4	4.8	0.4
AD-85446.5		15.9	1.5	5.9	1.1	6.8	0.6
AD-85435.13		18.4	2.2	5.4	1.3	7.2	0.1
AD-85438.5		10.8	1.7	10.2	2.1	14.2	2.5

TABLE 13

In vitro activity for various designs in Hep3B								
Target	Duplex ID	Parent			Duplex ID	D1		
		10 nM	1 nM	0.1 nM		10 nM	1 nM	0.1 nM
Marc1	AD-1531684.2	8 ± 1.2	18.4 ± 3.7	12 ± 1.8	AD-1636769.1	6.6 ± 0.8	11.4 ± 1.4	16.4 ± 1.3
Marc1	AD-1531719.2	26.8 ± 2.8	61.9 ± 6.6	69 ± 8.4	AD-1636805.1	10.1 ± 1.7	32.9 ± 5.2	53.2 ± 3.4
Marc1	AD-1531682.2	16.4 ± 2.8	42.4 ± 7.2	27.3 ± 2.4	AD-1636782.1	11.9 ± 1.1	25.9 ± 1.8	38.2 ± 5.1
PNPLA3	AD-75247.4	53.6 ± 9	88.8 ± 9.5	90.3 ± 10.4	AD-1636754.1	52.2 ± 3.3	71.2 ± 17.2	69.6 ± 13.7
C3	AD-1531655.2	14.3 ± 2.4	46 ± 2.2	54.8 ± 11.1	AD-1636695.1	14.6 ± 2.4	26.6 ± 4.6	30.5 ± 3.1
PNPLA3	AD-67589.6	40.1 ± 10.3	68.2 ± 16.9	57.6 ± 7.6	AD-1636733.1	29.9 ± 4.7	31 ± 5	37.9 ± 1.7
Marc1	AD-1531703.2	15.4 ± 1.5	47.9 ± 2.3	54.5 ± 6.6	AD-1636798.1	9 ± 1.8	16.6 ± 3.2	34.3 ± 3.5
C3	AD-1531665.2	46.4 ± 5.3	83.4 ± 12.2	94.3 ± 7	AD-1636720.1	35.7 ± 5	52.4 ± 3.4	71.4 ± 3.5
C3	AD-1531660.2	14.8 ± 4.1	39.2 ± 8.6	54.6 ± 8.7	AD-1636701.1	13.9 ± 4.4	30.3 ± 6.8	41.1 ± 4.1
C3	AD-1531672.2	25.6 ± 4.4	62.5 ± 3.8	82 ± 9.7	AD-1636716.1	19.2 ± 2.9	38.5 ± 5.4	55.1 ± 2.6
PNPLA3	AD-1010735.4	35.2 ± 8.8	55.1 ± 21.7	52.6 ± 7.4	AD-1636729.1	48.4 ± 3.4	45 ± 4.2	56.2 ± 11
Marc1	AD-1531721.2	38.2 ± 4.3	61.6 ± 5.7	69.9 ± 10.8	AD-1636806.1	16.7 ± 1.1	30.4 ± 2	53.6 ± 9.8
PNPLA3	AD-67605.10	30.7 ± 3.2	44.9 ± 3.4	42.2 ± 2.6	AD-1636735.1	41.9 ± 2.3	51.9 ± 11.3	64.6 ± 11.6
C3	AD-571552.3	33.6 ± 7.3	73.4 ± 4.7	87.3 ± 7.4	AD-1636714.1	25.1 ± 2.2	42.2 ± 7.6	50.4 ± 1.3
Marc1	AD-1531716.2	8 ± 1.4	20.3 ± 4.7	15.6 ± 2.3	AD-1636778.1	7.5 ± 0.8	10.5 ± 1.5	13.2 ± 1.1
C3	AD-569269.6	7.3 ± 1	12.3 ± 1.4	12.3 ± 3.2	AD-1479350.2	8.7 ± 1.2	10.5 ± 1.2	14.4 ± 1.7
Marc1	AD-1531718.2	9.1 ± 1.4	29.1 ± 5.1	23.1 ± 2.7	AD-1636785.1	7.6 ± 1.3	12.6 ± 1.9	24.1 ± 2.7
C3	AD-1531663.3	14.9 ± 1.6	26.9 ± 4.6	56.5 ± 2.7	AD-1636763.1	12.6 ± 2	14.5 ± 4.5	27.4 ± 3.6
SCN9A	AD-1531765.2	40.2 ± 2.8	55.3 ± 7.9	68.2 ± 1.9	AD-1636834.1	34.7 ± 3.3	38.6 ± 2.2	47.7 ± 10.4
SCN9A	AD-1531735.2	33.2 ± 4	46.1 ± 5.2	34.9 ± 4	AD-1636811.1	28.7 ± 3	42.5 ± 5.4	52.8 ± 4.3
PNPLA3	AD-519933.3	54.6 ± 8	99.2 ± 17.3	91.4 ± 18.6	AD-1636757.1	52 ± 6.2	56.2 ± 4.3	60.4 ± 2.7
PNPLA3	AD-519780.2	50.1 ± 3.9	80.1 ± 12.2	79.5 ± 10.2	AD-1636758.1	44.8 ± 7.9	55.5 ± 8	74.1 ± 6.4
Marc1	AD-1531679.2	28.2 ± 2.9	60.8 ± 10.5	61.4 ± 4.4	AD-1636791.1	16.4 ± 1.4	46.6 ± 8.1	59 ± 7.7
Marc1	AD-1531692.2	13.5 ± 0.7	23.7 ± 1.7	17.9 ± 2.9	AD-1636777.1	9.8 ± 2.7	20 ± 1.4	28.7 ± 3.8
SCN9A	AD-1531741.2	72.7 ± 1.3	83.5 ± 8.9	69.8 ± 2.9	AD-1636838.1	47.3 ± 10.9	76.9 ± 4	106.3 ± 11.5
C3	AD-572575.3	20.8 ± 4.2	57.3 ± 9.8	68.9 ± 11.1	AD-1636703.1	26.4 ± 6	38 ± 10.6	58.9 ± 14.5
SCN9A	AD-1531744.2	49.2 ± 3.9	65.5 ± 8.5	63.3 ± 7.5	AD-1636831.1	34.5 ± 5.7	66.5 ± 5.5	89.5 ± 7.2
PNPLA3	AD-519354.4	62.3 ± 16.1	77.2 ± 22.9	84.6 ± 15.6	AD-1636732.1	37.5 ± 7.3	44.5 ± 6.7	64.5 ± 13.5
PNPLA3	AD-67551.7	22.5 ± 1.9	38.4 ± 12	41 ± 4.2	AD-1636737.1	30.5 ± 4.4	39.2 ± 8.3	60 ± 7.1
Marc1	AD-1531687.2	14.7 ± 2.6	34.6 ± 3.9	21.1 ± 1.9	AD-1636779.1	14.3 ± 2.8	23.6 ± 1.9	32.5 ± 7.6
SCN9A	AD-1531736.2	73.3 ± 10.3	78.8 ± 7.7	68.3 ± 6.1	AD-1636843.1	82.1 ± 9.5	95.3 ± 12.7	107.5 ± 11.9
Marc1	AD-1531688.2	5 ± 1.1	13 ± 2.3	9.2 ± 1.3	AD-1636772.1	6.5 ± 1	10.1 ± 0.5	11.2 ± 1.7
SCN9A	AD-1531747.2	69.4 ± 2	77.8 ± 3.6	80 ± 5.3	AD-1636836.1	45.7 ± 2.9	60.4 ± 5.5	93.2 ± 3.3
SCN9A	AD-1531756.2	38.3 ± 4.9	46.5 ± 2.8	46.4 ± 5.2	AD-1636813.1	29.8 ± 1.1	46.7 ± 4.7	53.2 ± 2.7
Marc1	AD-1531696.2	9.4 ± 0.8	22.3 ± 4.3	18.5 ± 4.2	AD-1636783.1	7.5 ± 0.7	12.7 ± 1.3	25 ± 2.2
C3	AD-1531658.2	13.3 ± 1.8	30.7 ± 0.8	39.7 ± 3.7	AD-1636692.1	12.1 ± 1.2	24.2 ± 5.8	40.9 ± 10.6
SCN9A	AD-1531757.2	64.6 ± 4	75.8 ± 7.1	70.3 ± 3.8	AD-1636844.1	30.1 ± 2.4	46.6 ± 5.4	59.5 ± 3.3
PNPLA3	AD-67565.3	62.4 ± 17.9	69.5 ± 15	68.4 ± 8.1	AD-1636726.1	71.3 ± 5.6	50 ± 2.7	76.6 ± 18.7
PNPLA3	AD-1010714.3	52.6 ± 19	99.7 ± 26.3	85.9 ± 13	AD-1636760.1	28.4 ± 5.4	40 ± 6	54.8 ± 4.3
SCN9A	AD-1531728.2	60.8 ± 4.2	69.8 ± 9.1	76.8 ± 4.7	AD-1636841.1	33.6 ± 5.1	53.3 ± 8.2	65.3 ± 8.5
Marc1	AD-1531686.2	16.4 ± 2.6	52.5 ± 3.5	50.2 ± 8.3	AD-1636789.1	12.3 ± 2.3	25.2 ± 4.6	45.5 ± 4.1
PNPLA3	AD-67583.7	45.2 ± 4.6	81.4 ± 24.3	64.5 ± 13.5	AD-1636762.1	46.4 ± 12.4	48.8 ± 8.4	48.8 ± 10.2
C3	AD-572388.7	20.2 ± 1.8	49.7 ± 7.8	68.7 ± 5.5	AD-1636766.1	71 ± 9.6	81.7 ± 9.4	90.1 ± 12.4
C3	AD-1531662.2	37.3 ± 5.1	83.1 ± 8.3	91.9 ± 5.3	AD-1636718.1	31.5 ± 1.6	68.9 ± 9.3	77.7 ± 5.9
C3	AD-571901.2	26.2 ± 7.2	62.6 ± 5.1	78.2 ± 7.4	AD-1636707.1	32.1 ± 7.9	60 ± 7.6	61 ± 6.7
PNPLA3	AD-1010732.4	30.3 ± 6.7	63.6 ± 15	52.1 ± 4.7	AD-1636727.1	39.3 ± 3.8	55 ± 13.3	59.5 ± 11
PNPLA3	AD-520061.6	32.7 ± 8.1	49.5 ± 8.4	51.3 ± 5.3	AD-1636741.1	42.2 ± 5.7	44.4 ± 2.4	57.3 ± 4.9
SCN9A	AD-1531770.2	50.6 ± 8.6	74.1 ± 0.9	76.6 ± 3.5	AD-1636830.1	38.3 ± 1.7	59.3 ± 7.3	72.6 ± 7.7

TABLE 13-continued

In vitro activity for various designs in Hep3B								
Target	Parent				D1			
	Duplex ID	10 nM	1 nM	0.1 nM	Duplex ID	10 nM	1 nM	0.1 nM
Marc1	AD-1531705.2	5.8 ± 0.7	21.9 ± 2.7	11.9 ± 2.1	AD-1636780.1	6 ± 0.7	14.2 ± 5.2	27.9 ± 2.5
C3	AD-569266.3	7.5 ± 1.3	19 ± 4.3	24.5 ± 5.1	AD-1636689.1	12.2 ± 2.1	18.5 ± 3.2	22.2 ± 4.3
SCN9A	AD-1531740.2	68.1 ± 10.5	84.9 ± 5.5	66.9 ± 2.4	AD-1636839.1	41.3 ± 6	56.5 ± 9.4	72 ± 4.9
PNPLA3	AD-75265.5	48.9 ± 5.5	83.9 ± 20.7	82.7 ± 8.8	AD-1636753.1	36.1 ± 10.1	55.8 ± 8.4	67.9 ± 6.8
Marc1	AD-1531709.2	13.9 ± 3.1	37.2 ± 7.5	46.9 ± 13.3	AD-1636792.1	10.5 ± 0.7	20.9 ± 2.8	30.7 ± 6.5
C3	AD-569048.2	20.9 ± 4	56.4 ± 11.3	64.2 ± 11.2	AD-1636709.1	34.3 ± 0.8	50.6 ± 10	75.3 ± 7.9
C3	AD-571633.3	34.2 ± 5.7	72.1 ± 7.6	81.3 ± 7	AD-1636721.1	29.7 ± 2.1	51.3 ± 3	78.2 ± 9.8
C3	AD-569516.3	28.8 ± 2.2	76.2 ± 9.6	91.9 ± 10.4	AD-1636719.1	32.5 ± 5.2	51.9 ± 12.9	72.2 ± 11.7
PNPLA3	AD-519346.5	26.2 ± 4.5	53 ± 6.2	51 ± 4.9	AD-1636765.1	27 ± 5.3	28.3 ± 3.6	41.8 ± 5.1
C3	AD-571748.3	8.2 ± 2.8	26.5 ± 2.2	38.3 ± 5.8	AD-1636693.1	13.5 ± 2.4	32.2 ± 7.1	37.7 ± 6.8
Marc1	AD-1531708.2	27.5 ± 1.5	61.3 ± 6.1	70.9 ± 12.7	AD-1636807.1	18.9 ± 2.9	26.2 ± 4.7	49.5 ± 5.9
C3	AD-570712.7	7.3 ± 0.5	17.3 ± 1.6	28.1 ± 3.6	AD-1479283.2	10.2 ± 0.7	16.5 ± 1.3	27.9 ± 4.1
Marc1	AD-1531706.2	5.4 ± 0.4	14.1 ± 1.1	8.5 ± 1.1	AD-1636774.1	5.7 ± 0.9	10.8 ± 2.4	13.8 ± 1.1
PNPLA3	AD-67584.8	49.1 ± 8.5	57.2 ± 5.6	49.8 ± 13.5	AD-1636725.1	57.4 ± 13	57.4 ± 16.8	67.4 ± 9.5
C3	AD-1531668.2	17.2 ± 2.9	49.8 ± 8.7	61.8 ± 9.7	AD-1636715.1	13.8 ± 1.8	28.2 ± 4.4	45.9 ± 4.9
PNPLA3	AD-67577.7	27.2 ± 3.3	41.7 ± 6.5	44.1 ± 7.5	AD-1636750.1	30.7 ± 7.3	38.2 ± 4.1	58.8 ± 7.1
C3	AD-571753.4	11.7 ± 2.3	30 ± 6.6	47.8 ± 10	AD-1636702.1	12.3 ± 0.9	17.9 ± 3.4	31.6 ± 3.1
C3	AD-1531657.2	20.6 ± 5.8	58.1 ± 9.5	73 ± 7.9	AD-1636705.1	16 ± 2	30.2 ± 5.1	42 ± 4.2
C3	AD-571715.5	5.7 ± 0.6	11.7 ± 0.9	24.5 ± 4.2	AD-1636690.1	6.4 ± 1.5	13.3 ± 2.2	19.1 ± 2.2
C3	AD-1531659.2	12.1 ± 1.9	29.4 ± 3.5	38.2 ± 7.2	AD-1636694.1	12.8 ± 1.8	22.3 ± 6.6	25.6 ± 2.1
PNPLA3	AD-67560.7	52.7 ± 14.7	63.9 ± 17.1	68.5 ± 10	AD-1636743.1	43.9 ± 6.2	46.6 ± 7.3	51.7 ± 9.1
PNPLA3	AD-1531674.2	51.8 ± 5.6	55.2 ± 5.4	47.2 ± 8.1	AD-1636756.1	47.3 ± 8.3	61.1 ± 10.9	60.3 ± 6.7
PNPLA3	AD-519757.6	48.2 ± 8.9	43.1 ± 14.7	59.6 ± 5.1	AD-1636734.1	36.9 ± 8.1	65.6 ± 10.7	51.3 ± 10.8
SCN9A	AD-1531732.2	65.4 ± 1.9	43.7 ± 8.3	57.1 ± 4.6	AD-1636840.1	54.3 ± 6.5	63.4 ± 5.7	56.7 ± 12.5
C3	AD-1531656.2	57.5 ± 10.1	104.8 ± 19.6	90.9 ± 15.1	AD-1636712.1	44.6 ± 2.5	59.1 ± 11.9	94.7 ± 8.9
Marc1	AD-1531678.2	36.1 ± 3.9	66 ± 4.3	58.6 ± 11.2	AD-1636800.1	15.4 ± 3.1	22.7 ± 1.6	38.5 ± 5.7
Marc1	AD-1531701.2	15.3 ± 1.7	32.8 ± 2.4	31.4 ± 3.2	AD-1636790.1	16 ± 2.9	31.8 ± 3.5	49.4 ± 7.7
Marc1	AD-1531680.2	5.1 ± 0.6	16.4 ± 1.8	9.7 ± 1.1	AD-1636770.1	6.5 ± 0.8	14.5 ± 3.6	19.7 ± 5
SCN9A	AD-1531734.2	56.4 ± 4.3	67.5 ± 10.7	82 ± 4	AD-1636837.1	48.4 ± 6.4	78.6 ± 3.6	107.1 ± 13.3
C3	AD-568977.3	14.4 ± 1.7	38 ± 4.8	41.4 ± 11.1	AD-1636699.1	17.7 ± 1.1	24.9 ± 3.3	39.3 ± 2.8
C3	AD-569268.10	6.8 ± 0.4	10.2 ± 1.9	14.2 ± 1.9	AD-1479341.2	8.7 ± 1.6	12 ± 0.7	16.6 ± 2.5
C3	AD-572818.2	10.5 ± 1.5	27.2 ± 4.4	42 ± 7.4	AD-1636691.1	13.1 ± 1.8	23.3 ± 2.2	36.9 ± 6.1
SCN9A	AD-1531729.2	45 ± 3.7	60.9 ± 4.8	49.6 ± 7.2	AD-1636815.1	35.8 ± 5.6	49.7 ± 6	51.2 ± 1.5
Marc1	AD-1531689.2	36.6 ± 7.1	61.4 ± 7.9	69.7 ± 4.9	AD-1636795.1	22.5 ± 3.4	33.4 ± 1.5	51.1 ± 5.9
PNPLA3	AD-75275.3	56 ± 8.3	60.8 ± 11.8	44.6 ± 2.6	AD-1636731.1	48.6 ± 4.2	61 ± 8.4	70.7 ± 11.9
PNPLA3	AD-67568.3	33.8 ± 2.8	35 ± 4.9	50.8 ± 8.1	AD-1636744.1	36.2 ± 6.1	43.9 ± 5.5	48.5 ± 2.2
C3	AD-569494.2	98.1 ± 8	113.9 ± 3.9	105.2 ± 8.5	AD-1636717.1	105.6 ± 13.2	92.8 ± 7.2	90.9 ± 14.1
Marc1	AD-1531722.2	26.5 ± 0.9	54.4 ± 5	54.4 ± 6.6	AD-1636796.1	12.3 ± 2.2	21.8 ± 0.4	27.7 ± 3.4
C3	AD-571610.3	21.1 ± 7.8	49 ± 4.8	69.1 ± 9.5	AD-1636710.1	25.4 ± 1.7	45.3 ± 5.6	61.5 ± 8.6
SCN9A	AD-1531753.2	42.9 ± 1.7	59.3 ± 4.7	54.2 ± 5.1	AD-1636822.1	33 ± 2.8	53.9 ± 6.7	62 ± 7.7
SCN9A	AD-1531759.2	46.3 ± 3.4	50.6 ± 5.4	70.6 ± 2.3	AD-1636828.1	30.8 ± 2.4	41.1 ± 4.7	52.9 ± 7.6
C3	AD-572495.3	8.7 ± 0.9	10.7 ± 1.7	17 ± 2.5	AD-1636767.1	11.5 ± 1.2	15 ± 1.5	21.2 ± 5.2
PNPLA3	AD-1010719.3	30 ± 4.2	79.4 ± 24.3	28.9 ± 4	AD-1636739.1	20.9 ± 2.1	24.4 ± 5.8	33.7 ± 8.2
Marc1	AD-1531683.2	28.6 ± 5.8	64.7 ± 6.8	54.8 ± 3	AD-1636803.1	32.4 ± 4.6	58.5 ± 9.1	85.2 ± 5
Marc1	AD-1531711.2	22.6 ± 6.1	46.2 ± 9.7	41.4 ± 4.7	AD-1636801.1	11.7 ± 1	20.4 ± 5.1	40.3 ± 4.2
C3	AD-1531664.2	19.1 ± 5.3	50.6 ± 8.8	59.4 ± 2.3	AD-1636713.1	32.7 ± 5.9	47.5 ± 4.9	53.4 ± 2.6
SCN9A	AD-1531762.2	57.9 ± 4	75.4 ± 6.4	73.8 ± 5.9	AD-1636833.1	35.9 ± 3.5	49.3 ± 5.4	78.1 ± 13.9
PNPLA3	AD-67526.5	26.9 ± 8.8	41.8 ± 13.4	22 ± 2.2	AD-1636740.1	19.7 ± 3.7	19.4 ± 1.9	32.9 ± 4.9
SCN9A	AD-1531725.2	26.8 ± 4.4	32.5 ± 3.4	37.6 ± 3.4	AD-1636823.1	23.8 ± 2.4	32.5 ± 5.5	36.7 ± 4.8
SCN9A	AD-1531737.2	40.1 ± 2.1	43.3 ± 3.6	57.4 ± 10	AD-1636827.1	33.5 ± 2.8	38 ± 4.2	54.5 ± 6.9
PNPLA3	AD-1531673.2	42 ± 4.6	78.6 ± 23.6	65.1 ± 4.5	AD-1636749.1	40.7 ± 3.3	31.6 ± 5.5	35.8 ± 3.3
Marc1	AD-1531717.2	16.1 ± 2.2	38.5 ± 2.8	46.9 ± 5.2	AD-1636799.1	11 ± 1.2	13.8 ± 2.7	25.9 ± 7.2
C3	AD-570644.2	15.1 ± 3.2	43.5 ± 6.8	61.4 ± 10.6	AD-1636700.1	12.8 ± 1.7	30.7 ± 6	35.8 ± 7.3
C3	AD-568962.3	24 ± 4.2	57.1 ± 12.1	69.8 ± 7.9	AD-1636706.1	21.1 ± 4.3	35.1 ± 2.9	50.5 ± 4.3
Marc1	AD-1531690.2	6.7 ± 0.9	14.2 ± 2.8	8.7 ± 1.3	AD-1636771.1	6 ± 1.1	10.5 ± 1	10.8 ± 1.3
SCN9A	AD-1531750.2	46.9 ± 2.7	58.3 ± 6.8	55.3 ± 4.3	AD-1636826.1	41 ± 5.9	52.9 ± 9.3	66.5 ± 9.4
Marc1	AD-1531710.3	14.9 ± 2.2	34.3 ± 6.1	40.1 ± 7.6	AD-1636848.1	8.2 ± 1.1	25 ± 6	44.4 ± 3.3
PNPLA3	AD-67578.3	61 ± 15.4	54.9 ± 4.2	66.5 ± 4.9	AD-1636746.1	58.6 ± 4.5	60.5 ± 16.3	68.6 ± 14.1
PNPLA3	AD-67582.6	47.8 ± 7	71.9 ± 12.8	80.3 ± 12.1	AD-1636747.1	39.9 ± 1.7	51.8 ± 5.2	66.7 ± 12.8
C3	AD-1531666.2	39.7 ± 10.3	71.9 ± 4.8	85.9 ± 11.1	AD-1636723.1	15 ± 2.2	31.9 ± 6.2	67.3 ± 12.3
Marc1	AD-1531695.3	6.7 ± 0.6	11.8 ± 3.2	11 ± 1.4	AD-1636847.1	7.5 ± 0.5	13.4 ± 2.7	19.9 ± 3.2
SCN9A	AD-1531723.2	29.7 ± 5.8	37.8 ± 5.7	43.1 ± 3.3	AD-1636824.1	19.3 ± 1.5	28.9 ± 6.7	32.1 ± 3
PNPLA3	AD-75270.4	46.6 ± 12	62.6 ± 15.2	65.6 ± 12.2	AD-1636748.1	42.2 ± 1.7	39.2 ± 9.5	54.7 ± 2.5
C3	AD-1531671.2	11.8 ± 2.7	33.2 ± 4.2	49.2 ± 7.8	AD-1636704.1	24.1 ± 3.8	37 ± 8.3	56.6 ± 10.9
PNPLA3	AD-519350.6	35.8 ± 6.4	50.4 ± 4.8	49.4 ± 5	AD-1636724.1	45.8 ± 8.2	55 ± 8.8	50.7 ± 9
SCN9A	AD-1531764.2	63.2 ± 4.6	66 ± 3.1	57 ± 10	AD-1636832.1	57.3 ± 4.8	69 ± 7.9	62.6 ± 5.9
SCN9A	AD-1531739.2	39.7 ± 5.3	58.5 ± 5.4	60.7 ± 7.9	AD-1636835.1	24.5 ± 5.3	38.9 ± 1.9	57 ± 9.2
Marc1	AD-1531681.2	17.3 ± 3.6	55.2 ± 7.2	60.8 ± 11.6	AD-1636788.1	11.3 ± 2.6	21.4 ± 5.8	32.2 ± 9.2
Marc1	AD-1531704.2	7.4 ± 0.7	36.8 ± 5	26.5 ± 2.2	AD-1636787.1	8.6 ± 2.2	14.5 ± 3.1	23.5 ± 3.8
PNPLA3	AD-67564.3	71.1 ± 5.4	74.7 ± 9	76.7 ± 7.6	AD-1636761.1	56.3 ± 2.7	55.7 ± 4.5	63.4 ± 6.7
Marc1	AD-1531676.2	18.8 ± 4.6	57.6 ± 10	80 ± 12.9	AD-1636794.1	9.6 ± 1.9	22.2 ± 3.4	34.6 ± 7.4

TABLE 13-continued

In vitro activity for various designs in Hep3B								
Target	Parent				D1			
	Duplex ID	10 nM	1 nM	0.1 nM	Duplex ID	10 nM	1 nM	0.1 nM
Marc1	AD-1531700.2	30.1 ± 5.2	71.1 ± 6.8	76.9 ± 7.3	AD-1636804.1	12.8 ± 2.6	28.6 ± 2.7	51.1 ± 9.4
Marc1	AD-1531702.2	37.9 ± 6.4	72.7 ± 4.3	75.3 ± 5.5	AD-1636802.1	30.3 ± 4.9	55 ± 9.5	71.9 ± 20.4
SCN9A	AD-1531730.2	39.4 ± 2.8	47.3 ± 1.2	42.4 ± 6.1	AD-1636814.1	30.6 ± 1.6	46.2 ± 2.2	57.3 ± 0.4
SCN9A	AD-1531738.2	32.3 ± 5.3	39.4 ± 1.9	45 ± 5.5	AD-1636812.1	30.4 ± 2.2	38.5 ± 7.1	43.3 ± 8
SCN9A	AD-1531733.3	45.3 ± 2	53.3 ± 10.4	35.1 ± 6.9	AD-1636850.1	29.7 ± 3.6	44.5 ± 13.3	59.6 ± 9.9
PNPLA3	AD-67554.11	39.8 ± 4.4	45.5 ± 10	55.4 ± 7	AD-1636738.1	48.8 ± 4.6	56.6 ± 13.9	56 ± 9.1
Marc1	AD-1531685.2	25.3 ± 1.4	41.9 ± 6.6	51.9 ± 2.1	AD-1636797.1	6 ± 0.9	14.9 ± 2.9	23.6 ± 1.9
PNPLA3	AD-75269.3	49.5 ± 10.5	66.5 ± 13.9	44.8 ± 8.1	AD-1636728.1	44.3 ± 2.7	40.4 ± 4	57.7 ± 11.3
SCN9A	AD-1531752.2	87.9 ± 13.1	64.7 ± 10.1	70 ± 9.8	AD-1636842.1	62.9 ± 1.5	78.6 ± 8.9	87.1 ± 5.6
SCN9A	AD-1531754.3	42.1 ± 1.8	46.6 ± 3.8	39.8 ± 6.3	AD-1636852.1	25.4 ± 4	46 ± 2.2	52.6 ± 2.1
PNPLA3	AD-518942.2	59 ± 6.6	90.6 ± 20.3	106.6 ± 15.5	AD-1636764.1	37.6 ± 5.1	33.4 ± 4.9	40.6 ± 6.9
C3	AD-571932.2	22.1 ± 2.2	45.8 ± 6.9	62 ± 6.7	AD-1636698.1	17.8 ± 1.9	31 ± 5.9	49.4 ± 11.2
C3	AD-572022.2	14.4 ± 2.8	34 ± 3.8	44.4 ± 6.9	AD-1636697.1	15.9 ± 1.1	32.2 ± 5.4	52.1 ± 7.3
SCN9A	AD-1531760.2	46.7 ± 3.2	68.7 ± 6.3	58.3 ± 1.6	AD-1636816.1	29.1 ± 5.4	51.5 ± 7.6	60.9 ± 8.7
SCN9A	AD-1531766.2	50.2 ± 8.7	55.5 ± 5.8	48.9 ± 4.1	AD-1636846.1	55.2 ± 4.6	67.2 ± 14.9	87.1 ± 18.8
C3	AD-572577.3	35.8 ± 7.2	79.9 ± 14.7	95.7 ± 6.3	AD-1636711.1	51 ± 12.3	74.5 ± 14.2	89.2 ± 5.4
Marc1	AD-1531697.2	21.5 ± 4.6	41 ± 1.2	26.5 ± 0.5	AD-1636793.1	13.1 ± 1.9	17.1 ± 3.5	31.4 ± 5.3
PNPLA3	AD-67561.3	71.2 ± 14.1	81.5 ± 16.5	101.6 ± 7.1	AD-1636742.1	65.7 ± 4.6	93.1 ± 24.5	74.2 ± 12.7
PNPLA3	AD-67586.3	73.6 ± 11.5	87.6 ± 12.8	86.7 ± 6.3	AD-1636755.1	57.5 ± 5.8	61.7 ± 12.1	64.6 ± 9.3
C3	AD-569763.5	18.2 ± 4.8	45.4 ± 3.6	52.7 ± 2.7	AD-1636696.1	19.2 ± 6.9	34.1 ± 5.9	43.5 ± 5.2
SCN9A	AD-1531726.2	32.2 ± 2.5	42 ± 3.8	46.7 ± 4.1	AD-1636821.1	24.9 ± 1.7	39.4 ± 4.5	54.4 ± 2.4
SCN9A	AD-1531742.3	32.4 ± 5.7	52.4 ± 4.3	43.4 ± 4.2	AD-1636851.1	23.8 ± 4.1	56.9 ± 3.2	75.8 ± 13.1
PNPLA3	AD-67573.3	62.3 ± 10.5	74.5 ± 8.6	77.5 ± 10.2	AD-1636751.1	42.6 ± 2.7	41.8 ± 5.2	69.3 ± 10.7
SCN9A	AD-1531761.2	58.2 ± 5.9	72.5 ± 5.2	60.6 ± 8.9	AD-1636818.1	26.3 ± 4.7	47.6 ± 10	56.8 ± 6.5
Marc1	AD-1531712.2	8 ± 0.2	19.6 ± 2	14.5 ± 0.5	AD-1636781.1	11 ± 0.1	13.5 ± 0.7	19.5 ± 3.7
C3	AD-1531669.2	48.1 ± 3.2	87 ± 9.8	80.1 ± 5.5	AD-1636722.1	29.7 ± 3.1	50.1 ± 3.9	59.9 ± 5.4
SCN9A	AD-1531745.2	41.3 ± 3.1	50.1 ± 2.4	42 ± 1.9	AD-1636820.1	29.3 ± 2.2	38.5 ± 2.8	47.9 ± 3.8
SCN9A	AD-1531724.2	48.3 ± 2.7	55.4 ± 4.6	47.9 ± 0.9	AD-1636817.1	38 ± 6.1	50.9 ± 5	67.9 ± 5.4
SCN9A	AD-1531731.2	43.6 ± 5.2	52.8 ± 1.2	66.9 ± 7	AD-1636825.1	33.6 ± 4.3	49.6 ± 5.4	56.3 ± 2
C3	AD-570132.3	26.7 ± 3.7	60.8 ± 9.6	87.8 ± 11.1	AD-1636708.1	19.5 ± 3.6	31 ± 3.8	45.3 ± 5.5
SCN9A	AD-1531769.2	38.3 ± 4.6	53.1 ± 6.1	48.9 ± 3.6	AD-1636819.1	32.6 ± 2.1	43.7 ± 5.3	54.3 ± 1.1
Marc1	AD-1531694.2	6.4 ± 0.6	17.3 ± 2	11.6 ± 1.7	AD-1636775.1	8 ± 1.9	15.7 ± 2.8	19 ± 3.1
Marc1	AD-1531677.2	31.7 ± 3.8	59.5 ± 7.4	60.1 ± 15.2	AD-1636784.1	16.3 ± 1.8	25.4 ± 6.1	33 ± 4.8
SCN9A	AD-1531727.2	50.1 ± 2.8	61.8 ± 5.6	74.4 ± 6.6	AD-1636829.1	42.8 ± 3.9	59.8 ± 7.2	68.7 ± 9
PNPLA3	AD-67575.11	62.6 ± 13.6	56.2 ± 5.8	58.4 ± 4.3	AD-1636768.1	49.4 ± 9.5	50.9 ± 6.4	48.9 ± 7.4
PNPLA3	AD-75274.6	72.6 ± 6	74.9 ± 13.3	96.4 ± 13.1	AD-1636745.1	66 ± 8.7	68.9 ± 7.1	86.5 ± 7.2
SCN9A	AD-1531768.2	32.2 ± 1.8	46.7 ± 4.3	54.9 ± 5.9	AD-1636809.1	20 ± 0.9	30.9 ± 4.4	43.4 ± 4.9
C3	AD-570134.4	7.7 ± 1.1	12.8 ± 0.6	24.1 ± 7	AD-1636688.1	9 ± 1	20.9 ± 5.7	31.8 ± 8.4
Marc1	AD-1531707.2	6.6 ± 1.2	15.8 ± 0.7	9.2 ± 0.8	AD-1636773.1	7.9 ± 1	10 ± 0.4	12.1 ± 0.6
PNPLA3	AD-75272.3	56.9 ± 19.1	73.2 ± 12	68.6 ± 13.7	AD-1636752.1	41 ± 6	42 ± 7.5	69.6 ± 9.1
PNPLA3	AD-67567.3	82.7 ± 22.8	95.3 ± 15.3	85.4 ± 9.9	AD-1636759.1	56.3 ± 6.1	62.9 ± 15.9	72.3 ± 8.3
Marc1	AD-1531714.3	6.4 ± 0.9	18.1 ± 2.3	23.7 ± 5.3	AD-1636849.1	9 ± 1	16.6 ± 1.8	24.5 ± 1.2
Marc1	AD-1531720.2	5.4 ± 0.7	12.6 ± 1.7	10.8 ± 1.4	AD-1636776.1	5.3 ± 0.6	10.9 ± 1.2	12.6 ± 1.2
SCN9A	AD-1531746.2	73.3 ± 4.7	86.7 ± 2.2	71.3 ± 10.6	AD-1636845.1	43.2 ± 5.8	91.2 ± 8.6	86.4 ± 19.1
SCN9A	AD-1531755.2	37.8 ± 6.1	49.1 ± 5.4	46.9 ± 7.4	AD-1636810.1	36.2 ± 2.1	61.4 ± 3.2	73.5 ± 5.5
Marc1	AD-1531675.2	16.9 ± 3.3	32.5 ± 4	33 ± 5.7	AD-1636786.1	10.1 ± 1.6	16.9 ± 4.3	27 ± 4.3
SCN9A	AD-1531767.2	28.3 ± 1.2	44 ± 2.7	56.2 ± 8.9	AD-1636808.1	20.5 ± 1.4	32 ± 3.3	39.4 ± 5.1
PNPLA3	AD-520053.7	30.9 ± 5.9	34.7 ± 8.4	52.9 ± 8.6	AD-1636736.1	31.1 ± 6.6	44 ± 4.2	52.5 ± 4
PNPLA3	AD-1010734.3	42.8 ± 16.5	58.8 ± 17.2	49.6 ± 9.8	AD-1636730.1	50.8 ± 10.1	38.6 ± 7.4	53.7 ± 6.5

TABLE 14

In vitro activity for various designs targeting Agt in Hep3B							
Parent				D1			
Duplex ID	10 nM	1 nM	0.1 nM	Duplex ID	10 nM	1 nM	0.1 nM
AD-1632799.1	88 ± 15	105 ± 23	90 ± 21	AD-1657992.1	55 ± 11	91 ± 9	73 ± 8
AD-1632801.1	6 ± 1	23 ± 4	33 ± 3	AD-1657994.1	2 ± 1	12 ± 3	15 ± 4
AD-1632805.1	82 ± 7	95 ± 7	83 ± 12	AD-1657998.1	43 ± 4	65 ± 16	66 ± 6
AD-1632838.1	58 ± 5	94 ± 7	92 ± 10	AD-1658030.1	64 ± 5	84 ± 7	82 ± 8
AD-1684490.1	50 ± 11	83 ± 26	80 ± 18	AD-1684491.1	38 ± 12	67 ± 14	73 ± 17
AD-1684492.1	28 ± 4	68 ± 7	71 ± 20	AD-1684493.1	19 ± 2	47 ± 12	70 ± 11
AD-1684494.1	92 ± 19	101 ± 10	116 ± 6	AD-1684495.1	18 ± 4	54 ± 9	60 ± 6
AD-1684496.1	24 ± 5	72 ± 8	99 ± 10	AD-1684497.1	4 ± 1	15 ± 3	42 ± 7
AD-1684498.1	8 ± 2	31 ± 9	66 ± 9	AD-1684499.1	34 ± 8	53 ± 10	79 ± 8
AD-1684500.1	4 ± 1	16 ± 5	33 ± 6	AD-1684501.1	2 ± 0	7 ± 2	24 ± 2

TABLE 14-continued

In vitro activity for various designs targeting Agt in Hep3B							
Parent				D1			
Duplex ID	10 nM	1 nM	0.1 nM	Duplex ID	10 nM	1 nM	0.1 nM
AD-1684502.1	98 ± 12	107 ± 8	98 ± 3	AD-1684503.1	78 ± 15	102 ± 10	112 ± 14
AD-1684504.1	12 ± 2	56 ± 6	78 ± 10	AD-1684505.1	1 ± 0	7 ± 3	18 ± 5
AD-1684506.1	16 ± 5	53 ± 12	58 ± 11	AD-1684507.1	10 ± 3	46 ± 9	54 ± 6
AD-1684508.1	31 ± 8	87 ± 10	101 ± 15	AD-1684509.1	5 ± 1	17 ± 2	52 ± 3
AD-1684510.1	95 ± 8	119 ± 9	131 ± 11	AD-1684511.1	30 ± 6	67 ± 5	96 ± 13
AD-1684512.1	2 ± 1	9 ± 3	24 ± 3	AD-1684513.1	4 ± 2	28 ± 3	49 ± 9
AD-1684514.1	39 ± 12	67 ± 8	82 ± 10	AD-1684515.1	44 ± 9	88 ± 10	100 ± 4
AD-1632840.1	17 ± 4	62 ± 5	77 ± 7	AD-1658032.1	7 ± 2	34 ± 3	52 ± 3
AD-1632841.1	9 ± 2	45 ± 8	70 ± 7	AD-1658033.1	2 ± 0	21 ± 4	32 ± 6
AD-1632842.1	62 ± 11	60 ± 4	67 ± 13	AD-1658034.1	29 ± 6	77 ± 17	72 ± 12
AD-1632843.1	6 ± 3	38 ± 6	45 ± 5	AD-1658035.1	3 ± 1	9 ± 1	18 ± 3
AD-1632844.1	7 ± 1	28 ± 5	45 ± 6	AD-1658036.1	3 ± 0	12 ± 2	20 ± 4
AD-1632846.1	7 ± 3	30 ± 7	54 ± 10	AD-1658038.1	5 ± 2	18 ± 2	38 ± 1
AD-1632847.1	9 ± 4	35 ± 7	47 ± 4	AD-1658039.1	2 ± 0	5 ± 1	15 ± 7
AD-1632848.1	67 ± 11	92 ± 8	108 ± 8	AD-1658040.1	5 ± 1	18 ± 2	34 ± 5
AD-1632849.1	4 ± 1	13 ± 2	22 ± 1	AD-1658041.1	7 ± 2	16 ± 0	31 ± 4
AD-1632836.1	1 ± 0	3 ± 2	7 ± 1	AD-1658042.1	1 ± 0	1 ± 0	3 ± 1
AD-1632850.1	26 ± 2	62 ± 7	70 ± 9	AD-1658043.1	2 ± 0	7 ± 1	14 ± 2
AD-1632851.1	17 ± 5	53 ± 8	74 ± 5	AD-1658044.1	1 ± 0	6 ± 2	12 ± 2
AD-1632852.1	4 ± 2	23 ± 4	46 ± 10	AD-1658045.1	2 ± 1	6 ± 2	16 ± 2
AD-1632853.1	16 ± 7	55 ± 14	62 ± 7	AD-1658046.1	2 ± 1	6 ± 1	19 ± 5
AD-1632854.1	40 ± 12	68 ± 11	74 ± 9	AD-1658047.1	6 ± 3	17 ± 1	30 ± 7
AD-1632855.1	2 ± 1	9 ± 1	13 ± 1	AD-1658048.1	3 ± 1	9 ± 3	18 ± 4
AD-1632856.1	21 ± 6	47 ± 8	75 ± 9	AD-1658049.1	1 ± 0	7 ± 1	16 ± 3
AD-1632857.1	8 ± 3	40 ± 5	43 ± 11	AD-1658050.1	5 ± 1	20 ± 5	29 ± 4
AD-1632858.1	6 ± 1	23 ± 6	48 ± 4	AD-1658051.1	6 ± 2	14 ± 1	31 ± 2
AD-1632859.1	3 ± 2	12 ± 3	43 ± 8	AD-1658052.1	2 ± 0	6 ± 0	17 ± 3
AD-1632860.1	9 ± 3	27 ± 4	57 ± 5	AD-1658053.1	6 ± 1	22 ± 2	51 ± 4
AD-1632861.1	10 ± 2	38 ± 5	63 ± 8	AD-1658054.1	2 ± 1	8 ± 4	22 ± 3
AD-1632862.1	53 ± 10	69 ± 10	100 ± 11	AD-1658055.1	12 ± 4	40 ± 5	69 ± 6
AD-1632863.1	13 ± 2	29 ± 5	75 ± 14	AD-1658056.1	18 ± 5	51 ± 1	80 ± 5
AD-1632864.1	61 ± 11	74 ± 9	113 ± 24	AD-1658057.1	67 ± 6	66 ± 16	96 ± 18
AD-1632865.1	37 ± 4	63 ± 10	86 ± 17	AD-1658058.1	19 ± 5	40 ± 6	56 ± 12
AD-1632866.1	17 ± 1	42 ± 8	62 ± 8	AD-1658059.1	12 ± 4	27 ± 6	41 ± 7
AD-1632991.1	18 ± 5	50 ± 2	74 ± 10	AD-1658184.1	10 ± 2	38 ± 9	58 ± 9
AD-1632992.1	63 ± 9	100 ± 20	90 ± 11	AD-1658185.1	38 ± 6	67 ± 12	86 ± 13
AD-1632993.1	50 ± 4	87 ± 8	102 ± 20	AD-1658186.1	5 ± 1	29 ± 1	55 ± 9
AD-1632994.1	6 ± 1	28 ± 2	65 ± 14	AD-1658187.1	3 ± 0	11 ± 1	32 ± 4
AD-1632995.1	35 ± 9	47 ± 4	75 ± 17	AD-1658188.1	4 ± 2	11 ± 3	26 ± 4
AD-1632996.1	13 ± 2	42 ± 8	76 ± 6	AD-1658189.1	7 ± 2	27 ± 4	56 ± 12
AD-1632997.1	5 ± 1	23 ± 4	50 ± 6	AD-1658190.1	4 ± 1	19 ± 6	31 ± 6
AD-1632998.1	8 ± 2	23 ± 2	58 ± 4	AD-1658191.1	13 ± 4	23 ± 3	48 ± 11
AD-1632999.1	41 ± 5	75 ± 4	83 ± 16	AD-1658192.1	21 ± 6	46 ± 10	77 ± 9
AD-1633000.1	106 ± 9	126 ± 8	96 ± 10	AD-1658193.1	25 ± 3	60 ± 5	83 ± 11
AD-1633003.1	4 ± 1	33 ± 7	64 ± 9	AD-1658196.1	2 ± 0	6 ± 2	22 ± 6
AD-1633004.1	55 ± 10	83 ± 11	122 ± 18	AD-1658197.1	8 ± 1	24 ± 5	32 ± 3
AD-1633007.1	6 ± 1	19 ± 2	69 ± 8	AD-1658200.1	2 ± 0	7 ± 1	22 ± 4
AD-1633008.1	26 ± 7	53 ± 8	53 ± 3	AD-1658201.1	19 ± 3	63 ± 12	89 ± 9
AD-1633009.1	5 ± 1	12 ± 3	30 ± 5	AD-1658202.1	10 ± 1	25 ± 5	50 ± 9
AD-1633010.1	5 ± 0	9 ± 3	13 ± 2	AD-1658203.1	1 ± 0	1 ± 0	4 ± 1
AD-1633011.1	3 ± 1	10 ± 2	26 ± 1	AD-1658204.1	1 ± 0	5 ± 1	16 ± 3
AD-1633012.1	5 ± 1	33 ± 4	64 ± 9	AD-1658205.1	2 ± 0	4 ± 1	10 ± 2
AD-1633013.1	3 ± 1	10 ± 3	25 ± 5	AD-1658206.1	1 ± 0	5 ± 1	14 ± 3
AD-1633014.1	3 ± 1	23 ± 4	35 ± 5	AD-1658207.1	4 ± 1	12 ± 3	22 ± 5
AD-1633015.1	16 ± 4	58 ± 5	72 ± 7	AD-1658208.1	4 ± 0	26 ± 6	39 ± 6
AD-1633016.1	21 ± 5	90 ± 5	104 ± 13	AD-1658209.1	14 ± 3	51 ± 4	71 ± 4
AD-1633018.1	84 ± 5	118 ± 17	109 ± 7	AD-1658211.1	84 ± 18	105 ± 10	102 ± 18
AD-1633019.1	8 ± 1	28 ± 0	50 ± 5	AD-1658212.1	2 ± 0	4 ± 1	8 ± 3
AD-1633020.1	27 ± 5	70 ± 6	96 ± 13	AD-1658213.1	9 ± 2	28 ± 6	56 ± 6
AD-1633027.1	17 ± 2	46 ± 7	65 ± 12	AD-1658220.1	4 ± 1	6 ± 1	15 ± 4
AD-1633028.1	15 ± 2	49 ± 8	63 ± 7	AD-1658221.1	5 ± 1	28 ± 4	49 ± 7
AD-1633029.1	113 ± 4	120 ± 14	108 ± 15	AD-1658222.1	28 ± 2	47 ± 4	58 ± 9
AD-1633030.1	19 ± 3	70 ± 13	86 ± 7	AD-1658223.1	4 ± 1	25 ± 7	36 ± 3
AD-1633031.1	6 ± 3	14 ± 2	40 ± 7	AD-1658224.1	4 ± 1	4 ± 0	10 ± 2
AD-84731.2	8 ± 1	16 ± 1	28 ± 7	AD-1658225.1	3 ± 1	6 ± 1	12 ± 2
AD-1633032.1	6 ± 2	14 ± 4	29 ± 7	AD-1658226.1	4 ± 1	6 ± 1	19 ± 4
AD-1633033.1	6 ± 1	12 ± 2	29 ± 3	AD-1658227.1	3 ± 1	5 ± 1	10 ± 3
AD-1633034.1	82 ± 12	92 ± 11	95 ± 16	AD-1658228.1	42 ± 8	75 ± 17	95 ± 13
AD-1684516.1	42 ± 11	83 ± 18	77 ± 23	AD-1684517.1	10 ± 2	30 ± 9	70 ± 6
AD-1684518.1	80 ± 10	107 ± 22	108 ± 20	AD-1684519.1	100 ± 7	107 ± 21	111 ± 21
AD-1684520.1	24 ± 6	75 ± 13	87 ± 8	AD-1684521.1	6 ± 1	15 ± 5	21 ± 2

TABLE 14-continued

In vitro activity for various designs targeting Agt in Hep3B							
Parent				D1			
Duplex ID	10 nM	1 nM	0.1 nM	Duplex ID	10 nM	1 nM	0.1 nM
AD-1633048.1	46 ± 9	82 ± 8	98 ± 10	AD-1658242.1	20 ± 6	43 ± 6	65 ± 10
AD-1633049.1	10 ± 2	28 ± 6	39 ± 10	AD-1658243.1	9 ± 1	19 ± 2	45 ± 4
AD-1633094.1	19 ± 7	31 ± 3	56 ± 7	AD-1658288.1	24 ± 4	53 ± 10	63 ± 13
AD-1633095.1	48 ± 15	63 ± 13	66 ± 11	AD-1658289.1	16 ± 5	36 ± 7	55 ± 7
AD-1633119.1	43 ± 8	72 ± 17	96 ± 16	AD-1658313.1	25 ± 3	49 ± 8	54 ± 5
AD-1633121.1	18 ± 5	53 ± 10	74 ± 13	AD-1658315.1	3 ± 1	11 ± 2	34 ± 9
AD-1633122.1	66 ± 18	85 ± 17	96 ± 15	AD-1658316.1	13 ± 4	40 ± 8	63 ± 8
AD-1633254.2	11 ± 4	21 ± 6	32 ± 5	AD-1658448.2	7 ± 1	16 ± 4	21 ± 9
AD-1633257.2	10 ± 2	27 ± 4	39 ± 7	AD-1658451.2	9 ± 1	23 ± 3	28 ± 3
AD-1633269.2	10 ± 0	18 ± 4	22 ± 2	AD-1658463.2	7 ± 1	9 ± 2	11 ± 1
AD-1633270.2	47 ± 3	68 ± 2	75 ± 12	AD-1658464.2	48 ± 4	69 ± 8	81 ± 13
AD-1633271.2	12 ± 3	29 ± 8	44 ± 10	AD-1658465.2	8 ± 3	17 ± 3	22 ± 4
AD-1633272.2	27 ± 6	35 ± 6	57 ± 4	AD-1658466.2	20 ± 3	27 ± 4	43 ± 7
AD-1633273.2	76 ± 2	78 ± 2	98 ± 17	AD-1658467.2	83 ± 6	76 ± 11	101 ± 11
AD-1633290.2	42 ± 4	68 ± 11	75 ± 9	AD-1658484.2	5 ± 1	16 ± 6	28 ± 3
AD-1633291.2	30 ± 5	50 ± 13	64 ± 11	AD-1658485.2	73 ± 15	97 ± 8	93 ± 20
AD-1633324.2	6 ± 2	14 ± 3	22 ± 5	AD-1658519.2	2 ± 0	4 ± 2	4 ± 1
AD-84739.5	2 ± 0	3 ± 1	3 ± 1	AD-1658520.2	2 ± 0	4 ± 1	8 ± 1
AD-1633325.2	2 ± 0	3 ± 1	5 ± 2	AD-1658521.2	1 ± 1	4 ± 1	5 ± 1
AD-1633326.2	3 ± 1	7 ± 1	17 ± 1	AD-1658522.2	1 ± 0	2 ± 0	3 ± 1
AD-1633327.2	20 ± 4	38 ± 5	49 ± 3	AD-1658523.2	12 ± 1	29 ± 6	58 ± 3
AD-1633328.2	7 ± 1	16 ± 2	27 ± 3	AD-1658524.2	4 ± 1	11 ± 1	28 ± 2
AD-1633329.2	7 ± 1	21 ± 4	44 ± 7	AD-1658525.2	2 ± 1	6 ± 1	12 ± 4
AD-1633330.2	14 ± 3	34 ± 2	58 ± 1	AD-1658526.2	6 ± 1	24 ± 7	33 ± 6
AD-1633331.2	112 ± 5	115 ± 6	123 ± 19	AD-1658527.2	31 ± 2	59 ± 5	80 ± 9
AD-1633332.2	12 ± 3	40 ± 8	61 ± 6	AD-1658528.2	8 ± 2	25 ± 3	33 ± 3
AD-1633333.2	2 ± 0	8 ± 2	15 ± 1	AD-1658529.2	3 ± 0	6 ± 1	12 ± 2
AD-1633334.2	9 ± 1	24 ± 4	50 ± 3	AD-1658530.2	6 ± 1	11 ± 2	18 ± 4
AD-1633335.2	105 ± 18	127 ± 10	140 ± 12	AD-1658531.2	27 ± 2	50 ± 5	73 ± 8
AD-1633343.2	63 ± 5	87 ± 7	100 ± 14	AD-1658539.2	32 ± 4	44 ± 5	69 ± 4
AD-1633345.2	29 ± 3	56 ± 5	87 ± 11	AD-1658541.2	13 ± 3	27 ± 3	49 ± 5
AD-1633346.2	5 ± 1	13 ± 3	27 ± 6	AD-1658542.2	5 ± 1	16 ± 5	20 ± 4
AD-1633409.2	4 ± 0	12 ± 4	21 ± 4	AD-1658605.2	4 ± 1	13 ± 5	23 ± 1
AD-1633453.2	67 ± 7	81 ± 13	102 ± 6	AD-1658650.2	25 ± 1	53 ± 3	82 ± 6
AD-1633464.2	14 ± 1	38 ± 4	67 ± 3	AD-1658661.2	5 ± 1	14 ± 2	27 ± 3
AD-1633465.2	5 ± 1	15 ± 3	39 ± 3	AD-1658662.2	4 ± 3	10 ± 6	12 ± 2
AD-1633466.2	11 ± 2	34 ± 10	61 ± 5	AD-1658663.2	3 ± 1	10 ± 1	19 ± 0
AD-1633467.2	6 ± 2	22 ± 1	39 ± 6	AD-1658664.2	3 ± 1	8 ± 2	13 ± 2
AD-1633468.2	85 ± 8	104 ± 9	114 ± 3	AD-1658665.2	52 ± 6	67 ± 5	89 ± 7
AD-1633604.2	82 ± 13	103 ± 14	118 ± 17	AD-1658801.2	51 ± 6	75 ± 6	106 ± 12
AD-1633621.2	4 ± 1	8 ± 1	21 ± 2	AD-1658818.2	3 ± 1	7 ± 2	16 ± 2
AD-1633622.2	24 ± 5	53 ± 5	84 ± 10	AD-1658819.2	25 ± 4	48 ± 13	75 ± 7
AD-1633623.2	46 ± 4	75 ± 9	118 ± 6	AD-1658820.2	10 ± 2	35 ± 4	58 ± 6
AD-1633627.2	6 ± 1	20 ± 4	34 ± 1	AD-1658824.2	4 ± 1	8 ± 2	19 ± 3
AD-1633628.2	4 ± 1	11 ± 2	19 ± 2	AD-1658825.2	2 ± 1	7 ± 1	14 ± 3
AD-1633630.2	41 ± 7	56 ± 5	67 ± 15	AD-1658827.2	15 ± 2	20 ± 5	53 ± 9
AD-1633631.2	5 ± 1	11 ± 2	22 ± 4	AD-1658828.2	3 ± 1	6 ± 1	10 ± 2
AD-1633632.2	3 ± 1	8 ± 1	17 ± 2	AD-1658829.2	3 ± 1	9 ± 3	17 ± 5
AD-1633633.1	4 ± 1	9 ± 1	33 ± 9	AD-1658830.1	3 ± 1	9 ± 2	20 ± 6
AD-1633634.1	3 ± 1	5 ± 1	19 ± 4	AD-1658831.1	1 ± 0	2 ± 0	6 ± 2
AD-1633635.1	20 ± 3	36 ± 6	71 ± 8	AD-1658832.1	4 ± 1	10 ± 2	29 ± 4
AD-1633636.1	1 ± 0	2 ± 0	6 ± 1	AD-1658833.1	1 ± 0	1 ± 0	4 ± 1
AD-1633637.1	9 ± 1	25 ± 7	54 ± 17	AD-1658834.1	2 ± 1	2 ± 0	5 ± 2
AD-1633638.1	24 ± 5	36 ± 9	71 ± 10	AD-1658835.1	7 ± 1	13 ± 1	30 ± 6
AD-1633639.1	1 ± 0	2 ± 0	5 ± 1	AD-1658836.1	3 ± 1	5 ± 2	12 ± 4
AD-1633640.1	4 ± 1	6 ± 1	17 ± 1	AD-1658837.1	3 ± 0	5 ± 1	18 ± 5
AD-1633641.1	7 ± 3	17 ± 3	31 ± 8	AD-1658838.1	2 ± 0	3 ± 1	8 ± 1
AD-1633642.1	39 ± 7	65 ± 11	71 ± 12	AD-1658839.1	2 ± 1	4 ± 1	10 ± 2
AD-1633643.1	23 ± 1	39 ± 9	70 ± 14	AD-1658840.1	8 ± 2	10 ± 2	34 ± 8
AD-1633644.1	3 ± 0	8 ± 2	25 ± 4	AD-1658841.1	2 ± 1	8 ± 1	16 ± 4
AD-1633645.1	9 ± 2	20 ± 3	37 ± 5	AD-1658842.1	2 ± 1	3 ± 1	8 ± 2
AD-1633646.1	42 ± 6	48 ± 8	55 ± 8	AD-1658843.1	19 ± 5	28 ± 3	48 ± 4
AD-1633647.1	87 ± 9	78 ± 10	89 ± 9	AD-1658844.1	66 ± 12	81 ± 11	93 ± 8
AD-1633648.1	8 ± 3	13 ± 3	32 ± 4	AD-1658845.1	2 ± 0	5 ± 0	14 ± 1
AD-1633649.1	27 ± 3	48 ± 8	87 ± 3	AD-1658846.1	2 ± 1	7 ± 3	19 ± 5
AD-1633650.1	31 ± 8	54 ± 7	82 ± 8	AD-1658847.1	17 ± 8	30 ± 5	53 ± 4
AD-1633651.1	44 ± 6	63 ± 4	75 ± 8	AD-1658848.1	19 ± 3	32 ± 5	53 ± 4
AD-1633652.1	10 ± 1	18 ± 5	28 ± 3	AD-1658849.1	17 ± 3	25 ± 4	46 ± 6
AD-84707.2	2 ± 1	5 ± 2	12 ± 2	AD-1321390.2	2 ± 0	3 ± 1	8 ± 2
AD-1633653.1	6 ± 1	15 ± 3	28 ± 7	AD-1658850.1	5 ± 1	10 ± 3	16 ± 4
AD-1633678.1	113 ± 6	99 ± 13	96 ± 9	AD-1658875.1	67 ± 7	103 ± 12	95 ± 2

TABLE 14-continued

In vitro activity for various designs targeting Agt in Hep3B							
Parent				D1			
Duplex ID	10 nM	1 nM	0.1 nM	Duplex ID	10 nM	1 nM	0.1 nM
AD-1633683.1	114 ± 14	118 ± 4	110 ± 8	AD-1658880.1	88 ± 8	96 ± 11	98 ± 10
AD-1633732.1	1 ± 0	3 ± 1	5 ± 2	AD-1658929.1	2 ± 1	5 ± 1	6 ± 2
AD-1633733.1	6 ± 2	10 ± 2	20 ± 2	AD-1658930.1	4 ± 1	8 ± 3	15 ± 2
AD-1633734.1	20 ± 3	32 ± 4	51 ± 6	AD-1658931.1	2 ± 0	3 ± 0	5 ± 2
AD-1633735.1	12 ± 4	21 ± 1	38 ± 6	AD-1658932.1	3 ± 1	9 ± 1	21 ± 5
AD-1633736.1	5 ± 1	8 ± 1	20 ± 2	AD-1658933.1	1 ± 0	2 ± 1	4 ± 1
AD-1633737.1	27 ± 3	41 ± 2	63 ± 5	AD-1658934.1	3 ± 1	8 ± 2	19 ± 3
AD-1633738.1	36 ± 3	47 ± 2	66 ± 3	AD-1658935.1	9 ± 2	15 ± 2	31 ± 5
AD-1633739.1	2 ± 1	4 ± 1	8 ± 2	AD-1658936.1	1 ± 0	2 ± 1	4 ± 1
AD-1633740.1	1 ± 0	2 ± 0	4 ± 1	AD-1658937.1	2 ± 0	2 ± 0	3 ± 1
AD-1633741.1	5 ± 1	10 ± 2	22 ± 2	AD-1658938.1	2 ± 0	4 ± 0	6 ± 1
AD-1633742.1	12 ± 1	22 ± 2	37 ± 8	AD-1658939.1	5 ± 1	8 ± 1	16 ± 1
AD-1633743.1	5 ± 2	6 ± 2	14 ± 3	AD-1658940.1	4 ± 1	6 ± 1	8 ± 2
AD-1633759.1	1 ± 0	2 ± 1	4 ± 1	AD-1658954.1	1 ± 0	2 ± 1	3 ± 2
AD-1684522.1	4 ± 2	7 ± 2	13 ± 3	AD-1658955.1	3 ± 1	6 ± 1	9 ± 2
AD-1684523.1	3 ± 1	6 ± 1	7 ± 2	AD-1658956.1	3 ± 1	5 ± 1	7 ± 1
AD-1684524.1	2 ± 1	3 ± 1	8 ± 2	AD-1658957.1	2 ± 0	3 ± 1	5 ± 1
AD-1684525.1	15 ± 5	18 ± 4	34 ± 5	AD-1658958.1	2 ± 1	4 ± 2	5 ± 3
AD-1684526.1	2 ± 1	3 ± 1	7 ± 1	AD-1658959.1	1 ± 0	1 ± 0	2 ± 1
AD-1684527.1	1 ± 1	3 ± 1	4 ± 1	AD-1658960.1	1 ± 1	2 ± 1	3 ± 1
AD-1633777.1	14 ± 3	18 ± 3	38 ± 5	AD-1658992.1	2 ± 1	5 ± 1	16 ± 2
AD-1633779.1	11 ± 4	27 ± 7	41 ± 7	AD-1658994.1	3 ± 1	6 ± 1	19 ± 4
AD-1633780.1	30 ± 5	52 ± 8	61 ± 11	AD-1658995.1	51 ± 6	52 ± 7	55 ± 9
AD-1633840.1	66 ± 13	91 ± 7	99 ± 13	AD-1659055.1	7 ± 1	19 ± 3	25 ± 3
AD-1633841.1	96 ± 13	99 ± 25	88 ± 9	AD-1659056.1	93 ± 10	103 ± 9	92 ± 11
AD-1633842.1	4 ± 2	22 ± 6	49 ± 10	AD-1659057.1	2 ± 1	13 ± 3	31 ± 5
AD-1633843.1	6 ± 1	36 ± 2	55 ± 13	AD-1659058.1	3 ± 1	12 ± 2	37 ± 7
AD-1633844.1	22 ± 3	47 ± 10	81 ± 9	AD-1659059.1	5 ± 1	27 ± 6	54 ± 8
AD-1633845.1	28 ± 4	83 ± 9	71 ± 7	AD-1659060.1	14 ± 2	41 ± 5	65 ± 13
AD-1633846.1	10 ± 2	32 ± 6	58 ± 14	AD-1659061.1	1 ± 0	12 ± 3	22 ± 8
AD-84712.2	2 ± 0	8 ± 1	17 ± 4	AD-1659062.1	1 ± 0	6 ± 2	12 ± 3
AD-1633847.1	14 ± 3	69 ± 10	86 ± 5	AD-1659063.1	5 ± 1	28 ± 8	54 ± 4
AD-1633848.1	12 ± 4	40 ± 2	80 ± 20	AD-1659064.1	3 ± 1	22 ± 6	31 ± 1
AD-1633849.1	8 ± 1	34 ± 3	64 ± 6	AD-1659065.1	6 ± 2	21 ± 5	50 ± 12
AD-1633850.1	2 ± 0	12 ± 3	23 ± 1	AD-1659066.1	2 ± 1	9 ± 1	27 ± 2
AD-1633851.1	6 ± 2	37 ± 5	52 ± 7	AD-1659067.1	3 ± 0	24 ± 5	46 ± 8
AD-1633852.1	8 ± 2	39 ± 5	62 ± 8	AD-1659068.1	7 ± 1	20 ± 6	39 ± 4
AD-1633853.1	1 ± 0	7 ± 2	17 ± 4	AD-1659069.1	1 ± 0	3 ± 1	14 ± 3
AD-1633854.1	22 ± 3	59 ± 9	85 ± 9	AD-1659070.1	7 ± 2	27 ± 4	52 ± 12
AD-1633855.1	21 ± 3	82 ± 14	91 ± 12	AD-1659071.1	2 ± 0	13 ± 3	35 ± 5
AD-1684528.1	7 ± 2	30 ± 5	61 ± 8	AD-1684529.1	3 ± 0	12 ± 3	34 ± 7
AD-1684530.1	3 ± 1	18 ± 1	45 ± 4	AD-1684531.1	2 ± 0	9 ± 1	33 ± 5
AD-1684532.1	7 ± 1	28 ± 3	44 ± 5	AD-1684533.1	3 ± 1	19 ± 8	47 ± 11
AD-1684534.1	3 ± 1	15 ± 1	30 ± 3	AD-1684535.1	4 ± 3	4 ± 0	12 ± 2
AD-1633946.1	3 ± 1	10 ± 3	22 ± 3	AD-1659162.1	1 ± 0	6 ± 2	19 ± 4
AD-67328.2	1 ± 0	7 ± 1	20 ± 3	AD-1321384.2	3 ± 0	9 ± 1	22 ± 4
AD-1633947.1	3 ± 1	10 ± 2	23 ± 1	AD-1659163.1	2 ± 1	10 ± 1	22 ± 3
AD-1633948.1	2 ± 1	17 ± 9	32 ± 5	AD-1659164.1	1 ± 0	4 ± 1	11 ± 3
AD-1633949.1	3 ± 1	21 ± 1	51 ± 8	AD-1659165.1	3 ± 0	12 ± 2	23 ± 2
AD-84700.2	11 ± 2	37 ± 3	66 ± 4	AD-1659166.1	2 ± 0	12 ± 1	25 ± 4
AD-1633950.1	1 ± 0	9 ± 2	29 ± 5	AD-1659167.1	1 ± 0	6 ± 2	18 ± 2
AD-1633951.1	5 ± 1	29 ± 6	60 ± 10	AD-1659168.1	4 ± 1	15 ± 1	33 ± 7
AD-1633991.1	3 ± 1	19 ± 3	46 ± 4	AD-1659208.1	2 ± 0	11 ± 2	26 ± 4
AD-1633992.1	14 ± 1	45 ± 5	77 ± 9	AD-1659209.1	3 ± 1	8 ± 2	23 ± 5
AD-1633993.1	27 ± 5	57 ± 10	73 ± 3	AD-1659210.1	11 ± 1	43 ± 5	70 ± 8
AD-1634065.1	13 ± 3	48 ± 6	79 ± 1	AD-1659282.1	4 ± 1	18 ± 4	44 ± 6
AD-1634066.1	45 ± 5	77 ± 9	98 ± 12	AD-1659283.1	30 ± 2	69 ± 14	80 ± 11
AD-1634067.1	25 ± 5	68 ± 6	87 ± 11	AD-1659284.1	24 ± 4	60 ± 2	71 ± 3
AD-1634068.1	10 ± 2	38 ± 4	58 ± 3	AD-1659285.1	11 ± 1	38 ± 2	64 ± 5
AD-84730.2	7 ± 1	19 ± 4	30 ± 8	AD-1659286.1	5 ± 1	21 ± 7	41 ± 6
AD-1634069.1	5 ± 1	19 ± 4	34 ± 6	AD-1659287.1	2 ± 1	12 ± 2	22 ± 5
AD-1634070.1	2 ± 1	12 ± 4	24 ± 4	AD-1659288.1	3 ± 1	13 ± 3	22 ± 2
AD-1634071.1	29 ± 3	65 ± 6	98 ± 16	AD-1659289.1	24 ± 6	55 ± 8	79 ± 12
AD-1684536.1	54 ± 6	83 ± 5	84 ± 5	AD-1684537.1	75 ± 4	105 ± 5	109 ± 19
AD-1684538.1	64 ± 6	103 ± 6	102 ± 20	AD-1684539.1	61 ± 5	105 ± 8	83 ± 8
AD-1684540.1	73 ± 5	65 ± 4	87 ± 9	AD-1684541.1	57 ± 9	67 ± 10	96 ± 15
AD-1684542.1	89 ± 22	65 ± 5	97 ± 9	AD-1684543.1	85 ± 10	78 ± 1	104 ± 19
AD-1684544.1	98 ± 15	87 ± 8	117 ± 19	AD-1684545.1	56 ± 9	56 ± 6	94 ± 6
AD-1684546.1	79 ± 9	84 ± 9	94 ± 12	AD-1684547.1	23 ± 3	45 ± 3	64 ± 8
AD-1684548.1	58 ± 9	62 ± 12	86 ± 18	AD-1684549.1	16 ± 1	44 ± 8	57 ± 8
AD-1684550.1	47 ± 7	63 ± 15	100 ± 17	AD-1684551.1	37 ± 3	50 ± 7	80 ± 3

TABLE 14-continued

In vitro activity for various designs targeting Agt in Hep3B							
Parent				D1			
Duplex ID	10 nM	1 nM	0.1 nM	Duplex ID	10 nM	1 nM	0.1 nM
AD-1684552.1	55 ± 5	74 ± 14	112 ± 20	AD-1684553.1	31 ± 5	54 ± 7	76 ± 9
AD-1684554.1	75 ± 8	98 ± 17	107 ± 18	AD-1684555.1	25 ± 2	50 ± 9	53 ± 5
AD-1684556.1	76 ± 10	82 ± 12	115 ± 26	AD-1684557.1	26 ± 7	45 ± 8	75 ± 20
AD-1684558.1	57 ± 9	65 ± 7	101 ± 14	AD-1684559.1	21 ± 6	41 ± 5	69 ± 6
AD-1634072.1	8 ± 1	19 ± 3	32 ± 8	AD-1684560.1	6 ± 0	15 ± 3	25 ± 4
AD-1634073.1	5 ± 1	13 ± 3	18 ± 3	AD-1684561.1	3 ± 1	10 ± 1	11 ± 3
AD-1634074.1	3 ± 0	8 ± 1	15 ± 3	AD-1659290.1	3 ± 1	9 ± 1	15 ± 1
AD-1634075.1	5 ± 1	15 ± 2	32 ± 9	AD-1659291.1	5 ± 1	15 ± 2	19 ± 3
AD-1634076.1	11 ± 2	36 ± 5	45 ± 4	AD-1659292.1	9 ± 1	25 ± 5	37 ± 9
AD-1634077.1	12 ± 2	30 ± 5	59 ± 8	AD-1659293.1	3 ± 2	10 ± 2	16 ± 1
AD-1634078.1	11 ± 3	29 ± 7	55 ± 8	AD-1659294.1	5 ± 1	13 ± 3	24 ± 4
AD-1634079.1	3 ± 1	10 ± 3	21 ± 4	AD-1659295.1	2 ± 0	4 ± 0	11 ± 1
AD-1634080.1	10 ± 3	34 ± 2	48 ± 7	AD-1659296.1	6 ± 2	15 ± 2	35 ± 7
AD-1634081.1	5 ± 1	20 ± 2	38 ± 8	AD-1659297.1	14 ± 2	26 ± 2	52 ± 7
AD-1634082.1	27 ± 3	60 ± 4	79 ± 11	AD-1659298.1	17 ± 3	34 ± 6	60 ± 10
AD-1634105.1	2 ± 0	4 ± 1	12 ± 2	AD-1659321.1	2 ± 1	5 ± 0	15 ± 2
AD-1634107.1	6 ± 2	21 ± 2	37 ± 9	AD-1659323.1	6 ± 1	13 ± 3	25 ± 2
AD-1634109.1	17 ± 2	40 ± 5	66 ± 8	AD-1659325.1	7 ± 2	17 ± 3	39 ± 6
AD-1634110.1	9 ± 2	26 ± 5	42 ± 2	AD-1659326.1	6 ± 1	19 ± 1	41 ± 6
AD-1634111.1	5 ± 2	11 ± 3	25 ± 3	AD-1659327.1	3 ± 1	14 ± 1	18 ± 2
AD-1634112.1	10 ± 1	32 ± 2	50 ± 9	AD-1659328.1	3 ± 1	10 ± 1	24 ± 6
AD-1634113.1	7 ± 2	17 ± 3	37 ± 4	AD-1659329.1	4 ± 1	15 ± 1	29 ± 6
AD-1634114.1	17 ± 5	33 ± 1	68 ± 8	AD-1659330.1	2 ± 1	5 ± 1	11 ± 2
AD-1634115.1	35 ± 5	63 ± 11	110 ± 7	AD-1659331.1	7 ± 1	24 ± 4	43 ± 7
AD-84724.2	24 ± 2	45 ± 6	61 ± 8	AD-1659332.1	5 ± 1	11 ± 1	24 ± 2
AD-1634116.1	5 ± 2	15 ± 3	33 ± 5	AD-1659333.1	3 ± 2	9 ± 1	17 ± 3
AD-1634117.1	9 ± 1	32 ± 3	45 ± 6	AD-1659334.1	10 ± 3	29 ± 1	42 ± 6
AD-1634118.1	25 ± 4	46 ± 5	79 ± 4	AD-1659335.1	6 ± 1	20 ± 3	31 ± 4
AD-68579.2	17 ± 5	41 ± 5	69 ± 6	AD-1659336.1	3 ± 1	10 ± 2	17 ± 3
AD-1634119.1	4 ± 1	14 ± 1	33 ± 4	AD-1659337.1	3 ± 0	10 ± 2	21 ± 2
AD-1634120.1	11 ± 3	33 ± 6	80 ± 5	AD-1659338.1	3 ± 1	8 ± 3	20 ± 1
AD-1634121.1	8 ± 2	22 ± 6	44 ± 4	AD-1659339.1	2 ± 1	6 ± 1	18 ± 3
AD-1634122.1	18 ± 13	29 ± 2	60 ± 7	AD-1659340.1	2 ± 0	15 ± 3	24 ± 4
AD-1634123.1	4 ± 1	14 ± 3	41 ± 1	AD-1659341.1	6 ± 1	23 ± 3	46 ± 8
AD-1634124.1	5 ± 1	17 ± 6	32 ± 5	AD-1659342.1	3 ± 0	12 ± 2	19 ± 3
AD-1634125.1	4 ± 1	11 ± 2	22 ± 2	AD-1659343.1	4 ± 1	10 ± 3	26 ± 5
AD-1634126.1	6 ± 1	16 ± 4	30 ± 4	AD-1659344.1	2 ± 0	8 ± 1	16 ± 3
AD-1634127.1	3 ± 0	7 ± 2	16 ± 1	AD-1659345.1	3 ± 0	6 ± 1	12 ± 2
AD-1634128.1	1 ± 0	1 ± 1	3 ± 2	AD-1659346.1	1 ± 0	2 ± 1	5 ± 1
AD-1634129.1	2 ± 1	6 ± 3	15 ± 2	AD-1659347.1	1 ± 0	1 ± 0	5 ± 2
AD-1634130.1	4 ± 1	13 ± 1	24 ± 4	AD-1659348.1	1 ± 1	1 ± 0	3 ± 1
AD-1684562.1	16 ± 4	33 ± 3	54 ± 7	AD-1659349.1	9 ± 2	23 ± 2	40 ± 1
AD-1684563.1	1 ± 0	1 ± 1	3 ± 0	AD-1659350.1	1 ± 0	1 ± 1	2 ± 1
AD-1684564.1	1 ± 0	2 ± 1	2 ± 1	AD-1659351.1	1 ± 0	1 ± 0	1 ± 0
AD-1634135.1	2 ± 1	7 ± 1	13 ± 4	AD-1659371.1	1 ± 0	2 ± 1	4 ± 1
AD-1634136.1	2 ± 1	7 ± 3	12 ± 1	AD-1659372.1	3 ± 1	9 ± 1	21 ± 4
AD-1634137.1	1 ± 0	2 ± 0	2 ± 1	AD-1659373.1	1 ± 1	2 ± 1	3 ± 1
AD-1634146.1	1 ± 0	2 ± 0	3 ± 1	AD-1659382.1	1 ± 0	1 ± 0	1 ± 0
AD-1634147.1	1 ± 0	4 ± 1	10 ± 3	AD-1659383.1	1 ± 1	2 ± 0	2 ± 0
AD-68585.2	5 ± 2	20 ± 2	29 ± 4	AD-1659384.1	2 ± 1	7 ± 1	11 ± 5
AD-1634148.1	2 ± 0	5 ± 1	9 ± 2	AD-1659385.1	1 ± 0	1 ± 0	1 ± 0
AD-1634149.1	1 ± 0	1 ± 0	3 ± 1	AD-1659386.1	1 ± 0	1 ± 0	3 ± 0
AD-1634150.1	1 ± 0	1 ± 0	4 ± 1	AD-1659387.1	1 ± 0	1 ± 0	2 ± 1
AD-1634151.1	0 ± 0	1 ± 0	2 ± 0	AD-1659388.1	0 ± 0	1 ± 0	1 ± 0
AD-1634152.1	1 ± 0	1 ± 0	2 ± 1	AD-1659389.1	1 ± 0	1 ± 0	2 ± 0
AD-1634153.1	2 ± 1	2 ± 1	5 ± 1	AD-1659390.1	1 ± 0	2 ± 0	5 ± 2
AD-1634162.1	3 ± 0	5 ± 2	12 ± 2	AD-1659399.1	3 ± 1	5 ± 1	8 ± 1
AD-1634163.1	4 ± 0	11 ± 2	18 ± 3	AD-1659400.1	9 ± 2	18 ± 5	30 ± 3
AD-1634164.1	29 ± 6	35 ± 3	48 ± 7	AD-1659401.1	26 ± 1	23 ± 3	33 ± 4
AD-1634165.1	19 ± 3	19 ± 2	30 ± 5	AD-1659402.1	20 ± 3	20 ± 5	24 ± 3
AD-1634169.1	65 ± 4	70 ± 7	79 ± 5	AD-1659406.1	63 ± 18	59 ± 11	53 ± 6
AD-1634170.1	84 ± 13	74 ± 9	72 ± 1	AD-1659407.1	58 ± 5	69 ± 8	68 ± 6
AD-1634171.1	81 ± 15	75 ± 8	85 ± 7	AD-1659408.1	86 ± 25	91 ± 9	81 ± 5
AD-1634172.1	50 ± 13	53 ± 7	67 ± 8	AD-1659409.1	58 ± 4	68 ± 6	65 ± 7
AD-1634173.1	60 ± 5	61 ± 5	56 ± 7	AD-1659410.1	49 ± 6	49 ± 2	52 ± 7
AD-1634174.1	69 ± 2	73 ± 5	77 ± 9	AD-1659411.1	36 ± 3	37 ± 4	39 ± 2
AD-1634175.1	61 ± 9	70 ± 9	81 ± 8	AD-1659412.1	36 ± 4	41 ± 7	42 ± 5
AD-1634176.1	70 ± 5	82 ± 5	88 ± 7	AD-1659413.1	64 ± 9	68 ± 4	74 ± 6
AD-1634177.1	57 ± 12	65 ± 9	67 ± 4	AD-1659414.1	52 ± 10	67 ± 3	67 ± 3
AD-1634178.1	63 ± 8	76 ± 6	74 ± 4	AD-1659415.1	58 ± 5	76 ± 4	67 ± 9
AD-1634179.1	79 ± 11	83 ± 18	103 ± 9	AD-1659416.1	65 ± 6	86 ± 7	92 ± 7

TABLE 14-continued

In vitro activity for various designs targeting Agt in Hep3B							
Parent				D1			
Duplex ID	10 nM	1 nM	0.1 nM	Duplex ID	10 nM	1 nM	0.1 nM
AD-1634180.1	47 ± 7	67 ± 10	72 ± 14	AD-1659417.1	58 ± 11	74 ± 7	73 ± 15
AD-1634181.1	71 ± 11	79 ± 6	72 ± 16	AD-1659418.1	54 ± 9	69 ± 7	56 ± 13
AD-1634182.1	62 ± 5	77 ± 3	86 ± 4	AD-1659419.1	57 ± 17	68 ± 7	76 ± 7
AD-1634183.1	61 ± 11	70 ± 11	81 ± 11	AD-1659420.1	63 ± 8	79 ± 11	86 ± 5
AD-1634184.1	64 ± 12	74 ± 9	80 ± 16	AD-1659421.1	33 ± 3	56 ± 8	62 ± 10
AD-1634185.1	58 ± 9	71 ± 9	76 ± 12	AD-1659422.1	58 ± 7	72 ± 18	61 ± 6
AD-1634186.1	60 ± 9	86 ± 12	86 ± 10	AD-1659423.1	62 ± 11	81 ± 15	82 ± 9
AD-1634187.1	43 ± 10	51 ± 9	42 ± 8	AD-1659424.1	81 ± 9	92 ± 11	82 ± 27
AD-1634188.1	37 ± 5	53 ± 8	48 ± 10	AD-1659425.1	42 ± 8	52 ± 14	73 ± 4
AD-1634189.1	49 ± 5	70 ± 17	87 ± 8	AD-1659426.1	47 ± 9	51 ± 9	67 ± 5
AD-1634190.1	50 ± 9	56 ± 10	72 ± 5	AD-1659427.1	46 ± 6	59 ± 16	76 ± 9
AD-1634191.1	44 ± 3	56 ± 7	65 ± 7	AD-1659428.1	56 ± 6	50 ± 8	80 ± 15
AD-1634192.1	45 ± 10	49 ± 5	61 ± 5	AD-1659429.1	34 ± 5	42 ± 7	50 ± 5
AD-1634193.1	47 ± 10	42 ± 8	62 ± 10	AD-1659430.1	63 ± 6	47 ± 7	69 ± 11
AD-1634194.1	49 ± 2	53 ± 2	62 ± 7	AD-1659431.1	49 ± 8	50 ± 7	63 ± 11
AD-1634195.1	52 ± 6	59 ± 11	80 ± 7	AD-1659432.1	59 ± 5	73 ± 7	82 ± 13
AD-1634196.1	57 ± 7	49 ± 3	71 ± 11	AD-1659433.1	50 ± 12	43 ± 7	68 ± 11
AD-1634197.1	52 ± 3	62 ± 4	70 ± 11	AD-1659434.1	52 ± 5	63 ± 5	72 ± 8
AD-1634199.1	61 ± 8	51 ± 5	60 ± 13	AD-1659436.1	65 ± 5	53 ± 3	78 ± 8
AD-1634200.1	62 ± 8	50 ± 5	73 ± 15	AD-1659437.1	57 ± 4	61 ± 3	55 ± 5
AD-1634203.1	77 ± 3	63 ± 7	86 ± 6	AD-1659440.1	65 ± 4	54 ± 6	64 ± 13
AD-1634209.1	91 ± 15	74 ± 8	112 ± 17	AD-1659446.1	78 ± 8	69 ± 3	88 ± 12
AD-1634210.1	73 ± 6	65 ± 6	85 ± 9	AD-1659447.1	72 ± 5	68 ± 6	93 ± 10
AD-1634211.1	74 ± 13	69 ± 8	85 ± 8	AD-1659448.1	66 ± 12	50 ± 9	78 ± 9
AD-1634212.1	43 ± 3	45 ± 3	68 ± 8	AD-1659449.1	68 ± 3	61 ± 5	91 ± 9
AD-1634213.1	71 ± 2	77 ± 17	98 ± 21	AD-1659450.1	73 ± 2	61 ± 6	103 ± 10
AD-1634214.1	85 ± 7	59 ± 5	85 ± 15	AD-1659451.1	52 ± 9	49 ± 9	56 ± 7
AD-1634215.1	82 ± 5	79 ± 9	108 ± 9	AD-1659452.1	67 ± 8	59 ± 6	83 ± 6
AD-1634216.1	76 ± 9	70 ± 6	113 ± 7	AD-1659453.1	77 ± 10	64 ± 3	90 ± 12
AD-1634217.1	92 ± 6	83 ± 1	116 ± 15	AD-1659454.1	75 ± 4	70 ± 6	101 ± 6
AD-1684565.1	85 ± 7	104 ± 15	84 ± 14	AD-1659481.1	49 ± 5	49 ± 5	61 ± 5
AD-1684566.1	122 ± 27	74 ± 9	98 ± 5	AD-1659482.1	67 ± 6	51 ± 4	108 ± 15
AD-1684567.1	85 ± 10	74 ± 7	98 ± 7	AD-1659483.1	89 ± 2	77 ± 9	102 ± 5
AD-1684568.1	71 ± 6	82 ± 12	107 ± 14	AD-1659485.1	67 ± 4	60 ± 6	91 ± 11
AD-1684569.1	68 ± 2	80 ± 4	100 ± 5	AD-1659487.1	68 ± 7	88 ± 15	91 ± 13
AD-1684570.1	78 ± 7	83 ± 6	123 ± 12	AD-1659488.1	79 ± 7	87 ± 13	111 ± 9
AD-1634234.1	45 ± 8	73 ± 12	91 ± 5	AD-1659489.1	81 ± 5	75 ± 10	97 ± 7
AD-1634235.1	65 ± 9	72 ± 6	91 ± 4	AD-1659490.1	47 ± 2	68 ± 4	89 ± 15
AD-1634236.1	70 ± 8	75 ± 9	96 ± 15	AD-1659491.1	32 ± 6	39 ± 3	48 ± 5
AD-1634237.1	75 ± 7	86 ± 14	113 ± 11	AD-1659492.1	36 ± 7	40 ± 4	48 ± 4
AD-1634238.1	79 ± 13	91 ± 11	96 ± 12	AD-1659493.1	74 ± 3	78 ± 4	104 ± 5
AD-1634282.1	64 ± 7	58 ± 8	72 ± 6	AD-1659537.1	63 ± 9	51 ± 8	61 ± 6
AD-1634283.1	76 ± 8	66 ± 3	97 ± 18	AD-1659538.1	59 ± 2	55 ± 5	71 ± 4
AD-1634304.1	69 ± 7	68 ± 8	102 ± 16	AD-1659559.1	70 ± 7	61 ± 11	85 ± 14
AD-1634305.1	51 ± 3	44 ± 4	50 ± 12	AD-1659560.1	49 ± 3	59 ± 3	55 ± 4
AD-1634306.1	61 ± 5	65 ± 5	74 ± 4	AD-1659561.1	51 ± 5	39 ± 5	73 ± 5
AD-1634307.1	83 ± 7	80 ± 7	103 ± 2	AD-1659562.1	74 ± 5	66 ± 8	93 ± 7
AD-1634308.1	61 ± 4	73 ± 7	82 ± 7	AD-1659563.1	69 ± 3	72 ± 4	98 ± 8
AD-1634327.1	61 ± 8	64 ± 7	80 ± 15	AD-1659582.1	49 ± 9	52 ± 2	74 ± 10
AD-1634328.1	64 ± 6	64 ± 9	91 ± 4	AD-1659583.1	44 ± 3	56 ± 4	70 ± 5
AD-1634329.1	65 ± 3	64 ± 7	85 ± 5	AD-1659584.1	60 ± 5	61 ± 9	88 ± 11
AD-1634330.1	56 ± 2	50 ± 6	54 ± 11	AD-1659585.1	72 ± 2	65 ± 7	81 ± 8
AD-1634331.1	44 ± 10	31 ± 6	49 ± 9	AD-1659586.1	53 ± 7	51 ± 9	72 ± 7
AD-1634332.1	46 ± 10	49 ± 13	81 ± 3	AD-1659587.1	34 ± 6	57 ± 1	52 ± 17
AD-1634333.1	67 ± 10	59 ± 15	84 ± 13	AD-1659588.1	64 ± 3	62 ± 11	97 ± 13

In Vivo Mouse and Cyno Studies

[0441] All studies were conducted using protocols consistent with local, state and federal regulations as applicable and approved by the Institutional Animal Care and Use Committees (IACUCs) at Alnylam Pharmaceuticals.

[0442] In mouse pharmacodynamic studies, female C57BL/6 mice (Charles River Laboratories) were administered a single dose of a vehicle control (1× PBS or 0.9% sodium chloride) or siRNA subcutaneously in the upper back. Bleeds were collected by retro-orbital bleeding. Serum were separated by centrifuging at 13000 rpm at room temperature for 10 mins. Mouse livers were collected and immediately snap frozen in liquid nitrogen, and stored at -80° C. for mRNA and siRNA analysis.

[0443] As shown in FIGS. 2A-2D and 10A-10D, the results demonstrate an improved or similar target knock-down in mice and an improved or similar target knockdown and duration of silencing in cyno.

Serum Protein Quantification

[0444] TTR protein was quantified by ELISA from serum isolated from whole blood. ELISA was performed according to manufacturer protocol (ALPCO, 41-PALMS-EO1) after a 3025-fold dilution of the serum samples. Data were normalized to pre-bleed TTR levels. All samples were assayed in duplicate and each data point is the average of all the mice within each cohort (n=3).

[0445] In some embodiments, the dsRNA molecule is not a dsRNA molecule listed in any one of Tables 15-25.

TABLE 15

Exemplary dsRNA molecules					
Duplex ID	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:	
AD-1353484.1	asasaag (Ahd) gaAfAfGfuguuuuauasasa	2141	VPusdTsaudAadAacacdTuUfcucuuuuuscscu	2166	
AD-1353468.1	asasaga (Chd) ugAfUfAfcagaaacgasusa	2142	VPusdAsucdGudTcugudAuCfagucuuuscsc	2167	
AD-1353485.1	asasagagaaAfGfUfguuu (Uhd) auasusa	2143	VPusdAsuadTadAaacadCuUfucucuuuusc	2168	
AD-1353483.1	asascag (Uhd) gcUfAfAfuguuauugsgsa	2144	VPusdCscadAudAacaudTaGfcacuguuugsg	2169	
AD-1353486.1	asasgag (Ahd) aaGfUfGfuuuuauasasa	2145	VPusdTsaudAudAaacadAcUfuucucuuususu	2170	
AD-1353479.1	asasuuggauUfCfGfccau (Uhd) uuasusa	2146	VPusdAsuadAadAuggdGafuccaaauuscsc	2171	
AD-1353473.1	ascsaga (Ahd) caGfUfCfcuaauccsasa	2147	VPusdTsggdAudTaaggdAcUfgucuguscscg	2172	
AD-1353467.1	ascscaggaaAfGfAfcuga (Uhd) acasgsa	2148	VPusdCsugdTadTcagudCuUfuccuggusgsc	2173	
AD-1353482.1	asgsauu (Ahd) gaGfAfGfuuuuauuuscscsa	2149	VPusdGsaadAudAaacadTcUfcuaauucusc	2174	
AD-1353480.1	asusugg (Ahd) uuCfGfCfcuuuuuauasusa	2150	VPusdAsaudAadAauggdCgAfaucuaaususc	2175	
AD-1353463.1	csasuca (Chd) caUfGfCfagauuauugscsa	2151	VPusdGscadTadAucugdCaUfggugaugsusu	2176	
AD-1353478.1	cscsucu (Uhd) ggAfAfUfuggauucgscsa	2152	VPusdGscgdAadTccaadTuCfcaaggaggsgsc	2177	
AD-1353466.1	csusacagcaCfAfAfcuaa (Uhd) gugsasa	2153	VPusdTscadCadTuugudTgUfgcuguagsgsg	2178	
AD-1353488.1	gsasaag (Uhd) guUfUfUfaauauacggsusa	2154	VPusdAsccdGudAuauadAaAfcacuuucusc	2179	
AD-1353487.1	gsasgaa (Ahd) guGfUfUfuauauacggsa	2155	VPusdCsgudAudAuaaadAcAfcuuucucscsu	2180	
AD-1353481.1	gsasuucgccAfUfUfuuuu (Uhd) uuuscscsa	2156	VPusdGsaadAadAuaaadAuGfgcgaaucscscg	2181	
AD-1353477.1	gsusccu (Uhd) aaUfCfCfagaaaccusgsa	2157	VPusdCsagdGudTucugdGaUfuaaggacsusg	2182	
AD-1353489.1	gsusguu (Uhd) uaUfAfUfacgguacuscusa	2158	VPusdAsagdTadCcguadTaUfaaaacacsusu	2183	
AD-1353491.1	usasgac (Ahd) uuGfCfUfaucuguuuscusa	2159	VPusdAsaadCadGaaudGcAfaugcuuasusu	2184	
AD-1397058	csusacagcaCfAfAfcuaa (Uhd) gugaaL96	2160	VPusdTscadCadTuugudTgUfgcuguagsgsg	2178	
AD-1397059	asasaga (Chd) ugAfUfAfcagaaacgaauL96	2161	VPusdAsucdGudTcugudAuCfagucuuuscsc	2167	
AD-1397064	asusugg (Ahd) uuCfGfCfcuuuuuauaL96	2162	VPusdAsaudAadAauggdCgAfaucuaaususc	2175	
AD-1397065	gsasuucgccAfUfUfuuuu (Uhd) uuucalL96	2163	VPusdGsaadAadAuaaadAuGfgcgaaucscscg	2181	
AD-1397066	gsasgaa (Ahd) guGfUfUfuauauacgaL96	2164	VPusdCsgudAudAuaaadAcAfcuuucucscsu	2180	
AD-1397067	gsusguu (Uhd) uaUfAfUfacgguacuuaL96	2165	VPusdAsagdTadCcguadTaUfaaaacacsusu	2183	

TABLE 16

Exemplary dsRNA Molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID NO	Antisense sequence (5'→3')	SEQ ID NO:
AD-1397070.1	ascsgug (Ahd) ccCfAfAfgcucgcaugaL96	2185	VPusdCsaudGcdGagcudTgGfgucacgusgsa	2290
AD-1397073.1	usgsacc (Chd) aaGfCfUfcgcauggucaL96	2186	VPusdGsacdCadTgcgadGcUfugggucascsg	2291
AD-1397075.1	ascseca (Ahd) gcUfCfGfcauggucagaL96	2187	VPusdCsugdAcdCaugcdGaGfucuggguscsa	2292
AD-1397090.1	usgsgca (Ghd) caAfCfAfaaggauuugaL96	2188	VPusdCsaadAudCcuudGuUfgcugccascsu	2293
AD-1397091.1	gsgscag (Chd) aaCfAfAfaaggauuugaL96	2189	VPusdTscadAadTccuudTgUfugcugccsasc	2294
AD-1397094.1	asascaa (Ahd) ggAfUfUfugaacuugaL96	2190	VPusdCsaadGudTucaadAuCfcuuuguusgsc	2295
AD-1397095.1	ascseca (Ghd) gaUfUfUfgaaacuugaL96	2191	VPusdCscadAgdTucaadAuUfccuuugusug	2296
AD-1397096.1	csasaag (Ghd) auUfUfGfaaacuugguaL96	2192	VPusdAsccdAadGuuucdAaAfuccuuugsus u	2297
AD-1397097.1	asasagg (Ahd) uuUfGfAfaacuuggugaL96	2193	VPusdCsacdCadAguuudCaAfauccuuusgsu	2298
AD-1397098.1	asasgga (Uhd) uuGfAfAfacuugguguaL96	2194	VPusdAscadCcdAaguudTcAfaaaccuusug	2299
AD-1397102.1	asgsacg (Ahd) ugUfCfAfaccuuguguaL96	2195	VPusdAscadCadAgguudGaCfaucgucugsc	2300
AD-1397107.1	usasggg (Chd) uaAfCfCfaguucucuuaL96	2196	VPusdAsagdAgdAacugdGuUfagcccuasasa	2301
AD-1397108.1	gsgsgcu (Ahd) acCfAfGfuuucucuugaL96	2197	VPusdCsaadAgdAgaacdTgGfuuagcccsusa	2302
AD-1397109.1	gsgscua (Ahd) ccAfGfUfucucuugaL96	2198	VPusdAscadAadGagaadCuGfuuagcccsu	2303
AD-1397110.1	asascca (Ghd) uuCfUfCfuuuugaaggaL96	2199	VPusdCscudTadCaaagdAgAfacugguusasg	2304
AD-1397116.1	csuscuu (Uhd) guAfAfGfgacuugugcaL96	2200	VPusdGscadCadAguccdTuAfaaagagsasa	2305
AD-1397118.1	asusacu (Ghd) agGfGfUfgaaauaagaL96	2201	VPusdCsudAadTucaadCcCfucaguauusgsg	2306
AD-1397119.1	ascsuga (Ghd) ggUfGfAfaauuaaggaL96	2202	VPusdCsccdTudAauuudCaCfccucagusasu	2307
AD-1397120.1	csusgag (Ghd) guGfAfAfaauaagggaaL96	2203	VPusdTscddCudTaaudTcAfccucaggsusa	2308
AD-1397126.1	usasggg (Ghd) uuUfCfUfgccuuguugaL96	2204	VPusdCsaadCadAggcaadGaAfaccuagsgsg	2309
AD-1397127.1	asgsgug (Uhd) uuCfUfGfccuuguugaL96	2205	VPusdTscadAcadAaggcdAgAfaaacaccusag	2310
AD-1397132.1	asgscug (Ahd) acAfUfAfuaacauagauaL96	2206	VPusdAsucdTadTguuudAuGfuuacagcugsc	2311
AD-1397133.1	gsacsuga (Ahd) caUfAfUfacaugaugaL96	2207	VPusdCsaudCudAuguadTaUfguucagcsusg	2312
AD-1397135.1	gsasaca (Uhd) auAfCfAfaugauguugaL96	2208	VPusdCsaadCadTcuuudGuAfuauguucsasg	2313
AD-1397138.1	gsasguu (Ghd) uaGfUfUfggaauuuguaL96	2209	VPusdGsacdAadAuccadAcUfacaacucsasa	2314
AD-1397139.1	asgsuug (Uhd) agUfUfGfgauuugucuaL96	2210	VPusdAsgadCadAauccdAaCfuacaacuscsc	2315
AD-1397140.1	gsusugu (Ahd) guUfGfGfaauuugucuaL96	2211	VPusdCsagdAcadAaaucdCaAfcuacaacsusc	2316
AD-1397144.1	asgsuug (Ghd) auUfUfGfucuguuuuaL96	2212	VPusdAsuadAadCagacdAaAfucaacacsasc	2317
AD-1397145.1	ususgga (Uhd) uuGfUfCfuguuuuagcaL96	2213	VPusdGscadTadAacagdAcAfaaaucaacsu	2318
AD-1397150.1	ususguc (Uhd) guUfUfAfugcuuggauaL96	2214	VPusdAsucdCadAgcaudAaAfcagacaasasu	2319
AD-1397159.1	csacsaga (Ghd) ugAfCfUfugaauagugaL96	2215	VPusdCsacdTadTcauudGuCfacucuggsusg	2320
AD-1397160.1	csasgag (Uhd) gaCfUfAfugaauagugaL96	2216	VPusdCsaudGcdGagcudTgGfgucacgusgsa	2290
AD-1397163.1	csgscau (Ghd) uaUfCfUfugaauugcuaL96	2217	VPusdGsacdCadTgcgadGcUfugggucascsg	2291
AD-1397165.1	csasugu (Ahd) ucUfUfGfaaaugcuugaL96	2218	VPusdCsugdAcadCaugcdGaGfucuggguscsa	2292
AD-1397180.1	gsgsuuu (Ghd) ggUfAfCfaguuaaaggaL96	2219	VPusdCsaadAudCcuudGuUfgcugccascsu	2293

TABLE 16-continued

Exemplary dsRNA Molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID NO	Antisense sequence (5'→3')	SEQ ID NO:
AD-1397181.1	ususugg (Ghd) uaCfAfGfuuaaaggcaaL96	2220	VPusdTscadAadTccuudTgUfugcugccsasc	2294
AD-1397184.1	csasuua (Chd) ugCfCfAfacaguucgaL96	2221	VPusdCsaadGudTucaadAuCfcuuuguusgsc	2295
AD-1397185.1	usascug (Chd) caAfCfAfguuucggcuaL96	2222	VPusdCscadAgdTucadAaUfccuuugusug	2296
AD-1397186.1	gsusucc (Uhd) cuUfCfCfugaaguucuaL96	2223	VPusdAsccdAadGuuucdAaAfuccuuugsusu	2297
AD-1397187.1	ususccu (Chd) uuCfCfUfgaaguucuaL96	2224	VPusdCsacdCadAguuudCaAfauccuuusgsu	2298
AD-1397188.1	uscscuc (Uhd) ucCfUfGfaaguucugaL96	2225	VPusdAscadCcdAaguudTcAfaaaccuusug	2299
AD-1397191.1	uscscuuc (Chd) ugAfAfGfuucugugcaL96	2226	VPusdAscadCadAgguudGacfaucgucugsgsc	2300
AD-1397196.1	csascgc (Uhd) ggCfUfUfgugaucuuaL96	2227	VPusdAsagdAgdAacugdGuUfagcccuasasa	2301
AD-1397197.1	usgsggc (Uhd) agAfUfAfggauuacuaL96	2228	VPusdTscadCudAucaudAgUfcacucugsgsu	2321
AD-1397200.1	usasgau (Ahd) ggAfUfAfuacuguaL96	2229	VPusdAsgcdAudTucaadGaUfacaugcgsusc	2322
AD-1397205.1	ascscuu (Ahd) ucAfAfUfaguuccauuaL96	2230	VPusdCsucdTudTacaadGcAfuuucaagsasu	2323
AD-1397206.1	asusagu (Uhd) ccAfUfUfuaauugacaL96	2231	VPusdCsucdCudTuacadAgCfauuucagsgsa	2324
AD-1397211.1	asgsacu (Ghd) uaUfCfCfuguuugcuaL96	2232	VPusdAsgadAadCcucudTuAfcagcaususu	2325
AD-1397212.1	csusgua (Uhd) ccUfGfUfugcuauugaL96	2233	VPusdTsgadAadAccucdTuUfacaagcasusu	2326
AD-1397217.1	asusgga (Chd) auCfUfGfguugcuuugaL96	2234	VPusdCsucdTudAacugdTaCfccaaccsasc	2327
AD-1397219.1	csusucu (Ghd) auUfUfCfucucagcuaL96	2235	VPusdTscudCudAaccadCcAfcacaaucssusa	2328
AD-1397221.1	usgsauu (Uhd) cuCfUfUfcagcuuugaaL96	2236	VPusdCsgadAadCuguudGgCfaguaaugsasg	2329
AD-1397222.1	gsasuuu (Chd) ucUfUfCfagcuuugaaaL96	2237	VPusdAsgcdCgdAaacudGuUfggcaguuasasu	2330
AD-1397224.1	csusugc (Ahd) agUfCfCfcaugauuucalL96	2238	VPusdAsagdAadCuucadGgAfagaggaascsc	2331
AD-1397225.1	usgscac (Ghd) ucCfCfAfuuuuucuaL96	2239	VPusdCsaadGadAcuucdAgGfaagaggasasc	2332
AD-1397226.1	csasagu (Chd) ccAfUfGfauuucucgaL96	2240	VPusdAscadAgdAacuudCaGfagagaggsasa	2333
AD-1397227.1	asgsucc (Chd) auGfAfUfucucucgguaL96	2241	VPusdCsacdAadGaacudTcAfaggaagagsgsa	2334
AD-1397228.1	gsusccc (Ahd) ugAfUfUfucucugguuaL96	2242	VPusdGscadCadAgaacdTuCfaggaagagsgsg	2335
AD-1397230.1	cscscu (Ghd) auUfUfCfucugguuaL96	2243	VPusdCsuaadAadCcaugdAuCfuuagggcugsgsg	2336
AD-1397231.1	cscscu (Ahd) uuUfCfUfucugguuaL96	2244	VPusdCsucdAadAccaudGaUfcuuaggcsusg	2337
AD-1397232.1	asgsaga (Chd) auGfAfAfaucacuuuaL96	2245	VPusdAsaudTudAucugdCcAfgcacugasusc	2338
AD-1397234.1	gsasaca (Uhd) gaAfAfUfcaucuuagcaL96	2246	VPusdAsgudAudAuccudAuCfuagccascsc	2339
AD-1397235.1	gsasacu (Ghd) aaAfUfCfauucuuagcuaL96	2247	VPusdCsagdTadTaucdTaUfcuagcccsasc	2340
AD-1397237.1	asusgaa (Ahd) ucAfUfCfuucguuugaL96	2248	VPusdCsaudAcadAguuudAuCfcuauucagsgsc	2341
AD-1397238.1	gsasaa (Chd) auCfUfUfagcuuagcuaL96	2249	VPusdGsaadCudAuugadTaAfgagugaguscsc	2342
AD-1397243.1	gsuscu (Uhd) auAfGfUfguauuguguaL96	2250	VPusdGsucdAadTuuaadAuGfagacuausug	2343
AD-1397245.1	usasuu (Ahd) guGfUfAfuuguguguaL96	2251	VPusdCsaadAcadAggaudAcAfgucucacscsc	2344
AD-1397246.1	asusaa (Ghd) ugUfAfUfuguguguaL96	2252	VPusdGscadAadCaggadTaCfagucucacscsc	2345
AD-1397247.1	csasaa (Ghd) auUfUfAfcacugacugaL96	2253	VPusdAsgcdAadAcaggdAuAfcagucucacsc	2346

TABLE 16-continued

Exemplary dsRNA Molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID NO	Antisense sequence (5'→3')	SEQ ID NO:
AD-1397248.1	asasuga (Uhd) uuAfCfAfcugacuguuaL96	2254	VPusdTsagdCadAacagdGaUfacagucusc	2347
AD-1397249.1	gsasaa (Ahd) aaGfUfUfaauacucugaL96	2255	VPusdCsadTadGcaadCaGfgauacagsusc	2348
AD-1397070.2	ascsgug (Ahd) ccCfAfAfgcucgcaugaL96	2185	VPusdCsadGcdGagcudTgGfgucacgusgsa	2290
AD-1397073.2	usgsacc (Chd) aaGfCfUfcgcauggucaL96	2186	VPusdGsacdCadTgcgadGcUfugggucascsg	2291
AD-1397075.2	ascscga (Ahd) gcUfCfGfcauggucagaL96	2187	VPusdCsugdAcCdCaugcdGaGfcaugggusc	2292
AD-1397252.1	gscsucg (Chd) auGfGfUfcaguaaaagaL96	2256	VPusdCsudTudAcugadCcAfugcgagcsusu	2349
AD-1397253.1	csuscgc (Ahd) ugGfUfCfaguaaaagcaL96	2257	VPusdGscudTudTacugdAcCfauugcgagcsu	2350
AD-1397261.1	gsuscag (Uhd) aaAfAfGfcaagacgggaL96	2258	VPusdCsagdTcdTuugcdTuUfuacagacscsa	2351
AD-1397262.1	uscsgau (Ahd) aaAfGfCfaagacgggaL96	2259	VPusdCsacdGudCuugcdCuUfuauacagasc	2352
AD-1397263.1	csasgua (Ahd) aaGfCfAfaagacgggaL96	2260	VPusdTsacdCgdTcuudGcUfuuuacagsasc	2353
AD-1397266.1	asusaa (Ahd) ucAfAfAfcacgucgggaL96	2261	VPusdCsagdGadCgugudTuGfauuuuasc	2354
AD-1397268.1	asasuau (Chd) aaAfCfAfcgucgggaL96	2262	VPusdCsacdGgdGacgudGuUfugauuuas	2355
AD-1397270.1	usasua (Ahd) acAfCfGfuccgggagaL96	2263	VPusdCsudCcdGggacdGuGfuugauuas	2356
AD-1397271.1	asusaa (Ahd) caCfGfUfcccgggaggaL96	2264	VPusdCsudCcdCgggadCgUfuuuagausa	2357
AD-1397277.1	csascgu (Chd) ccGfGfGfaggcggaL96	2265	VPusdCsugdCcdGccudCcGfggacgususu	2358
AD-1397281.1	uscscgg (Ghd) gaGfGfCfggcagugugaL96	2266	VPusdCsacdAcCdTgcccGcUfcccgggascsg	2359
AD-1397291.1	gscsaaa (Uhd) agUfCfUfacaaccagaL96	2267	VPusdCsugdGudTugadGaCfuauuugcsa	2360
AD-1397294.1	asasuag (Uhd) cuAfCfAfaaccaguugaL96	2268	VPusdCsadCudGguudGuAfgacuaauus	2361
AD-1397295.1	usasguc (Uhd) acAfAfAfcagugagaL96	2269	VPusdGsudAadCuggudTuGfuagacuas	2362
AD-1397303.1	csasaca (Uhd) ccAfUfCfauaaaccagaL96	2270	VPusdCsugdGudTaugdAuGfgauguugsc	2363
AD-1397306.1	asuscca (Uhd) caUfAfAfaaccaggaggaL96	2271	VPusdCsudCcdTggudTaUfaguggausgsu	2364
AD-1397308.1	cscauc (Ahd) uaAfAfCfcaaggaggugaL96	2272	VPusdCsacdCudCuggdTuUfauaguggasu	2365
AD-1397309.1	csasuca (Uhd) aaAfCfCfaggaggugaL96	2273	VPusdCsacdCcdTccugdGuUfauaguggsa	2366
AD-1397317.1	ascscag (Ghd) agGfUfGfgccagguggaL96	2274	VPusdCsacdCcdTggcdAcCfuccuggususu	2367
AD-1423244.1	asgsaua (Ahd) uuAfAfUfaagaagcugaL96	2275	VPusdCsagdCudTcuudTuAfaauuacugsc	2368
AD-1423250.1	ususaau (Ahd) agAfAfGfcugaucuuaL96	2276	VPusdAsagdAudCcagcdTuCfuauuuasusu	2369
AD-1423253.1	asusaag (Ahd) agCfUfGfgaucuuagcaL96	2277	VPusdGscudAadGaucdAgCfuuuuauus	2370
AD-1423259.1	asgsucg (Ghd) auCfUfUfagcaacgucaL96	2278	PusdGsacdGudTgcudAgAfuccagcususc	2371
AD-1423260.1	gscsugg (Ahd) ucUfUfAfgcaacguccaL96	2279	VPusdGsgadCgdTugudAaGfauccagcsusu	2372
AD-1423262.1	usgsgau (Chd) uuAfGfCfaacguccagaL96	2280	VPusdCsugdGadCguudCuAfagauccagsc	2373
AD-1423271.1	gscsaac (Ghd) ucCfAfGfuccaaguguaL96	2281	VPusdAscadCudTggacdTgGfacguugcsusa	2374
AD-1423272.1	csasacg (Uhd) ccAfGfUfccaagugugaL96	2282	VPusdCsacdAcCdTuggadCuGfgacguugscsu	2375

TABLE 16-continued

Exemplary dsRNA Molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID NO	Antisense sequence (5'→3')	SEQ ID NO:
AD-1423273.1	asascgu (Chd) caGfUfCfcaaguguggaL96	2283	VPusdCscadCadCuuggdAcUfggacguusgsc	2376
AD-1423281.1	gsuscca (Ahd) guGfUfGfgcucaaaggaL96	2284	VPusdCscudTudGagccdAcAfcuuggacsusg	2377
AD-1423282.1	uscscaa (Ghd) ugUfGfGfcucaaaggaaL96	2285	VPusdTscddTudTgagcdCaCfacuuggacsu	2378
AD-1423283.1	cscsaag (Uhd) guGfGfCfucaaaggauaL96	2286	VPusdAsucdCudTugagdCcAfcacuuggsasc	2379
AD-1423289.1	gsusggc (Uhd) caAfAfGfgauauaucaL96	2287	VPusdGsaudAudTaucddTuUfgagccacsasc	2380
AD-1423291.1	gsgscuc (Ahd) aaGfGfAfuaauaaucaaaL96	2288	VPusdTsgdAudAuauadCcUfuugagccsasc	2381
AD-1423299.1	gsgsaua (Ahd) uaUfCfAfaacacguccaL96	2289	VPusdGsgadCgdTguuudGaUfauuauccsus u	2382
AD-1397266.2	asusaaU (Ahd) ucAfAfAfcacgucccggaL96	2261	VPusdCsggdGadCgugudTuGfauauuasc c	2354
AD-1397268.2	asasuau (Chd) aaAfCfAfcgucccggaL96	2262	VPusdCscddGgdGacgudGuUfugauuusas u	2355
AD-1397270.2	usasuca (Ahd) acAfCfGfucccgggagaL96	2263	VPusdCsucdCcdGggacdGuGfuugauuas u	2356
AD-1397271.2	asuscaa (Ahd) caCfGfUfcccgggaggaL96	2264	VPusdCscudCcdCgggadCgUfguugausasu	2357
AD-1397277.2	csascgu (Chd) ccGfGfGfaggcggaL96	2265	VPusdCsugdCcdGccucdCcGfggacgugsusu	2358
AD-1397294.2	asasuag (Uhd) cuAfCfAfaaccaguugaL96	2268	VPusdCsaadCudGguuudGuAfgacuaus g	2361
AD-1397306.2	asuscca (Uhd) caUfAfAfacaggaggaL96	2271	VPusdCscudCcdTgguudTaUfgaugausgsu	2364
AD-1397308.2	cscsauc (Ahd) uaAfAfCfcaggaggaL96	2272	VPusdCsacdCudCcuggdTUfauauggsasu	2365
AD-1397073.3	usgsacc (Chd) aaGfCfUfcgcauggucaL96	2186	VPusdGsacdCadTgagadGcUfugggucascsg	2291
AD-1397252.2	gscsucg (Chd) auGfGfUfcaguaaaagaL96	2256	VPusdCsudTudAcugadCcAfugcgagsusu	2349
AD-1397263.2	csasgua (Ahd) aaGfCfAfaagcgggaL96	2260	VPusdTscddCgdTcuuudGcUfuuuacugsasc	2353
AD-1397309.2	csasuca (Uhd) aaAfCfCfaggaggugaL96	2273	VPusdCscadCcdTccugdGuUfuaugaugsgsa	2366

TABLE 17

Exemplary dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1251268.1	csasuga (Uhd) cuUfCfUfuugucguagaL96	2383	VPusdCsudCgdAcaaadGadAgdAuc augusg	2470
AD-1251270.1	csasuga (Uhd) cuUfCfUfuugucguagaL96	2383	VPusdCsudCgdAcaaadGadAgdAuc augscsc	2471
AD-1251275.1	csasuga (Uhd) cuUfCfUfuugucguagaL96	2383	VPudCuadCgdAcaaadGadAgdAucau gsusg	2472
AD-1251275.2	csasuga (Uhd) cuUfCfUfuugucguagaL96	2383	VPudCuadCgdAcaaadGadAgdAucau gsusg	2472
AD-1251282.1	usgsucgaguAfCfAfcuuu (Uhd) acugaL96	2384	VPusdCsagdTadAaagudGudAcdTcga casusu	2473

TABLE 17-continued

Exemplary dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1251357.1	gsasgaa (Uhd) UfcaCfUfuuuuucguaL96	2385	VPudAcgdAadGaaaadGudGadAuuc ucscsu	2474
AD-1251385.1	asusgau (Chd) UfuCfUfUfugucguaguaL96	2386	VPudAcudAc dGacaadAgdAadGauca usgsu	2475
AD-1251386.1	usgsauc (Uhd) UfcUfUfUfgucguagugaL96	2387	VPudCacdTadCgacadAadGadAgauc asusg	2476
AD-1251459.1	asusaua (Uhd) UfuUfaCfaacauccguaL96	2388	VPudAcgdGadTguugdTadAadAuaua uscsg	2477
AD-1331348	asusgau (Chd) UfuCfUfUfugucguagsusa	2389	VPudAcudAc dGacaadAgdAadGauca usgsu	2475
AD-1479607	usgsauc (Uhd) UfcUfUfUfgucguagusgsa	2390	VPudCacdTadCgacadAadGadAgauc asusg	2476
AD-1481956	asusgau (Chd) UfuCfUfUfugucguagsusa	2389	VPudAscudAc dGacaadAgdAadGau causgsu	2478
AD-1481957	usgsauc (Uhd) UfcUfUfUfgucguagusgsa	2390	VPudCsacdTadCgacadAadGadAgauc asusg	2479
AD-1251472.1	usasuuu (Uhd) acdAaCfauccguuauaL96	2391	VPudAuadAc dGgaugdTudGudAaaa uasusg	2480
AD-1251272.1	usgsaucuUfCfUfuugu (Chd) guagaL96	2392	VPudCsudCgdAcaaadGadAgdAuc asusg	2481
AD-1251358.1	cscsugaagcAfUfAfaaug (Uhd) uuucaL96	2393	VPudGsaadAadCauuudAudGcUfuc aggsusu	2482
AD-1251378.1	usasaauUfuUfuCf gaaa (Uhd) ucacaL96	2394	VPudGugdAadTuucgdAadAaCfauuu asusg	2483
AD-1251405.1	usc suuugUfcgUfAfguga (Uhd) uuucaL96	2395	VPudGaadAadTcacudAc dGacfaaaga sgsg	2484
AD-1251476.1	ususua (Chd) aacaUfCfcguuuuacaL96	2396	VPudGuadAudAacggdAudGuUfguaa asgsu	2485
AD-1251266.1	ususgauaguUfaCfcuag (Uhd) uugcaL96	2397	VPudGcadAadCuaggdTadAcUfauca asgsu	2486
AD-1251259.1	ususgauaguUfaCfcuag (Uhd) uugcaL96	2397	VPudGscadAadCuaggdTadAcUfauc aasgsu	2487
AD-1479603	usasaauUfuUfuCf gaaa (Uhd) ucascsa	2398	VPudGugdAadTuucgdAadAaCfauuu asusg	2483
AD-1479612	usc suuugUfcgUfAfguga (Uhd) uuusc sa	2399	VPudGaadAadTcacudAc dGacfaaaga sgsg	2484
AD-1479603	usasaauUfuUfuCf gaaa (Uhd) ucascsa	2398	VPudGugdAadTuucgdAadAaCfauuu asusg	2483
AD-1479612	usc suuugUfcgUfAfguga (Uhd) uuusc sa	2399	VPudGaadAadTcacudAc dGacfaaaga sgsg	2484
AD-1481954	usasaauUfuUfuCf gaaa (Uhd) ucascsa	2398	VPudGsugdAadTuucgdAadAaCfau uasusg	2488
AD-1481961	usc suuugUfcgUfAfguga (Uhd) uuusc sa	2399	VPudGsaadAadTcacudAc dGacfaaa gasgsu	2489
AD-1251262.1	gsasuaguUfaCfcuag (Uhd) uugcaL96	2400	VPudGscadAadCuaggdTadAcUfauc sgsg	2490

TABLE 17-continued

Exemplary dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1251279.2	csasaca (Chd) aaUfUfUfcuucuagcaL96	2401	VPudGcudAadGaagadAaUfudGuguugsusu	2491
AD-1251278.1	ascsacaaUfUfUfcuuc (Uhd) uagcaL96	2402	VPusdGscudAadGaagadAaUfudGugusug	2492
AD-1251319.1	asgsggaaaaCfadAucuu (Chd) cguuaL96	2403	VPudAacdGgdAagaudTgUfuUfucccususu	2493
AD-1251348.1	usgsuaggagdAaUfucuu (Uhd) uuuaL96	2404	VPusdGsaadAadAugaadTuCfuCfcuacascsg	2494
AD-1251354.1	asgsgagaauUfcdAcuuu (Uhd) cuucaL96	2405	VPudGaadGadAaagudGaAfuUfcuccusgsc	2495
AD-1251365.1	asgsca (Uhd) aaaUfgUfuuucgaaauaL96	2406	VPudAuudTcdGaaaadCaUfuUfaugcususc	2496
AD-1251419.1	gsusaga (Uhd) CfuUfgCfaauuaccuaL96	2407	VPudAugdGudAauugdCaAfgAfucua cs gsg	2497
AD-1251474.1	asusuuuacacAcdaucgg (Uhd) uauuaL96	2408	VPudAaadAadCggaudGuUfgUfaaaa usgsu	2498
AD-1251419.2	gsusaga (Uhd) CfuUfgCfaauuaccuaL96	2407	VPudAugdGudAauugdCaAfgAfucua cs gsg	2497
AD-1479622	gsusaga (Uhd) CfuUfgCfaauuaccasusa	2409	VPudAugdGudAauugdCaAfgAfucua cs gsg	2497
AD-1479593	asgsca (Uhd) aaaUfgUfuuucgaaasusa	2410	VPudAuudTcdGaaaadCaUfuUfaugcususc	2496
AD-1479622	gsusaga (Uhd) CfuUfgCfaauuaccasusa	2409	VPudAugdGudAauugdCaAfgAfucua cs gsg	2497
AD-1481938	gsusaga (Uhd) CfuUfgCfaauuaccasusa	2409	VPusdAsugdGudAauugdCaAfgAfucua cs gsg	2499
AD-1481952	asgsca (Uhd) aaaUfgUfuuucgaaasusa	2410	VPusdAsuudTcdGaaaadCaUfuUfaugcususc	2500
AD-1481938	gsusaga (Uhd) CfuUfgCfaauuaccasusa	2409	VPusdAsugdGudAauugdCaAfgAfucua cs gsg	2499
AD-1479586	usgsuaggagdAaUfucuu (Uhd) uuuscsa	2411	VPusdGsaadAadAugaadTuCfuCfcuacascsg	2494
AD-1251304.1	ascsaaaaggAfAfAfacaa (Uhd) cuucaL96	2412	VPudGaadGadTuguuuuCfcCfuuuugusgu	2501
AD-1251305.1	csasaagggaAfAfAfcaau (Chd) uucaL96	2413	VPudGgadAgdAuuguuuUfcCfcuuugusug	2502
AD-1251321.1	gsgsaaaa (Chd) aaUfCfuuccguuuaL96	2414	VPudGaadAcdGgaagauUfgUfuuuccscsu	2503
AD-1251345.1	usgsuaggAfAfAfUfucuu (Uhd) uuuaL96	2415	VPusdGsaadAadAugaauUfcCfcuacascsg	2504
AD-1251351.1	gsusaggagaaUfUfcacu (Uhd) uucuaL96	2416	VPudAgadAadAgugaauUfcUfccuacsg	2505
AD-1251353.1	usasggagaaUfUfcfacuu (Uhd) ucuaL96	2417	VPusdAsagdAadAgugaaUfuCfuccuascsg	2506
AD-1251381.1	usasca (Uhd) gauCfUfUfcuuugucguaL96	2418	VPudAscgdAcdAaagaagAfuCfaugua scsc	2507

TABLE 17-continued

Exemplary dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1251265.1	ususgauaguUfAfCfcuag (Uhd) uugcaL96	2419	VPudGcadAadCuagguaAfcUfaucaasgsg	2508
AD-1251257.1	ususgau (Ahd) guUfAfCfcuaguuugcaL96	2420	VPusdGscadAadCuagguaAfcUfaucaasgsg	2509
AD-1251264.1	ususgauaguUfAfCfcuaa (Uhd) uugcaL96	2421	VPusdGscadAadTuagguaAfcUfaucaasgsg	2510
AD-1251258.1	ususgauaguUfAfCfcuag (Uhd) uugcaL96	2419	VPusdGscadAadCuagguaAfcUfaucaasgsg	2509
AD-1251261.1	gsasuaguUfAfCfcuag (Uhd) uugcaL96	2422	VPusdGscadAadCuagguaAfcUfaucsgsg	2511
AD-1479593	asgsca (Uhd) aaaUfgUfuucgaaasusa	2410	VPudAuudTcdGaaaadCaUfuUfaugc ususc	2496
AD-1251283.1	usgsucgaguAfCfAfcuuu (Uhd) acugaL96	2384	VPusCfsagdTadAaagudGudAcdTcga casusu	2512
AD-1251384.1	csasuga (Uhd) CfUfCfUfuugucguagaL96	2423	VPuCfuadCgdAcaaadGadAgdAucau gsusg	2513
AD-1251274.2	csasuga (Uhd) cuUfCfUfuugucguagaL96	2383	VPuCfuadCgdAcaaadGadAgdAucau gsusg	2513
AD-1251269.1	csasuga (Uhd) cuUfCfUfuugucguagaL96	2383	VPusCfsuadCgdAcaaadGadAgdAuca ugsusg	2514
AD-1251274.1	csasuga (Uhd) cuUfCfUfuugucguagaL96	2383	VPuCfuadCgdAcaaadGadAgdAucau gsusg	2513
AD-1251271.1	csasuga (Uhd) cuUfCfUfuugucguagaL96	2383	VPusCfsuadCgdAcaaadGadAgdAuca ugsusc	2515
AD-1251274.3	csasuga (Uhd) cuUfCfUfuugucguagaL96	2383	VPuCfuadCgdAcaaadGadAgdAucau gsusg	2513
AD-1331354	csasuga (Uhd) cuUfCfUfuugucguasgsa	2424	VPuCfuadCgdAcaaadGadAgdAucau gsusg	2513
AD-1479606	csasuga (Uhd) CfUfCfUfuugucguasgsa	2425	VPuCfuadCgdAcaaadGadAgdAucau gsusg	2513
AD-1331354	csasuga (Uhd) cuUfCfUfuugucguasgsa	2424	VPuCfuadCgdAcaaadGadAgdAucau gsusg	2513
AD-1479606	csasuga (Uhd) CfUfCfUfuugucguasgsa	2425	VPuCfuadCgdAcaaadGadAgdAucau gsusg	2513
AD-1481942	csasuga (Uhd) cuUfCfUfuugucguasgsa	2424	VPusCfsuadCgdAcaaadGadAgdAuca ugsusg	2514
AD-1481955	csasuga (Uhd) CfUfCfUfuugucguasgsa	2425	VPusCfsuadCgdAcaaadGadAgdAuca ugsusg	2514
AD-1251273.1	usgsaucuUfCfUfuugu (Chd) guagaL96	2392	VPusCfsuadCgdAcaaadGadAgdAuca susg	2516
AD-1251320.1	gsgsgaaaAfcAfaUfcuuc (Chd) guuuuL96	2426	VPusAfsaadCgdGaagadTudGuUfuuc ccsusu	2517
AD-1251325.1	asasaacaauCfUfUfccgu (Uhd) ucaaaL96	2427	VPuUfugdAadAcggadAgdAuUfguuu uscsc	2518
AD-1251356.1	gsgsagaaUfuCfaCfuuuu (Chd) uucgaL96	2428	VPuCfgadAgdAaaagdTgdAaUfucucc susg	2519

TABLE 17-continued

Exemplary dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1251418.1	usgsuagaUfcUfudGcaau (Uhd) accaaL96	2429	VPuUfggdTadAuugcdAadGaUfcuaca sgs	2520
AD-1251457.1	gsasua (Uhd) aUfuUfudAcaacaucgaL96	2430	VPuUfggdAudGuugudAadAaUfauau csgsc	2521
AD-1251485.1	csasagugUfuCfCfUfacug (Uhd) caugaL96	2431	VPuUfsgadGadCaguadGgdAaCfacu ugs	2522
AD-1251325.1	asasaacaauCfUfUfcgu (Uhd) ucaaaL96	2427	VPuUfugdAadAcggadAgdAuUfguuu uscsc	2518
AD-1331350	asasaacaauCfUfUfcgu (Uhd) ucasasa	2432	VPuUfugdAadAcggadAgdAuUfguuu uscsc	2518
AD-1331350	asasaacaauCfUfUfcgu (Uhd) ucasasa	2432	VPuUfugdAadAcggadAgdAuUfguuu uscsc	2518
AD-1479588	gsgsagaaUfuCfaCfuuuu (Chd) uucsgsa	2433	VPuUfgadAgdAaaagdTgdAaUfucucc susg	2519
AD-1481945	asasaacaauCfUfUfcgu (Uhd) ucasasa	2432	VPuUfsgdAadAcggadAgdAuUfguu uscsc	2523
AD-1481948	gsgsagaaUfuCfaCfuuuu (Chd) uucsgsa	2433	VPuUfsgadAgdAaaagdTgdAaUfucu ccsg	2524
AD-1251284.1	usgsucgaguAfcAfcuuu (Uhd) acugaL96	2384	VPuUfsgadTadAaagudGuAfcdTcgac asusu	2525
AD-1251286.1	usgsucgaguAfcAfcuuu (Uhd) acugaL96	2384	VPuUfsgadTadAaagudGuAfcdTcgac ascsc	2526
AD-1251288.1	uscsgaguAfcAfcuuu (Uhd) acugaL96	2434	VPuUfsgadTadAaagudGuAfcdTcgac csg	2527
AD-1251290.1	usgsucgaguAfcAfcuuu (Uhd) acugaL96	2384	VPuUfsgadTadAaagudGuAfcdTcgaca susu	2528
AD-1251471.1	asusuu (Uhd) UfaCfadAcauccguuaaL96	2435	VPuUfaadCgdGaugudTgUfadAaaua usgsu	2529
AD-1251475.1	ususuaa (Chd) aaCfaUfcguuuuaaL96	2436	VPuUfaadTadAcggadTgUfudGuaaaa usg	2530
AD-1251284.2	usgsucgaguAfcAfcuuu (Uhd) acugaL96	2384	VPuUfsgadTadAaagudGuAfcdTcgac asusu	2525
AD-1331352	usgsucgaguAfcAfcuuu (Uhd) acusgsa	2437	VPuUfsgadTadAaagudGuAfcdTcgac asusu	2525
AD-1251322.1	gsasaaa (Chd) aaUfCfUfuccauuucaL96	2438	VPuUfgadAadTggaagaUfudGuuuucsc	2531
AD-1479581	gsasaaa (Chd) aaUfCfUfuccauuuacsasa	2439	VPuUfgadAadTggaagaUfudGuuuucsc	2531
AD-1479581	gsasaaa (Chd) aaUfCfUfuccauuuacsasa	2439	VPuUfgadAadTggaagaUfudGuuuucsc	2531
AD-1481943	gsasaaa (Chd) aaUfCfUfuccauuuacsasa	2439	VPuUfsgadAadTggaagaUfudGuuuu cscsc	2532
AD-1479588	gsgsagaaUfuCfaCfuuuu (Chd) uucsgsa	2433	VPuUfgadAgdAaaagdTgdAaUfucucc susg	2519
AD-1251303.1	csascaaaggAfaAfaa (Uhd) cuuaL96	2440	VPuUfsgadAudTguuuucCfcUfuugu gsusu	2533

TABLE 17-continued

Exemplary dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1251306.1	asasagggAfaAfAfCfaauc (Uhd) uccgaL96	2441	VPusCfsggdAadGauuguuUfuCfccuu usgsu	2534
AD-1251307.1	asasagggAadAaCfaauc (Uhd) uccgaL96	2442	VPuCfsggdAadGauugdTuUfuCfccuuu sgsu	2535
AD-1251318.1	asgsggaaAfaCfAfAfucuu (Chd) cguuaL96	2443	VPusAfsacdGgdAagauugUfuUfuccc ususu	2536
AD-1251324.1	asasaacaAfuCfUfUfccgu (Uhd) ucaaaL96	2444	VPusUfsugdAadAcggaagAfuUfguuu uscsc	2537
AD-1251249.1	usgsucgaguAfCfAfccuu (Uhd) acugaL96	2384	VPusCfsagdTadAaguguAfcUfcgaca susu	2538
AD-1251254.1	usgsucgaguAfCfAfccuu (Uhd) acugaL96	2384	VPuCfagdTadAaguguAfcUfcgacasc sc	2539
AD-1251250.1	usgsucgaguAfCfAfccuu (Uhd) acugaL96	2384	VPusCfsagdTadAaguguAfcUfcgaca scsc	2540
AD-1251251.1	uscsgaguAfCfAfccuu (Uhd) acugaL96	2434	VPusCfsagdTadAaguguAfcUfcgascsg	2541
AD-1251350.1	gsusaggagaAfUfUfcacu (Uhd) uucuaL96	2445	VPusAfsagdAadAgugaaUfcUfccuac sgsc	2542
AD-1251352.1	usasggagAfaUfUfcfacuu (Uhd) ucuuaL96	2446	VPusAfsagdAadAagugaaUfuCfuccua scsg	2543
AD-1251355.1	gsgsagaaUfuCfAfCfuuuu (Chd) uucgaL96	2447	VPusCfsgadAgdAaaagugAfaUfuccc susg	2544
AD-1251359.1	csusgaagCfaUfAfAfaugu (Uhd) uucgaL96	2448	VPusCfsgadAadAcauuuaUfgCfuucag sgsu	2545
AD-1251363.1	gsasagcauadAaUfguuu (Uhd) cgaaaL96	2449	VPuUfucdGadAaacadTuUfaUfgcuuc sasg	2546
AD-1251362.1	gsasagcaUfaAfAfUfguuu (Uhd) cgaaaL96	2450	VPusUfsucdGadAaacuuUfaUfgcuu csasg	2547
AD-1251369.1	asgscauaaaUfgUfuuuu (Uhd) gaaauaL96	2451	VPusAfsuudTcdAaaaadCaUfuUfaug cususc	2548
AD-1251376.1	asusaaa (Uhd) guUfUfUfcgaaaucaaL96	2452	VPusUfsgadAudTucgaaaAfcAfuuuau sgsc	2549
AD-1251377.1	asusaaa (Uhd) guUfUfUfcgaaaucaaL96	2452	VPusUfsgadAudTucgaaaAfcAfuuuau sgsu	2550
AD-1251380.1	usasca (Uhd) gAfuCfUfUfcuuugcguuaL96	2453	VPusAfsagdAcdAaagaagAfuCfaugua sgsg	2551
AD-1251383.1	csasuga (Uhd) CfUfCfUfuugucguagaL96	2423	VPusCfsuadCgdAcaaagaAfgAfucaug susg	2552
AD-1251431.1	usasugugAfaAfCfAfaacu (Uhd) uacgaL96	2454	VPusCfsgudAadAguuguUfuCfacau asgsu	2553
AD-1251447.1	gsusgaaaCfadAaCfcuua (Chd) gugaaL96	2455	VPuUfcadCgdTaaggdTuUfgUfuucac sgsu	2554
AD-1251446.1	gsusgaaaCfaAfAfCfcuua (Chd) gugaaL96	2456	VPusUfscadCgdTaaggguUfgUfuucac sgsu	2555
AD-1251456.1	gsasua (Uhd) aUfuUfUfAfcaacauccgaL96	2457	VPusCfsggdAudGuuguaaAfaUfauau csgsc	2556

TABLE 17-continued

Exemplary dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1251458.1	asusaua (Uhd) UfuUfAfCfaacauccguaL96	2458	VPusAfscgdGadTguuguaAfaAfuaua uscs g	2557
AD-1251470.1	asusauu (Uhd) UfaCfAfAfauccguuaaL96	2459	VPusUfsaadCgdGauguugUfaAfaaua usgsu	2558
AD-1251473.1	asusuuuaCfaAfCfAfuccg (Uhd) uauuaL96	2460	VPusAfsaudAadCggauguUfgUfaaaa usgsu	2559
AD-1251484.1	csasagugUfuCfCfUfacug (Uhd) caugaL96	2431	VPusCfsaudGadCaguaggAfaCfacuug sgsg	2560
AD-1251249.1	usgsucgaguAfCfAfcuuu (Uhd) acugaL96	2384	VPusCfsagdTadAaguguAfcUfcgaca susu	2538
AD-1251318.1	asgsaggaaAfaCfAfAfcuuu (Chd) cguuaL96	2443	VPusAfsacdGgdAagauguUfuUfuccc ususu	2536
AD-1251363.1	gsasagcauadAaUfguuu (Uhd) cgaaL96	2449	VPuUfucdGadAaacadTuUfaUfgcuuc sasg	2546
AD-1251377.2	asusaaa (Uhd) guUfUfUfcgaaaucaaL96	2452	VPusUfsgadAudTucgaaaAfcAfuuuau sgsu	2550
AD-1331355	usgsucgaguAfCfAfcuuu (Uhd) acusgsa	2437	VPusCfsagdTadAaguguAfcUfcgaca susu	2538
AD-1479582	asasaacaAfuCfUfUfccgu (Uhd) ucasasa	2461	VPusUfsugdAadAcggaagAfuUfguuu uscs c	2537
AD-1479587	gsgsagaaUfuCfAfCfuuuu (Chd) uucsgsa	2462	VPusCfsgadAgdAaaagugAfaUfucucc susg	2544
AD-1479589	csusgaagCfaUfAfAfaugu (Uhd) uucsgsa	2463	VPusCfsgadAadAcauuuaUfgCfuucag sgsu	2545
AD-1479591	gsasagcaUfaAfAfUfguuu (Uhd) cgasasa	2464	VPusUfsucdGadAaacauUfaUfgcuu csasg	2547
AD-1479596	asgscauaaaUfgUfuuu (Uhd) gaaasusa	2465	VPusAfsuudTcdAaaaadCaUfuUfaug cususc	2548
AD-1479601	asusaaa (Uhd) guUfUfUfcgaaauucsasa	2466	VPusUfsgadAudTucgaaaAfcAfuuuau sgsc	2549
AD-1479602	asusaaa (Uhd) guUfUfUfcgaaauucsasa	2466	VPusUfsgadAudTucgaaaAfcAfuuuau sgsu	2550
AD-1479604	usasca (Uhd) gAfuCfUfUfcuuugucgsusa	2467	VPusAfscgdAcdAaagaagAfuCfaugua sgsg	2551
AD-1479605	csasuga (Uhd) CfuUfCfUfuugucguasgsa	2425	VPusCfsuadCgdAcaaagaAfgAfucaug susg	2552
AD-1331349	gsasagcauadAaUfguuu (Uhd) cgasasa	2468	VPuUfucdGadAaacadTuUfaUfgcuuc sasg	2546
AD-1481950	gsasagcauadAaUfguuu (Uhd) cgasasa	2468	VPusUfsucdGadAaacadTuUfaUfgcu ucsasg	2561
AD-1251251.1	uscs gaguAfCfAfcuuu (Uhd) acugaL96	2434	VPusCfsagdTadAaguguAfcUfcgasc s g	2541
AD-1479580	uscs gaguAfCfAfcuuu (Uhd) acusgsa	2469	VPusCfsagdTadAaguguAfcUfcgasc s g	2541
AD-1331349	gsasagcauadAaUfguuu (Uhd) cgasasa	2468	VPuUfucdGadAaacadTuUfaUfgcuuc sasg	2546

TABLE 18

Exemplary dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1395794.1	csuscuccaagUfAfUfgaucgucuuL96	2562	asdAsgadCgdAucaudAcUfuggagagsgcs	2567
AD-1395803.1	csasgugaugCfUfCfuccaaguauuL96	2563	asdAsuadCudTggagdAgCfaucacugsusg	2568
AD-1395805.1	csaagauGfAfUfCfugucgucaguL96	2564	asdCsugdCadGacgadTcAfuacuuggsasg	2569
AD-1395816.1	gsgsguuuguUfudGUfuucuuuL96	2565	asdGsaaudAudGaaacdAaAfcaaacccsusg	2570
AD-1395823.1	gsasguuuuUfUfCfagcaucaaL96	2566	asdTsugdAadTgcugdAgAfaauacucscsc	2571

TABLE 19

Exemplary dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1465901.1	csasgcuaagGfCfAfguucuauguL96	2572	asdAscgdTadGaacudGcCfuagcugsusg	2690
AD-1465906.1	ususcugggCfCfUfacuuaauuL96	2573	asdAsuadTadAaguadGgCfccaagaasgsu	2691
AD-1465907.1	usascuuauAfUfGfCfugaagucguL96	2574	asdCsgadCudTcagcdAuAfuaaaguasgs	2692
AD-1465908.1	ascsuuauaUfGfCfugaagucguL96	2575	asdCscgdAcdTucagdCaUfaaaagusasg	2693
AD-1465909.1	asgsuaauuAfUfCfagaaggugcuL96	2576	asdGscadCcdTucugdAuAfauuacugsu	2694
AD-1465911.1	asuscagaagGfUfGfCuucuuaccuL96	2577	asdGsgudAadGaagcdAcCfuucugausasa	2695
AD-1465912.1	uscsagaaggUfGfCfCuucuuaccuL96	2578	asdAsggdTadAgaagdCaCfcuucugasusa	2696
AD-1465913.1	csasgaaggUfCfUfCuucuuaccuL96	2579	asdAsagdGudAagaadGcAfccuucugsasu	2697
AD-1465915.1	ascscuaugaAfUfGfGfagauaucaguL96	2580	asdCsugdAudAcuccdAuUfcauaggusgsu	2698
AD-1465916.1	csaagaaUfGfGfagauaucaguL96	2581	asdAscudGadTacudCaUfcauaggusgsu	2699
AD-1465917.1	asusgaaggAfGfUfaucaugugaguL96	2582	asdCsucdAcdTgauadCuCfcauucausasg	2700
AD-1465918.1	asusgccucaCfAfCfacaucuuuuL96	2583	asdAsaudAgdAugugdTgUfgaggcausgs	2701
AD-1465919.1	usgscucacAfCfAfcaucuuuuL96	2584	asdTsaaudTadGaugudGuGfugaggcasusg	2702
AD-1465920.1	gscscucacaCfAfCfaucauuuacuL96	2585	asdGsuaudAudAgaugdTgUfgugaggcsasu	2703
AD-1465921.1	csaascacacAfCfAfucuuuacuL96	2586	asdAsgudAadTagaudGuGfugugaggscsa	2704
AD-1465922.1	csuscacacaCfAfUfcauuuacucuL96	2587	asdGsagdTadAuagadTgUfgugugagsgsc	2705
AD-1465927.1	gsasgcuuugAfCfAfagcaaaucguL96	2588	asdCsgadTudTgcuudGuCfaaacgucsusu	2706
AD-1465928.1	gscscagucaUfCfAfuccuaaanguL96	2589	asdCsaudTadGggaudGaUfgacuggsusc	2707
AD-1465932.1	asasuguacaCfAfGfcauaggauuL96	2590	asdAsucdCadTugacdTgUfguacauusasg	2708
AD-1465935.1	gscscagauaUfAfAfcauuuuguguL96	2591	asdCsacdAadAcugudTaUfauucggcsasu	2709
AD-1465936.1	csaagauaUfAfCfaguuuugugcuL96	2592	asdGscadCadAacugdTuAfauucggscsa	2710
AD-1465937.1	gsasgcagaaCfCfAfucuaaagguuL96	2593	asdAsccdTudAugaudGgUfucugcucscsa	2711
AD-1465942.1	csasuccacuAfCfCfcaaaauuguL96	2594	asdCsaudAudTugcgdGuAfguggauggsusa	2712
AD-1465945.1	gscsugggauGfCfAfggcuuacauuL96	2595	asdAsugdTadAgccudGcAfuccagcsusu	2713
AD-1465946.1	csusgggaugCfAfGfGfcauuuacuL96	2596	asdAsaudGudAagccdTgCfaucccagcsu	2714

TABLE 19-continued

Exemplary dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1465947.1	usgsggaugcAfGfGfcuuacauuguL96	2597	asdCsaadTgdTaagcdCuGfcaucccasgsc	2715
AD-1465949.1	gsgsaugcagGfCfUfuacauugacuL96	2598	asdGsucdAadTguaadGcCfugcauccscsa	2716
AD-1465951.1	asusgcaggcUfUfAfcuugacauuL96	2599	asdAsugdTcdAaugudAaGfccugcauscsc	2717
AD-1465955.1	asgsgcuuacAfUfUfgacauuaauL96	2600	asdTsuudAadTgucdAuGfuaagccsgsc	2718
AD-1465956.1	gsgscuuacaUfUfGfacaauaaaauL96	2601	asdTsuudTadAugudAaUfguaagccsusg	2719
AD-1465957.1	gscsuuacauUfGfAfcuaaaaaacuL96	2602	asdTsuudTudAaugudCaAfuguaagcscsu	2720
AD-1465958.1	csusuacauUfGfAfcuaaaaaacuL96	2603	asdGsuidTudTaaugdTcAfauguaagcsc	2721
AD-1465959.1	ususacauugAfCfAfuuaaaaaacuL96	2604	asdAsugdTudTuaudGuCfaauguaagsgsc	2722
AD-1465960.1	usascuugaCfAfUfuaaaaaacuguL96	2605	asdCsagdTudTuuaadTgUfcaauguasasg	2723
AD-1465961.1	ascsauugacAfUfUfuaaaaaacugcuL96	2606	asdGscadGudTuuaadAuGfuaaugusasa	2724
AD-1465968.1	csasccuugaAfUfAfccagcgaaauL96	2607	asdAsuudCgdCuggudAuUfacaggugscsa	2725
AD-1465969.1	usgsuauacCfAfGfcgaauaugguL96	2608	asdCscadTadTucgdcTgGfuaauacsgsg	2726
AD-1465970.1	gsusauuaccAfGfCfgaauauggauL96	2609	asdTscddAudAuucgdCuGfgauuacsasg	2727
AD-1465972.1	gsusuauguaCfAfCfacagacgauL96	2610	asdTscgdTadCugugdTgUfacauaacsusu	2728
AD-1465973.1	ususauguacAfCfAfcagucgaauL96	2611	asdTsucdGudAcugudGuGfuacauaascsu	2729
AD-1465976.1	gsusacacacAfGfUfagcagauguL96	2612	asdCsaudCudTcguadCuGfuguguaacsasu	2730
AD-1465986.1	asasaccuauAfCfUfuaauaagugguL96	2613	asdCscadCudTaaadGuAfuagguuuscsc	2731
AD-1465992.1	asusgaaucaAfAfCfaucaugagcuL96	2614	asdGscudCadTgaugdTuUfgauucausasa	2732
AD-1465997.1	csagsagggcAfCfUfucugugaguguL96	2615	asdCsacdTadCagaadGuGfccacuggsasc	2733
AD-1465998.1	asgsuggcacUfUfCfugugaguguguL96	2616	asdCsacdAcdTacagdAaGfugccacugsg	2734
AD-1465999.1	csusggggcacUfCfAfuucaucuaauL96	2617	asdAsuadGadTgaudGaGfugcccagsusg	2735
AD-1466002.1	ususaacuucCfAfUfgaaauucaguL96	2618	asdCsudGadAuucadTgGfaaguuascsa	2736
AD-1466004.1	usgsaugaagAfCfUfcauauagagauL96	2619	asdTscudCadTaugadGuCfuucaucasusc	2737
AD-1466005.1	asasacucaCfAfUfugaauacagguL96	2620	asdCscudGadTucadTgAfugaguuuuscg	2738
AD-1466006.1	asasacacagAfUfAfuauuuguuguL96	2621	asdCsaadCadAuuaudAuCfuguguuugsa	2739
AD-1466007.1	csascagauaUfAfAfuuguugguuuL96	2622	asdAsacdCadAcaudTaUfaucugugusu	2740
AD-1466009.1	asusucugaaGfAfCfccuauagaguL96	2623	asdCsucdTadTagggdTcUfucagaasasu	2741
AD-1466010.1	csgsucuaCuUfUfCfacuuggugcuL96	2624	asdGscadCcdAagudAaAfugagacgsusa	2742
AD-1466012.1	asasuacuaGfCfAfcuaaaaguuuL96	2625	asdAsacdTudTaugudGcUfaguaauususc	2743
AD-1466013.1	usascuagcaCfAfUfaaaguuggguL96	2626	asdCscddAadCuuaadTgUfgcuaguasasu	2744
AD-1466014.1	gsasgauggcAfUfUfuggcuucuguL96	2627	asdCsagdAadGccaadAuGfccaucucscsc	2745
AD-1466019.1	asasguuuccUfAfGfaguuaagacuL96	2628	asdTsgudCudAacudTaGfgaaacuusug	2746
AD-1466020.1	cscsuagaguUfAfGfacaauaaacuL96	2629	asdGsaudTudAugudTaAfcucuaggsasa	2747
AD-1466021.1	usascaaguaAfGfAfcaggauggauL96	2630	asdTscddAudCcugudCuUfacuuguasgsa	2748
AD-1466024.1	asgsgaccuuUfCfAfccucuaaguL96	2631	asdCsuidAgdAgggudGaAfagguccuscsg	2749
AD-1466026.1	usgscuucAuAfAfAfuuccaagaauL96	2632	asdTsucdAudTggudTuAfugaagcascsc	2750

TABLE 19-continued

Exemplary dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1466027.1	cscsaaugaaAfCfAfucucuuccuL96	2633	asdGsggdAadGagaudGuUfucauuggsasu	2751
AD-1466034.1	gsasccgaagUfCfAfcaaguccuuL96	2634	asdAsagdGadCuugudGaCfuucggucsasu	2752
AD-1466039.1	ususcuaccUfUfCfugaaucauguL96	2635	asdCsudGadTucagdAaGfgguagaasusa	2753
AD-1466040.1	csasuccuAfCfUfCfcaaugauL96	2636	asdAsudAudTgagadGuAfggagauasasa	2754
AD-1466043.1	ususuaauccAfCfUfCfuuuuaguL96	2637	asdAscudAudAaccadGuGfgauuaasusu	2755
AD-1466044.1	usuaauccaCfUfGfguuuaguL96	2638	asdCsacdTadTaaccdAgUfggauuaasasu	2756
AD-1466046.1	asusgguaCaGfAfUfcauugaguL96	2639	asdCsudAadTguadTcUfguaccuascu	2757
AD-1466047.1	ascsugauguUfAfGfgacaaauL96	2640	asdAsugdTudTguccdTaAfcuacagususu	2758
AD-1466050.1	usgsaagacuCfUfGfauauuuL96	2641	asdGsadTadTcaudAgAfgucucasasu	2759
AD-1466051.1	gsusaagaAfGfCfauucggaL96	2642	asdTsudCgdAgaugCuCfuucacuascu	2760
AD-1466052.1	asgsagcaucUfCfGfgaaauuL96	2643	asdCsadGadAuuccdGaGfaugcucusc	2761
AD-1466055.1	asasuucugGfUfCfcauuuauL96	2644	asdTsgadTadAuaggdAcCfaagaauscsc	2762
AD-1466058.1	usgsaaguggAfUfGfauuuauL96	2645	asdGsgadTadAcaudAuCfcacuucascg	2763
AD-1466059.1	gsasaguggaUfGfAfuguuauL96	2646	asdTsggdAudAcaudCaUfccacuucascg	2764
AD-1466060.1	asuscagaggGfAfAfagacuauL96	2647	asdCsaudAadGucuudTcCfcucugausgsa	2765
AD-1466062.1	asgsccaaAfGfCfaguauauL96	2648	asdGsgudAudAacugdCuAfuuugcuscgsa	2766
AD-1466063.1	asgsagauaUfAfCfcauacguL96	2649	asdCsaudAcdGuaggdTaUfaacucuscasu	2767
AD-1466070.1	gsasgagaauUfUfGfucuuauL96	2650	asdAsuadGudAagacdAaAfuucucuscasu	2768
AD-1466081.1	gsasaaagaaGfAfGfCfugguacuL96	2651	asdTsagdTadCcagcdTcUfucuuucscasu	2769
AD-1466083.1	asasagaagaGfCfUfCfuuuauL96	2652	asdCsaudAgdTaccadGcUfcuucuuusc	2770
AD-1466085.1	asgsaagagcUfGfGfuaucuaL96	2653	asdTsudAudAguacdCaGfcucucuscusu	2771
AD-1466089.1	gsasgcugguAfCfUfauaaaagL96	2654	asdTsudTudTcaudGuAfccagcuscusu	2772
AD-1466091.1	gscsugguacUfAfUfGfaaaagL96	2655	asdCsudCudTuucadTaGfuaccagcusc	2773
AD-1466095.1	ususucacgcCfAfUfuaaugggaL96	2656	asdAsudCcdAuuaadTgGfcgugaaascu	2774
AD-1466096.1	asusuaauggGfAfUfGfauuacaguL96	2657	asdCsugdTadGaucadTcCfcauuuagsg	2775
AD-1466097.1	gscsucccaaGfAfCfauucacguL96	2658	asdCsacdGudGaaugdTcUfugggagcscg	2776
AD-1466100.1	asusgcaaacGfCfCfauuucuuauL96	2659	asdAsuadAgdAaagdGcGfuuugcauscsc	2777
AD-1466101.1	gscsaaacgcCfAfUfucuuuauL96	2660	asdTsgadTadAgaadTgGfcguuugcscasu	2778
AD-1466104.1	usasagcacuGfGfUfaucauauL96	2661	asdAsgadTadTgauadCcAfgugcuasgsu	2779
AD-1466108.1	asasacaagGfUfGfauuauauL96	2662	asdTsaudAadGauccdAcCfauuuguasasa	2780
AD-1466109.1	csasauggugGfAfUfCfuuuaaguL96	2663	asdCsaudTadTaagadTcCfaccuugscusu	2781
AD-1466110.1	gsgsuggaucUfUfAfuuaugcuL96	2664	asdCsadGcdAuuaudAaGfauccaccscasu	2782
AD-1466111.1	asuscuaaAfUfGfCfuuuggagL96	2665	asdCsacdTcdCaagcdAuUfauaagauscsc	2783
AD-1466115.1	usasuaccacAfGfAfguucuauguL96	2666	asdAscadTadGaacudCuGfugguauasgs	2784
AD-1466116.1	usasccacagAfGfUfucuauguL96	2667	asdCsudCadTagaadCuCfugguauasusa	2785

TABLE 19-continued

Exemplary dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1466118.1	csascagaguUfCfUfaguagcuuuL96	2668	asdAsagdCudAcauadGaAfcucugugsgsu	2786
AD-1466120.1	asgsuucuauGfUfAfgcuuacaguL96	2669	asdAscudGudAagcudAcAfuagaacuscus	2787
AD-1466126.1	uscsaguuuGfCfCfaccuauuguL96	2670	asdCsaadTadGguggdGuCfaaacugasusu	2788
AD-1466127.1	csasguuugaCfCfCfaccuauuguL96	2671	asdAscadAudAggugdGgUfcaaacugsasu	2789
AD-1466128.1	csusauugugGfCfUfagauauuuL96	2672	asdAsaudAudAucudGcCfacaauagsgsu	2790
AD-1466131.1	asgscaaaucAfCfAfgcuucuucguL96	2673	asdCsgadAgdAagcudGuGfauuugcusug	2791
AD-1466133.1	asasauugauCfUfAfcucaagacuL96	2674	asdGsaudCudTgagudAgAfucaauuuscus	2792
AD-1466139.1	uscscaugguGfGfAfaagauuuuuL96	2675	asdAsaadAudCuugudCcAfccauggasgsg	2793
AD-1466140.1	cscsauggugGfAfaCfaagauuuuuL96	2676	asdAsaadAadTcuugdTcCfaccauggsasg	2794
AD-1466141.1	csasuggugGfAfaCfagauuuuuL96	2677	asdCsaadAadAucudGuCfcaccaugsgsa	2795
AD-1466142.1	asusgguggaCfAfaGfauuuuugauL96	2678	asdTscadAadAaucudTgUfccaccausgsg	2796
AD-1466143.1	usgsguggacAfAfaGfauuuuugaauL96	2679	asdTsucdAadAaaucdTgUfuccaccasusg	2797
AD-1466144.1	gsgsuggacaAfGfAfuuuuugaaguL96	2680	asdCsuudCadAaaudCuUfguccaccsasus	2798
AD-1466145.1	gsusggacaaGfAfaUfuuuugaagguL96	2681	asdCscudTcdAaaaadTcUfuguccacscsa	2799
AD-1466152.1	asgsauuuuuGfAfaGfgaaaauacuL96	2682	asdAsgudAudTuccudTcAfaaaaucusug	2800
AD-1466153.1	gsasuuuuugAfAfaGfgaaaauacuL96	2683	asdTsagdTadTuuccdTcCfaaaaaucsusu	2801
AD-1466157.1	ususugaaggAfAfaUfuacuauacuL96	2684	asdGsuadTudAguadTuCfcuucuaasasa	2802
AD-1466158.1	ususgaaggaAfAfaUfacuaauaccuL96	2685	asdGsgudAudTaguadTuUfccuucuaasasa	2803
AD-1466160.1	csusaaauaccAfAfaGfgacauuguguL96	2686	asdCsacdAudGuccudTuGfguauuagsusa	2804
AD-1466161.1	usasauaccaAfAfaGfgacauugauL96	2687	asdTscadCadTguccdTuUfgguauuasgsu	2805
AD-1466162.1	csasaucauuUfCfCfagguuuuacuL96	2688	asdGsaudAadAccugdGaAfaugauugsgsg	2806
AD-1466163.1	asuscauuucCfAfaGfguuuauccguL96	2689	asdCsggdAudAaaccdTgGfaaaugausug	2807

TABLE 20

Exemplary dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1395757.1	csasggu (Chd) cuCfAfCfuuuuauccsusa	2808	VPusdAsggdAudTaaagdTgdAgdGaccugscsg	2811
AD-1395758.1	csasggu (Chd) cuCfaCfUfuuuuauccsusa	2809	VPusdAsggdAudTaaagdTgdAgdGaccugscsg	2811
AD-1395763.1	csascuu (Uhd) aaUfCfCfucuauccasgsa	2810	VPusdCsugdGadTagagdGadTudAagugsasg	2812
AD-1395756.1	csasggu (Chd) cuCfAfCfuuuuauccsusa	2808	VPusdAsggdAudTaaagdTgAfggaccugscsg	2813
AD-1395762.1	csascuu (Uhd) aaUfCfCfucuauccasgsa	2810	VPusdCsugdGadTagagdGaUfuuaagugsasg	2814

TABLE 21

Exemplary dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1331279.1	asasuguucaCfAfAfuaagcuccuL96	2815	asdGsgadGcdTuaudTgUfgaacausususu	2839
AD-1331280.1	asusuugcuaUfGfUfuagacgauguL96	2816	asdCsaudCgdTcuaadCaUfagcaaauscsu	2840
AD-1331281.1	usugscuaugUfUfAfgacgauguL96	2817	usdTsaadAudCgucudAaCfauagcaasasu	2841
AD-1331282.1	usgscuauguUfAfGfagcauguL96	2818	usdTsuadCadTcgucdTaaCfauagcasasa	2842
AD-1331283.1	asascugagaAfGfAfacuacauaL96	2819	usdAsuadTgdTaguudCuUfcucaguuscsc	2843
AD-1331284.1	asasccaacaGfCfAfuaugcaaaL96	2820	usdAsuudTgdAcuaudGcUfguugguususa	2844
AD-1331285.1	cscscacagaAfAfUfuucucuacuL96	2821	asdGsaudAgdAgaaadTuUfcugugggsusu	2845
AD-1331286.1	csasgguaugCfCfAfuggacauuaL96	2822	usdTsaadTgdTccaudGgAfcauaccgsasu	2846
AD-1331287.1	gsgsuaguccAfUfGfgacauuaL96	2823	asdAsuudAadTguccdTaaGfgacuaccsusg	2847
AD-1331288.1	asgsuuggaaGfAfCfuggaagacaL96	2824	usdGsucdTudTccagdTcUfuccaacuscsa	2848
AD-1331289.1	usgsgaaagaCfAfAfcaaacauuaL96	2825	asdTsaadTgdTuugudTgUfcuuuccasgsu	2849
AD-1331290.1	ususuacuugGfGfAfaucacgaaaL96	2826	usdTsuadGudGauuudCcCfaaguaasasa	2850
AD-1331291.1	gsgsgaaauCfAfCfGfaaaccaacuL96	2827	usdAsuadTgdGuuudGuGfauuucccsasa	2851
AD-1331292.1	gsasaaucacGfAfAfaccaacuauaL96	2828	usdAsuadGudTggudTcGfugauuucscsc	2852
AD-1331293.1	csasaaaccaAfCfUfauacgcuacaL96	2829	usdGsuaadGcdGuuadGuUfgguuucgsusg	2853
AD-1331294.1	asuscaaccaAfAfAfugugaucacL96	2830	usdGsgadTcdAacaudTuUfgguugaususu	2854
AD-1331295.1	ususaaaacuUfUfAfaacuugacuL96	2831	usdAsuadCadAguuudTgAfguuuuaascsa	2855
AD-1331296.1	csasaaacuGfAfAfagccuccuauL96	2832	asdTsagdTgdGgcudTcAfaguuuugsasg	2856
AD-1331297.1	uscsaaacuGfAfAfuauggaauL96	2833	asdAsuadCadTcuaudTcGfauuggasasu	2857
AD-1331298.1	csasaaacuUfAfAfugaacguguL96	2834	asdCsadGudTcuaudTgAfaguuuugsusg	2858
AD-1331299.1	asasucacgaAfAfCfcaacuauacuL96	2835	asdGsuaadTadGuugdTgUfcugauuususc	2859
AD-1331300.1	gsgsgaaauCfAfUfuaugagguuL96	2836	asdAsuadAudCuaadAuUfgauuuccsasc	2860
AD-1331301.1	csasaaauUfGfAfuccauccaauL96	2837	asdTsugdTgdTggadCaAfcauuuugsusu	2861
AD-1331302.1	usgsgacauUfAfUfuaacacgaL96	2838	usdCsadTgdTugaadTuAfauguccasusg	2862
AD-1331328.1	asasuguucaCfAfAfuaagcuccuL96	2815	asdGsgadGcdTuaudTgdTgdAacausususu	2863
AD-1331329.1	asusuugcuaUfGfUfuagacgauguL96	2816	asdCsaudCgdTcuaadCadTadGcaaauscsu	2864
AD-1331330.1	usugscuaugUfUfAfgacgauguL96	2817	usdTsaadAudCgucudAadCadTagcaasasu	2865
AD-1331330.1	usgscuauguUfAfGfagcauguL96	2818	usdTsuadCadTcgucdTadAcdaugcasasa	2866
AD-1331331.1	asascugagaAfGfAfacuacauaL96	2819	usdAsuadTgdTaguudCudTcdTcaguuscsc	2867
AD-1331332.1	asasccaacaGfCfAfuaugcaaaL96	2820	usdAsuudTgdAcuaudGcdTgdTugguususa	2868
AD-1331333.1	cscscacagaAfAfUfuucucuacuL96	2821	asdGsaudAgdAgaaadTudTcdTgugggsusu	2869
AD-1331334.1	csasgguaugCfCfAfuggacauuaL96	2822	usdTsaadTgdTccaudGgdAcdTaccugsasu	2870
AD-1331311.1	gsgsuaguccAfUfGfgacauuaL96	2823	asdAsuudAadTguccdTaaGgdAcuaccsusg	2871
AD-1331335.1	asgsuuggaaGfAfCfuggaagacaL96	2824	usdGsucdTudTccagdTcdTudCcaacuscscsa	2872
AD-1331336.1	usgsgaaagaCfAfAfcaaacauuaL96	2825	asdTsaadTgdTuugudTgdTcdTuuccasgsu	2873
AD-1331314.1	ususuacuugGfGfAfaucacgaaaL96	2826	usdTsuadGudGauuudCcdCadAguaaasasa	2874

TABLE 21-continued

Exemplary dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1331337.1	gsgsgaaaacAfCfGfaaaccaacuL96	2827	usdAsgudTgdGuuucdGudGadTuucccsasa	2875
AD-1331316.1	gsasaaucacGfAfAfaccaacuauaL96	2828	usdAsuadGudTgguidTcdGudGauuucscsc	2876
AD-1331338.1	csgsaaaccaAfCfUfauacgcuacaL96	2829	usdGsuadGcdGuauadGudTgdGuuucgsusg	2877
AD-1331339.1	asuscaaccaAfAfAfuguugaucuaL96	2830	usdGsgadTcdAacaudTudTgdGuugaususu	2878
AD-1331340.1	ususaaaacuCfUfAfaacuugacuaL96	2831	usdAsgudCadAguuudTgdAgdTuuuaascsa	2879
AD-1331320.1	csasaaacuGfAfAfagccuccuauL96	2832	asdTsagdGadGccuudTcdAadGuuugsasg	2880
AD-1331341.1	uscsaacaucGfAfAfuaugauggauL96	2833	asdAsucdCadTcuadTcdGadTguugasasu	2881
AD-1331322.1	csasaaacuCfAfAfugaaacguguL96	2834	asdCsacdGudTucadTgdAadGuuugsusg	2882
AD-1331342.1	asasucacgaAfAfCfcaacuauacuL96	2835	asdGsuadTadGuuggdTudTcdGugauususc	2883
AD-1331343.1	gsgsgaaucaAfUfUfuugaugguuL96	2836	asdAsccdAudCuuaadAudTgdAuucccsasc	2884
AD-1331325.1	csasaaauguUfGfAfuccauccaauL96	2837	asdTsugdGadTggadCadAcdAuuuugsusu	2885
AD-1331344.1	usgsgacauuAfAfUfucaacaucaL96	2838	usdCsgadTgdTugaadTudAadTguccasusg	2886

TABLE 22

Exemplary dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	ISEQ ID NO:
AD-1554875	gscscugugaGfGfAfcuccaagaguL96	2887	asdCsucdTudGgagudCcUfcacaggcscsu	3013
AD-1554909	gsgsugcuacUfCfUfggauuuuccuL96	2888	asdGsgadAadTaccadGaGfuagcaccscsc	3014
AD-1554910	gsusgcuacuCfUfGfguauuuuccuL96	2889	asdAsggdAadAuaccdAgAfguagcacscsc	3015
AD-1554911	usgscuacucUfGfGfuauuuuccuauL96	2890	asdTsagdGadAauacdCaGfaguagcascsc	3016
AD-1554912	gscsuacucUfGfUfauuuuccuaguL96	2891	asdCsuadGgdAaauadCcAfgaguagcsasc	3017
AD-1554913	csusacucugGfUfAfuuuccuagguL96	2892	asdCsudAgdGaaudAcCfagaguagcsa	3018
AD-1554914	usascucuggUfAfUfuuccuagggguL96	2893	asdCsccdTadGgaaadTaCfcagaguasgsc	3019
AD-1554915	ascscucuggAfUfUfuccuagggguL96	2894	asdAsccdCudAggaadAuAfccagagusasg	3020
AD-1554916	csuscugguaUfUfUfccuagggguauL96	2895	asdTsacdCcdTaggadAaUfaccagagsusa	3021
AD-1554917	uscsugguauUfUfCfcuagggguacuL96	2896	asdGsuadCcdCuaggdAaAfuaccagagsgu	3022
AD-1554923	asusuuccaGfGfGfuacaaggcguL96	2897	asdCsgcdCudTguacdCcUfaggaaausasc	3023
AD-1554951	gsgsucagccAfGfGfuguacucaguL96	2898	asdCsugdAgdTacacdCuGfgcugaccsas	3024
AD-1554955	asgscagggaGfUfAfcucaggcaguL96	2899	asdCsugdCcdTgagudAcAfcucuggcusgsa	3025
AD-1554992	gscscacuucUfCfCfcaggauuuuL96	2900	asdAsagdAudCcuggdGaGfaagggcsgsa	3026
AD-1554997	ususcuccaGfGfAfucuuaccguL96	2901	asdCsggdGudAagaudCcUfgggagaasgsu	3027
AD-1555000	uscsccagggaUfCfUfuaccggccguL96	2902	asdCsggdCgdGguaadGaUfccugggasgsa	3028
AD-1555030	gscscuuccGfAfGfugaaacgcuL96	2903	asdGscgdGudTucadTgCfggaggcsasc	3029
AD-1555106	csasacuccaGfCfUfccgucuauuuL96	2904	asdAsaudAgdAcggadGcUfggaguugsusa	3030

TABLE 22-continued

Exemplary dsRNA molecules					
Duplex ID	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	ISEQ ID NO:	
AD-1555112	csasgcuccgUfCfUfauuccuuuguL96	2905	asdCsaadAgdGaauadGaCfaggagcugsgsa	3031	
AD-1555114	csuscaccugCfUfUfucucugguuuL96	2906	asdAsacdCadGaagadAgCfaggugagsgsg	3032	
AD-1555115	uscsaccugcUfUfCfuucugguucuL96	2907	asdGsaadCcdAgaagdaAGfcaggugagsgsg	3033	
AD-1555117	ascscugcuuUfUfUfcugguucauuL96	2908	asdAsugdAadCcagadAgAfcagcaggusgsa	3034	
AD-1555118	cscsugcuuUfUfCfugguucauuL96	2909	asdAsaudGadAccagdAaGfaagcaggusug	3035	
AD-1555120	usgscuucuuCfUfGfguucuuuuL96	2910	asdAsgadAudGaaccdAgAfaagcagsgsg	3036	
AD-1555121	gscsuucuuUfGfGfguucuuuL96	2911	asdGsagdAadTgaacdCaGfaagaagcsasg	3037	
AD-1555122	csusucuuUfGfGfUfucuuuccuL96	2912	asdGsgadGadAugaadCcAfgaagaagcsa	3038	
AD-1555123	ususcucugGfUfUfcauuccauL96	2913	asdTsggdAgdAaugadAcCfagaagaasgsc	3039	
AD-1555128	csusggguuUfUfCfuccaaucuuL96	2914	asdGsgadTudTggagdAaUfgaaccagsasa	3040	
AD-1555184	ascsagggccGfAfGfGfuacgauguL96	2915	asdCsacdTudCguacdTcGfgccugusasg	3041	
AD-1555185	csasggggccGfAfGfUfacaaguguL96	2916	asdCscadCudTcguadCuCfggccugusasa	3042	
AD-1555212	cscsagugugAfAfAfGfacaugcuL96	2917	asdAsgcdTadTgucudTuCfacacugsgcsu	3043	
AD-1555213	csasgugugaAfAfGfacaugcuguL96	2918	asdCsagdCudAugucdTuUfcacacugsgsc	3044	
AD-1555234	asusugaauUfCfCfGfcgugguuuL96	2919	asdAsacdCcdAgcgudGgAfaaucaausgsc	3045	
AD-1555235	ususgaauUfCfCfGfcgugguuuL96	2920	asdCsaadCcdCagcgdTgGfaaucaausug	3046	
AD-1555236	usgsaaucUfCfCfGfcgugguuuL96	2921	asdAscadAcCdCagcdGuGfgaauucasasu	3047	
AD-1555238	asasuuccacGfCfUfGfgguuguuuL96	2922	asdTsaadCadAcccadGcGfuggaauuscsa	3048	
AD-1555241	uscscacgcuGfGfGfuguuaccguL96	2923	asdCsaggdTadAcaacdCcAfgcguggasasu	3049	
AD-1555242	cscsacgcuGfGfUfuguuaccguL96	2924	asdGscgdGudAacaadCcCfagcguggsasa	3050	
AD-1555243	csascgcuGfUfUfGfuuaccguL96	2925	asdAsgcdGgdTaacadAcCfcagcgugsgsa	3051	
AD-1555247	csusggguUfUfAfccgcuacaguL96	2926	asdCsugdTadGcgudAaCfaaccagsgsg	3052	
AD-1555342	gsgsgaccgaCfUfGfGfcaugauuL96	2927	asdAsuadCadTggcdAgUfcgguccsgsg	3053	
AD-1555343	gsgsaccgacUfGfGfcaugauuL96	2928	asdCsaudAcdauggcdCaGfucgguccsgsg	3054	
AD-1555345	ascscgacugGfCfCfauaugacuL96	2929	asdGsucdAudAcaugdGcCfagucgguscsc	3055	
AD-1555346	cscsgacugCfCfAfugauagacuL96	2930	asdCsgudCadTacaudGgCfcagucggusc	3056	
AD-1555348	gsascuggccAfUfGfuaugacguL96	2931	asdCsacdGudCauacdAuGfgccagucsgsg	3057	
AD-1555349	ascsuggccaUfGfUfauagcugguL96	2932	asdCscadCgdTcauadCaUfggcccaguscsg	3058	
AD-1555350	csusggccaGfUfAfugacgugguL96	2933	asdGscddAcGducaudAcAfuggccagsusc	3059	
AD-1555366	asgsgcucaUfAfCfCfcugguuL96	2934	asdTsacdAcCdGaggdTgAfugagccuscsc	3060	
AD-1555428	gscscugcacAfGfCfuacuacgacuL96	2935	asdGsucdGudAgaugdCuGfugcaggcscsc	3061	
AD-1555429	cscsugcacaGfCfUfacuacgacuL96	2936	asdGsgudCgdTaguadGcUfugcaggscsc	3062	
AD-1555535	cscsucucugGfAfCfCfuacggcuuL96	2937	asdCsaadGcdCguagdTcCfagagaggsgsc	3063	
AD-1555537	uscscuggaCfUfAfCfcgguugguL96	2938	asdGsccdAadGccgudAgUfcagagagsgsg	3064	
AD-1555546	usascggcuGfGfCfcucugguuuL96	2939	asdAsacdCadGagggdCcAfcagccguasgsu	3065	

TABLE 22-continued

Exemplary dsRNA molecules					
Duplex ID	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:	ISEQ ID NO:
AD-1555547	ascsggcucugGfCfCfcucuguuuuL96	2940	asdAaadCcdAgaggdGcCfaagccgusag	3066	
AD-1555548	csgsgcuuggCfCfCfcucugguuuL96	2941	asdCsaadAcdCagagdGgCfcaagccgsusa	3067	
AD-1555549	gsgscuuggCfCfUfcugguuuL96	2942	asdTscadAadCagagdGgGfccaagccgsu	3068	
AD-1555581	gsasggaggCfGfAfagauuuL96	2943	asdAaadCadTacuudCuGfccucccsag	3069	
AD-1555583	gsgsaggcagAfAfGfauuuL96	2944	asdCsaadAudCauacdTuCfugccuccsusc	3070	
AD-1555584	gsasggcagaAfGfUfauuuL96	2945	asdGscadAadTcaudCuUfcugccucscsu	3071	
AD-1555585	asgsgcagaaGfUfAfauuuL96	2946	asdGsgcdAadAucaudAcUfucugccuscsc	3072	
AD-1555586	gsgscagaagUfAfUfauuuL96	2947	asdCsggdCadAaucadTaCfuucgcccusc	3073	
AD-1555587	gscsagaagUfUfGfauuuL96	2948	asdAscgdGcdAaaudAuAfcuucgcsusc	3074	
AD-1555588	csasgaagUfGfAfauuuL96	2949	asdCsacdGgdCaaudCaUfacuucgscsc	3075	
AD-1555589	asgsaagauGfAfUfugccgugL96	2950	asdGscadCgdGcaadTcAfuacuucgsc	3076	
AD-1555590	gsasagauGfUfUfugccgugL96	2951	asdTsgcdAcdGgcaudAuCfauacuucsg	3077	
AD-1555615	csasgugagGfAfUfcccagacL96	2952	asdCsugdTudCuggadTcGfuccacugsgsc	3078	
AD-1555616	asgsugagGfUfCfcccagacL96	2953	asdCscudGudTcuggdAuCfuccacugsgsc	3079	
AD-1555626	cscsagaacGfGfAfggcugugL96	2954	asdCsacdAcdAgccudCcUfguucggsasu	3080	
AD-1555628	asgsaacaggAfGfGfcugugL96	2955	asdGscddAcdAcagcdCuCfcugucgsgsc	3081	
AD-1555706	usgsugcggGfUfGfCfauagccuL96	2956	asdAsgcdCadTagugdCaCfccgcacascsc	3082	
AD-1555707	gsusgcggGfCfAfcauagccuL96	2957	asdAsagdCcdAuagudGcAfcgccacascsc	3083	
AD-1555709	gscsgggGfCfUfauagccuL96	2958	asdAscadAgdCcaudGuGfcaaccgcsasc	3084	
AD-1555711	gsgsgugcGfAfUfGgcuuL96	2959	asdGsuadCadAgccadTaGfugcaccgscsc	3085	
AD-1555717	ascsuagGfUfGfuaaaccagL96	2960	asdCsugdGudTguacdAaGfccauagsgsc	3086	
AD-1555723	gscsuuagGfAfCfCfagcggL96	2961	asdGsuadCgdAcuggdTuGfuaaagcscsa	3087	
AD-1555725	csusgcccGfAfGfaguuccuL96	2962	asdAsgadGgdAacudTcCfagggcagsgsc	3088	
AD-1555768	gscscuggGfAfGfagaacugL96	2963	asdGscadGudTucudTcAfuccagcscsc	3089	
AD-1555771	usgsaugagGfGfAfaacugcuL96	2964	asdAsacdGcdAguudCuCfucauccagsgsc	3090	
AD-1555772	gsgsaugagGfAfAfacugcuL96	2965	asdAaadCgdCaguudTcUfcuauccsag	3091	
AD-1555776	gsasgagaaCfUfGfGcuuL96	2966	asdCsugdCadAacgdAgUfuucucscsu	3092	
AD-1555789	ususugcagGfCfCfcauuccagL96	2967	asdCsugdGadAugudGcUfcugcaaacscg	3093	
AD-1555894	gsusgggacUfUfCfaccuuccagL96	2968	asdCsugdGadAggudAaUfguccacscsu	3094	
AD-1555895	usgsaggacUfCfAfccuuccagL96	2969	asdAscudGgdAagudGaAfuguccacscsa	3095	
AD-1555897	gsgsacauGfCfCfauuccagL96	2970	asdAscadCudGgaagdGuGfaauggcscsa	3096	
AD-1555898	gsascauGfCfUfuccagugL96	2971	asdCsacdAcdTggaadGgUfgaagucscsc	3097	
AD-1555899	ascsaucacCfUfUfCfagugL96	2972	asdTscadCadCuggadAgGfugaagucscsc	3098	
AD-1555900	csasuucacUfUfCfCfagugL96	2973	asdCsucdAcdAcuggdAaGfgugaugsusc	3099	
AD-1556052	asuscgcGfCfGfCfugugL96	2974	asdAsucdAcdCagcdGgUfcagcgausga	3100	
AD-1556057	usgsaccgGfGfGfGfauaacagL96	2975	asdCsugdTudAucadCcAfcggucagsgsc	3101	

TABLE 22--continued

Exemplary dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	ISEQ ID NO:
AD-1556126	csgsuguuccUfGfGfGcaagguguuL96	2976	asdAscadCcdTugccdCaGfgaacacsgsgu	3102
AD-1556127	gsusguuccuGfGfGfcaagguguguL96	2977	asdCsacdAcdCuugcdCcAfggaaacacsgsg	3103
AD-1556137	gscsaaggugUfGfGfcagaacucguL96	2978	asdCsgadGudTcugcdCaCfaccuugcscsc	3104
AD-1556139	asasggugugGfCfAfgaacucgcguL96	2979	asdCsgcdGadGuucudGcCfacaccuusgsc	3105
AD-1556163	csusggagagGfUfGfuccuuaaguL96	2980	asdCsuudGadAggacdAcCfucuccagsgsc	3106
AD-1556164	usgsgagaggUfGfUfccuuaagguL96	2981	asdCscudTgdAaggadCaCfcuccacsgsg	3107
AD-1556166	gsasgaggugUfCfCfuuaagguguL96	2982	asdCsacdCudTgaagdGaCfaccucucscsa	3108
AD-1556167	asgsagguguCfCfUfucaaggugauL96	2983	asdTscadCcdTugaadGgAfcaccucuscsc	3109
AD-1556319	asuscccacaGfGfAfccugugcaguL96	2984	asdCsgudCadCaggudCcUfgugggauscsa	3110
AD-1556359	usgsacgcccaCfGfCfauuguguguL96	2985	asdAscadCadGcaugdCgUfggcgucascsc	3111
AD-1556360	gsascgccacGfCfAfugcuguguguL96	2986	asdCsacdAcdAgaudGcGfuggcgucsasc	3112
AD-1556382	gscsuaccgcAfAfGfggcaagaaguL96	2987	asdCsuudCudTgccccUfGfcgguagcscsg	3113
AD-1556383	csusaccgcaAfGfGfGcaagaagguL96	2988	asdCscudTcdTugccdCuUfgcgguagcscsc	3114
AD-1556465	gsgsccuacUfAfCfuucggcgucuL96	2989	asdGsacdGcdCgaagdTaGfuuaggccsgsg	3115
AD-1556466	gscscuaacuAfCfUfucggcgucuuL96	2990	asdAsgadCgdCgaadCuAfguuaggcscsg	3116
AD-1556484	csusacaccccGfCfAfucacagguguL96	2991	asdCsacdCudGugaudGcGfgguguagcsasc	3117
AD-1556510	gscsuggaucCfAfGfcaagguguguL96	2992	asdCsacdCadCuugcdTgGfauccagcsusg	3118
AD-1556584	usgsgcaggaGfGfUfggcaucuuguL96	2993	asdCsaadGadTgccadCcUfccugccascsc	3119
AD-1556585	gsgscaggagGfUfGfgcaucuuguuL96	2994	asdAscadAgdAugccdAcCfuccugccsasc	3120
AD-1556586	gscsaggaggUfGfGfcaucuugucuL96	2995	asdGsacdAadGaugcdCaCfuccugcscsa	3121
AD-1556587	csasggagguGfGfCfauucuugucuL96	2996	asdAsgadCadAgaugdCcAfccuccugscsc	3122
AD-1556613	usgsaugucuGfCfUfccagugauguL96	2997	asdCsaudCadCuggadGcAfagacaucsgsg	3123
AD-1556677	csasaauucUfCfUfccuccguccuL96	2998	asdGsgadCgdGaggadGaGfagaauugsgsg	3124
AD-1556709	gsgscucagcAfGfCfaagaauvcuuL96	2999	asdAsgcdAudTcuugdCuGfcugagccsasc	3125
AD-1556710	gscsucagcaGfCfAfagaauvcuguL96	3000	asdCsagdCadTucuudGcUfgcugagcscsa	3126
AD-1556789	csusggucuaAfCfUfuggaucuguL96	3001	asdCsagdAudCcaadGuUfagaccagsgsg	3127
AD-1556790	usgsgucuaaCfUfUfgggaucugguL96	3002	asdCscadGadTcccadAgUfuagaccasgsg	3128
AD-1556791	gsgsucuaacUfUfGfggaucuggguL96	3003	asdCsccdAgdAucccdAaGfuuagaccsasg	3129
AD-1556795	usasacuuggGfAfUfcugggaauguL96	3004	asdCsaudTcdCcgadTcCfcaaguuasgsa	3130
AD-1556799	ususgggaucUfGfGfgaauggaaguL96	3005	asdCsuudCcdAuuccdCaGfaucccaasgsu	3131
AD-1556802	gsgsaucuggGfAfAfuggaagguguL96	3006	asdCsacdCudTccaudTcCfcagaucscsa	3132
AD-1556908	usgsagcucaGfCfUfgcccuuugguL96	3007	asdCscadAadGggcadGcUfgagcucascsc	3133
AD-1556909	gsasgcucagCfUfGfccccuuggauL96	3008	asdTsccdAadAgggcdAgCfugagcucsasc	3134
AD-1556911	gscsucagcuGfCfCfcuuuggaauuL96	3009	asdAsuudCcdAaaggdGcAfgcugagcsusc	3135
AD-1556915	asgscugcccUfUfUfggaauaaaguL96	3010	asdCsuudTadTuccadAaGfggcagcusgsa	3136

TABLE 22-continued

Exemplary dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID	Antisense sequence (5'→3')	ISEQ ID
		NO:		NO:
AD-1556917	csusgccccuuUfGfGfaauaaagcuuL96	3011	asdAsgcdTudTauucdCaAfagggcagscsu	3137
AD-1556918	usgsgccuuUfGfGfaauaaagcuguL96	3012	asdCsagdCudTuauudCcAfaagggcasgsc	3138

TABLE 23

Exemplary dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID	Antisense sequence (5'→3')	SEQ ID
		NO:		NO:
AD-1181513.1	ascsagacAfaGfAfCfcaucuaacauL96	3139	asdGsugdTadGauggucUfudGucugusgsc	3153
AD-1181527.1	cscsgagcCfGfUfCfucuaaauuL96	128	asdAsaudTgdTagagaaCfGdGcucggsasu	3154
AD-1181564.1	gsasugccAfaGfAfCfcaucaugauuL96	3140	asdAsucdAudAguguucUfudGgcaucscsu	3155
AD-1181612.1	ascsagagdAaaUfUfCfcaucauauL96	3141	asdAsugdTadGuagaaUfuCfucugusgsg	3156
AD-1181618.1	gsasaaUfCfUfCfcaucauauL96	3142	asUfsaudAgdAuguadGudAgdAauuucsusc	3157
AD-1181537.1	csgsagccGfUfCfCfcauauauL96	3143	asUfsaadTudGuagagadAcGgucgsgsc	3158
AD-1181552.1	csasagccUfuGfGfCfcauauccauL96	3144	asUfsggdTadTugagccAdGgcuugsgsc	3159
AD-1181601.1	asgsaaaUfCfUfCfcaucauauL96	84	asAfsuadGadTguagdTadGaAfuuucscsu	3160
AD-1181614.1	gsasgaaaUfuCfUfCfcaucauauL96	3145	asUfsagdAudGuagudAgdAaUfuucuscusg	3161
AD-1181524.1	asgsauccdGagCfCfcaucaugaauL96	3146	asUfsucdAudAguaggcUfcdGgaucususc	3162
AD-1181528.1	cscsgagcCfGfUfCfucuaaauuL96	128	asAfsaudTgdTagagaaCfGdGcucggsasu	3163
AD-1181529.1	cscsgagcCfGfUfCfucuaaauuL96	128	asAfsaudTgdTagagaaCfGdGcucgsgsc	3164
AD-1181598.1	asgsagaaAfuUfCfCfcauauauL96	3147	asAfsagdTadTaguagaAfuUfucuscgsu	3165
AD-1181604.1	csasagguCfuUfCfCfucuggcuguL96	3148	asCfsagdCcdAgagagaAfgAfccuugsgsc	3166
AD-1181609.1	csucuacGagAfaAfaucuaauL96	3149	asUfsagdTadGaauuucUfCfGuaggscsu	3167
AD-1181611.1	ascsagagAfaAfuUfCfcaucauauL96	3150	asAfsugdTadGuagaaUfuCfucugusgsg	3168
AD-1181617.1	gsasaaUfCfAfuUfCfcaucauauL96	3151	asUfsaudAgdAuguagaAfgAfaauucsusc	3169
AD-1181619.1	ascsccuacfuCfUfCfGuuguucgaauL96	3152	asUfsucdGadAcaacagAfgUfagggusgsg	3170

TABLE 23-continued

Exemplary dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID	Antisense sequence (5'→3')	SEQ ID
		NO:		NO:
AD-1181604.2	csasagguCfuUfCfUfcucuggcuguL96	3148	asCfsagdCcdAgagagaAfgAfccuugsgsc	3166

TABLE 24

Exemplary dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID	Antisense sequence (5'→3')	SEQ ID
		NO:		NO:
AD-1143243.1	csusuaaaAfgGfGfAfcagauuucuaL96	3171	usdAsgadAudAcuguccCfuUfuaagscsc	3179
AD-1143252.1	gscsuuaaAfaGfGfGfacagauuucL96	3172	asdGsaadTadCuguccCfuUfuaagcsasa	3180
AD-1143254.1	gscsuuaaAfaGfGfGfacagauuucL96	3172	asdGsaadTadCuguccCfuUfuaagcsgsc	3181
AD-1143278.1	csusuaaaAfgGfGfAfcagauuucL96	3173	asdAsgadAudAcuguccCfuUfuaagscsc	3182
AD-1143245.1	usasaaAfgGfGfAfcagauuucuaL96	3174	usdAsgadAudAcuguccCfuUfuuasasg	3183
AD-1143256.1	ususaaAfaGfGfGfacagauuucL96	3175	asdGsaadTadCuguccCfuUfuaagsgsc	3184
AD-1143280.1	usasaaAfgGfGfAfcagauuucL96	3176	asdAsgadAudAcuguccCfuUfuuasasg	3185
AD-1143286.1	asasaaggGfaCfAfGfuauucucaguL96	3177	asCfsugdAgdAauacugUfcCfcuuuusasa	3186
AD-1143287.1	asasaaggGfaCfAfGfuauucucaguL96	3177	asCfsugdAgdAauacugUfcCfcuuuugsgc	3187
AD-1143288.1	asasaaggGfaCfAfGfuauucucaguL96	3177	asCfsugdAgdAauacugUfcCfcuuuuscs	3188
AD-1143289.1	asasggGfaCfAfGfuauucucaguL96	3178	asCfsugdAgdAauacugUfcCfcuususu	3189

TABLE 25

Exemplary dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID	Antisense sequence (5'→3')	SEQ ID
		NO:		NO:
AD-1231468	csusuc (Uhd) gGfuCfUfAfaacaaacusasa	3190	VPusUfsagdTudTguuuagAfcCfagaagsasu	3212
AD-1231470	ususaau (Chd) UfcAfCfAfuaguaaucsusa	3191	VPusAfsagdTudAcuauguGfaGfauuaasasg	3213
AD-1231471	usasacu (Uhd) UfaAfUfCfucauauagscsa	3192	VPusGfscudAudAugagauUfaAfaguuasasc	3214
AD-1231472	cscsuga (Uhd) CfuUfCfUfgguuuuaascsa	3193	VPusGfsuudTadAaccagaAfgAfucaggsasa	3215

TABLE 25-continued

Exemplary dsRNA molecules				
Duplex ID	Sense sequence (5'→3')	SEQ ID NO:	Antisense sequence (5'→3')	SEQ ID NO:
AD-1231473	usgsua (Chd) aGfuAfAfGfugauaacasgsa	3194	VPusCfsugdTudAucacuuAfcUfguacasaasg	3216
AD-1231478	ususcgugGfuAfUfUfcuua (Chd) uagsusa	3195	VPusAfs cudAgdTaagaauAfcCfcagcaasasg	3217
AD-1231479	uscsuuc (Uhd) GfgUfCfUfaaaugaacsusa	3196	VPusAfsugdTcdAuuuagaCfcAfgaagasusc	3218
AD-1231485	uscsugug (Chd) UfaCfAfAfcuuucucasasa	3197	VPusUfsugdAgdAaguugUfaGfcacgasusu	3219
AD-1231492	gsusugugUfaCfAfCfacau (Uhd) ggusasa	3198	VPusUfsacdCadAugugugUfaCfacaacsasu	3220
AD-1231496	csgsugc (Uhd) AfcAfAfCfuucuucaasgsa	3199	VPusCfsuudGadAgaaguUGfuAfgcacgsasu	3221
AD-1231497	usgsaaa (Uhd) UfgUfUfGfacauugugsasa	3200	VPusUfscadCadAugucaaCfaAfuucasgsc	3222
AD-1231498	ascsaaug (Uhd) uGfCfUfuuuuaaacsusa	3201	VPusAfsugdTudAaaaagCfaCfauugususa	3223
AD-1231499	asasaugUfuGfAfCfacua (Uhd) gagsusa	3202	VPusAfs cudCadTagugucAfaCfaauuuscsc	3224
AD-1231502	gscsugaaAfuUfGfUfugaua (Chd) ugsusa	3203	VPusAfs cadGudAucaacaAfuUfucagcsasg	3225
AD-1231505	usgsuugaCfaCfUfGfugaa (Uhd) gcususa	3204	VPusAfsagdCadTucacagUfgUfcaacasasu	3226
AD-1231507	ascsauugCfcAfAfAfggu (Uhd) ucsasa	3205	VPusUfsagdAadAccuuuUGfgCfaaugusug	3227
AD-1231513	ususcuug (Chd) UfaAfUfCfuuguugcusasa	3206	VPusUfsagdCadAcaagauUfaGfcagaasgsc	3228
AD-1231519	asuscuugUfuUfUfCfucua (Uhd) ucsasa	3207	VPusUfsugdAadTagagaaAfaCfaagaugsgsa	3229
AD-1231520	csasgucaUfaAfUfCfuua (Uhd) uaasasa	3208	VPusUfsuudAadTauagauUfaUfgacugsug	3230
AD-1231521	usgsguaaCfaCfUfAfauaa (Uhd) aaasasa	3209	VPusUfsuudTadTuauuagUfgUfuaccascsc	3231
AD-1523180.1	cscsucauGfgAfAfGfagaagcagauL96	3210	asUfscudGcdTucucuuCfcAfugaggscsu	3232
AD-1498528	cscsggu (Chd) AfgGfUfUfcugcuuuusasa	3211	VPusUfsaadAadGcagaacCfudGaccggscsc	3233

[0446] All of the U.S. patents, U.S. patent application publications, foreign patents, foreign patent applications and non-patent publications referred to in this specification are incorporated herein by reference, in their entirety. Aspects of the embodiments can be modified, if necessary to employ concepts of the various patents, applications and publications to provide yet further embodiments.

[0447] These and other changes can be made to the embodiments in light of the above-detailed description. In

general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

SEQUENCE LISTING

The patent application contains a lengthy sequence listing. A copy of the sequence listing is available in electronic form from the USPTO web site (<https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US20250283071A1>). An electronic copy of the sequence listing will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

1. A dsRNA agent comprising a sense strand and antisense, each strand independently having a length of 15-35 nucleotides wherein each nucleotide is independently modified or unmodified, wherein

the sense strand comprises a 2'-fluoro nucleotide at position 10, counting from 5'-end of the sense strand, and the antisense strand comprises a 2'-deoxy nucleotide at positions 2, 5, 7 and 12, counting from 5'-end of the antisense strand, and

wherein:

- (i) the antisense strand comprises a 2'-fluoro nucleotide at position 14 and a nucleotide other than a 2'-deoxy or 2'-fluoro nucleotide at position 16, counting from the 5'-end of the antisense strand; or
- (ii) the antisense strand comprises a 2'-deoxy nucleotide at position 14 or 16, counting from the 5'-end of the antisense strand, and the sense strand comprises a nucleotide other than a 2'-fluoro nucleotide at position 7, counting from the 5'-end of the sense strand.

2. The dsRNA agent of claim 1, wherein the sense strand further comprises a 2'-fluoro nucleotide at position 9 or 11, counting from 5'-end of the sense strand.

3. (canceled)

4. The dsRNA agent of claim 1, wherein the sense strand further comprises a 2'-fluoro nucleotide at positions 9 and 11, counting from 5'-end of the sense strand.

5. The dsRNA agent of claim 1, wherein the sense strand comprises a 2'-fluoro nucleotide at positions 8 and 9 or positions 11 and 12, counting from 5'-end of the sense strand.

6. (canceled)

7. The dsRNA agent of claim 1, wherein the sense strand comprises a nucleotide other than a 2'-fluoro at position 7, counting from the 5'-end of the sense strand.

8. The dsRNA agent of claim 1, wherein the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, and a nucleotide other than a 2'-fluoro at position 7, counting from the 5'-end of the sense strand.

9. The dsRNA agent of claim 1, wherein the sense strand comprises at least one 2'-OMe nucleotide.

10. The dsRNA agent of claim 1, wherein the sense strand comprises a 2'-OMe nucleotide at position 7, counting from the 5'-end of the sense strand.

11. The dsRNA agent of claim 1, wherein the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, and a 2'-OMe nucleotide at position 7, counting from the 5'-end of the sense strand.

12. (canceled)

13. (canceled)

14. The dsRNA agent of claim 1, wherein the antisense strand comprises a 2'-OMe nucleotide at position 16, counting from the 5'-end of the antisense strand.

15. (canceled)

16. (canceled)

17. (canceled)

18. (canceled)

19. (canceled)

20. (canceled)

21. (canceled)

22. (canceled)

23. (canceled)

24. (canceled)

25. (canceled)

26. The dsRNA agent of claim 1, wherein the dsRNA agent comprises a ligand.

27. The dsRNA agent of claim 1, wherein the sense strand comprises a ligand.

28. The dsRNA agent of claim 27, wherein the ligand is at 3'-end of the sense strand.

29. The dsRNA agent of claim 27, wherein the ligand is at 5'-end of the sense strand.

30. The dsRNA agent of claim 27, wherein the ligand comprises an ASGPR ligand.

31. The dsRNA agent of claim 27, wherein the ligand is lipophilic group.

32. The dsRNA agent of claim 31, wherein the ligand is a C₁₀₋₃₀aliphatic group.

33. The dsRNA agent of claim 32, wherein the C₁₀₋₃₀aliphatic group is a C₁₀₋₃₀alkyl group.

34. The dsRNA agent of claim 33, wherein the C₁₀₋₃₀alkyl group is a straight-chain or branched tetradecyl, hexadecyl, octadecyl, icosyl, docosyl, or tetracosyl group.

35. The dsRNA agent of claim 23, wherein the ligand is conjugated to a non-terminal nucleotide of the sense strand.

36. The dsRNA agent of claim 35, wherein the ligand is conjugated to the 2'-position of a non-terminal nucleotide of the sense strand, optionally conjugated to one of positions 5, 6, 7, or 8 of the sense strand, counting from the 5' end).

37. The dsRNA agent of claim 26, wherein the ligand comprises an abasic nucleotide, optionally the abasic nucleotide is an inverted nucleotide and linked via a 5'->5' or a 3'->3' linkage to a strand of the dsRNA agent.

38. (canceled)

39. (canceled)

40. (canceled)

41. (canceled)

42. (canceled)

43. (canceled)

44. The dsRNA agent of claim 1, wherein the sense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 5' end of the sense strand.

45. The dsRNA agent of claim 1, wherein the antisense strand comprises at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 5' end of the antisense strand and at least two phosphorothioate internucleotide linkages between the first five nucleotides counting from the 3' end of the antisense strand.

46. The dsRNA agent of claim 1, wherein the dsRNA has a duplex region of from 18 to about 25 basepairs.

47. (canceled)

48. (canceled)

49. A dsRNA agent comprising a sense strand and an antisense strand,

wherein the sense strand is 18-23 nucleotides in length and comprises a 2'-fluoro nucleotide at position 10, counting from 5'-end of the sense strand and a 2'-fluoro nucleotide at position 9 or 11, counting from 5'-end of the sense strand, and the antisense strand is 18-25 nucleotide in length and comprises a 2'-deoxy nucleotide at positions 2, 5, 7, and 12, counting from 5'-end of the antisense strand,

wherein:

the antisense strand comprises a 2'-fluoro nucleotide at position 14 and a nucleotide other than a 2'-deoxy or 2'-fluoro nucleotide at position 16, counting from the 5'-end of the antisense strand.

50. The dsRNA agent of claim 49, wherein the sense strand comprises a 2'-fluoro nucleotide at positions 9, 10 and 11, counting from 5'-end of the sense strand.

51. The dsRNA agent of claim 1, comprising a phosphate mimic at the 5'-end of the antisense strand.

52. The dsRNA agent of claim 51, wherein the phosphate mimic is a 5'-E-vinyl phosphonate.

53. (canceled)

54. The dsRNA agent of claim 49, wherein remaining nucleotides in the sense strand are unmodified nucleotides or modified nucleotides, optionally selected from the groups consisting of 2'-OMe, 2'-F, 2'-H, and an 2'-O—C₁₀₋₃₀aliphatic group, provided no more than one modified nucleotide is an 2'-O—C₁₀₋₃₀aliphatic group; and the remaining nucleotides in the antisense strand are unmodified nucleotides or modified nucleotides, optionally selected from the group consisting of 2'-OMe, 2'-F, 2'-H, GNA and 3'-RNA, the 3'-RNA being optionally 3'-OH, provided no more than one modified nucleotide is GNA or 3'-RNA.

55. The dsRNA agent of claim 54, wherein remaining nucleotides (i.e., nucleotides at positions not otherwise defined) in the sense strand are modified nucleotides selected from the group consisting of 2'-OMe, 2'-F, 2'-H, and an 2'-O—C₁₀₋₃₀aliphatic group, provided no more than one modified nucleotide is an 2'-O—C₁₀₋₃₀aliphatic group; and the remaining nucleotides in the antisense strand are modified nucleotides selected from the group consisting of 2'-OMe, 2'-F, 2'-H, GNA, and 3'-RNA, the 3'-RNA being optionally 3'-OH, provided no more than one modified nucleotide is GNA or 3'-RNA.

56. (canceled)

57. (canceled)

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